



Valmet DNA Operate DNA Operate Manual

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1. Desktop User Manual

1.1 Introduction

This user's guide is intended for all users of the DNA Operate user interface. The first part of the guide covers the main components of the desktop. The latter part concentrates on the desktop operations. The structure of the user's guide is as follows:

- Chapter *Introduction* presents a general outline of the user's guide structure and the terms used.
- Chapter *Basics of the DNA Operate User Interface* provides an introduction into the function and features of the user interface.
- Chapter *Desktop Structure* presents an overview of the central components of the desktop.
- Chapter *Controlling the User Interface* gives basic information about the use of a keyboard and a mouse in the user interface.
- Chapter *Control Panel* covers the contents and use of the DNA Operate control panel.
- Chapter *Pictures and Picture Windows* describes the use of pictures and picture windows.
- Chapter *Operations* covers the most important operations of the desktop.

Terms Used in the User's Guide:

Active picture window is the picture window to which user selections and commands are applied. The toolbar of the active picture window is highlighted. A picture window becomes active when it is opened or when it or one of its objects is clicked on. Only one picture window at a time can be active.

Area buttons for different areas and parts of process are located on the left side of the control panel. By clicking an area button, you can select a process to monitor as the overview picture is updated to match your selection.

Control panel is the window which by default is located on the topmost part of the desktop. It contains the main controls of the user interface, area buttons, overview picture, event management functions as well as date and time display.

Desktop refers to all the windows and their functions displayed on the monitors of a workstation.

Favorites List contains links to the most frequently used pictures. You can add links to pictures or saved desktops to it. The order of the links can be modified and unnecessary links can be removed.

Overview picture is located on the control panel. It displays an overview of the area selected with an area button. The overview picture often contains links from which more detailed pictures can be opened in picture windows. Area buttons and overview picture enable quick location-based picture selection.

Picture window is a window in which a picture is opened. It contains more detailed information required in controlling the process and the process can be operated through it.

Working area refers to the empty area on the desktop. It is designated for picture windows and different tools and dialogs.

1.2 Basics of the DNA Operate User Interface

DNA Operate is a graphical user interface designed for process control and monitoring.

Correctly performed tasks, early detection of disturbances, and a quick recovery from them form the basis for a profitable operation. DNA Operate user interface provides the operators with efficient tools to control large entities and complex situations. The illustrative displays with easy-to-use analysis tools and quick-to-access support information also form a versatile learning environment.

The most important components of the desktop are the control panel, Favorites List, and various picture windows. The control panel includes an overview picture of the process and a toolbar. With the toolbar function buttons you can perform various general desktop operations most of which are connected with picture selection.

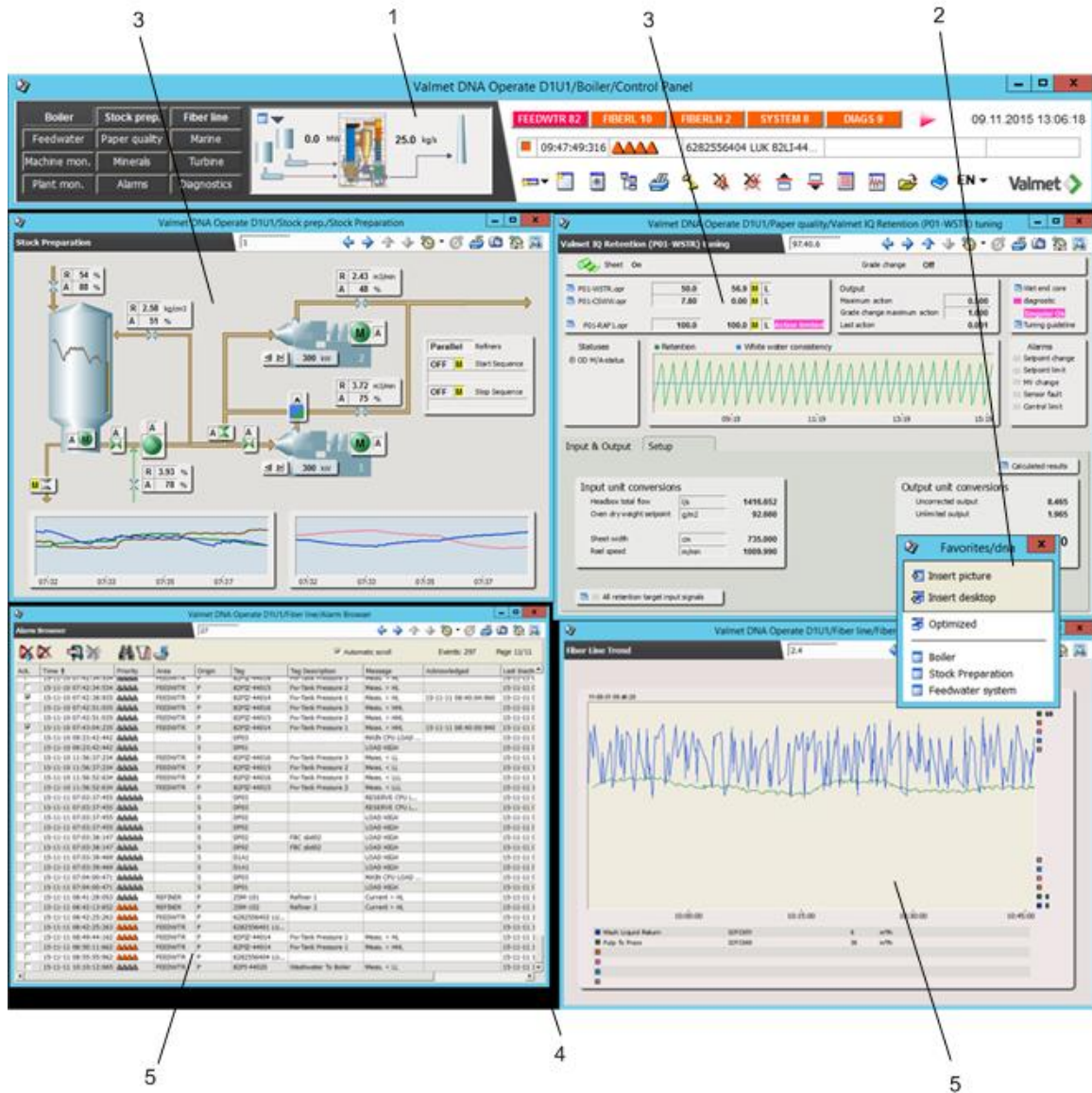
Process data can be examined through pictures opened on the desktop. The pictures can contain up-to-date information, history data, plans, and predictions. Pictures are opened on the desktop in picture windows. There can be several picture windows open on the desktop simultaneously. Important pictures are easily accessible from the Favorites List and they can be quickly opened with a single mouse click.

The Key Features of the DNA Operate User Interface are:

- Support for total view
 - Overview pictures of processes and subprocesses
 - Quick and versatile browsing capabilities
- Quick direct access to information and knowledge with easy linking
 - Easy-to-learn navigation from the overview picture to more detailed picture
 - One-click access to the alarming process graphics
 - One-click access to the most frequently used pictures
 - Drag-and-drop comparison of trend curves
 - Illustrative presentation of history data
 - Support information linked to the object
- Adaptability to the tasks
 - Efficient windowing
 - Modifiable favorite picture collection
- Excellent usability
 - Self-guiding
 - Only a few rules to memorize
 - Consistent principles

1.3 Desktop Structure

Desktop refers to all the windows and their functions displayed on the monitors of a workstation. A desktop comprises of the following elements:



1 Control Panel

Control panel displays an overview picture of an area and its processes. The overview picture can be changed by clicking one of the area buttons. From the control panel you can also perform general operations regarding the whole desktop. In disturbance situations, troubleshooting is started with the event management functions located in the event management panel. There can be several control panels simultaneously open on the desktop.

2 Favorites List

The Favorites List contains selection buttons to the most frequently used pictures. With a single mouse click, you can open the desired picture. You can easily modify the contents of the Favorites List.

3 Picture Windows

More detailed data needed for the process control is displayed in the picture windows and in the tools which can be opened on the desktop. To enable the efficient use of the desktop in all situations, there are several ways to open a picture.

4 Working Area

The empty area on the desktop is called the working area which is designated for picture windows and various tool windows and dialogs. By default, the control panel is located on the topmost part of the monitor. When a desktop of two or more monitors is used, windows can be moved from one monitor to another. You can thus freely arrange the desktop for different conditions or to meet the needs of different users.

5 Desktop Tools

Desktop tools include the trend analysis tool and different maintenance tools. In the event browser, you can view process events and their statuses in chronological order. With drag and drop operations, you can move measurements from the pictures to the trend analysis tool to view their graphs.

1.4 Controlling the User Interface

1.4.1 Keyboard and Shortcut Keys

A Valmet DNA workstation has a standard keyboard which is used in entering data in DNA Operate dialogs.

In addition, the keys in the numeric keypad can be used as shortcut keys which activate the following functions in the user interface:

0	Opens a more general picture in the picture directory.
1	Opens the previous picture from the picture history list.
2	Browses forwards the events in the event stepper.
3	
4	Opens the previous picture on the same directory level.
5	Activates the directory number text box in the active picture window.
6	Opens the next picture on the same directory level.
7	Opens the next picture from the picture history list.
8	Browses backwards the events in the event stepper.
9	
/	Acknowledges the alarm signal.
*	Acknowledges all events.
-	Opens the event browser.
+	

1.4.2 Mouse

Concepts

The following concepts are fundamentally related to mouse use on DNA Operate desktop:

Clicking	The pointer is moved on top of an object and the mouse button is pressed down and immediately released. If no further data is given, the button refers to the left mouse button.
Double-clicking	Two quick successive mouse clicks
Grabbing	The pointer is moved on top of an object and the mouse button is pressed down. If no further data is given, the button refers to the left mouse button.
Dragging	After grabbing, the pointer (and the object) is moved by keeping the button pressed down.
Dropping	After dragging, the button is released when the pointer (and the object) is in the desired location.

Operation

The mouse is used in moving the pointer to the desired location on the desktop. The left mouse button activates the typical function of the target, while the right mouse button opens the [action menu](#) related to the target. With the wheel button event browser pages can be browsed.

The pointer is normally arrow-shaped but its shape changes to a hand when it is on top of an operable target.

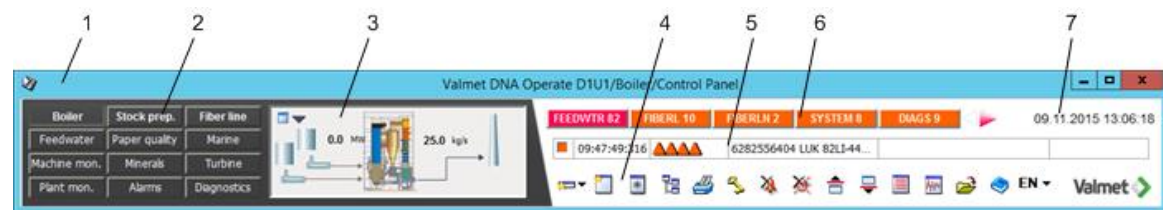
1.5 Control Panel

1.5.1 Function of the Control Panel

Control panel provides general information of the process as well as tools for performing the essential operations. These operations include logging into DNA Operate, opening new picture windows, and starting troubleshooting in disturbance situations. A desktop may simultaneously contain several control panels each designated to control the processes of one area.

1.5.2 Elements of the Control Panel

Control panel contains the following elements:

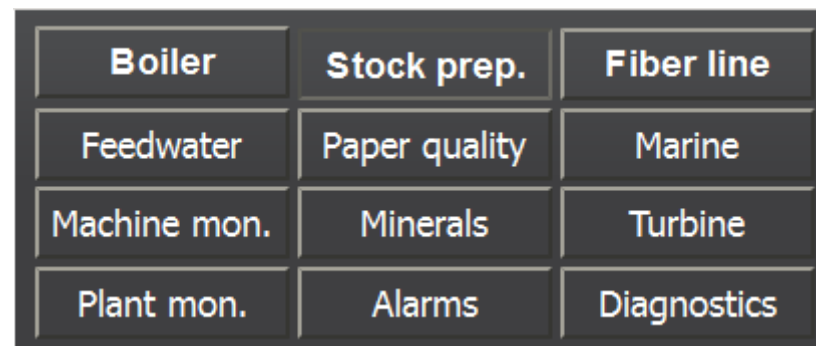


1. Title bar
2. Area buttons
3. Overview picture
4. Toolbar
5. Event stepper
6. Event counter
7. Date and time

1.5.3 Title Bar

The title bar of the control panel displays the name of the selected area. It also contains the standard buttons for [minimizing](#), [maximizing](#), or [closing](#) the control panel.

1.5.4 Area Buttons

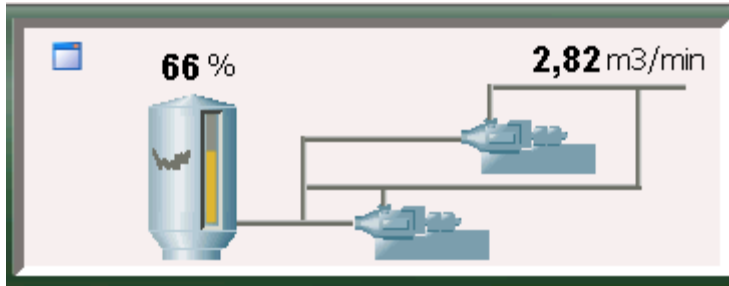


By clicking an area button, you select the area to want to control. When an area button is clicked, the corresponding overview picture representing a process or subprocess is opened in the control panel. All processes and subprocesses have their own buttons. For example, a control room might have the following area buttons: material storage, reactor 1, reactor 2, and product storage. When required, area buttons can also be defined to refer to other plants.

Depending on the needs of the control room, there can be up to 16 buttons on the control panel.

When an area button appears to be pressed down, the corresponding area is selected. The overview picture will always display information concerning the selected area.

1.5.5 Overview Picture



The overview picture contains links to more detailed pictures of the area in question. Its purpose is also to present general information of the selected area. Subprocesses and subfunctions are represented in groups. The overview picture will always portray the selected area. With the area buttons and the overview picture you can quickly navigate to a detailed process picture.

Overview pictures display information about process measurements and statuses in the form of numbers, bars, symbols, and trends, for example. Except for its size, the overview picture is similar to other picture windows. You can operate the tools and buttons of the overview picture with the mouse in the same manner you operate picture windows.

1.5.6 Toolbar



The toolbar contains buttons for general desktop operations. You can perform the following functions with the toolbar buttons:

	Open a new control panel
	Open a new picture window
	Open Favorites List
	Open picture browser
	Print the contents of a picture window
	Log in and log out
	Acknowledge alarm signal
	Acknowledge or mute alarm signals
	Acknowledge events
	Browse the event stepper forward and backward
	Open the event browser
	Open trend analysis tool



[Open picture](#)



[Open Diary](#)



[Open DNA Report](#)

NOTE! The contents of the toolbar can be modified and may vary.

1.5.7 Event Stepper

☒	13:25:10:911		25FC-100	Ref. 1 outlet valve	Meas. > HHL
-------------------------------------	--------------	---	----------	---------------------	-------------

When a disturbance occurs, the corresponding disturbance data is displayed in the event stepper even if the event browser were not open. From the event browser you can also quickly access the picture of the alarming subprocess or acknowledge the event.

When new events which require acknowledgement have been generated from the process, the event stepper always shows the details of the oldest unacknowledged event. You can browse the unacknowledged events with the tool bar buttons and so follow the chain of events.

NOTE! The order in which the events are presented in the event stepper may vary between applications due to configuration.

1.5.8 Event Counter

The function of the event counter is to present an overview of the events currently active in the control room and to provide a quick access to the information connected to the events. The event counter presents the advancing of a disturbance in a process from one event area to another. A disturbance is usually an alarm or a group of alarms requiring an acknowledgement.



The fields in the event counter refer to the process areas which have unacknowledged events. Up to 5 fields can be displayed in the event counter at a time. If there are more than 5 fields, the field list can be scrolled in the event counter in the groups of five by clicking the arrow buttons. The name of the process area and the number of events active in the area are displayed in each field. The color of the field represents the highest event priority in the area as follows:

- Red = high alarm priority
- Dark orange = medium alarm priority
- Light orange = low alarm priority
- Gray = notification

Similarly, the color of the arrow button represents the highest event priority in the group of the next five fields.

Event areas appear in the fields of the event counter at the same pace in which a disturbance occurs in the area in question. Therefore, the operator can see (in a chronological order) how a disturbance has advanced in the process from one event area to another, and find out the causes and solve the situation.

Shelved alarms are not included in the number of events.

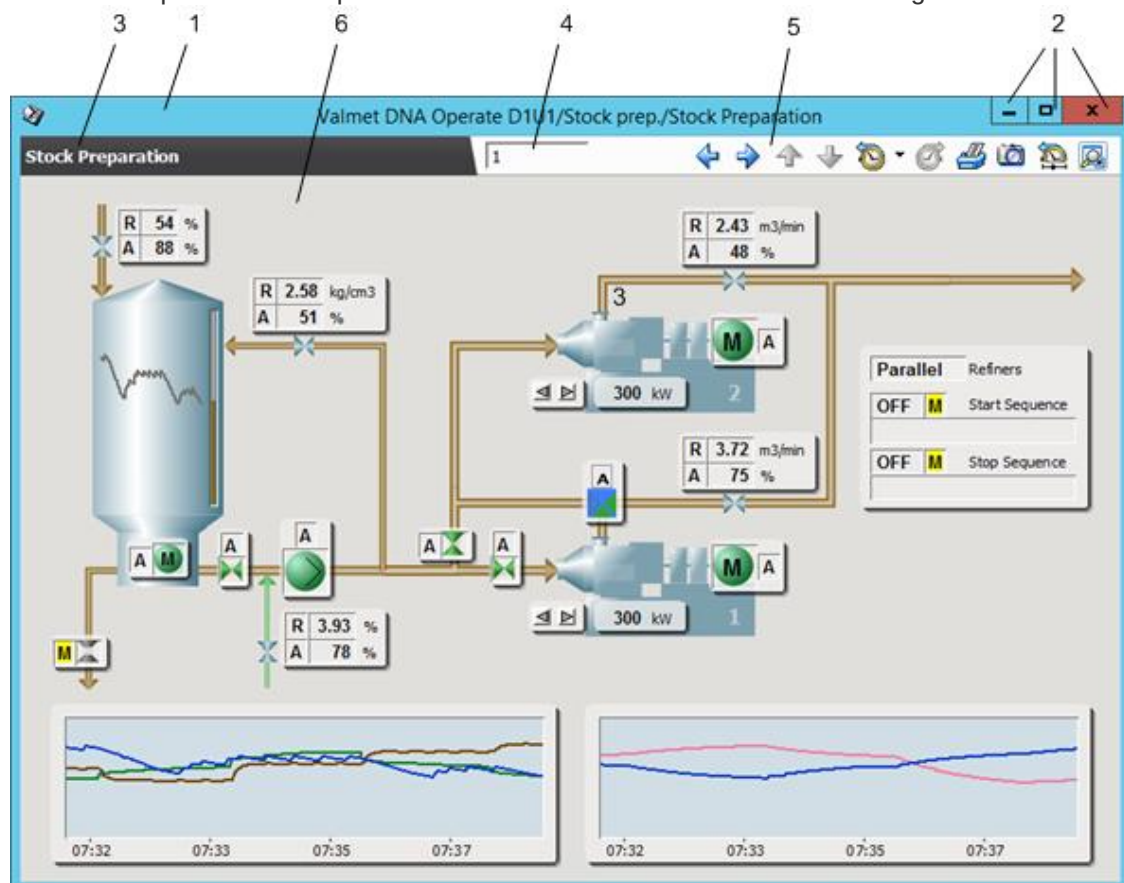
1.6 Pictures and Picture Windows

1.6.1 Function of the Pictures

Extensive processes can contain tens of thousands of measurements, and the information provided by them is usually further processed into several parallel displays of information such as history, statistics, and key figures. Pictures allow you to pick out the essential information from this vast amount of data. Pictures also make it possible for you to select different approaches to the information depending on your needs. The process is controlled through pictures.

1.6.2 Elements of the Picture Window

Pictures are presented in picture windows which contain the following elements:



1. Area name
2. Minimize, Maximize and Close buttons
3. Picture title
4. [Directory number](#) text box
5. Toolbar
6. Picture area

1.6.3 Contents of the Pictures

The pictures are composed of various components, which can be divided into static and dynamic as well as operable and non-operable components.

Static component data does not change while the picture is in the picture window. Typical examples of static data are various frames and title texts as well as process equipment that are drawn into the picture.

Dynamic component data changes either due to a change in the process or as a consequence of an operation. Examples of dynamic data are, for instance, the symbol representing the status of a motor, the trend of a measurement, the numerical value of a temperature measurement, and the bar depicting a surface level.

Through the operable components, you can modify process data, open new pictures as well as open various kinds of windows. Typical operations are, for example, starting and stopping a motor, changing the set-point value of a controller, and selecting a picture attached to a tank. The operable components can be distinguished by their embedded appearance and different buttons.

Pictures can be of different sizes and shapes to match the natural proportions of the process.

1.6.4 Picture Directory

Each area has its own picture directory. The picture directory determines the order (next/previous) and the location (more specific/more general) of the pictures in the area in relation to each other. Picture location in the directory is specified with a hierarchical number like 1.1 and 2.2.3. The directory number of a picture is displayed on the tool bar of the picture window. The tree structure of a picture directory can be examined with the picture browser. For more information, see chapter [How to Open Pictures from the Picture Browser](#).

1.6.5 Toolbar



The toolbar of a picture window contains command buttons with which the following picture and picture window related operations can be executed:



[Browse pictures in the picture directory](#)



[Browse recently used pictures](#)



[Open the list of recently used pictures](#)



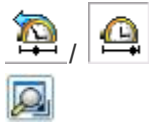
[Open the picture browser](#)



[Print the contents of the picture window](#)



[Take snapshot picture](#)



Toggle [history](#) and [real-time mode](#)

[Crop the picture to the picture window](#)

1.7 Operations

1.7.1 Handling Control Panels

Several control panels can be open on the desktop simultaneously. A new control panel can be used to take the operation responsibility of another process (control room) or to just view its pictures. Operations on the pictures of another control room may have been prevented, however. Note that when another control panel is opened, the events of the related area will now appear on the desktop and activate the alarm horn. Similarly, when a control panel is closed, the events of the corresponding area will stop appearing. It is also possible to define that alarms originating from additional control panels will not activate the alarm horn of the control room.

A control panel can be [moved](#) or [resized](#) similarly to a picture window.

How to Open a New Control Panel

To open a new control panel, proceed as follows:

- Open the list of available control areas by clicking  button on the control panel toolbar. Select the desired area from the list. The new control panel will appear on the desktop. The contents of the drop-down list is control room-specific and can be modified.

How to Close a Control Panel

To close a control panel, proceed as follows:

- Click the Close button  on the control panel title bar. The control panel will disappear from the desktop.

Normally it is not possible to close the control panel of the local control room.

1.7.2 Handling Picture Windows

New picture windows can be opened on the desktop according to needs. Correspondingly, unnecessary picture windows can be closed and thus free some space on the desktop. Window size can be changed when necessary and the picture can be scrolled in the window.

How to Open a New Picture Window

You can open a new picture window on the desktop by clicking the new window button on the control panel. A new picture window will also open automatically on the desktop when a new picture is opened and there are no picture windows on the desktop.

To open a new picture window, proceed as follows:

- Click the  button on the control panel toolbar. A new picture window will appear on the desktop displaying the first picture of the selected area.

How to Move a Picture Window

The location of a picture window on the desktop can be freely changed.

To move a picture window, proceed as follows:

- Click the title bar of the picture window and drag the picture window to the desired location while holding down the mouse button.

How to Minimize and Restore a Picture Window

You can minimize a picture window to a taskbar button by clicking the  button in the title bar.

When you want to restore the picture window back to the desktop, click its taskbar button.

How to Maximize and Restore a Picture Window

You can maximize a picture window to fill the whole display by clicking the  button in the title bar. Maximization will not change the picture ratio and it will take pictures of different shapes into account.

When you want to restore the picture window back to its original size, click the  button in the title bar.

How to Resize a Picture Window

A picture window can be resized. By dragging the vertical or horizontal border of the window, the size of the window can be changed in the corresponding direction. By dragging a window corner, the size of the window can be changed simultaneously in both vertical and horizontal direction.

To resize a picture window, proceed as follows:

1. Point the border of the picture window. The pointer changes to a two-headed arrow.
2. Grab the border by clicking and holding the left mouse button and drag the border in the desired direction.
3. Release the button when the size of the window is appropriate.

By default, the picture is scaled to fit in the window while maintaining its picture ratio. Alternatively, you can disable the scaling by clicking the  button on the picture window toolbar. When the scaling has been disabled, the picture size will not change when the window is resized and you can crop the picture to view only a certain part of it. If necessary, the rest of the picture can be viewed by scrolling the picture with the scrollbars or alternatively by holding down the mouse middle button and dragging the picture in the picture window. You can restore the whole picture to the window by reclicking the  button.

As the window size is reduced, the rightmost buttons of the toolbar may be cut invisible. When this occurs, you can still access the buttons by clicking the double arrowhead button .

How to Close a Picture Window

Picture windows can be removed from the desktop when they are no longer required.

To close a picture window, proceed as follows:

- Click the Close button  on the title bar of the picture window. The window will close.

1.7.3 How to Open Pictures

You can open pictures in the picture windows that are on the desktop by a variety of ways depending on each situation. For example, from the Favorites List, you can quickly open the most frequently used, important pictures. When you are following the course of the entire process, you may want to browse the pictures with various browser functions. All pictures, including infrequently needed, can be found using the picture browser.

When a picture is opened to a picture window which is already open on the desktop, the shape of the window may change.

Picture Window Activity

When a picture window is active, its toolbar appears highlighted. A picture window is activated when it is opened or when the window or an object in it is clicked. Only one picture window at a time can be active. All operations are always directed at the active picture window. A picture window does not need to be separately activated in order to operate one of its objects. If there are no picture windows open on the desktop, opening a picture opens a window as well.

On the desktop, you can perform the following operations for opening and browsing windows:

- [Open a picture from the overview picture of the control panel.](#)
- [Open important and frequently used pictures from the Favorites List.](#)
- [Open a picture from the picture browser.](#)
- [Browse picture directories.](#)
- [Open previously viewed pictures.](#)
- [Open a picture from a picture window by entering a directory number.](#)
- [Open a picture from another picture.](#)
- [Open a picture from the event stepper on the control panel.](#)
- [Open a picture from the event browser.](#)
- [Open a picture by its tag.](#)

How to Open a Picture with the Open Picture Function

The open picture function can be used in both overview pictures and normal process pictures.



When the open picture icon is clicked, a pre-defined picture is opened. The arrow opens a drop-down list from which the desired picture can be selected. The contents of the list can be modified by adding and removing picture links. Open picture icons may appear with or without drop-down lists.

How to Open a Picture from Open Picture Icon

Proceed as follows:

- Click the open picture icon on the picture. A new picture linked to the icon will open in the topmost window. If there are no picture windows open on the desktop, the picture will open in a new picture window.

How to Open a Picture From the Picture List

Proceed as follows:

1. Click the arrow  to open the picture list.
2. Click the desired picture name. The selected picture will open in the topmost picture window and the list will close.

How to Edit Picture Lists

The contents of a picture list can be modified with the action menu or by dragging pictures to it from the desktop. Content modification may also have been prevented.

How to Add a Picture to the Picture List

You can add a picture to a picture list from, for example, a picture window or the Favorites List.

Proceed as follows:

1. Grab the picture name in the picture window.
2. Drag it on top of the arrow  next to the open picture icon.
3. Release the mouse button. The name of the picture will appear last on the list.

A picture can be added to the picture list also with the action menu.

Proceed as follows:

1. Click the arrow  to open the picture list.
2. Right-click the picture list and select Insert on the action menu. The active picture will be added to the bottom of the list.

How to Remove a Picture from the Picture List

A picture can be removed from the picture list.

Proceed as follows:

1. Click the arrow  to open the picture list.
2. Right-click a picture name and select Remove on the action menu. The selected picture name will disappear from the list.

How to Rearrange the Picture List

The order of the pictures on the picture list can be rearranged.

Proceed as follows:

1. Click the arrow  to open the picture list.
2. Right-click a picture name and select Move on the action menu.
3. Drag the picture name to the desired place on the list and click the list. The picture name will be moved to the selected place and the list will close.

How to Open Pictures and Desktops from the Favorites List

Important and frequently used pictures can be quickly opened from the Favorites List. The list can hold a maximum of 40 picture links. By clicking a picture link in the Favorites List you can open a new picture in the topmost picture window or even switch to a previously saved desktop configuration.

How to open a picture from the Favorites List

To open a picture from the Favorites List on the topmost picture window on the desktop, proceed as follows:

- Click the picture name or symbol on the Favorites List. The selected picture will open in the topmost picture window. If there are no picture windows open on the desktop, the picture will open in a new picture window.

How to open a desktop from the Favorites List

To open a desktop from the Favorites List, proceed as follows:

- Click the desktop name or symbol in the Favorites List. The layout and windowing of the desktop will change to match the saved desktop.

Incorrect links and error messages

If the title of a picture has been changed, the picture link may refer to a wrong target. If the picture cannot be found in its referred location in the picture directory or the referred location is missing, the following error message will be displayed on the screen: *The picture selected is not available.*

If the Favorites List contains an incorrect link, proceed as follows:

1. Remove the incorrect picture link from the Favorites List. For more information, see chapter [How to Edit the Favorites List](#).
2. Add the link to the required picture. For more information, see chapter [How to Edit the Favorites List](#).

How to Open Pictures from the Picture Browser

Proceed as follows:

- Click the desired picture name in the [picture browser](#). The selected picture will open in the topmost picture window and the corresponding picture name will become highlighted in the picture browser.

OR

- Grab the desired picture name in the picture browser, drag it to a picture window already open on the desktop and then drop it in there. The corresponding picture will now open in the chosen picture window.

How to Browse Pictures in a Directory

The pictures of a directory can be browsed in a picture window using the toolbar arrow buttons or the shortcut keys of the numeric keyboard. With the left and right buttons you can browse pictures on the same directory level, whereas the up and down arrows are used for moving from a directory level to another. By browsing a picture directory you can follow the course of the process from one picture to another.

How to Browse Pictures in One Directory Level

To browse pictures on one directory level, proceed as follows:

- Click the left  or right  arrow button on the picture window toolbar.

OR

- Click the 4 or 6 key on the numeric keyboard.

The picture displayed in the picture window will be replaced by the previous or next picture in the picture directory respectively. For example, picture 2.3 will be replaced by picture 2.2 or 2.4 depending on the clicked button.

When you reach the beginning or the end of a branch on a directory level (for example, picture 2.16 or picture 2.1) the next click will take you to the opposite end of the level. You cannot jump from the end of one directory level to the beginning of another. For example, browsing will not take you from picture 2.16 to picture 3.1, or from picture 2.1 to picture 1.10. Any missing directory slots are skipped. If a directory slot is empty (i.e. the picture is missing), the following error message will be displayed on the screen: *The picture selected is not available.*

How to Browse Pictures in Different Directory Levels

To browse pictures in different directory levels, proceed as follows:

- Click the up  or down  arrow button on the picture window toolbar. The picture displayed in the picture window will be replaced by the first picture on the higher or lower directory level respectively. For example, picture 2.3 will be replaced by picture 2 or 2.3.1 depending on the clicked button.

The 0 key on the numeric keyboard has the same functionality as the  button.

When you reach either end of the picture directory (e.g. picture 0 or 2.16.1), you cannot continue browsing the same direction. If there are any missing levels in the picture directory, they will be skipped (for example, picture 2.3.11 can be followed by picture 2).

How to Browse Recently Used Pictures

The desktop keeps a record of the most recently used pictures so you can easily access them by either browsing pictures with the back and forward buttons, or by bringing the picture history list on the screen. You can choose any picture on the list by clicking the picture title. The picture history list is desktop-specific, i.e. all pictures that have been viewed in the picture windows of the desktop will be displayed on the list.

The history list can contain up to 20 most recently used pictures. A picture will be added to the list after it has been closed from the desktop. The pictures are shown in chronological order, the latest picture being the topmost and the oldest used picture being the bottommost picture on the list. A new picture that is not already on the list will remove the oldest (or bottommost) picture from the list, and the new picture will be placed as the topmost picture. When a picture that is already on the list is opened, it will be moved to the topmost position. A picture can appear on the list only once, even if it have been displayed several times.

How to Select a Picture from the History List

The button for opening the history list is located on the toolbar of a picture window,

between the clock buttons  and . By clicking the button you can open the drop-down list of the recently used pictures. The names on the list are links which open the corresponding picture on the picture window.

To open a picture from the picture history list, proceed as follows:

1. Click the arrowhead button  on the picture window toolbar to open the history list.
2. On the drop-down list, click the desired picture name. The picture will open in the picture window.

How to Browse Picture History

The pictures recently used in a picture window can be browsed using the clock

buttons   on the picture window toolbar. Alternatively, you can use the shortcut keys 7 and 1 on the numeric keyboard.

To browse the last used pictures in the picture window, proceed as follows:

1. Click the left clock button  on the picture window toolbar OR press the 7 key on the numeric keyboard. The most recently used picture will open in the picture window.
2. Reclick the same button or repress the 7 key. A picture which had originally been used before the currently shown picture will be opened in the picture window.

The picture history can also be browsed forwards back to the most recent pictures by

clicking the right clock button  or by pressing the 1 key on the numeric keyboard.

When you reach either end of the picture history, i.e. the latest or the oldest picture, the browsing will continue from the other end. In other words, if you browse the picture history backwards after you have already reached the oldest picture, the latest picture will be opened next.

How to Open a Picture by Its Directory Number

A picture can be opened by entering its directory number in the text box on the picture window toolbar. The picture search will be targeted at the picture directory of the area currently shown in the overview picture, even though the picture currently displayed in the active picture window belonged to another directory.

Proceed as follows:

1. Click the text box on the picture window toolbar OR press the 5 key on the numeric keyboard.
2. Enter the directory number, e.g. 1.3.7, in the text box and press the line feed key. A picture corresponding with the entered number will be opened in the picture window.

You can also edit the directory number by clicking the text box and moving the cursor to the desired point.

If you entered an incorrect directory number or the directory slot does not exist, the following error message will be displayed: *The picture selected is not available.*

How to Open a Picture from the Event Stepper

A picture related to an event can be opened directly from the event stepper.

Proceed as follows:

- Click the event displayed in the event stepper.

OR

- Right-click the event stepper and select Picture with Event on the action menu.

In both cases, the following dialog will open to the desktop. In it, you can select whether the related picture is opened in a new picture window or in the topmost picture window on the desktop.



The picture will open in the chosen picture window.

How to Open a Picture from the Event Browser

You can open a picture related to an event presented in the event browser.

Proceed as follows:

- In the event browser, double-click an event to open a picture related to it.

OR

- Right-click the target event in the event browser and select Picture with Event on the action menu.

OR

- When the target event in the event browser is already selected (indicated by a frame) press Enter.

In all cases, the following dialog will open to the desktop. In it, you can select whether the related picture is opened in a new picture window or in the topmost picture window on the desktop.



The picture will open in the chosen picture window.

How to Open a Picture by a Tag

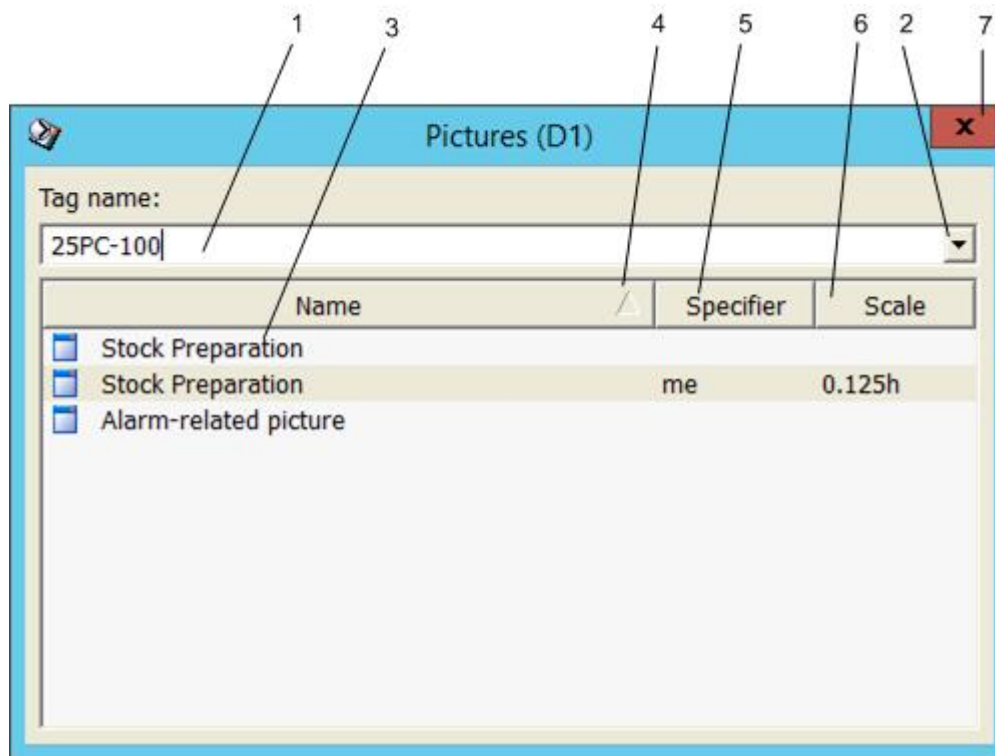
You can open a picture by selecting it from a tag-specific picture list which is opened from the action menu. This function is useful for finding the trends in which the tag in question is presented.

Proceed as follows:

1. Right-click the desired tag in a picture window and select Related Pictures... on the appearing action menu.
2. In the Pictures dialog, click the name of the picture you want to open. The selected picture will open in the topmost picture window and the picture name will become highlighted on the list.

If the selected picture cannot be found, the following error message will be displayed: *The picture selected is not available.*

The Pictures dialog contains the following elements:



1. Selected tag
2. Arrow button for opening the tag list
3. List of tag-related pictures
4. An arrow symbol indicating the column based on which the pictures are sorted. To change the sorting criterion, click the desired column.
5. Trend specifier
6. Scale of trend
7. Close button

If the Pictures dialog is opened from the event browser at the bottom of the list is Alarm-related picture or Alarm-related action. If you click the action, for example a new monitor window opens.

1.7.4 How to Select an Area

Area selection takes place through the area buttons on a control panel. After a new area has been selected, the overview picture on the control panel will be updated to display the process information of the area. Each area has a separate picture directory.

To select a new area for the overview picture, proceed as follows:

- Click the desired area button on the control panel. The button will press down to indicate the selected area and the previously selected button will rise up correspondingly. The overview picture on the control panel will be updated to match the selected area.

1.7.5 How to Print Pictures

With DNA Operate, you can print the contents of a picture window.

To print a picture in a picture window, proceed as follows:

- Click the print button  on the control panel or picture window toolbar. The contents of the picture window will be printed to the default printer. The contents of the picture window can also be specified to print to a pdf or png file.

1.7.6 How to Acknowledge Alarm Signal

An alarm signal indicates that there is a disturbance in the process which has caused a new alarm. The alarm signal can be acknowledged from the control panel.

Proceed as follows:

- Click the acknowledge button  on the control panel toolbar.

OR

- Click the  button on the control panel toolbar to get the list of possible commands. Select the command according to the desired function. It is possible to acknowledge all the alarm signals at the same time (Acknowledge all horns) or just one group of alarm signals at a time (Acknowledge horn_1 etc.). If you just want to mute, not acknowledge, all the alarm signals, select the Mute horns command.

OR

- Press the / key on the numeric keyboard.

The alarm signal will stop.

1.7.7 How to Acknowledge Events

How to Acknowledge All Events Simultaneously

All events can be acknowledged simultaneously from the control panel if this has been enabled in DNA Operate configuration.

Proceed as follows:

- Click the acknowledge button  on the control panel toolbar.

OR

- Press the * key on the numeric keyboard.

All events will be acknowledged.

If events can be acknowledged only from the event browser, the acknowledgement button has been located in picture windows. The button appears dimmed unless the event browser is open in the picture window.

How to Acknowledge a Single Event

Single events can be acknowledged through the [event stepper](#) or the [event browser](#).

If you want to acknowledge a single event through the event stepper or event browser, see chapters [How to Acknowledge the Event Stepper](#) and [How to Acknowledge an Event in the Event Browser](#) for more information.

How to Acknowledge All Events Related to a Certain Tag

All events related to a certain tag can be acknowledged simultaneously either through the process picture, the event browser or the event stepper by using the action menu.

Proceed as follows:

1. Right-click the tag you want to acknowledge.
2. Click Acknowledge Tag on the appearing action menu.
3. A dialog will open to the desktop. In it, you can confirm whether all the events related to the tag in question are acknowledged simultaneously or not.
4. Click the OK button. The dialog will close and the events will be acknowledged.

How to Acknowledge an Event Area in the Event Counter

You can also simultaneously acknowledge all events of one area in the event counter.

Proceed as follows:

1. Click the desired event area button on the control panel.



The acknowledgement dialog will open on the desktop:



2. Click the OK button. The dialog will close and all events in the area will be acknowledged.

1.7.8 How to Shelf Alarms

By shelving the alarm, the alarm can be temporarily prevented from being displayed, if the alarm is unnecessary and a nuisance to the operator. This can be due to an instrument malfunction, lack of maintenance or a fault in the process causing repetitive alarms.

A shelved alarm is removed from the event browser and the shelved event points are not displayed. The shelved alarm should be labeled with shelving expiry date and time and possible user comments. The state of shelved event points will be displayed only after the expiration of the shelving.

When shelving begins, the alarm is acknowledged and no audible alarm indication is given. A shelved alarm will not give an audible alarm indication either when alarm becomes active.

An alarm with any priority can be shelved.

To shelve an alarm, proceed as follows:

- On the event browser, right-click the alarm and select Shelf... on the action menu.

Note that shelving an alarm affects only the selected event point, not all event points of tag.

Time-based Shelving:

Shelving of event points is released when the preset time for shelving expires.

The default length of shelving is 8 hours. Other options are 1, 2, 4 or 12 hours. The options can be configured.

For more information, see section [How to Shelf an Alarm in the Event Browser](#).

State-based Shelving:

Event points are released from shelving when their state becomes passive. One shot shelving is possible when the alarm status is fully understood (e.g. device is under maintenance) and status is expected to last for some time until the situation normalizes (alarm becomes inactive).

For more information, see section [How to Shelf an Alarm in the Event Browser](#).

1.7.9 How to Open Picture Editor

You can open the picture editor (Picture Designer) directly from the control panel.

Proceed as follows:

- Click the picture editor button  on the control panel toolbar. The picture editor will open on the desktop in a dedicated window.

NOTE! Picture Designer is a supplement for Valmet DNA and not included in the basic configuration.

1.7.10 How to Open Diary Tool

The diary tool (Diary) is designed for reporting and commenting disturbances and other events. It uses a web browser as an interface.

To start Diary, proceed as follows:

- Click the diary button  on the control panel toolbar. Diary will open on the desktop in a dedicated window.

Diary's pre-filled [shift](#) and [disturbance entry](#) functions as well as its [search](#) function can also be executed through the action menu.

NOTE! Diary is a supplement for Valmet DNA and not included in the basic configuration.

1.7.11 How to Open DNA Operate User Manual

DNA Operate user manual can be opened on the desktop from the control panel toolbar.

Proceed as follows:

- Click the user manual button  on the control panel toolbar. DNA Operate user manual will open on the desktop in a dedicated window.

1.7.12 How to Open Web Reporting Tool

The web reporting tool (DNA Report) is an application used for viewing reports and Tracer pictures. DNA Report is opened from the toolbar of the control panel.

Proceed as follows:

- Click the web reporting button  on the control panel toolbar. DNA Report will open on the desktop in a dedicated window.

NOTE! DNA Report is a supplement for Valmet DNA and not included in the basic configuration.

1.7.13 How to Open Process Help

The contents of the process help can be freely modified. It may include, for example, special instructions related to the controlled area or process. The process help is opened from the control panel toolbar.

Proceed as follows:

- Click the process help button  on the control panel toolbar. The process help will open on the desktop in a dedicated window.

NOTE! Process help is a supplement for Valmet DNA and not included in the basic configuration.

1.7.14 How to Open Trend Analysis Tool

With the trend analysis tool, you can examine the process behaviour during exceptional states.

To open the trend analysis tool, proceed as follows:

- Click the trend analysis tool button  on the control panel toolbar. The trend analysis tool will open on the desktop in a new picture window.

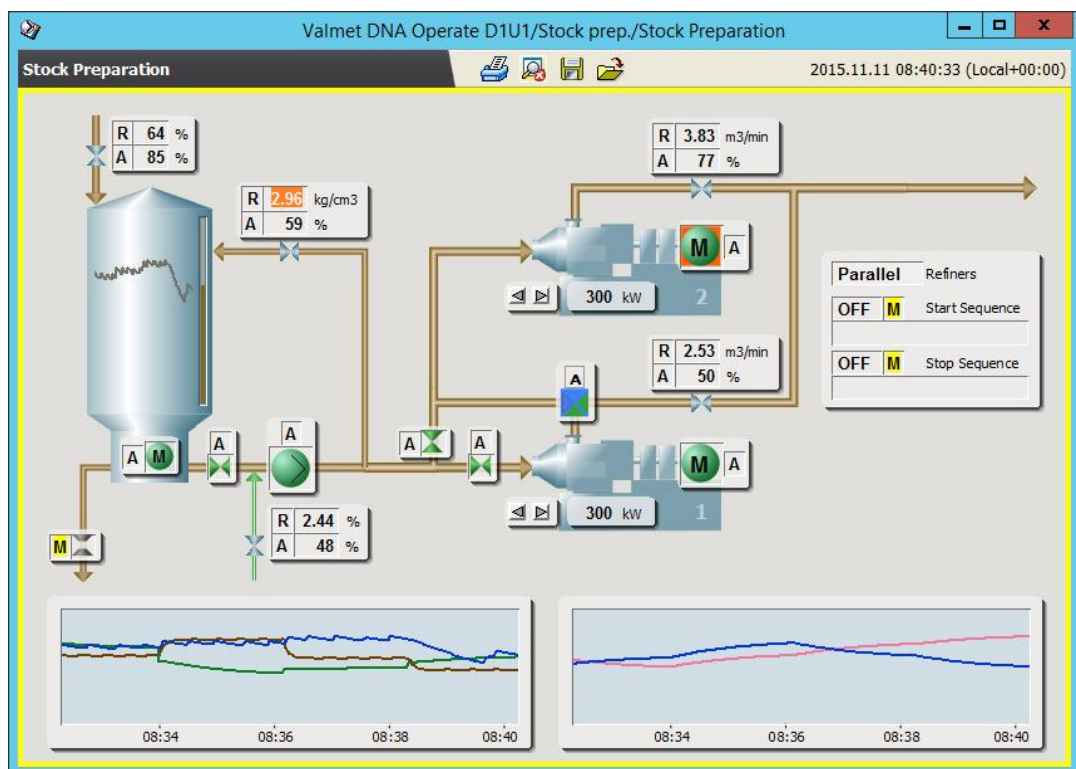
NOTE! The use of the trend analysis tool has been covered in the [Trend Picture User Manual](#).

1.7.15 How to Take Snapshot Picture

The contents of a picture window are usually dynamic and changing according to process status. However, you can take static snapshot copies of the contents. Each copy reflects the situation on only one instant. By examining copies taken in the course of time, you can, for example, compare cause and consequence relationships in loops' behavior. There can be several snapshot pictures open on the desktop at the same time.

To take a snapshot of the contents of a picture window, proceed as follows:

- Click the snapshot button  on the picture window toolbar. The snapshot will open on the desktop in a new picture window.



The snapshot picture window differs clearly from the ordinary picture window because the picture area of the picture window will be highlighted in yellow attention color.

Note that you cannot take a snapshot of a zoomed trend picture or profile. Zoomed trends and profiles always return to the original situation before zooming.

A [snapshot can be saved](#) for later examination.

NOTE! It is not possible to perform any process variable operations in a snapshot picture window. Only possible operations in a snapshot picture window are saving, printing, scaling and opening a picture, zooming a trend, and browsing event browser pages.

1.7.16 How to Use the Event Stepper

How to Browse the Event Stepper

The events presented in the event stepper can be browsed with the dedicated control panel buttons or with the shortcut keys of the numeric keyboard.

Proceed as follows:

- Click the  button (OR press the 2 key on the numeric keyboard) when you want to browse the events forwards in chronological order
- Click the  button (OR press the 8 key on the numeric keyboard) when you want to browse the events backwards in chronological order.

When an event is browsed on the event stepper, the related picture will be opened in the topmost picture window on the desktop.

How to Acknowledge an Event in the Event Stepper

When you want to acknowledge an event presented in the event stepper, proceed as follows:

1. Right-click the event stepper and select Acknowledge Event on the action menu. Depending on the configuration, a confirmation for the event acknowledgement might be asked. If the confirmation is not required, the event presented in the event stepper is acknowledged after the given action menu command. However, if the confirmation is required, the following Confirm dialog will open on the desktop:



2. Then click the OK button of the dialog. The dialog will close and the event presented in the event stepper is acknowledged.

How to Open a Picture from the Event Stepper

A picture related to an event can be opened directly from the event stepper.

Proceed as follows:

- Click the event displayed in the event stepper.

OR

- Right-click the event stepper and select Picture with Event on the action menu.

In both cases, the following dialog will open to the desktop. In it, you can select whether the related picture is opened in a new picture window or in the topmost picture window on the desktop.



The picture will open in the chosen picture window.

1.7.17 How to Log In and Out

User Privilege Groups

Users can be divided into three groups each of which has different access privileges. The system functions in a mode corresponding to the user group. The modes are as follows:

- Display mode
- Control mode
- Service mode

Display mode

In the display mode you can only open and view pictures. All process operations have been disabled.

Control mode

In the control mode all the so-called basic operations have been enabled. It is designed for normal production personnel. Almost all operations have been enabled with some exceptions, such as controller parameter tuning and changing of event limits and event masks, for example.

Service mode

In the service mode all operations are enabled. It is designed for personnel with an intimate knowledge of process and automation, such as maintenance and system engineers.

Each user can be assigned a user group, username, password, and password validity. User groups are assigned user privileges and a delay time which may elapse after the last operation before the user interface is reset to its default mode. DNA Operate will also be in the default mode after its start-up. The default mode is determined in user interface configuration.

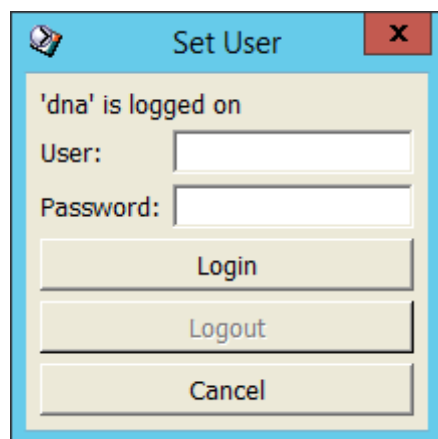
How to Log In

Some operations require higher level user privileges. Privileges are used to limit users' access to only certain functions or data. During login the user identifies himself to DNA Operate which responds by granting him his user privileges.

If the user tries to perform operations to which he has no privileges, the operations will not be executed and DNA Operate will respond with the corresponding error message.

To log into the system, proceed as follows:

1. Click the  button on the control panel toolbar to open the Set User dialog.



2. Enter your username and password in the corresponding fields.
3. Click the Login button. The dialog will close and the user privileges will change accordingly.

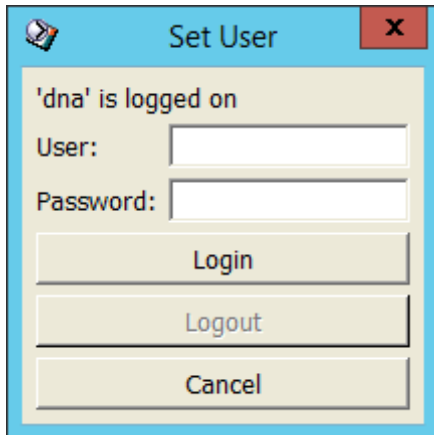
If you enter an invalid username or password, you will get the following error message: *Invalid user id.*

If user identification has not been activated or it is not used at all, Valmet DNA will accept any username and password. In this case all users have full privileges.

How to Log Out

To log out and restore access privileges to the default mode, proceed as follows:

1. Click the  button on the control panel toolbar to open the Set User dialog.



2. Click the Logout button. The dialog will close and access privileges will be restored to the default mode.

1.7.18 How to Use the Favorites List

Function of the Favorites List

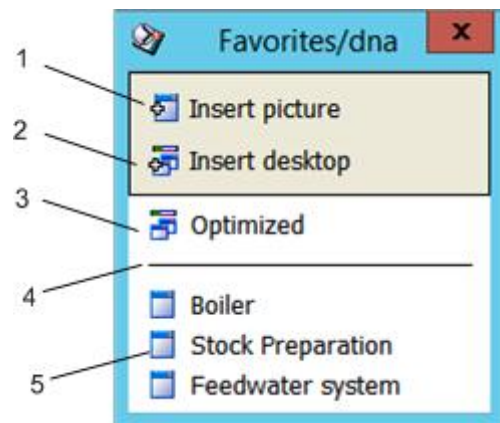
Favorites List contains a user-specific list of the most frequently used important pictures. The list functions as a collection of links through which you can easily open the desired picture by clicking the name or symbol of the picture. You can also add the configuration of the whole desktop to the list for different conditions or shifts. When a desktop configuration is restored, all windows including possible monitor windows will be opened as they were when the desktop configuration was stored. The contents of the Favorites List can be modified according to users' needs.

How to Open the Favorites List

To open the Favorites List, proceed as follows:

- Click the Favorites List  button on the control panel toolbar. The Favorites List will by default open on the middle of the desktop but it can also be defined to open elsewhere on the desktop.

The Favorites List can be freely moved on the desktop and you can close it by clicking the  button on the title bar.



1. Insert picture button
2. Insert desktop button
3. Desktop symbol and name
4. Separator
5. Picture symbol and name

How to Open Pictures and Desktops from the Favorites List

Important and frequently used pictures can be quickly opened from the Favorites List. The list can hold a maximum of 40 picture links. By clicking a picture link in the Favorites List you can open a new picture in the topmost picture window or even switch to a previously saved desktop configuration.

How to open a picture from the Favorites List

To open a picture from the Favorites List on the topmost picture window on the desktop, proceed as follows:

- Click the picture name or symbol on the Favorites List. The selected picture will open in the topmost picture window. If there are no picture windows open on the desktop, the picture will open in a new picture window.

How to open a desktop from the Favorites List

To open a desktop from the Favorites List, proceed as follows:

- Click the desktop name or symbol in the Favorites List. The layout and windowing of the desktop will change to match the saved desktop.

Incorrect links and error messages

If the title of a picture has been changed, the picture link may refer to a wrong target. If the picture cannot be found in its referred location in the picture directory or the referred location is missing, the following error message will be displayed on the screen: *The picture selected is not available.*

If the Favorites List contains an incorrect link, proceed as follows:

1. Remove the incorrect picture link from the Favorites List. For more information, see chapter [How to Edit the Favorites List](#).
2. Add the link to the required picture. For more information, see chapter [How to Edit the Favorites List](#).

How to Edit the Favorites List

You can make the following changes in the Favorites List:

- Add a new link ([picture](#) or [desktop](#)) or [separator](#).
- [Move](#) a link or separator.
- [Remove](#) a link or separator.
- [Rename](#) a link.

How to add a new picture to the Favorites List

To add a new picture to the Favorites List, proceed as follows:

- Right-click a row in the Favorites List and select Insert Picture on the appearing action menu. A link (picture title) to the topmost picture window will appear above the selected row in the Favorites List.

OR

- Click the Insert Picture button in the Favorites List. A link to the topmost picture window will appear in the lowest row in the Favorites List.

OR

- Grab the picture name in a picture window or the picture browser, drag it to the Favorites List and drop it in the desired location.

How to add a new desktop to the Favorites List

To add a new desktop to the Favorites List, proceed as follows:

1. Set the layout of the desktop according to your needs.
2. Right-click a row in the Favorites List and select Insert Desktop on the appearing action menu.
3. In the appearing dialog, enter a name for the desktop link.
4. Click the OK button. The dialog will close and a new desktop link will appear above the selected row in the Favorites List.

OR

1. Set the layout of the desktop according to your needs.
2. Click the Insert Desktop button in the Favorites List.
3. In the appearing dialog, enter a name for the desktop link.
4. Click the OK button. The dialog will close and a new desktop link will appear above the selected row in the Favorites List.

How to add a separator to the Favorites List

To add a separator to the Favorites List, proceed as follows:

- Right-click a row in the Favorites List and select Insert Separator on the appearing action menu. A new separator will appear above the selected row in the Favorites List.

How to move a link or separator in the Favorites List

To move a link or separator in the Favorites List, proceed as follows:

1. Move the pointer on top of the link or separator to be moved and press the left mouse button.
2. While holding down the left mouse button, move the target to the desired location in the Favorites List.
3. When the pointer is in the desired location, release the mouse button. The change will be updated in the Favorites List.

How to remove a link or separator from the Favorites List

To remove a link or separator from the Favorites List, proceed as follows:

1. Right-click the link or separator to be removed and select Remove... on the appearing action menu.
2. In the appearing dialog, click the OK button to confirm. The dialog will close and the target will disappear from the Favorites List.

OR

1. In the Favorites List, select the link or separator to be removed and click the Delete button.
2. In the appearing dialog box, click the OK button to confirm. The dialog will close and the target will disappear from the Favorites List.

NOTE! Some of the links in the Favorites List may have been configured so that they cannot be deleted.

How to rename a link in the Favorites List

To rename a link in the Favorites List, proceed as follows:

1. Right-click the link to be renamed and select Rename... on the appearing action menu.
2. In the appearing dialog, enter a new name for the link. You can also correct a part of the name by moving the cursor to the desired point first.
3. Click the OK button. The dialog will close and the name of the link will be updated in the Favorites List.

NOTE! In editing the Favorites List, you can also use the Cut, Copy and Paste commands found on the action menu.

NOTE! Some of the links in the Favorites List may have been configured so that they cannot be cut or renamed.

1.7.19 How to Use the Picture Browser

Function of the Picture Browser

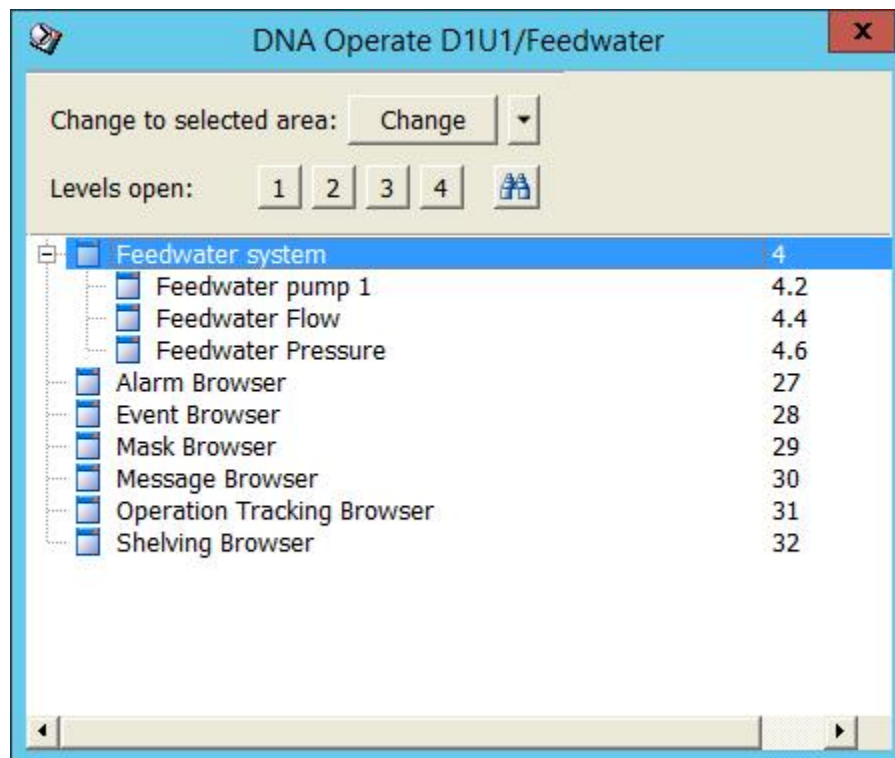
Pictures can be browsed and opened using the picture browser which is primarily designed for seeking infrequently used pictures. The picture browser can be opened from the picture window toolbar. Only one picture browser can be open at a time.

In the picture browser, the picture directory is presented in a tree structure where directory levels are clearly visible. Each picture is referred to with its name and directory number which function as links and open the picture in question when clicked.

How to Open the Picture Browser

Proceed as follows:

- Click the picture browser button  on the picture window toolbar. The picture browser will open on the desktop. The directory that is opened in the picture browser is determined by the picture shown in the picture window. The name of the picture shown in the topmost window is displayed highlighted.



Handling the Picture Browser

The picture browser can be moved on the desktop as any other window by grabbing its title bar. Similarly, its size can be changed by grabbing and dragging its borders or corners. The contents of the picture browser will not scale if the window size is changed. If the contents of the picture directory will not fit in the picture browser, it can be scrolled with the scrollbar. The picture browser is closed by clicking the  button in the browser title bar.

How to Show and Hide Directory Levels

Since a picture directory may contain hundreds of pictures, it may be convenient to hide part of it and view, for example, only one branch at a time. A plus sign (+) in front of a picture name indicates that it has subpictures which are not visible. Similarly, a minus sign (-) in front of a picture name indicates that all lower level pictures connected to it are visible. By clicking a plus or minus sign you can show or hide, respectively, lower or higher level pictures in the picture directory.

To show the pictures of the next level in the picture browser, proceed as follows:

- Click the plus sign (+). The picture names of the next level will become visible and the plus sign (+) will change to a minus sign (-).

To hide the pictures of the next level in the picture browser, proceed as follows:

- Click the minus sign (-). The picture names of the next level will disappear and the minus sign (-) will change to a plus sign (+).

You can also use the arrow keys of the keyboard to move in the picture browser. The UP and DOWN ARROW keys move the highlighted selector in the browser, and the contents of the topmost picture window are updated correspondingly. The LEFT ARROW key moves the selector to the first picture in the directory level or, if the selector is already at the first picture of the level, hides the directory level. The RIGHT ARROW key opens the next directory level if there is a plus sign (+) in front of the selected highlighted picture name.

How to Use the Level Buttons

The levels of a picture directory can be opened and closed one level at a time using the numbered level buttons in the picture browser. The numbers on the level buttons refer to the number of the picture levels they open.

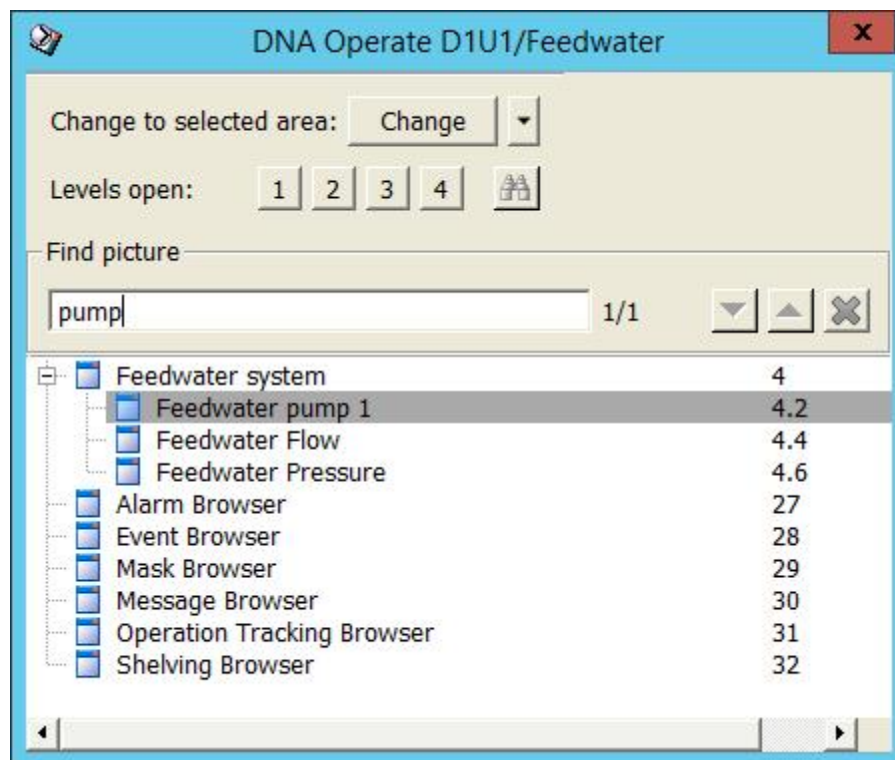
For example, to browse all first and second level pictures, proceed as follows:

- Click the number 2 level button in the picture browser. Picture levels lower than two will close and only the picture names from level one and two will be displayed.

How to Search Pictures in the Picture Browser

Picture directory can contain hundreds of pictures, and it is possible to search pictures in the picture browser.

1. Click  icon to open the search toolbar:



Toolbar can be opened also with F3 key or Ctrl + F or Ctrl + B keys.

2. Write keyword in the field and press Enter. You can browse the items with the arrow buttons on the toolbar or the following keys:

- Next item: Enter, F3 or Ctrl + F
- Previous item: Ctrl + B

You can close the toolbar with  button or Esc key. If you press Esc twice, picture browser will also be closed.

How to Open Pictures from the Picture Browser

Proceed as follows:

- Click the desired picture name in the [picture browser](#). The selected picture will open in the topmost picture window and the corresponding picture name will become highlighted in the picture browser.

OR

- Grab the desired picture name in the picture browser, drag it to a picture window already open on the desktop and then drop it in there. The corresponding picture will now open in the chosen picture window.

How to Update the Picture Browser

You can update the picture browser to display the picture directory of the area selected with the control panel area button by clicking the Change button.

If a picture from another picture directory has been opened in the active picture window, the picture browser does not automatically update to display the directory related to the picture. In this case, you can update the contents of the picture

browser by clicking the picture browser button  on the picture window toolbar.

How to Change the Picture Directory in the Picture Browser

You can change the picture directory displayed in the picture browser.

Proceed as follows:

1. Click the arrow  in the picture browser to open the list of the selectable areas.
2. Click the desired area. The list will close and the picture directory of the selected area will be displayed in the picture browser.

1.7.20 How to Use the Event Browser

Function of the Event Browser

The event browser contains a list of alarms and notifications generated by the process or system.

Events are used to monitor the running of the process, to find out the causes of disturbances, and to assess different process situations later. An event refers to a change in a particular state. Events are divided to alarms, notifications, operation tracking events and mask events. An alarm is an exceptional event and indicates an abnormal process status. A notification is an event which belongs to the normal operation of the process. An operation tracking event is an operation which the operator performs to a certain tag. If necessary, the operator can prevent a single event of a certain tag from triggering an alarm (mask events) in which case the event will not be displayed in the event browser.

One line in the event browser represents a single event. The events are displayed in chronological order. In the event browser, the maximum number of events per page is 27. When this number is exceeded, the next events will start a new page. The latest events are always presented on the topmost page. The contents of the older pages can be browsed with the scroll bar or with the wheel button of the mouse.

The maximum number of events presented in the event browser can be configured event-browser-specific. When the event browser list becomes full, old resolved events are deleted automatically. The contents of the event browser can also be [compressed](#) by deleting all resolved events on the current page. Active and unacknowledged events will remain in the browser list.

A new event which requires an acknowledgement is displayed with blinking acknowledgement box. This kind of events are alarms and notifications. When an event is acknowledged, the acknowledgement box stops blinking. The events presented in the event browser can be acknowledged an event, a page or all events at a time.

The priority level of the event is indicated with the number (1-5) and color of symbols in the Priority column:

	5 symbols	High alarm priority
	2-4 symbols	Medium alarm priority
	1 symbol	Low alarm priority

NOTE! Although priority levels 2-4 are by default presented with one color only, plant-specific configurations may have dedicated colors for each priority level.

Priority value (1-1000) is directly proportional to the priority level of the alarm.

When an alarm is active, the symbols of the alarm are red. When the alarm becomes resolved, the symbols of the alarm turn gray.

The symbol of a notification is a thick exclamation point . Similarly to alarms, the priority level of a notification is indicated with the number of notification symbols in the Priority column. Notifications are always displayed in gray.

How to Open the Event Browser

To open the event browser, proceed as follows:

- Click the event browser button  on the control panel toolbar.

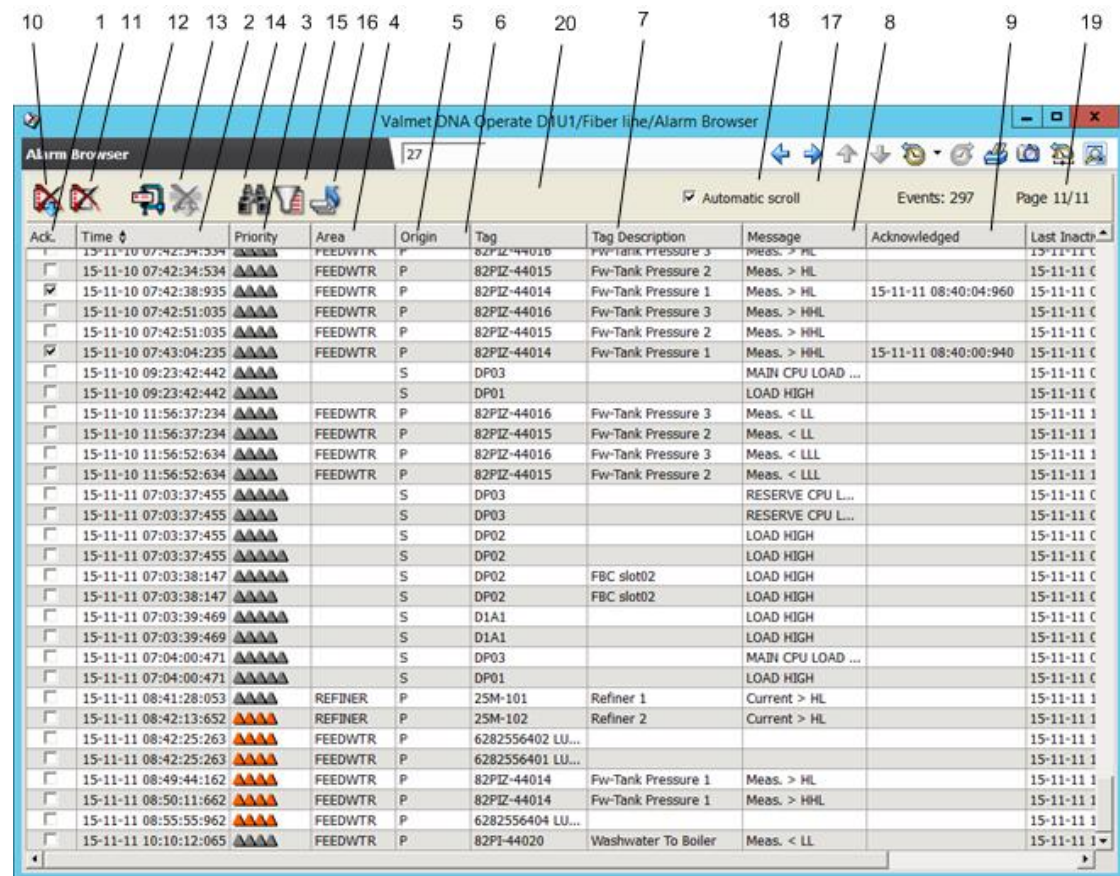
OR

- Press the - key in the numeric keyboard.

The event browser will open in the topmost (active) picture window on the desktop.

Elements of the Event Browser

The event browser contains the following elements:



- Acknowledgement box
- Date and time of event
- Priority symbols
- Event area
- Event type

P	Process event
S	System event
Ext	External system event
- Event tag identifier
- Tag description
- Event data
- Acknowledgement time
- Acknowledgement button for all events at the same time
- Page acknowledgement button
- Hide reset events button
- Button for deleting connection fault events
- Button for opening the Find dialog
- Button for opening the Filter dialog
- Button for opening the sorting criteria definition dialog
- Time display check box (local time/Coordinated Universal Time)
- Automatic scroll check box (on/off)
- Current page number of the event browser
- If events in the event browser are not up to date, a symbol indicating it will be shown here.

In addition to the above mentioned default columns, also some other columns can be configured to be displayed in the event browser, for example Last Inactivated, Priority value, Measurement, Limit, and Process State.

How to Acknowledge an Event in the Event Browser

Similarly to the event stepper, you can acknowledge a single event in the event browser.

Proceed as follows:

- Click the acknowledgement box next to the target event in the event browser.

OR

- Right-click the target event on the event browser and select Acknowledge Event on the action menu.

The acknowledgement box will be checked and it will stop blinking.

How to Acknowledge All Events in the Event Browser

You can acknowledge all events presented in the event browser at one go.

Proceed as follows:

- Click the Acknowledge list button  on the event browser toolbar.

OR

- Right-click the event browser and select Acknowledge List on the action menu.

All events presented in the event browser will be acknowledged.

How to Acknowledge a Page in the Event Browser

You can acknowledge all events presented on one page of the event browser at one go.

Proceed as follows:

- Click the Acknowledge current page button  on the event browser toolbar.

OR

- Right-click the event browser and select Acknowledge Page on the action menu.

All events presented on the page will be acknowledged.

How to Compress the Event Browser

The contents of the event browser can be compressed by removing all resolved events displayed in the event browser so that only active and unacknowledged events will remain.

Proceed as follows:

- Click the Hide reset events button  on the event browser toolbar.

OR

- Right-click the event browser and select Hide reset events on the action menu.

Only active and unacknowledged events will remain on the page. In addition, to indicate that the event browser is compressed, the button  on the event browser toolbar will be replaced by the  button.

Hiding Events with Connection Faults

In the event browser, it is possible to hide all events which are caused by a connection fault, for example, a connection break to the process control server, and thus are faulty and unreliable information. When such events are displayed in the event browser, the Remove unreliable events button  on the event browser toolbar becomes active () indicating that the button can be used. In addition, priority symbols of those faulty event lines are highlighted with the fault color (violet).

Proceed as follows:

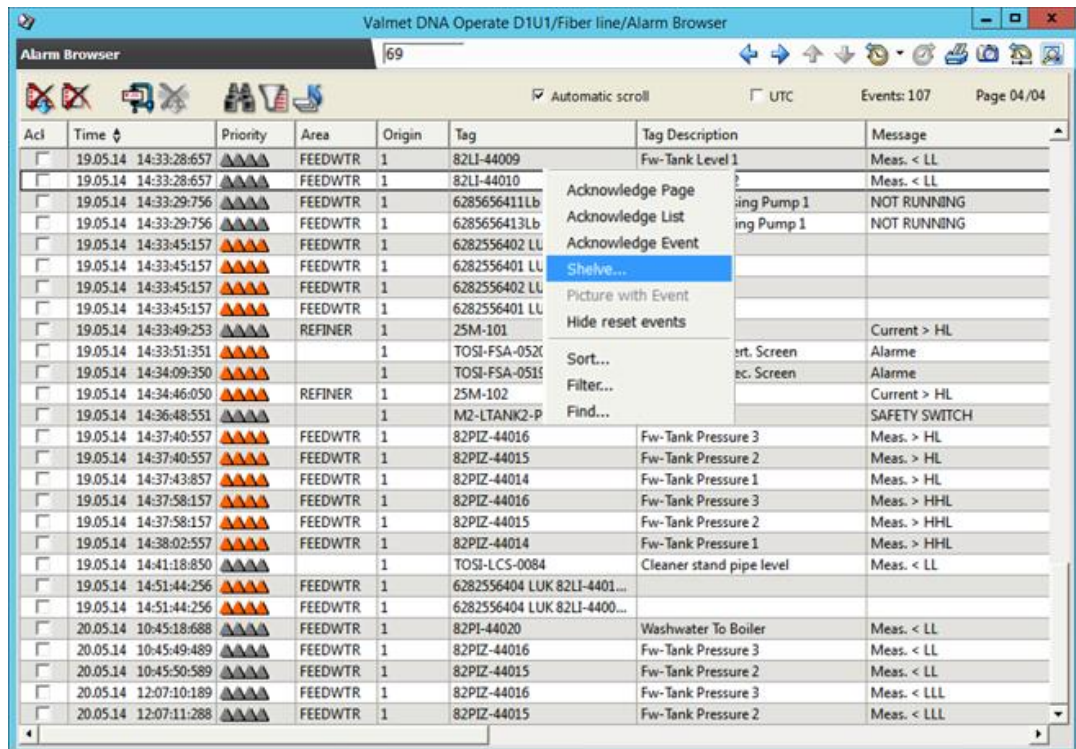
- Click the Remove unreliable events button  on the event browser toolbar.

All events caused by a connection fault and displayed in the event browser are hidden. Also the remove button appears dimmed again () and cannot be selected.

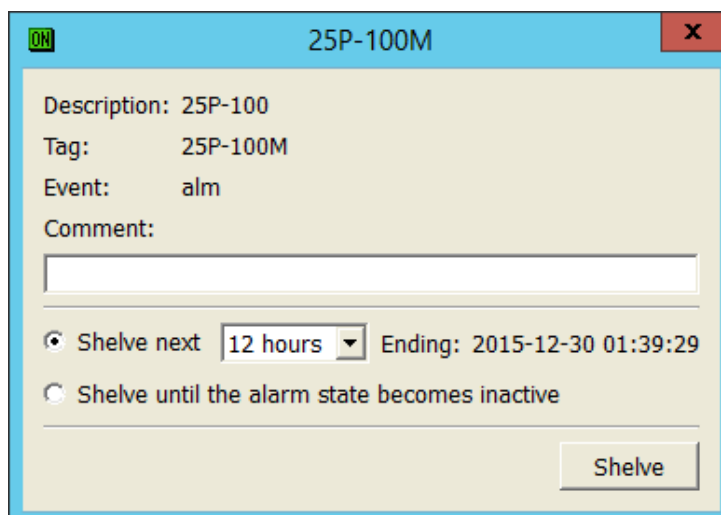
How to Shelf an Alarm in the Event Browser

To shelve an alarm, proceed as follows:

1. Right-click on the alarm in the event browser and select **Shelve...** on the action menu.



Shelving dialog opens to the desktop.

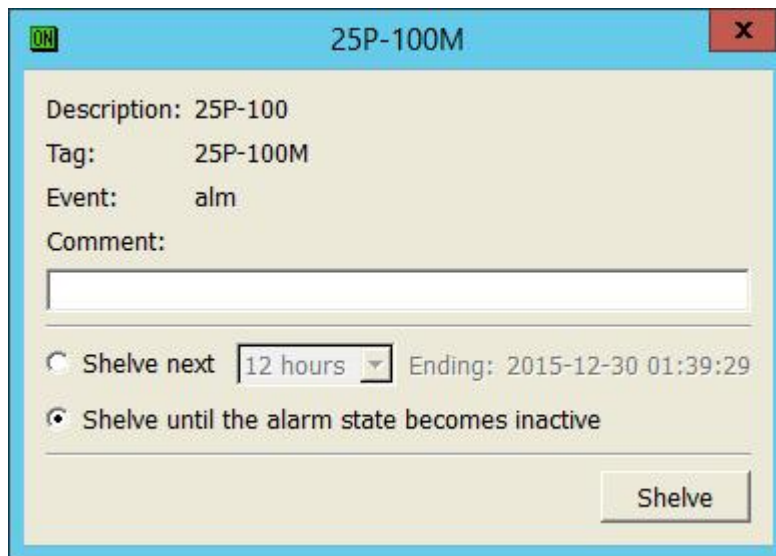


2. Add more details about the shelving to the Comment field when necessary.
3. Define the duration of shelving as follows:
 - Select the duration of shelving by clicking **Shelve next** and choose duration from the drop-down list. Options are 1, 2, 4, 8 or 12 hours.

The dialog shows the expiry date and time of shelving.

OR

- Select one shot shelving by clicking Shelf until the alarm state becomes inactive.



The screenshot shows a dialog box titled '25P-100M' with a blue header bar. In the top left corner of the header is a green 'ON' indicator, and in the top right is a red 'X' close button. The main content area has a light beige background and contains the following fields: 'Description: 25P-100', 'Tag: 25P-100M', 'Event: alm', and 'Comment:' followed by an empty text input box. Below these fields are two radio button options: 'Shelve next' with a dropdown menu set to '12 hours' and 'Ending: 2015-12-30 01:39:29', and 'Shelve until the alarm state becomes inactive' which is currently selected. A 'Shelve' button is located in the bottom right corner of the dialog.

4. Accept changes by clicking Shelf.

NOTE! Shelving functions are not in use when the event browser is in history mode.

The shelved alarms can be found in the [shelved alarms list](#), where the shelving can also be manually deleted.

Shelved Alarms List

The list contains the following elements:

Tag	Tag Description	Message	Pri. Lev	Area	Shelving Time	Unshelving Time	Acto	Comment	Time
2SP-100N	Feed pump	Release on 2	🔴🔴🔴	REFINER	15-12-30 06:58:19:179	Next Inactivation	dna	maintenance ongoing	15-12-30 06:47:26:617
2SP-100	Feed pump	Release on 1	🔴🔴🔴	REFINER	15-12-30 07:01:06:192	15-12-30 19:00:47:381	dna	controller maintenance	15-12-30 06:47:26:515

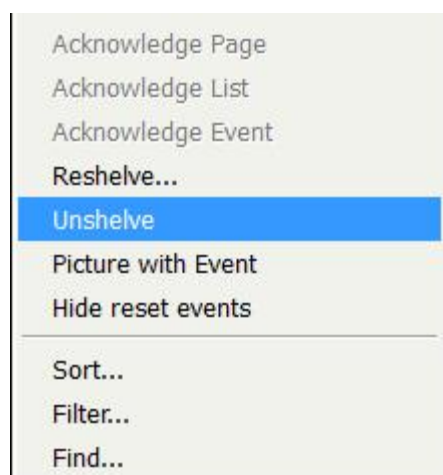
1. Shelving time
2. Unshelving time (in the case of one shot shelving, the field has the text Next Inactivation)
3. User who shelved the alarm
4. User comment (ie. the reason for shelving)
5. Date and time of alarm

Sorting, filtering and search are done with the event browser functions.

Unshelving an Alarm

To unshelve an alarm, proceed as follows:

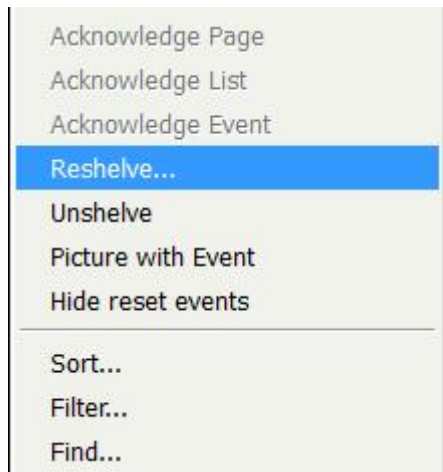
- Right-click the alarm in the event browser and select Unshelve in the action menu.



Reshelving an Alarm

To reshelve and alarm, proceed as follows:

- Right-click the alarm in the event browser and select Reshelve... in the action menu.



Searching for Events in the Event Browser

You can search for particular events in the event browser based on different search conditions.

Proceed as follows:

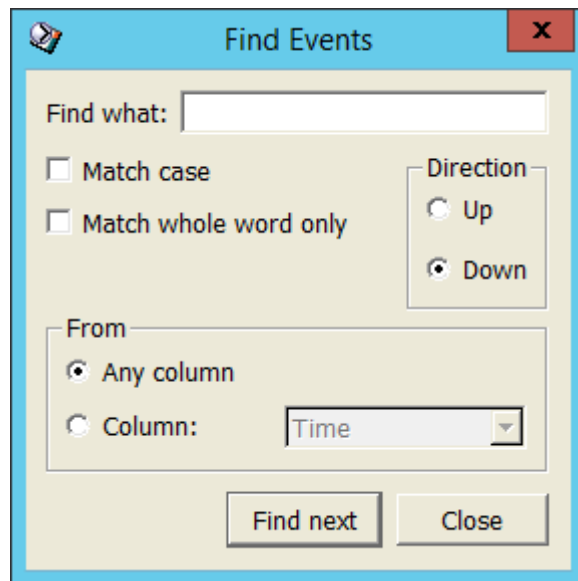
1. Open the Find Events dialog in one of the following ways:

- Click the Find button  on the event browser toolbar.

OR

- Right-click the event browser and select Find... on the action menu.

Find Events dialog will open on the desktop.



2. In the Find what field, enter the text to be searched or a part of it. Wildcard characters (? and *) can be used in the field. The asterisk (*) denotes an unknown or possibly missing part of the value. The question mark (?) denotes any single character.
3. If you want to search for events which have data that contains the entered text (or a part of it) typed exactly the same way as in the search condition including small and capital letters, click the Match case check box. If if you want to search for events that have data which contains the entered text as a whole word, click the Match whole word only check box.
4. In the Direction frame, select the search direction, i.e. whether to search for events downwards (Down) or upwards (Up) in the event browser. Click the desired option button.
5. If you want to search for the entered text in any column of the event browser, click the Any column option button in the From frame. However, if you want to search for the entered text only in a certain column of the event browser, click the Column option button and select the desired column from the drop-down list by clicking it.
6. Start the search by clicking the Find next button. The cursor of the event browser moves to the column the event of which matches the search conditions and the column will be highlighted by blue background color.
7. You will find the next event matching the search conditions by clicking the Find next button again. Repeat this step as many times as you wish.

Filtering Events in the Event Browser

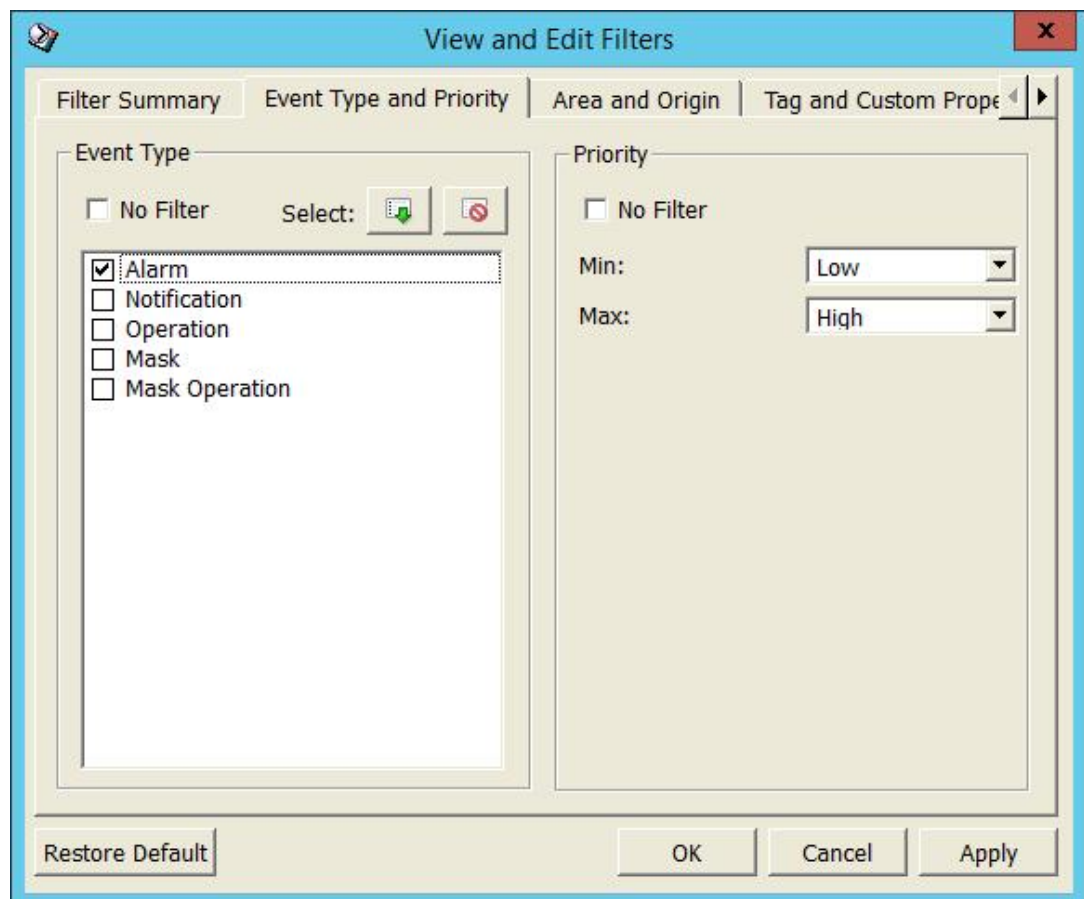
In the View and Edit Filters dialog, you can set different filtering conditions according to which you can limit the events shown in the event browser. To indicate that the events shown in the event browser are filtered, i.e. the event list of the

event browser does not show all the available events, the Edit filters  button will be highlighted by yellow background color.

Proceed as follows:

1. Open the View and Edit Filters dialog in one of the following ways:
 - Click the Edit filters button  on the event browser toolbar.
- OR
- Right-click the event browser and select Filter... on the action menu.

View and Edit Filters dialog will open on the desktop.



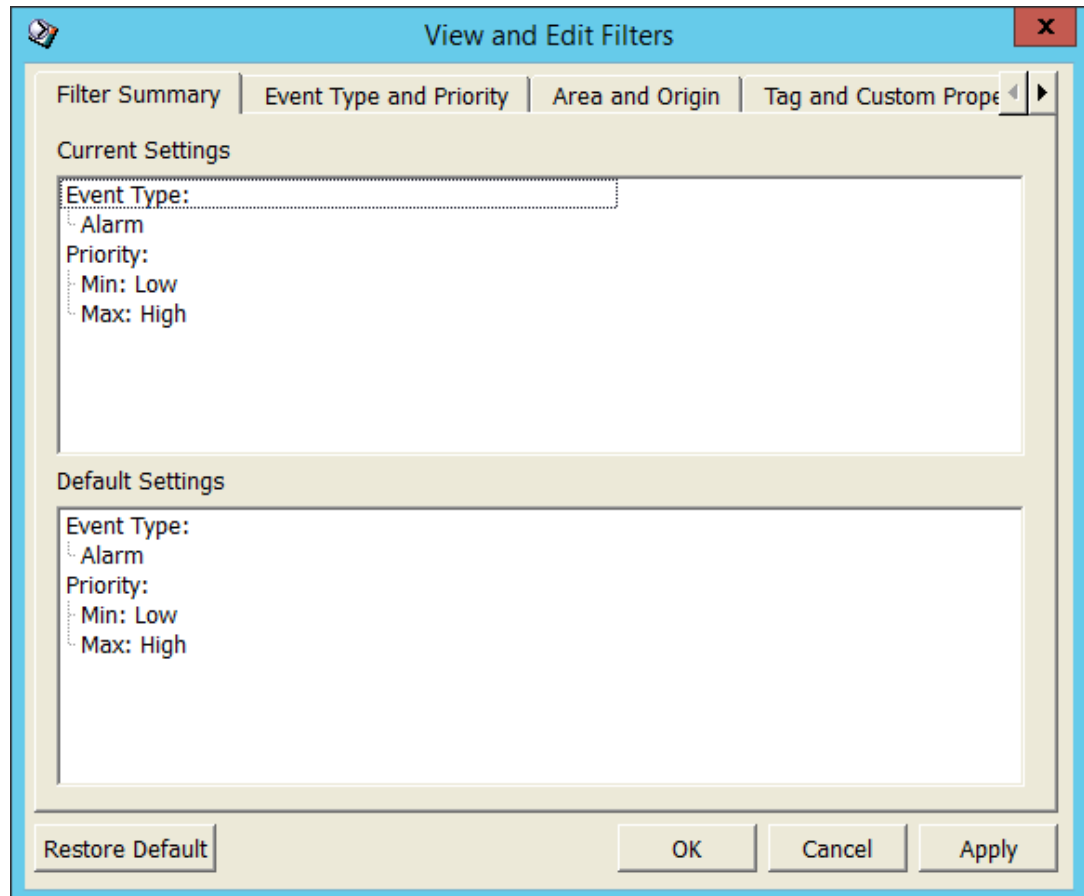
By default, events shown in the event browser are limited only according to the event type and priority as shown in the previous figure. If necessary, this default state can be restored later by clicking the dialog's Restore Default button.

Clicking Apply will apply the changed settings without closing the dialog.

View and Edit Filters dialog consists of the following four tabs:

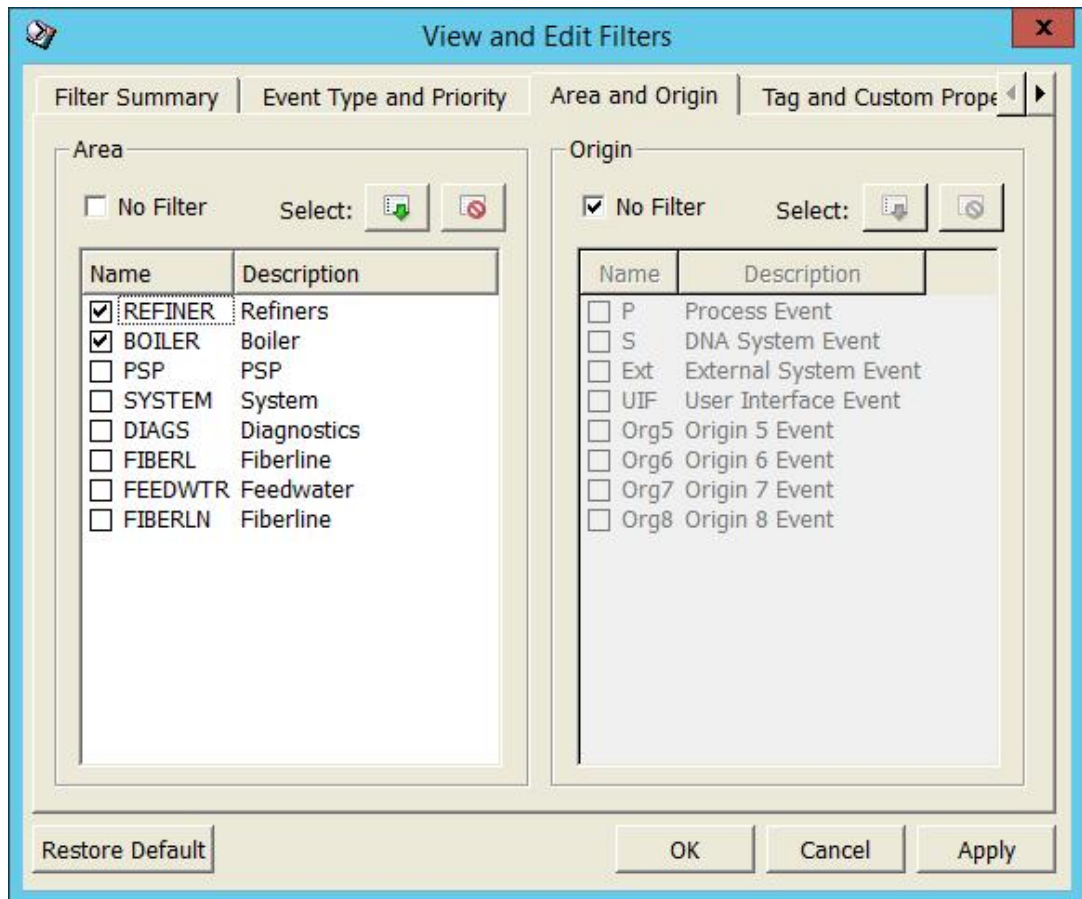
- Event Type and Priority
- Area and Origin
- Tag and Free Criteria
- Filter Summary

In the three first tabs, you can define the used filters, and in the fourth, Filter Summary tab, you can see a summary of valid filters and their filtering conditions as shown in the following figure. In Default Settings field the original filtering conditions are shown.

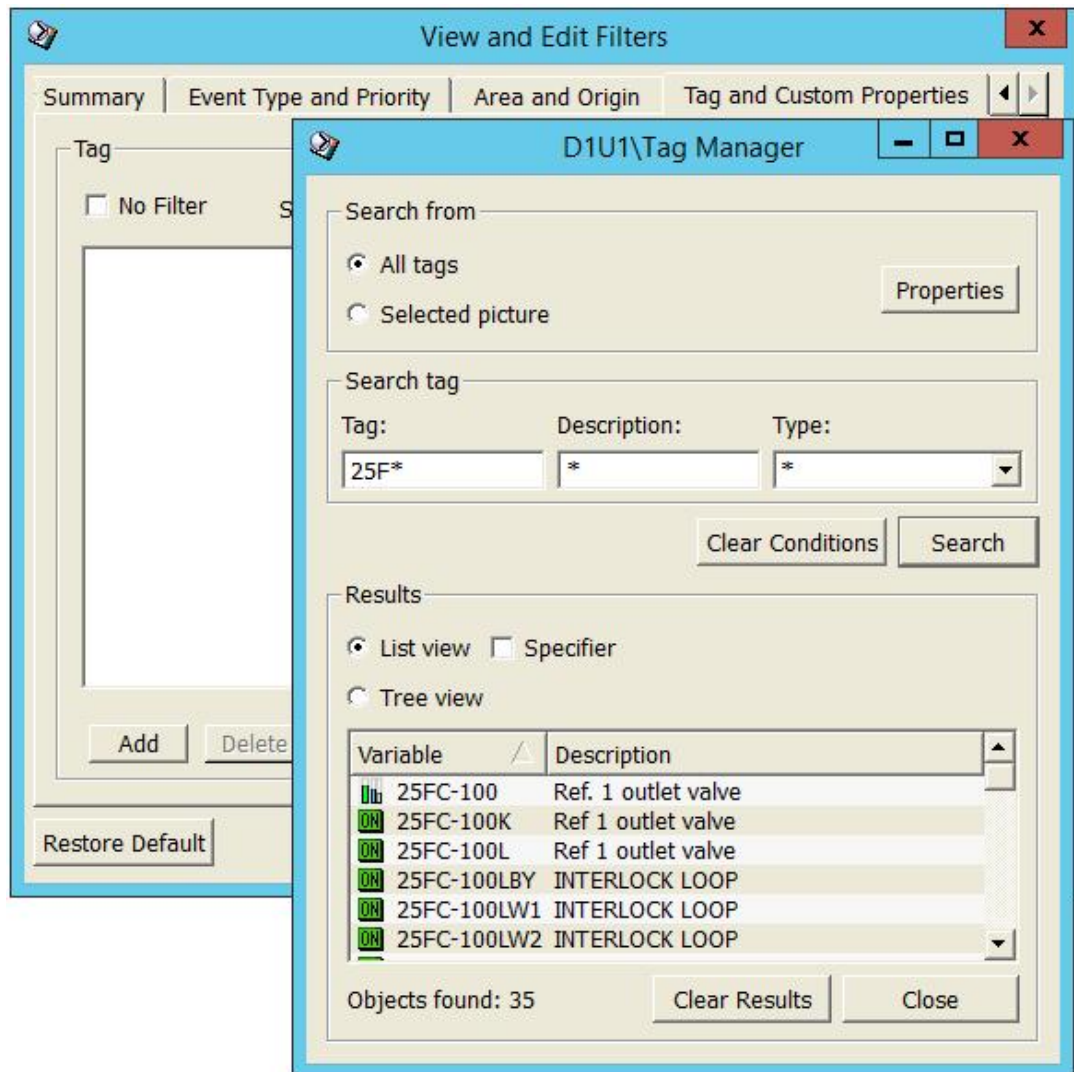


2. Click to open the tab the filters of which you want to use. Events can be filtered according to the following filters: event type, event priority, event area, event origin, tag or free group.

3. Select the desired items from the list of each used filter by clicking the corresponding check boxes. For example, by clicking the Refiner and Boiler check boxes in the list of the Area filter, all events which match the filtering conditions in those areas are shown in the event browser. If you want to select all list items at once, click the Select all button (). If you want to deselect all selections, click the Select none () button.

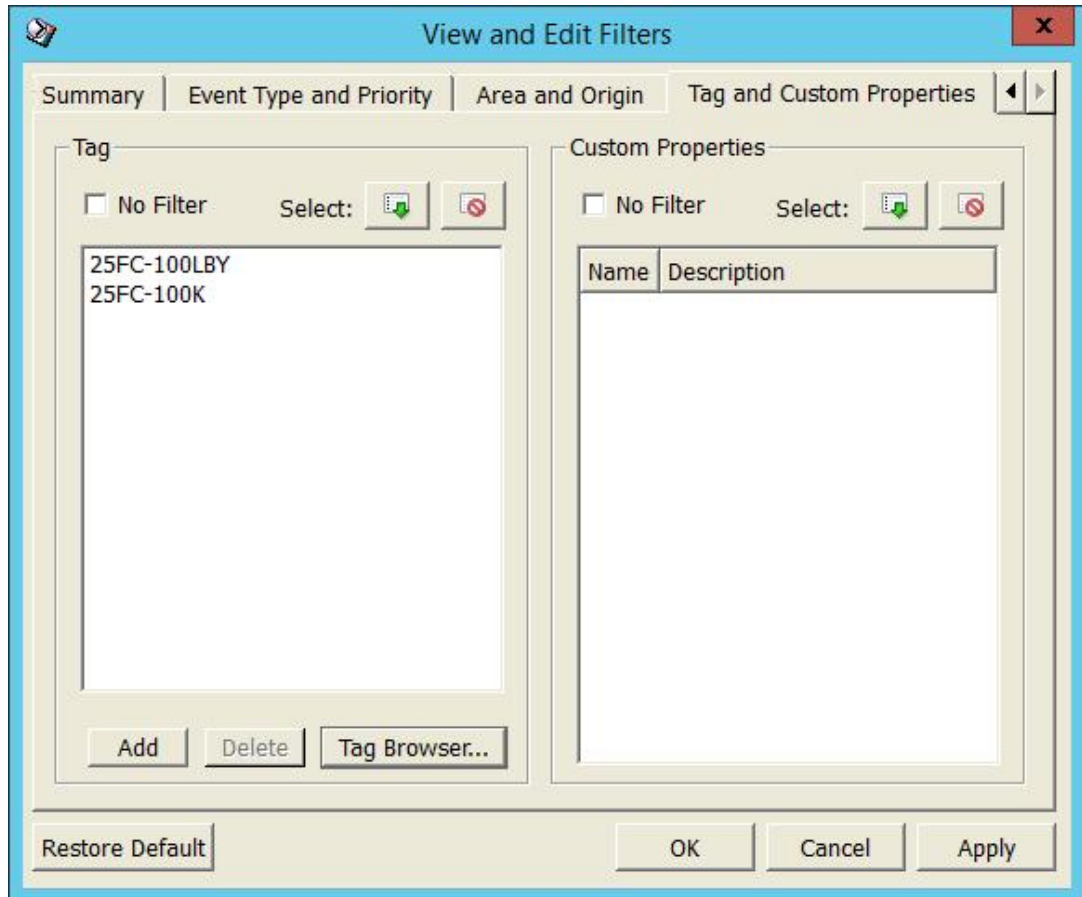


If you want to filter according to the tag, notice that you must add the tags whose events you want show in the event browser to the Tag list through Tag Browser. For more information on tag browser, see chapter [How to Use the Action Menu -> Picture -> Tag Manager](#).

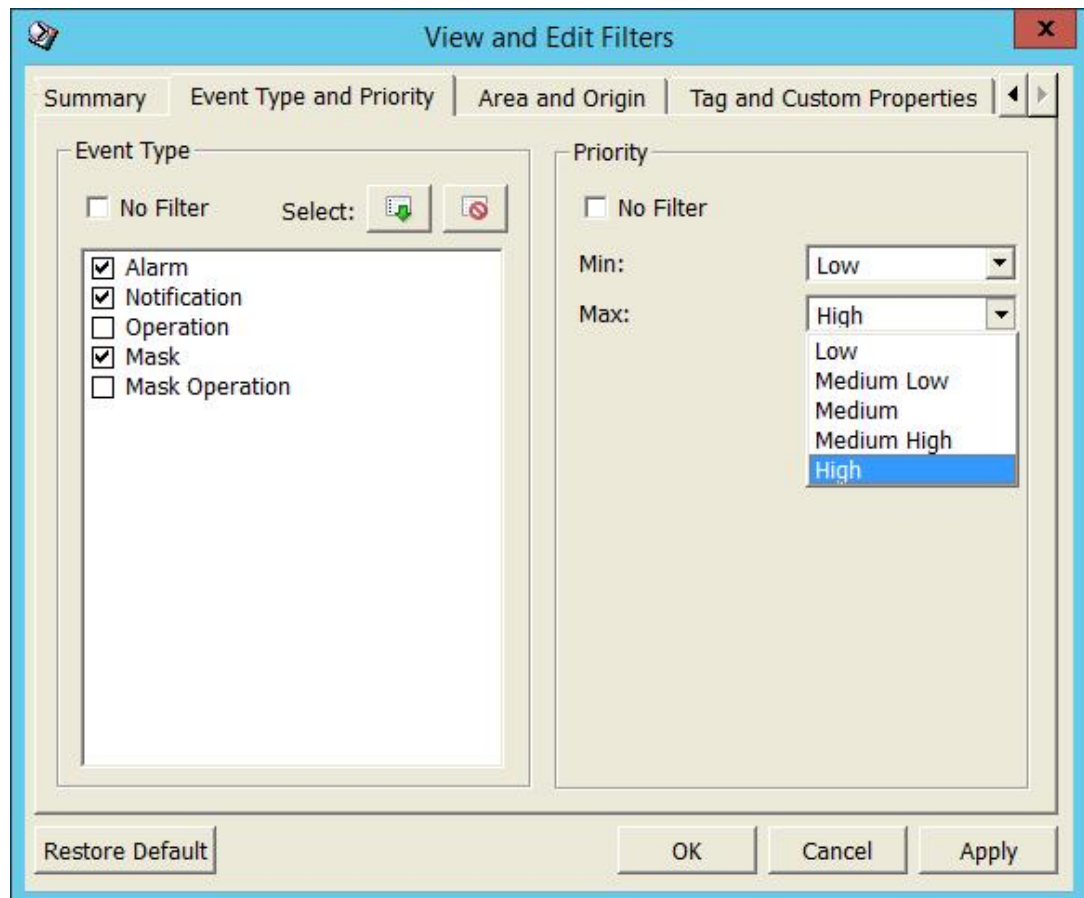


Tags can be added to the list by dragging and dropping from the tag manager to the filter dialog.

You can also manually add tags to the list by clicking Add and the cursor appears in Tag list and you can write the name on the list. The name can be modified by selecting it on the list. Delete button removes the tag from the list. If you want to select all list items at once, click the Select All button (). If you want to deselect all selections, click the Select none () button.



If you want to filter according to the priority, select the event priority limits through Min and Max drop-down lists.



To indicate that the used filters differ from the default ones, a small yellow symbol will appear next to the tab name.

Note that when the No Filter check box is selected, all the corresponding filter items are disabled.

4. Finally, click the dialog's OK button. The event browser will be updated to show only the events that match all the conditions defined in the previous step (3).

Sorting Events in the Event Browser

There are two ways to sort events in the event browser:

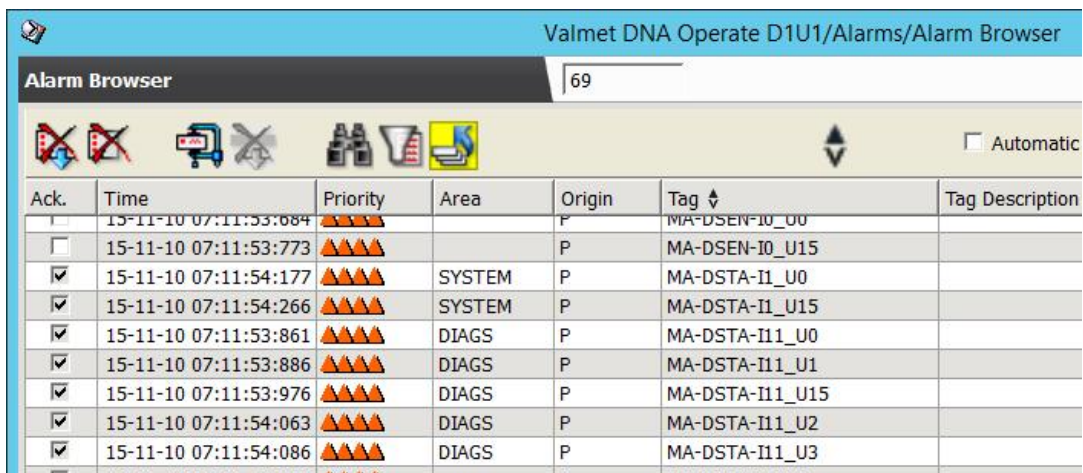
- [the use of quick sorting](#) (sorting according to only one column)
- [the use of the Sort... dialog](#) (sorting according to one or several columns)

Quick Sorting

If there is no need to use several sorting conditions (columns) at the same time, the easiest way to sort the events in the event browser is to use the quick sorting.

Proceed as follows:

1. Click the column header according to which you want to sort the events. In the column header line, a symbol  will appear and the events in the event browser are sorted in ascending order. Also, a symbol  will appear on the event browser toolbar, and the Edit sort conditions button  will be highlighted by yellow background color to indicate that events are sorted.



Ack.	Time	Priority	Area	Origin	Tag	Tag Description
☐	15-11-10 07:11:53:684	▲▲▲▲		P	MA-DSEN-I0_U0	
☐	15-11-10 07:11:53:773	▲▲▲▲		P	MA-DSEN-I0_U15	
☒	15-11-10 07:11:54:177	▲▲▲▲	SYSTEM	P	MA-DSTA-I1_U0	
☒	15-11-10 07:11:54:266	▲▲▲▲	SYSTEM	P	MA-DSTA-I1_U15	
☒	15-11-10 07:11:53:861	▲▲▲▲	DIAGS	P	MA-DSTA-I11_U0	
☒	15-11-10 07:11:53:886	▲▲▲▲	DIAGS	P	MA-DSTA-I11_U1	
☒	15-11-10 07:11:53:976	▲▲▲▲	DIAGS	P	MA-DSTA-I11_U15	
☒	15-11-10 07:11:54:063	▲▲▲▲	DIAGS	P	MA-DSTA-I11_U2	
☒	15-11-10 07:11:54:086	▲▲▲▲	DIAGS	P	MA-DSTA-I11_U3	

2. However, if you want to sort the events in descending order, click the same column header again. Now a symbol  will appear in the column header line and the events in the event browser are sorted in descending order.
3. To remove the selected sorting condition and to restore the event browser in the default state, click the column header for the third time. By default, events in the event browser are shown in chronological order (from the oldest to the newest).

The Use of the Sort Dialog

In the Sort... dialog, you can set several different sorting conditions according to which you can organize the events in the event browser. To indicate that some

sorting conditions used for sorting are not shown in the event browser, a symbol  will appear on the event browser toolbar, and the Edit sort conditions button  will be highlighted by yellow background color.

Proceed as follows:

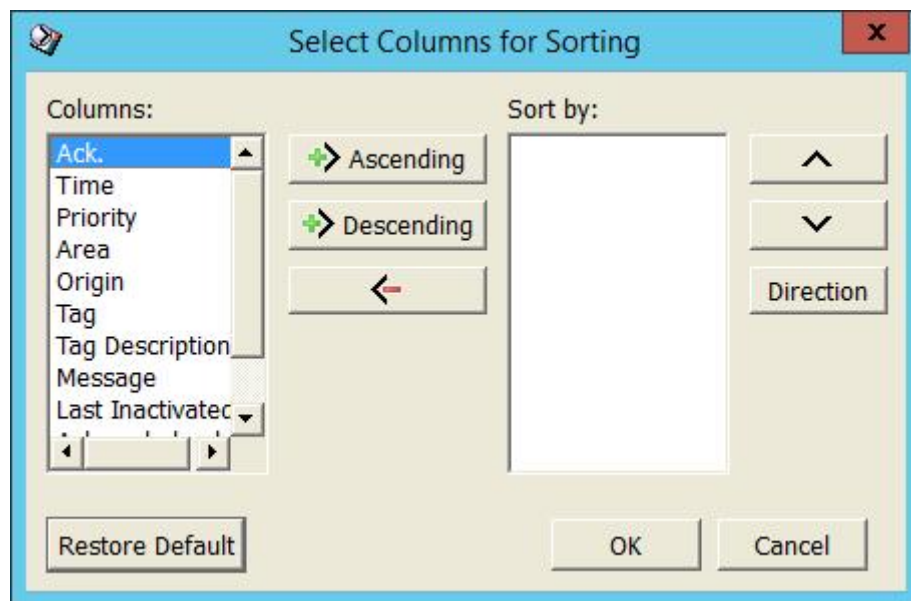
1. Open the Select Columns for Sorting dialog in one of the following ways:

- Click the Edit sort conditions button  on the event browser toolbar.

OR

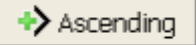
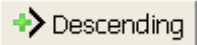
- Right-click the event browser and select Sort... on the action menu.

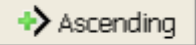
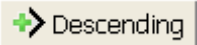
Select Columns for Sorting dialog will open on the desktop.

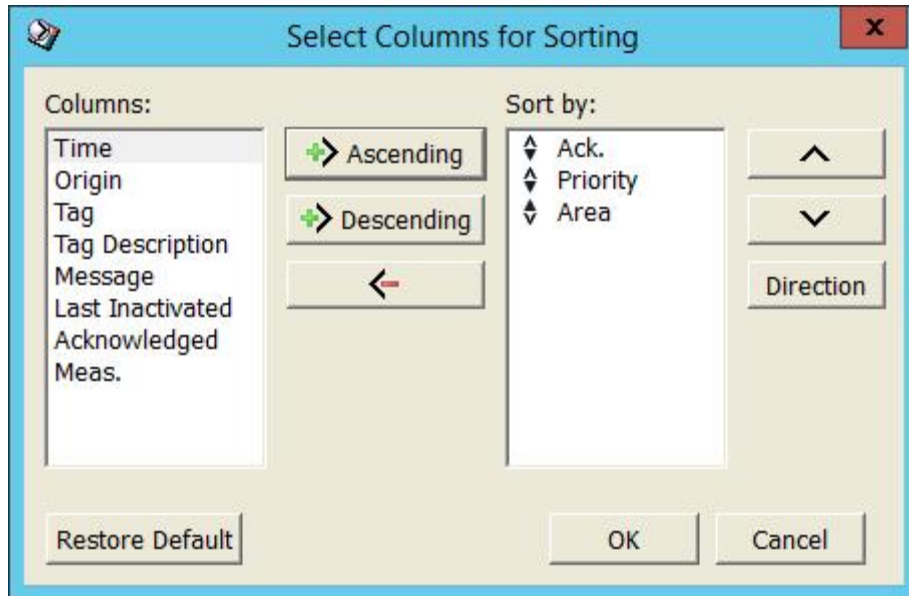


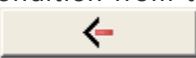
By default, events in the event browser are shown in chronological order (from the oldest to the newest). If necessary, this default state can be restored later by clicking the dialog's Restore Default button.

The Columns list contains all the sorting conditions (columns) according to which events in the event browser can be sorted. Sort by list, instead, shows the sorting conditions (columns) according to which events in the event browser are sorted. The uppermost condition in the Sort by list is the primary sorting condition, the second condition is the secondary condition, etc.

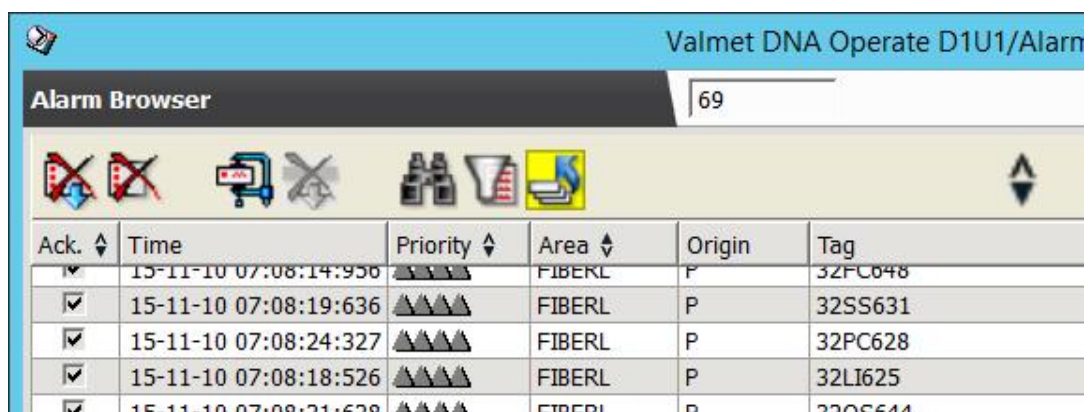
2. From the Columns list, select a column according to which you want to sort the events in the event browser. Click the  Ascending or  Descending button to move the selected column to the Sort by list.

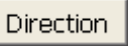
Clicking either the  or  button depending on the fact whether you want to sort the column in ascending or descending order. Repeat this until you have moved all the desired sorting conditions to the Sort by list. The following figure shows how the dialog could look like:



3. To remove a sorting condition from the Sort by list, first click the condition to be removed and then click the  button. To restore the default state, click the dialog's Restore default button.
4. Organize the sorting conditions in the Sort by list according to their priority by using the  and  buttons. The uppermost condition in the Sort by list is the primary sorting condition, the second condition is the secondary condition, etc. The sorting order is displayed in the event browser with

different-sized arrow symbols  located in the column header line. The bigger the symbol is, the more important the sorting condition is. The direction of the arrow point also indicates whether the sorting order is ascending or descending.



5. In the Sort by list, change the direction of the sorting condition (ascending / descending) with  button. The sorting direction can be changed also in the event browser by clicking the desired column header.
6. Start the sorting by clicking the OK button.

How to Change the Column Width

You can change the column width on the event browser, for example, to hide the data in an unnecessary column or to view all the information on a certain column.

Proceed as follows:

1. Point to the boundary on the right side of the column heading. The pointer becomes a .
2. Grab the boundary and drag it until the column is the width you want.
3. Drop the boundary. The width of the column will change accordingly.

You can also directly adjust the column width to fit the contents.

Proceed as follows:

- Double-click the boundary to the right of the target column heading. The width of the column will change to match the information displayed on the column.

How to Open a Picture from the Event Browser

You can open a picture related to an event presented in the event browser.

Proceed as follows:

- In the event browser, double-click an event to open a picture related to it.

OR

- Right-click the target event in the event browser and select Picture with Event on the action menu.

OR

- When the target event in the event browser is already selected (indicated by a frame) press Enter.

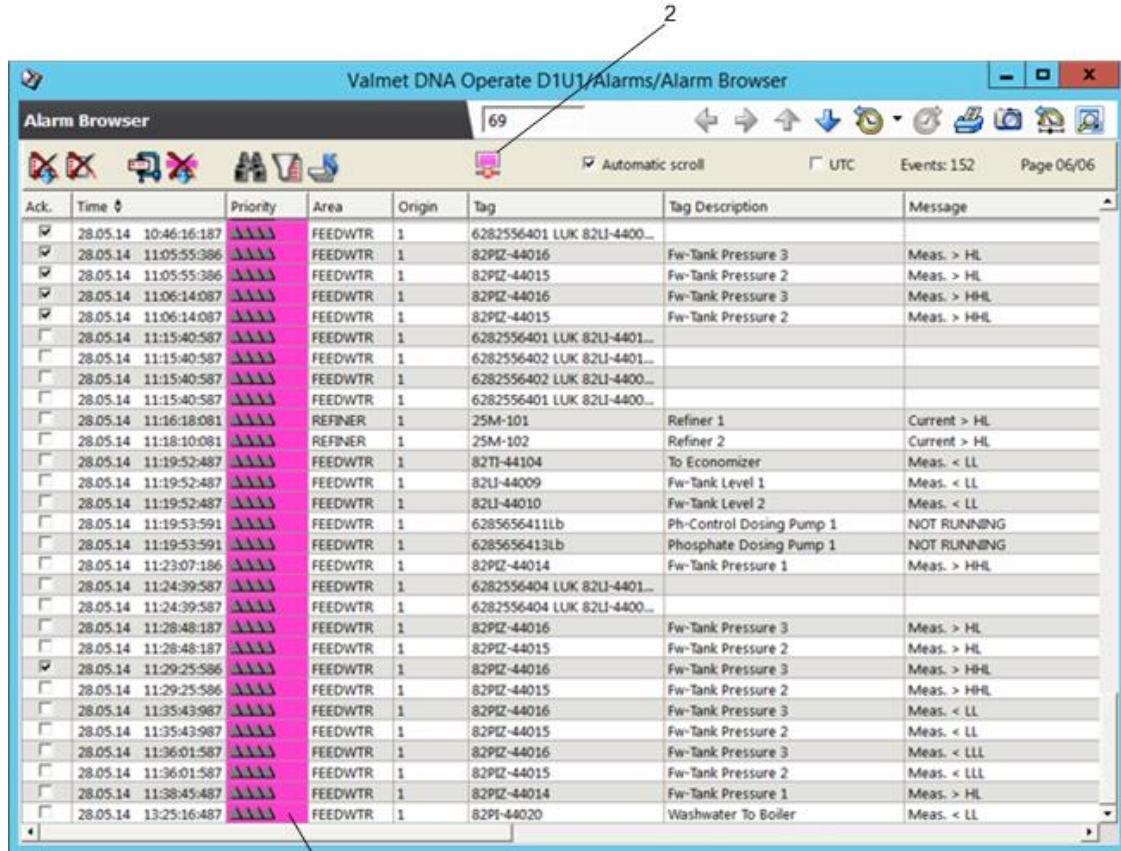
In all cases, the following dialog will open to the desktop. In it, you can select whether the related picture is opened in a new picture window or in the topmost picture window on the desktop.



The picture will open in the chosen picture window.

Events in the Event Browser Not Up to Date

If the connection to alarm server is disconnected, the events in the event browser are not up to date. The fault is indicated with violet color and symbols in control panel (1) and the event browser (2 and 3).



1.7.21 History Mode

Function of the History Mode

In the history mode, the run of the process can be examined and analyzed retrospectively. The history mode provides exact information on the run of the process in the desired interval which enables efficient later analysis. It can also be used as a training aid in demonstrating the development of disturbances. Potential problem situations can thus be learned to identify in time which decreases the number of breaks in production.

History mode operations are executed in the Timeline dialog.



How to Enter the History Mode

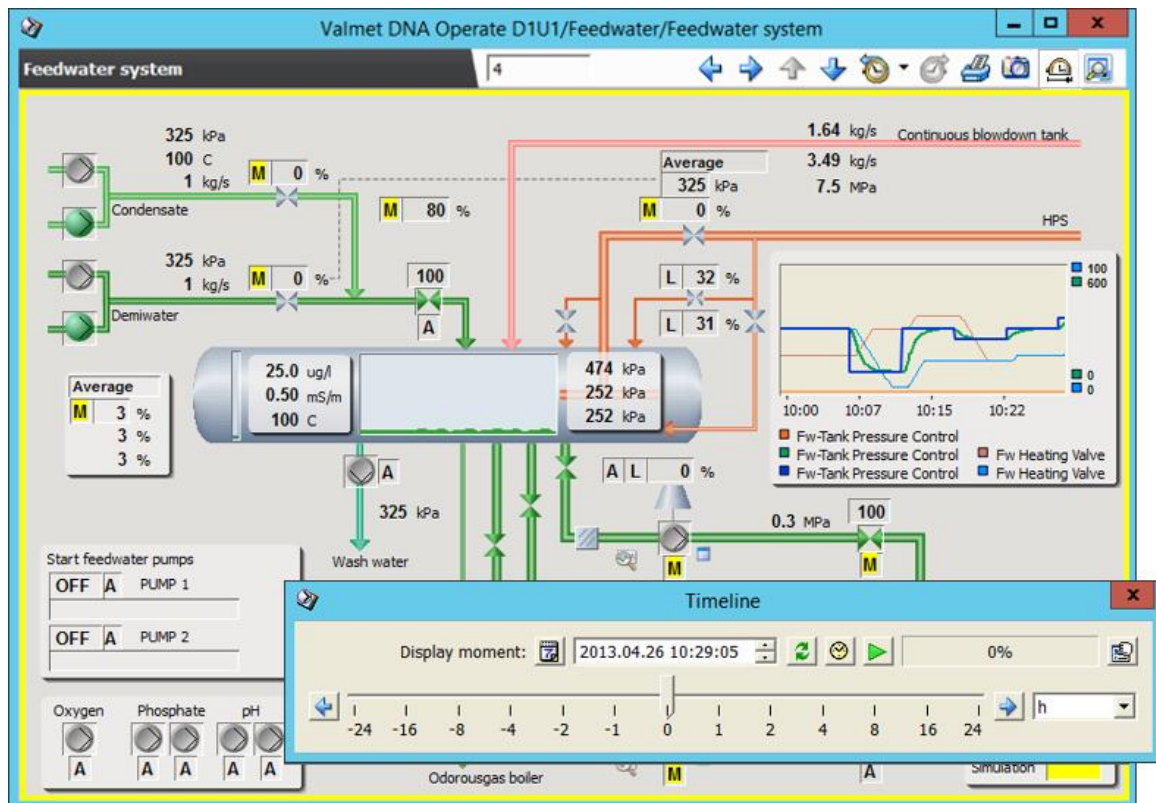
To enter the history mode, proceed as follows:

- Click the history mode button  on the picture window toolbar.

OR

- Right-click the picture window and select History Mode on the submenu of the Picture... command.

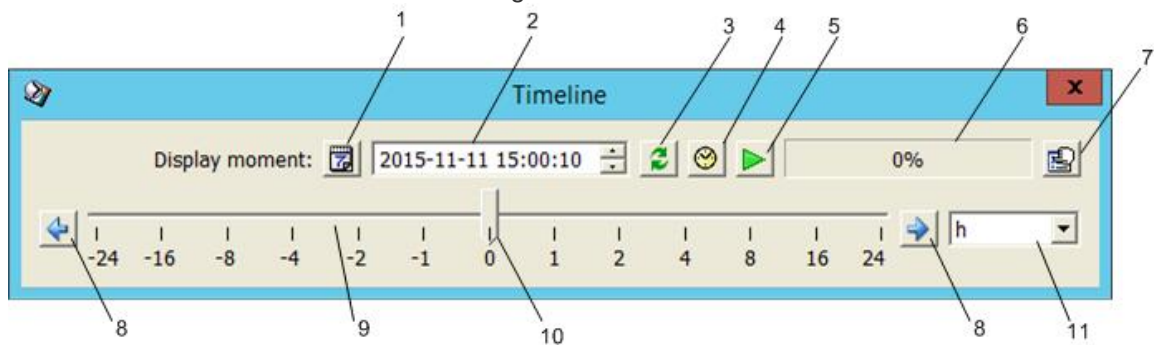
The timeline dialog will open on the desktop and the picture area of the picture window will be highlighted in yellow attention color.



Timeline

Elements of the Timeline

The Timeline consists of the following elements:



1. Display moment calendar button
2. Date and time box
3. Refresh button
4. Current time button
5. Replay button
6. Replay progress as per cent
7. Properties
8. Timeline step buttons
9. Timeline gauge
10. Slider
11. Slider unit

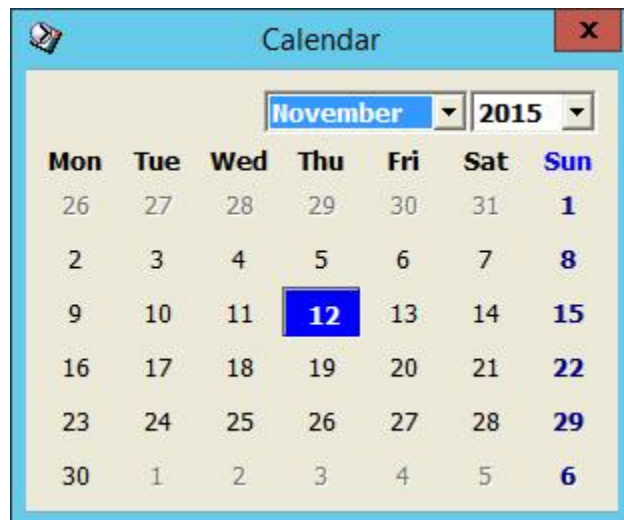
How to Set the Display Moment

Display moment defines the exact time of the data presented in the picture window. The date can be set either by selecting it with the calendar function or by defining it in the date box. The time is defined in the same box.

How to set the date with the calendar

Proceed as follows:

1. Click the display moment calendar button . The Calendar dialog will open on the desktop.



2. Select the desired month and year in the corresponding box.
3. Click to select the day.
4. Click the  button to close the dialog.
5. Click the Refresh button  if you do not want to change the time.

How to define the display moment in date and time box

Proceed as follows:

1. Click the value to be changed in the date and time box.
2. Do one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.
3. Click the Refresh button . The picture windows presented in history mode will be updated to match the new display moment.

Alternatively, you can set the display moment to the current time by clicking the clock button .

How to Browse History Data

History data can be quickly browsed with the timeline slider both forward and backward from the display moment.

Proceed as follows:

1. Click the slider unit and select the desired unit (s/min/h/d/step) from the list. The scale of the gauge will change accordingly.
2. Grab the slider and move it on the desired location on the gauge. The time box is updated with each step the slider moves.
3. Release the slider. It returns back to the center of the gauge, and the picture windows presented in history mode are updated to match the new display moment.

Alternatively, you can browse the close proximity of the current display moment.

Proceed as follows:

- Click the left step button  to decrease or the right step button  to increase the value of the display moment. Each click alters the display moment by the smallest possible step determined by the scale.

Replay

How to Set the Start and End Time

The start time is used together with the time span to define the interval in which the replay moment can be selected. The interval can be defined with the accuracy of 5 seconds.

When the history mode is entered, the transition moment determines the values of the end time and the replay moment. The time span value is locked and at its default value of 8 hours. The value of the start time is determined by the end time and the time span (i.e. end time - span value).

To set the start time value in the date and time box, proceed as follows:

1. Click the value to be changed in the date and time box.
2. Do one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.

The date can also be set by the use of the [calendar function](#).

3. Define the duration of replay in Timeline Properties dialog by setting the values in Shift frequency and Shift by fields and define step limit. See [How to Set the Replay Properties](#).

How to Set the Time Span

Time span is used together with the start time to define the interval in which the replay moment can be selected.

Proceed as follows:

1. Open Timeline Properties dialog by clicking . Click the value in the Time span box indicating the time in hours (h) and give the desired time span in one of the following ways:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.
2. Click the Apply button. The changes will become valid.

The scale of the gauge may change when the time span is changed.

How to Set the Replay Moment

With the Replay moment settings, you can define the event time for the data presented in the picture window.

Proceed as follows:

1. Click the value to be changed in the date and time box.
2. Do one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.
3. Click the Refresh button . The picture windows presented in history mode will be updated to display the new replay moment.

You can define the date also with the [calendar function](#).

The replay moment can also be changed by grabbing the slider and moving it to the desired location on the timeline gauge, or by clicking either step button which changes the display moment by the shift value defined in the [Timeline Properties](#).

Changing the replay moment settings will directly affect the location of the slider on the timeline gauge and vice versa.

How to Replay the Event History

Changes that have taken place in process during a specific interval can be retrospectively replayed and examined. This enables, for example, the analysis of the changes occurred in device statuses and their interrelations before the actual disturbance.

To start the replay from the [replay moment](#), proceed as follows:

- Click the Replay button  in the replay view. It will change to the Pause button  and the replay will begin in picture windows which are in history mode.

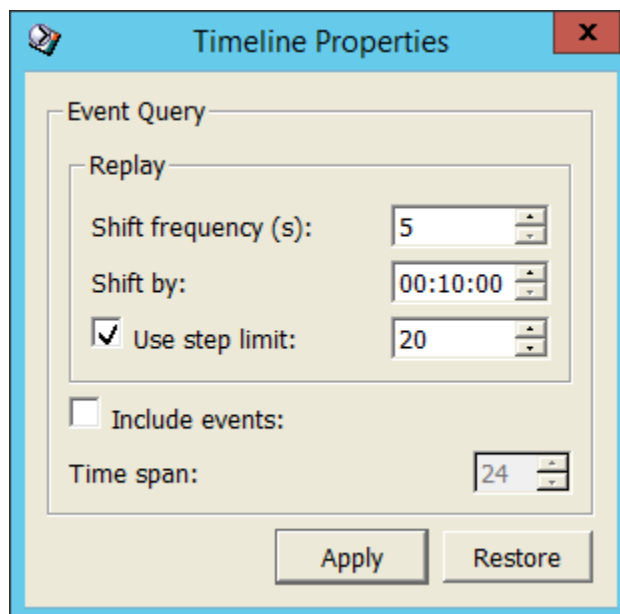
To stop the replay, proceed as follows:

- Click the Pause button  in the replay view. It will change to the Replay button  and the replay will stop.

Timeline Properties

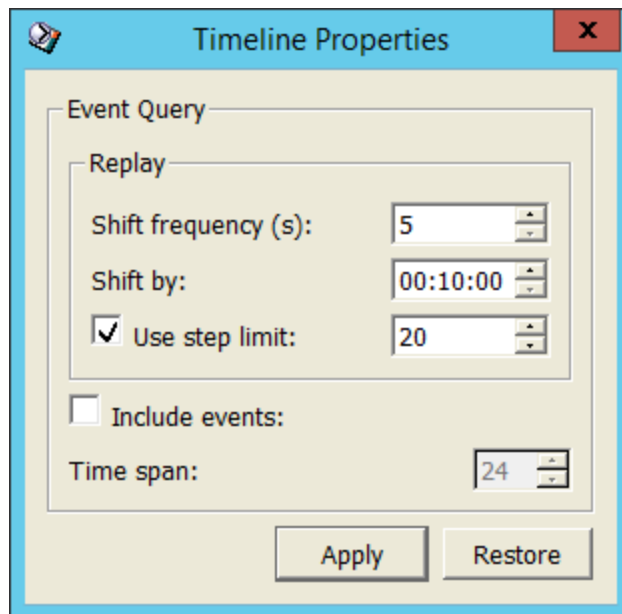
Clicking  on the Timeline dialog, Timeline Properties dialog opens.

In the Timeline Properties dialog, it is possible to define properties related to the replay view and event queries.



How to Set the Replay Properties

In the Timeline Properties dialog, you can define the frequency and magnitude of the shift, as well as optionally limit the number of steps.



Proceed as follows:

1. Click the Shift frequency box and select one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.

The minimum and maximum values are 5 s and 60 s respectively.

2. In the Shift by box, click the value (h:min:sec) to be changed and do one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.
3. Click the step limit box and do one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.

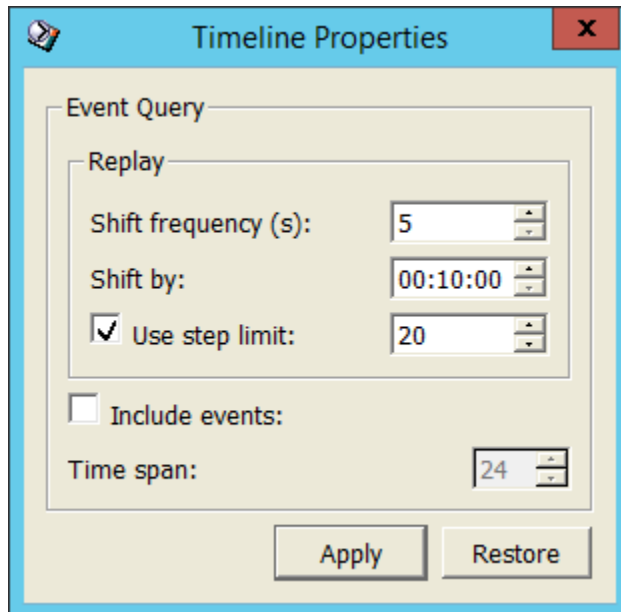
The minimum and maximum values of the number of steps are 1 and 100 respectively.

4. By default, the step limit is in use and the corresponding check box is selected. If you don't want to limit the length of the replay, click to clear the check box.
5. Click the Apply button. The changes will be updated in the Replay function.

To cancel the changes, click the Restore button.

How to Set Properties Related to the Event Queries

In the Timeline Properties dialog, you can also define whether the event data is fetched from the history database in the history mode or not, and for how long a period (time span).



Proceed as follows:

1. As a default, the Include events check box is not checked. This means that event queries of the history session in question are prevented and the event data will not be available in the history mode. If you want the event history to be fetched, check the Include events check box. In that case, the event queries are made in the history mode.
2. As a default, the time span of the event queries is 24 hours. If you want to change it, click the value in the Time span box indicating the time in hours (h) and give the desired time span in one of the following ways:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.

Notice that the time span is also defined in the event browser specific configuration. If the values of these two definitions are different, the longer value will be selected to be the time span of the event queries.

3. Click the Apply button. The changes will become valid.

To cancel the changes, click the Restore button.

Trend

How to Set the Start and End Time

The start time is used together with the time span to define the interval in which the trend can be examined. The interval can be defined with the accuracy of 5 seconds.

When the history mode is entered, the transition moment determines the value of the end time. The time span value is locked and at its default value of 8 hours. The value of the start time is determined by the end time and the time span (i.e. end time - span value).

Proceed as follows:

1. Click the value to be changed in the date and time box.
2. Do one of the following:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.

Alternatively, the current time can be set as the end time by clicking the clock button



The date can also be set by the use of the [calendar function](#).

3. Click the Refresh button . The picture windows presented in history mode will be updated to display the changed values.
4. Define the duration of replay in Timeline Properties dialog by setting the values in Shift frequency and Shift by fields and define step limit. See [How to Set the Replay Properties](#).

Changes in the start time value will also directly affect the location of the slider on the timeline gauge.

How to Set the Time Span

Time span is used together with the start time to define the interval for which the trend is displayed.

Proceed as follows:

1. Open Timeline Properties dialog by clicking . Click the value in the Time span box indicating the time in hours (h) and give the desired time span in one of the following ways:
 - Type a new value.
 - Select a new value by clicking the up and down arrow buttons.
 - Select a new value by pressing the UP and DOWN ARROW keys.
2. Click the Apply button. The changes will become valid.

The scale of the gauge may change when the time span is changed.

How to Browse the Trend

Trend history can be quickly browsed with the timeline slider both forward and backward.

Proceed as follows:

1. Click the slider unit and select the desired unit (s/min/h/d/step) from the list. The scale of the gauge will change accordingly.
2. Grab the slider and move it on the desired location on the gauge.
3. Release the slider. It returns back to the center of the gauge, and the picture windows presented in history mode are updated to match the new display moment. The values of the start time and the time span are updated accordingly.

Alternatively, you can browse the close proximity of the displayed interval.

Proceed as follows:

- Click the left step button  to move backward or the right step button  to move forward in time. Each click changes the interval by the smallest possible step determined by the scale. The values of the start time and the time span are updated accordingly.

How to Return to the Real-Time Mode

To return to the real-time mode from the history mode, proceed as follows:

- Click the real-time mode button  on the picture window toolbar.

OR

- Right-click the picture window and select Real-Time on the submenu of the Picture... command.

OR

- Close the Timeline dialog by clicking the  button its upper right corner.

OR

- Press Esc key.

NOTE! If you have zoomed in a trend picture in history mode, first pressing of Esc key will remove zoom and the second will return to real-time mode.

1.7.22 How to Change the Language of the DNA Operate User Interface

To change the language of the DNA Operate user interface, proceed as follows:

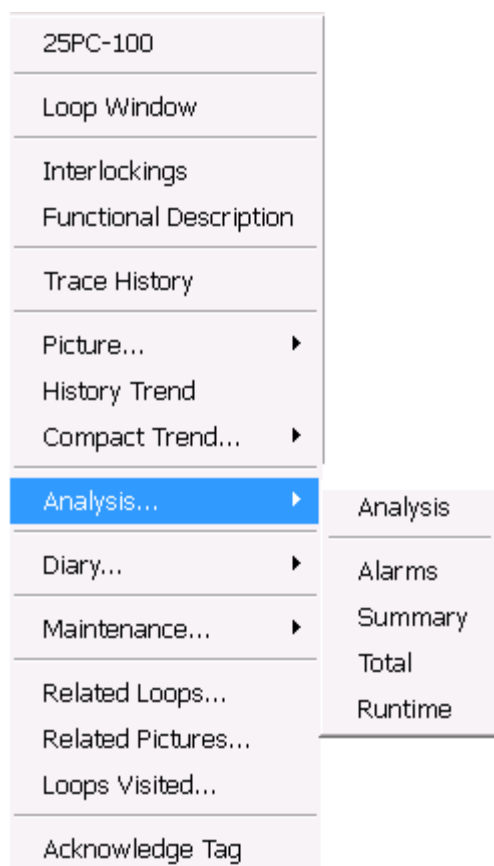
- Open the list of available languages by clicking the  button on the control panel toolbar. Select the desired language from the list. The language of the DNA Operate user interface changes.

1.7.23 How to Use the Action Menu

Function of the Action Menu

Many of the DNA Operate tools are opened from the action menu. You can open the action menu by clicking the target with the right mouse button. The contents of the action menu varies somewhat between different windows and targets. When clicking objects in windows of the same type (in picture windows, for example), the contents of the action menu is always the same but the available commands and actions vary according the target. If some of the menu commands or actions are not applicable to the target, they appear dimmed and cannot be selected.

The contents of the action menu can be modified and may vary from plant to plant.



Loop Window

The Loop Window command opens the loop window of the target on the desktop. The appearance and functions of the loop windows vary between different kinds of loops. Loop window descriptions and instructions on their use are included in component-specific user guides.

Interlockings

The Interlockings command opens a dedicated tool for presenting the current interlockings and interlocking statuses of the loop.

Functional Description

The command opens a functional description which contains information about the tag, for example, loop operation, interlockings, and instructions for disturbance situations. The functional description may be dynamic in which case the current statuses of the process are updated in it. Some functional descriptions may have the option for adding comments.

Trace History

This command starts the Tracer tool. Tracer is a tool that can be used for displaying history and event information in various forms.

The use of the Tracer tool is described in more detail in a separate manual.

Picture

History

The command changes the active picture window to the history mode.

Real-Time

The command restores the active picture window to the real-time mode.

Save As...

The command saves the contents of the active picture window for later examination.

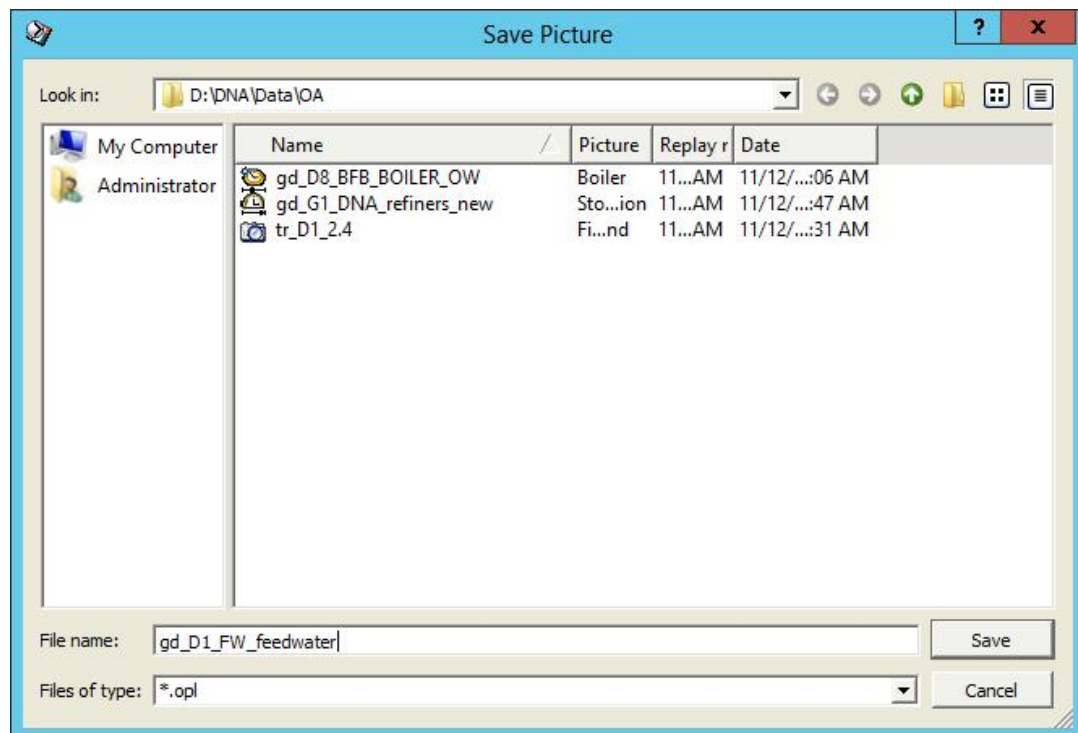
Proceed as follows:

1. Right-click the picture window to be saved and select Save As... on the submenu of the Picture... command.

OR

In case of a snapshot picture window, click the  button on the window toolbar.

The save dialog will open on the desktop.



The details of the picture windows saved in directories are visible in the save dialog. The symbol presented in front of a picture name refers to the type of the saved picture window as follows:

	Real-time picture
	History mode picture
	Snapshot picture

Saved picture windows can be renamed or deleted normally in the save dialog.

2. Select the directory in which you wish to save the picture window and click the Name field to enter a name for the picture window.
3. Select a suitable file type for the picture window from the drop-down list.

4. Click the Save button. The dialog will close and the picture window will be saved in the selected directory.

Open Picture...

The command opens a previously saved picture window on the desktop.

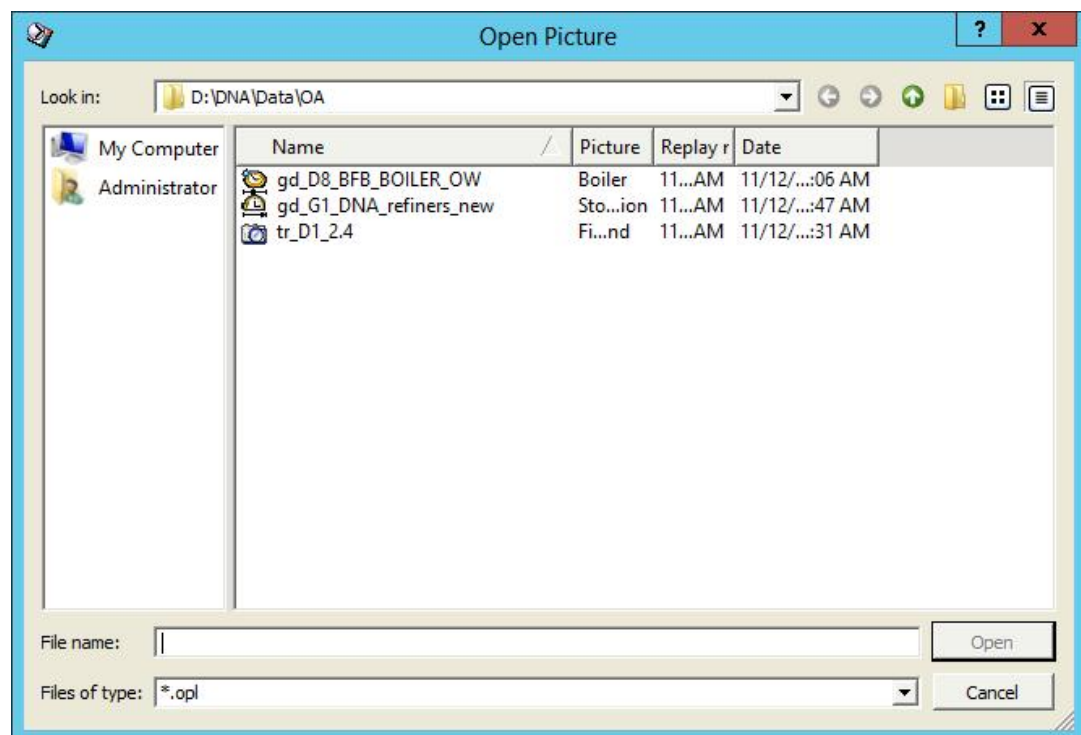
Proceed as follows:

1. Click the  button on the control panel toolbar.

OR

Right-click a picture window and select Open Picture... on the submenu of the Picture... command.

The open dialog will open on the desktop.



The details of the picture windows saved in directories can be seen in the open dialog. The symbol presented in front of a picture name refers to the type of the saved picture window as follows:

	Real-time picture
	History mode picture
	Snapshot picture

Saved picture windows can be renamed or deleted normally in the open dialog.

2. Select the directory from which you wish to open a picture window.
3. Click the name of the picture window to be opened and click the Open button.

OR

Double-click the name of the picture window to be opened.

The dialog will close and the selected picture window will open on the desktop.

Start History Collection

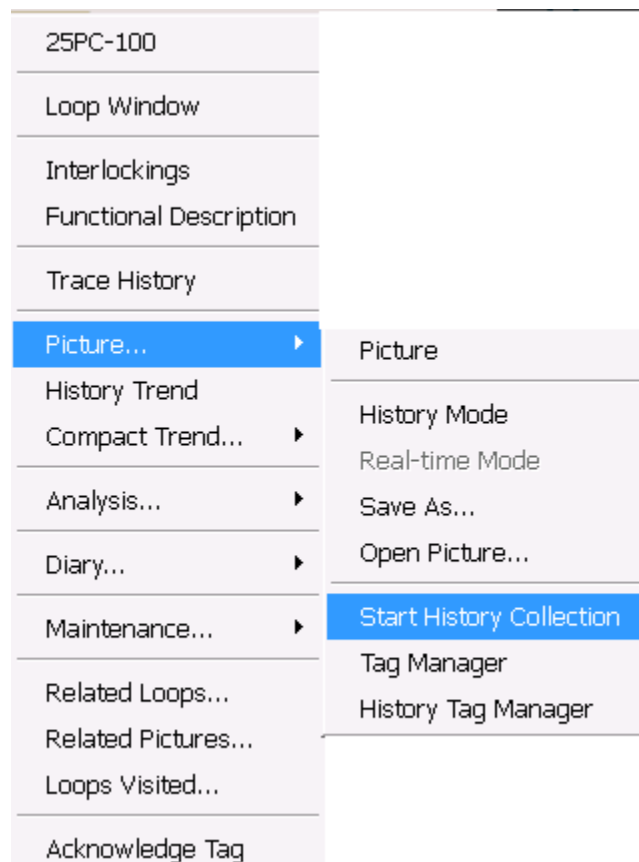
According to the default settings, not all tags in the system are included under the scope of history collection. You can check the status of the history collection storage capacity in the [History Database Properties](#) window.

You can check the status of tag history collection with the [History Tag Manager](#) by searching for the tag you are interested in. If a tag cannot be found, you must start collecting history information for it.

NOTE! Management of data collection is usually done with engineering tools and this function is rarely needed.

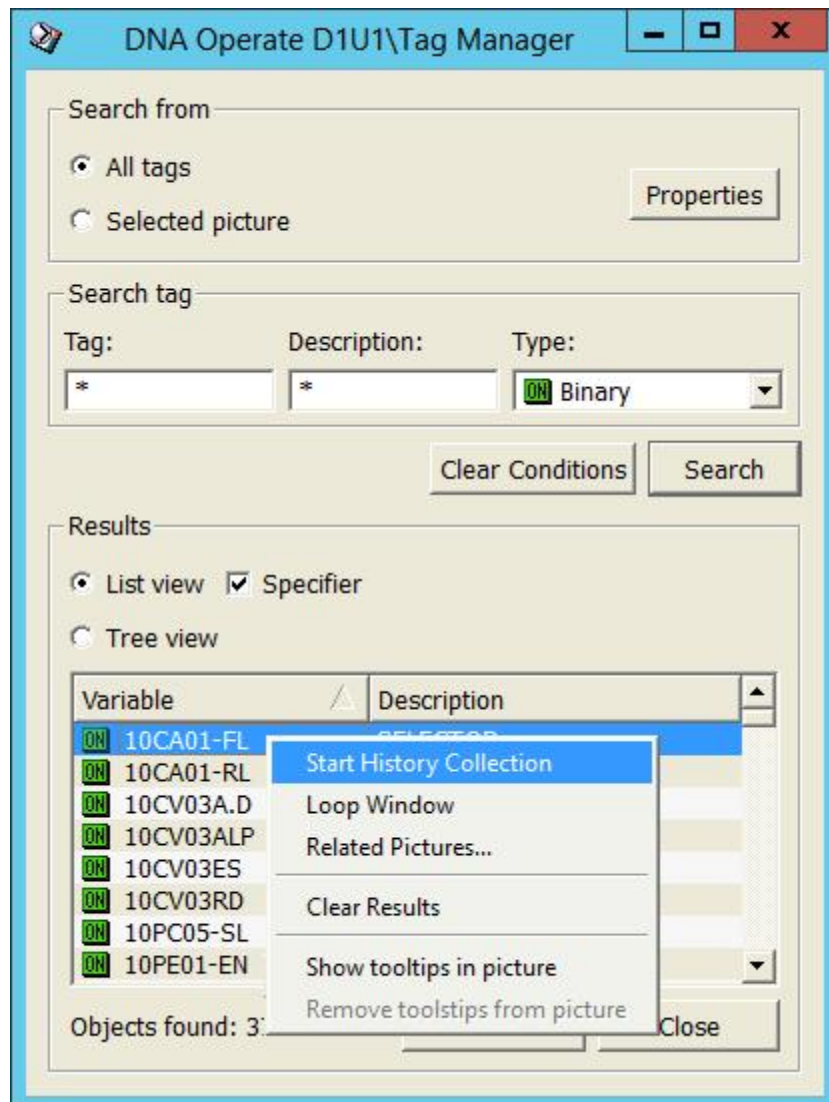
To start collecting history data:

1. From the picture window, select the component whose history information you want to start collecting.
2. Open the action menu by right-clicking.
3. Select Picture... and then select Start History Collection from the menu.



OR

1. Open the Tag Manager from the action menu.
2. Find the tag based on name, description, or type.
3. Select the tag whose information you want to start collecting.
4. Right-click the tag.



5. Select Start History Collection from the pop-up menu.

You can stop collecting history information for a tag by using the History Tag Manager. For more detailed instructions, see [Removing a tag from history collection](#).

Tag Manager

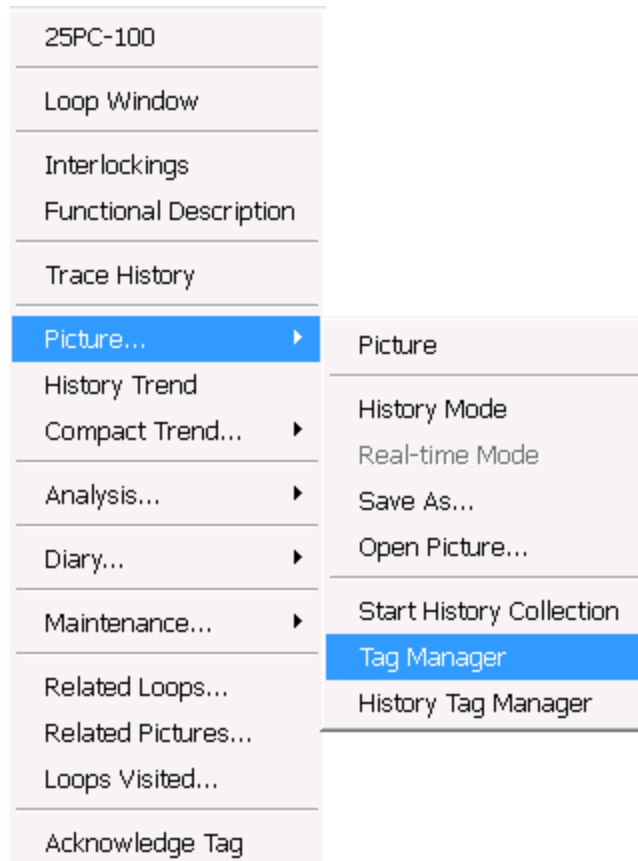
This command starts the Tag Manager that you can use to search for tags in DNA Operate based on tag IDs, descriptions, or types.

The Tag Manager is normally used together with trends. You can drag and drop tags from the search results directly into trends, if the trends support adding graphs.

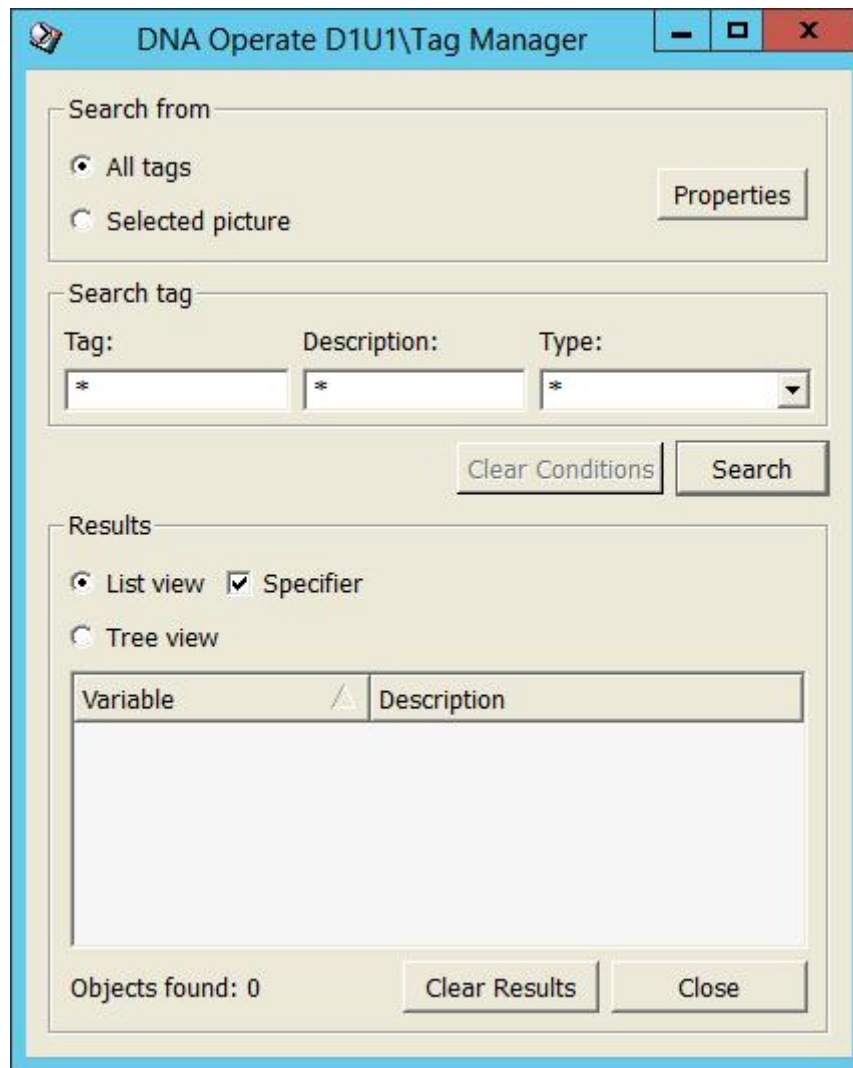
Searching for tags

Do the following:

1. Right-click the picture window and select Tag Manager from the Picture submenu.



The Tag Manager will open on your desktop:

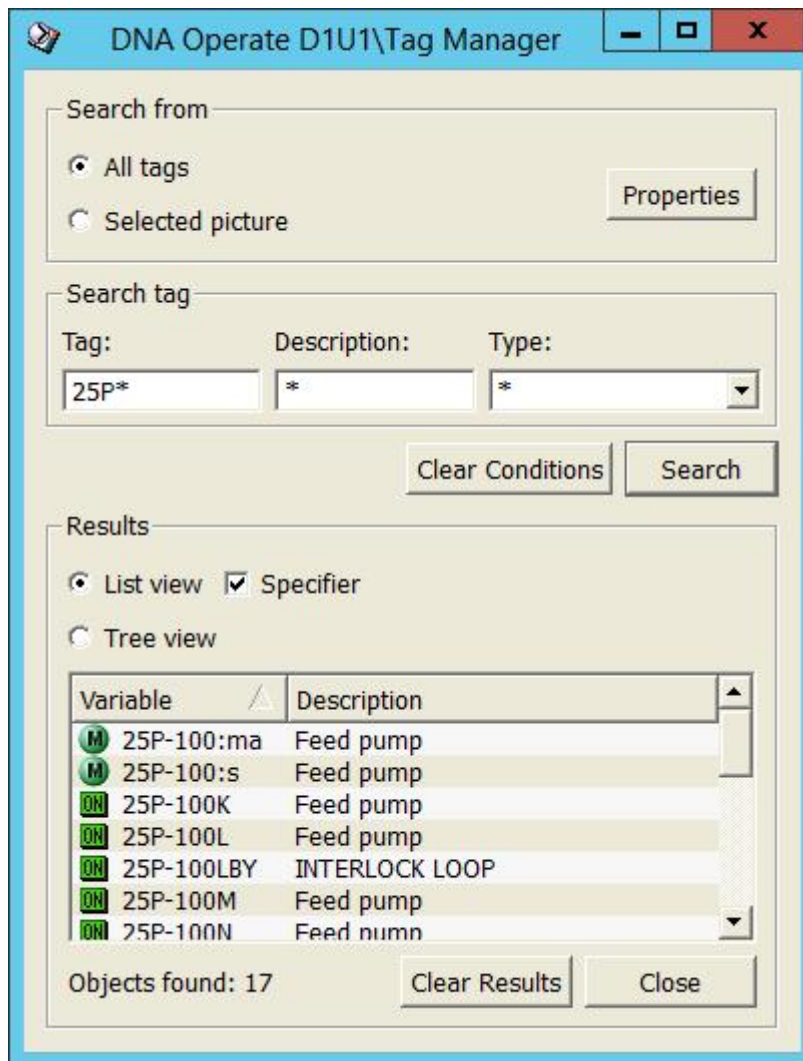


2. First select, if you are searching from all tags or the selected picture. Click the Tag, Description, or Type text box and enter your search condition. Wildcards (? and *) can be used in searches.

From the Type text box menu, you can select the following options:

- Analog
- Binary
- Control
- Counter
- Integer
- Measurement
- Motor
- Motor Actuator
- Profile
- Sequence

3. Click the Search button. The search results are updated in the results field.



DNA Operate D1U1\Tag Manager

Search from:

- ☒ All tags
- ☐ Selected picture

Properties

Search tag:

Tag: 25P* Description: * Type: *

Clear Conditions Search

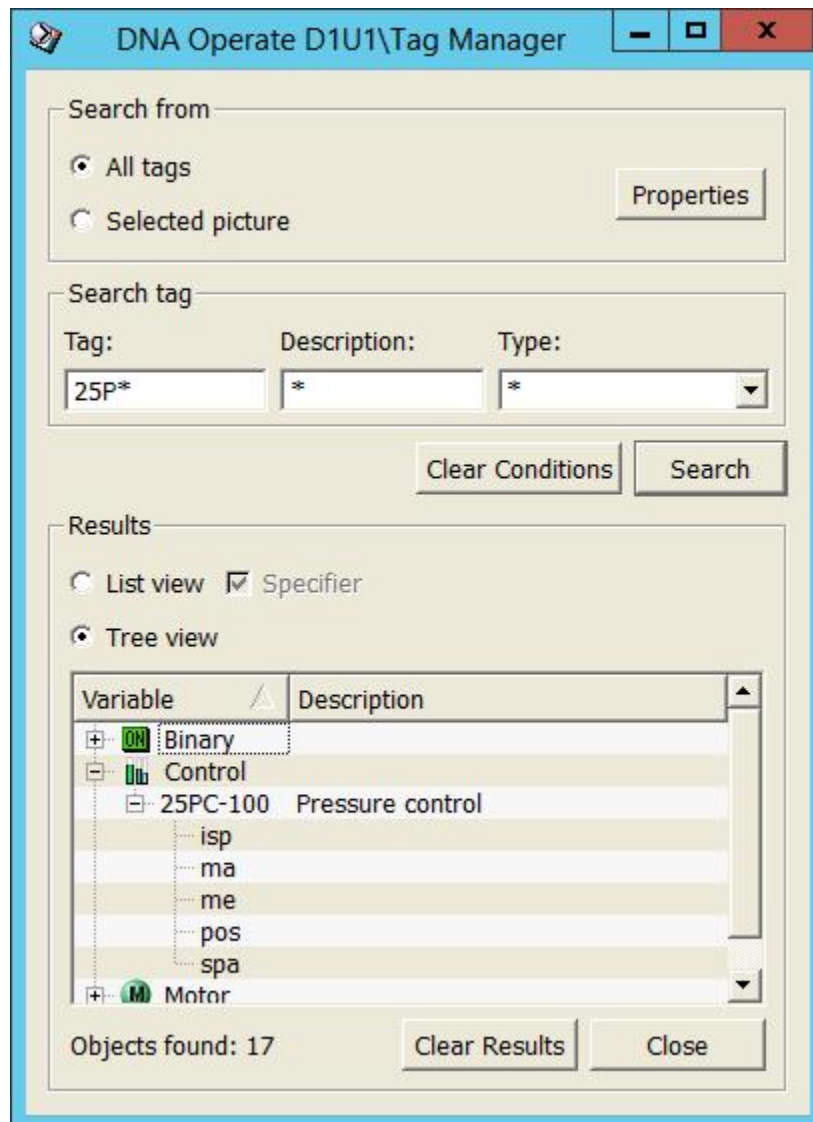
Results:

- ☒ List view ☒ Specifier
- ☐ Tree view

Variable	Description
 25P-100:ma	Feed pump
 25P-100:s	Feed pump
 25P-100K	Feed pump
 25P-100L	Feed pump
 25P-100LBY	INTERLOCK LOOP
 25P-100M	Feed pump
 25P-100N	Feed numn

Objects found: 17 Clear Results Close

To view the search results in a tree view, select Tree view. Viewing many search results may be easier in the tree view.

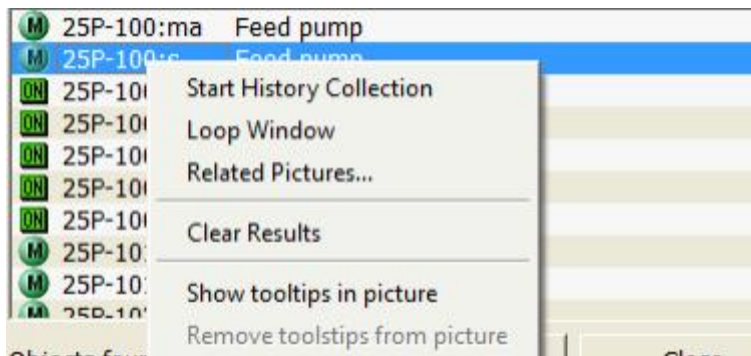


Selecting List view will return back to initial listing.

To perform a new search, click the Clear Conditions button to clear the search condition fields. Similarly, you can click the Clear Results button to clear the results field.

By right-clicking a tag in the Search window's results field, you can open a menu where you can:

- start history collection for a tag
- open a loop window for a tag
- open a dialog which contains a list of the pictures related to the tag.
- clear the results
- add tooltips to picture
- remove tooltips from picture

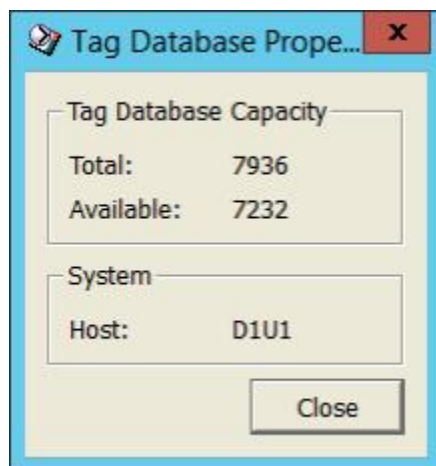


You can select multiple tags in the following ways:

- Select the first tag you want on the list. Hold down Shift key and select the last tag you want on the list. All the tags between the two tags will also be selected.
- You can select multiple single tags by holding down Ctrl key while selecting.
- You can toggle the current selection on the list with Ctrl + arrow up/down.
- Select all tags with Ctrl + A.

By clicking Properties button, you can open the Tag Database Properties window. This window displays the following information:

- Total capacity
- Available capacity
- Host server of the tag database

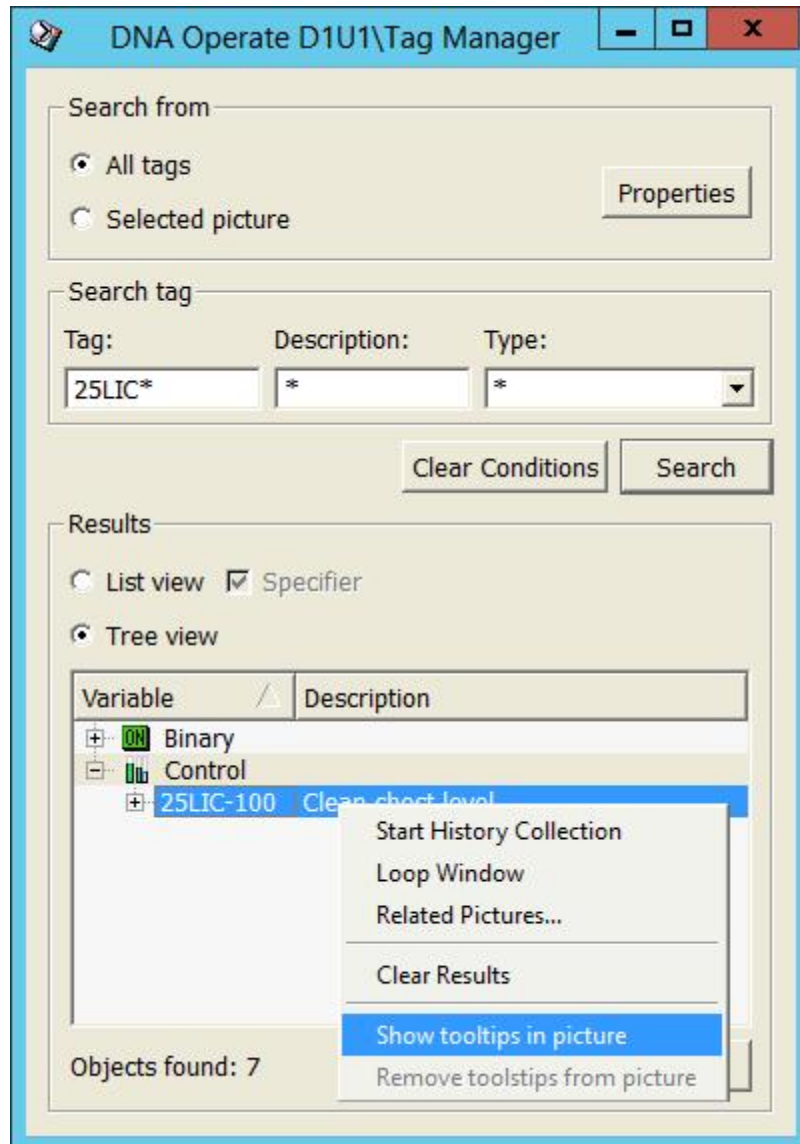


You can close the window by clicking the Close button.

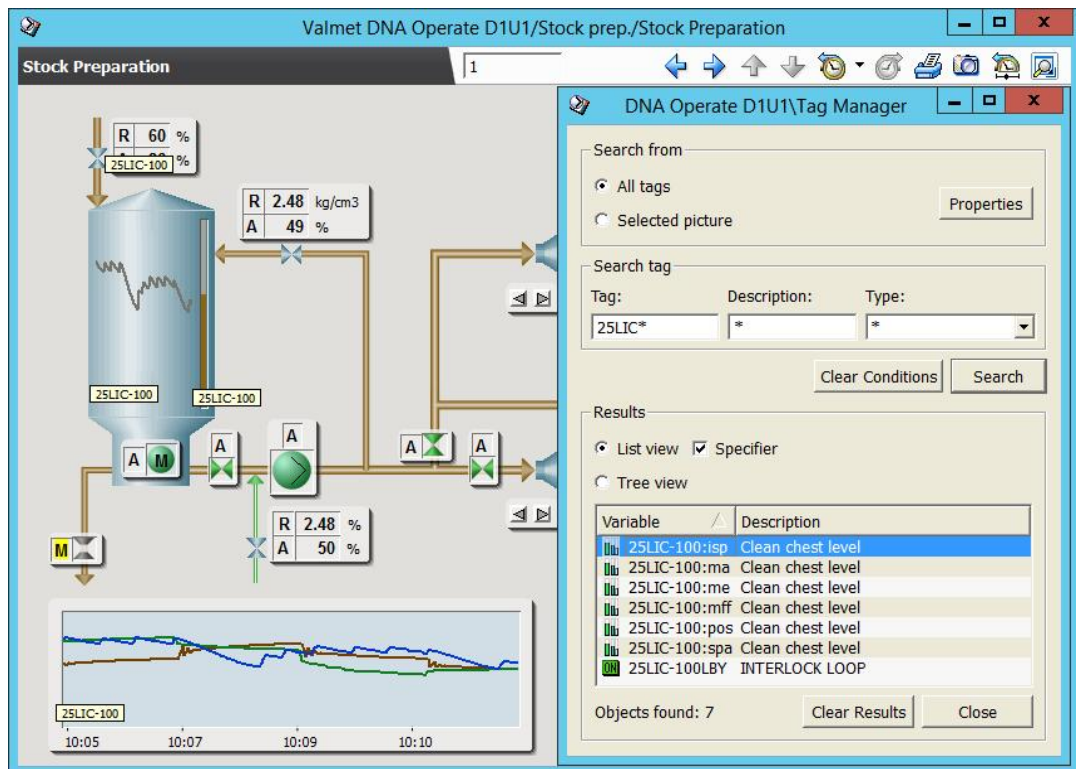
Tooltips

To add tooltips to the picture, proceed as follows:

1. Right-click the picture window and select Tag Manager from the Picture submenu.
2. Search for the desired tag by entering search conditions to Tag, Description or Type text box.
3. Click Search.
4. Right-click the tag in the results field and select Show tooltips in picture from the menu.



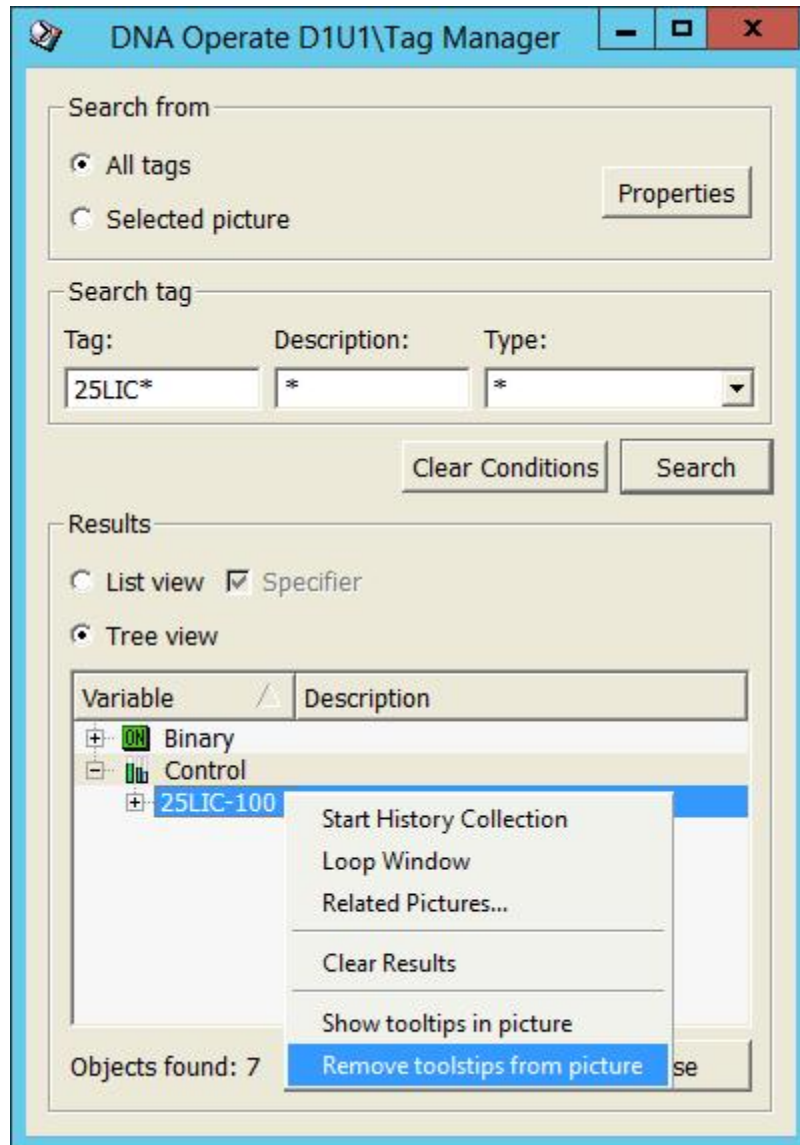
Tooltips are added in the picture:



Removing tooltips from the picture

To remove tooltips from the picture, proceed as follows:

- Right-click the tag in the results field and select Remove tooltips from picture from the menu.



Tooltips will also be removed from the picture when the tag manager is closed by clicking Close.

History Tag Manager

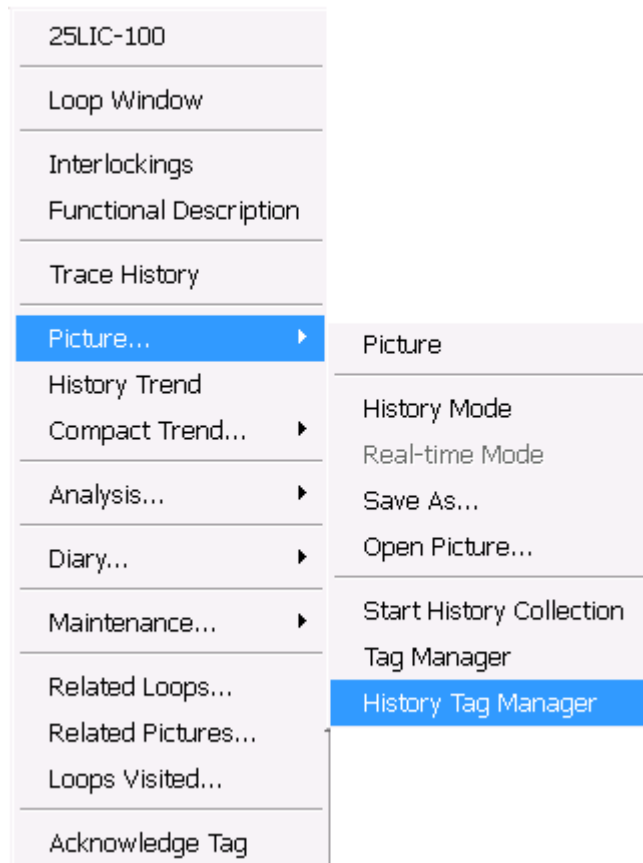
This command starts the History Tag Manager that you can use to search for tags in the history collection based on tag IDs, tag descriptions, or types.

The History Tag Manager is normally used together with trends. You can drag and drop tags from the search results directly into trends, if the trends support adding graphs.

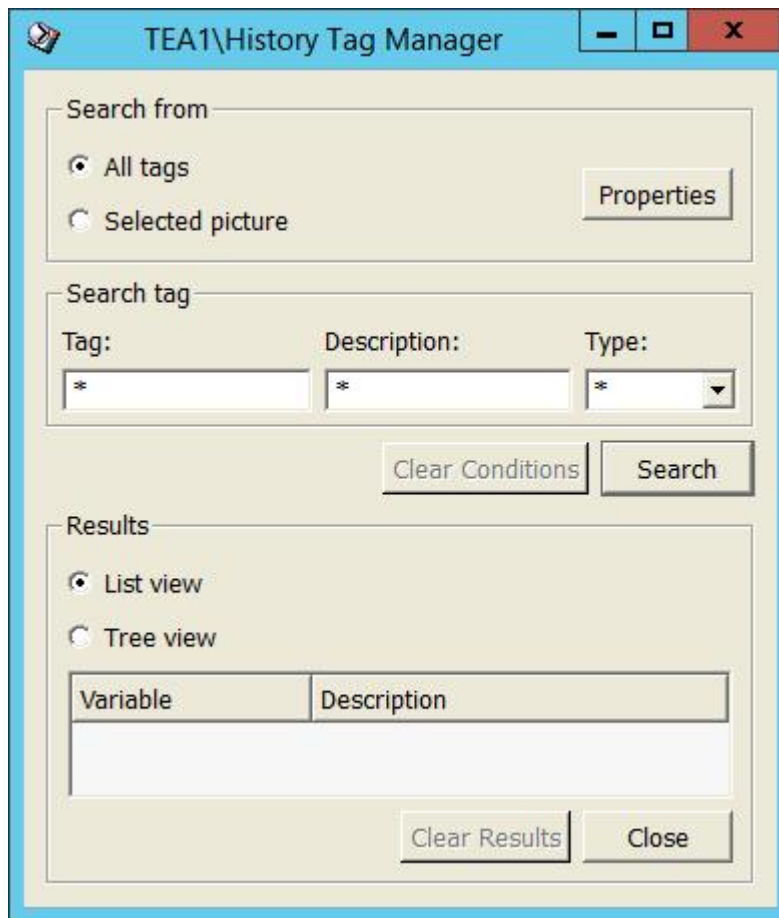
Searching for tags

Do the following:

1. Right-click the picture window and select History Tag Manager from the Picture submenu.

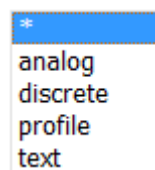


The History Tag Manager will open on your desktop:



2. First select, if you are searching from all tags or the selected picture. Click the Tag, Description, or Type text box, and enter a search condition. Wildcards (? and *) can be used in searches.

From the Type text box menu, you can select the following options:



3. Click the Search button. The search results are updated in the results field.

TEA1\History Tag Manager

Search from

☒ All tags

☐ Selected picture

Properties

Search tag

Tag: LIC-100 Description: * Type: *

Clear Conditions Search

Results

☒ List view

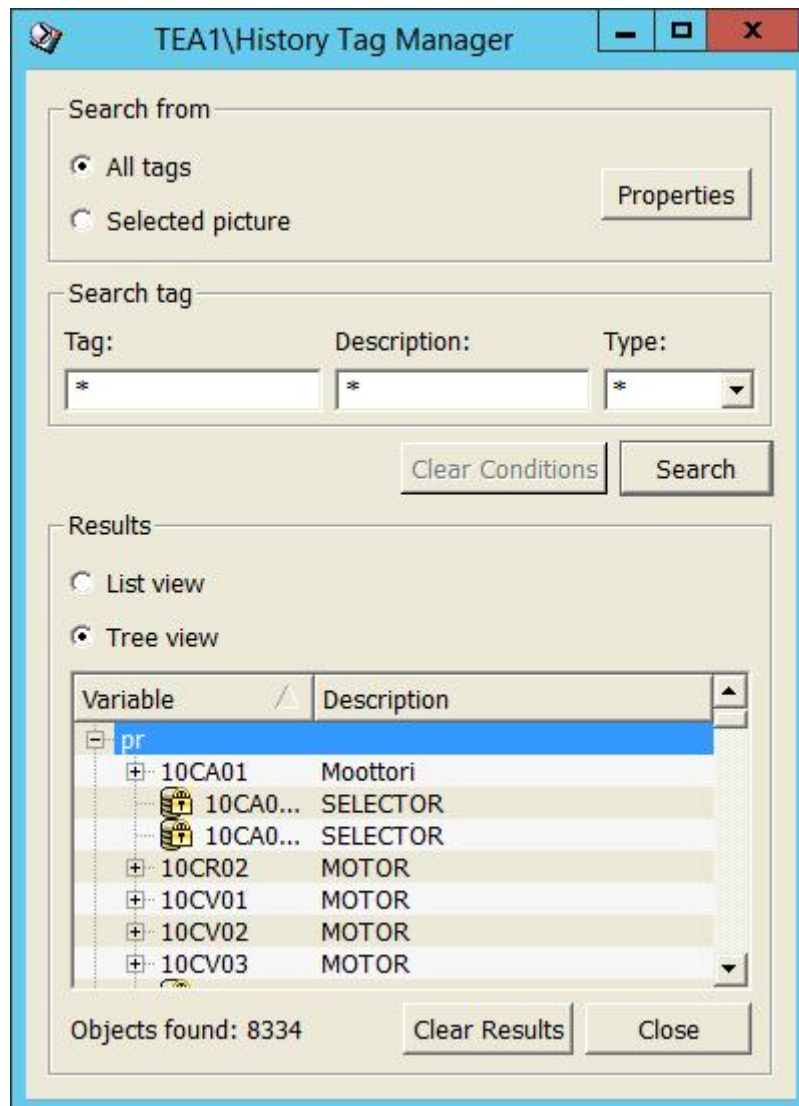
☐ Tree view

Variable	Description
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100:...	Clean chest level
pr:25LIC-100...	INTERLOCK LOOP_en
pr:LT-LIC-100...	PID BASIC

Objects found: 24

Clear Results Close

To view the search results in a tree view, click Tree View. Viewing many search results may be easier in the tree view.



Selecting List view will return back to initial listing.

To perform a new search, click the Clear Conditions button to clear the search condition fields. Similarly, you can click the Clear Results button to clear the results field.

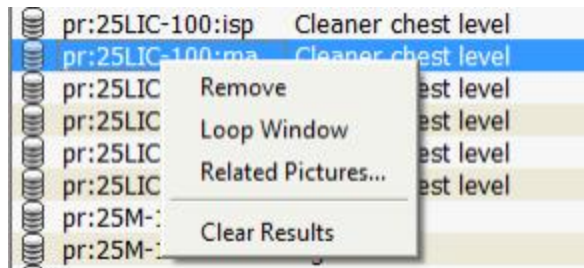
Variables marked with the  icon have been added to history collection using the engineering environment. Icons marked with the  icon have been added using operator interface (DNA Operate).

Removing a tag from history collection

NOTE! Management of data collection is usually done with engineering tools and this function is rarely needed.

To remove a tag from history collection:

1. Open the History Tag Manager.
2. Select the tag you want to remove from history collection.
3. Right-click the tag. The following menu will be displayed:



You can select multiple tags by holding down Shift key while making the selection.

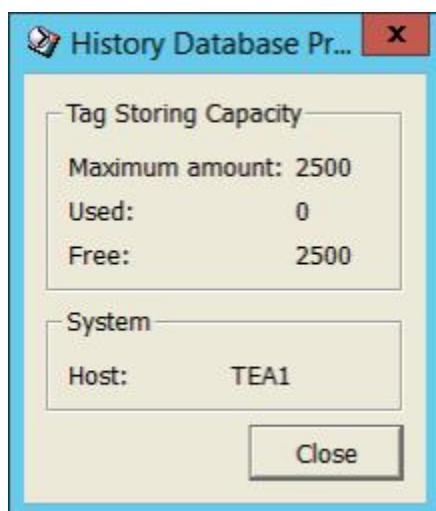
4. Click Remove to remove a tag from history collection.

NOTE! You can only remove tags marked with the  icon.

History Database Properties

Click the Properties button to open the History Database Properties window. This window displays the following history database information:

- Total capacity
- Used capacity
- Free capacity
- History database host server

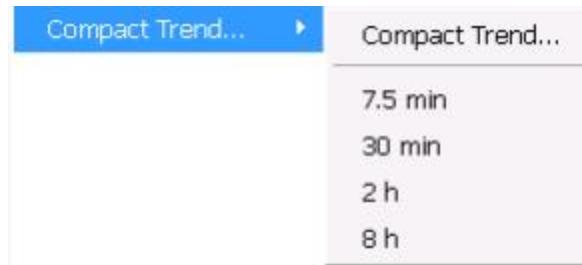


You can close the window by clicking the Close button.

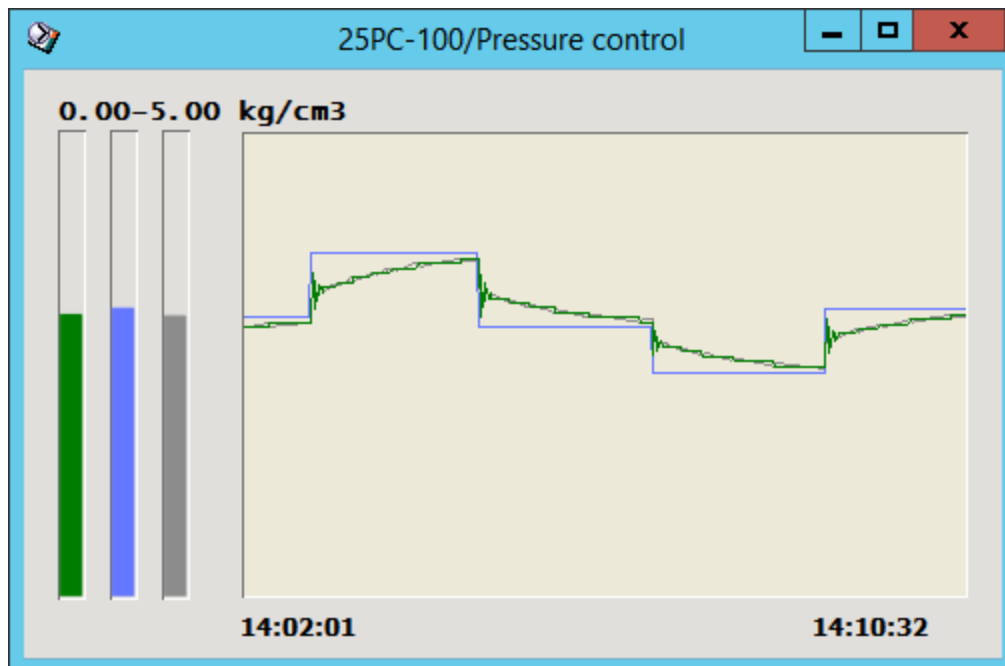
History Trend

The command opens a history trend of the selected tag. The history trend contains certain pre-configured graphs depending on the selected tag type. For more information, see chapter [History Trend -> Properties](#).

Compact Trend...



The command has a submenu in which the span of the presented trend is selected. It opens a small trend window which displays the trend as well as the current measurement, setpoint, and output values of the tag as bar diagrams. The trend curves are drawn from the moment the window is opened. Right-clicking the bar diagrams opens the normal action menu of the tag, while right-clicking the trend opens the trend's action menu. The contents of the trend action menu and the operation of the trend have been covered in [Trend Picture User's Guide](#).



NOTE! Compact trend can be opened only from a controller or analog measurement.

Analysis...

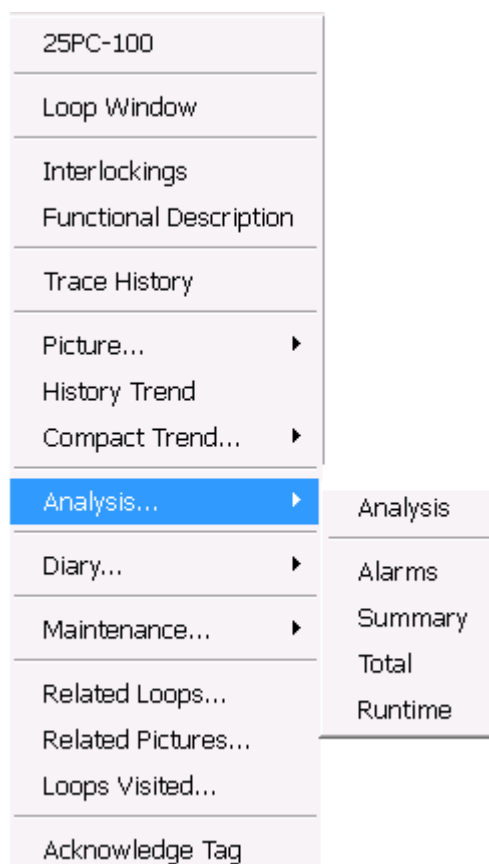
Properties

You can get summaries of multiple tags at the same time using Statistical reports and Alarms and Events Analyzing reports. The reports can also be designed to include graphs that describe selected tags. Reports and any parameters that have been defined for them can be saved for later use.

Opening the analysis tool

To open the analysis tool:

1. Right-click the measurement component.
2. Select the Analysis... command from the action menu.



3. From the menu, select the analysis you want to work with and click it with the left mouse button.

You can select the following:

- Alarms
- Summary report
- Total report
- Runtime report

Alarms

This command starts the Alarms and Events Analyzing tool. Alarms and Events Analyzing is a tool you can use to analyze alarm and event information collected to the history database from various sources.

Summary

Opens up the Summary report part of the Statistical Reports tool.

Total

Opens up the Total report part of the Statistical Reports tool.

Runtime

Opens up the Runtime report part of the Statistical Reports tool.

Diary...

Shift Entry to Diary

The command opens a diary tool (Diary) for a shift entry filled beforehand with target-specific information.

Exception Entry to Diary

The command opens a diary tool (Diary) for a disturbance entry.

Search from Diary

The command opens a diary tool (Diary) for searching entries related with the target.

Maintenance

The maintenance menu contains plant-specific tools for maintenance.

The content of the menu is configurable and plant-specific. The operation of the tools which open from the menu have been covered in separate user guides.

Loop Diagnostics

The command opens a dedicated tool (Web Maintenance) for viewing and analyzing the loop information. The operation of Web Maintenance has been covered in a separate user guide.

View Loop Diagram

The command opens a dedicated tool for viewing a loop diagram.

Edit Loop Diagram

The command opens a dedicated tool (Function Block CAD) for editing the loop diagram. The operation of Function Block CAD has been covered in a separate user guide.

Field Device Status

The command opens the Profibus diagnostics window of a loop-related field device in a maintenance tool (Web Maintenance). The operation of Web Maintenance is covered in a separate user guide.

Field Device Report

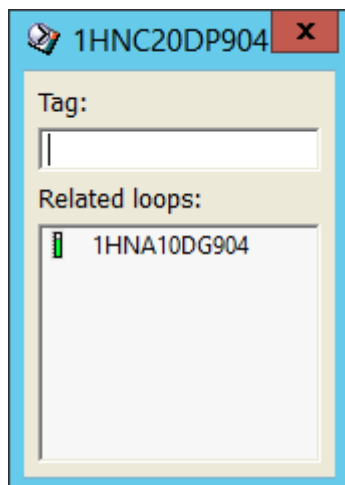
The command opens a report window of a loop-related field device in a maintenance tool (FieldCare). The operation of FieldCare is covered in a separate user guide.

Plant Wide Report

The command opens the general display of a maintenance tool (FieldCare) from which you can easily navigate to more specific displays. The operation of FieldCare is covered in a separate user guide.

Related Loops...

The Related Loops... command opens a dialog which contains a list of loops related to the target tag. You can click a name on the list to open the corresponding picture in the active picture window.

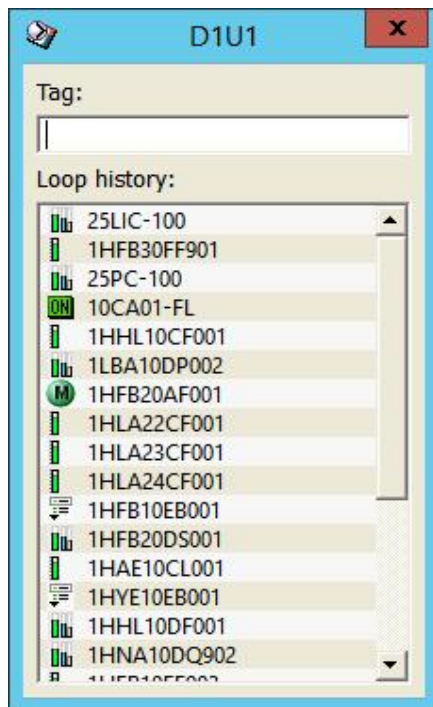


Related Pictures...

The command opens the Pictures dialog which contains a list of the pictures related to the tag. The operation of the dialog has been covered in the chapter [How to Open a Picture by a Tag](#).

Loops Visited...

The command opens a dialog from which you can open a previously displayed loop window by clicking the desired loop name on the list.



You can also open the loop window of a specific tag by entering the tag identifier in the Tag field and pressing the return key.

Acknowledge Tag

Acknowledge Tag command acknowledges all events of the target tag simultaneously. Before the acknowledgement is executed, a confirmation is asked from the user in a dialog.

2. Data Presentation

2.1 Introduction

This guide goes through the principles of data presentation and operation in the Valmet DNA user interface, DNA Operate.

Even though this guide is primarily intended for novice operators with no previous experience on Valmet DNA user interface, it is recommended that also the experienced operators would browse through it to familiarize themselves with the updated user interface features.

2.2 Basics

In DNA Operate pictures, the process status information is displayed as numbers, symbols, bars, texts, and diagrams. Whenever any changes occur in the process, the data is automatically updated in the pictures.

2.3 Process and Equipment Statuses

Processes and equipment are either in the normal status or in the exception status which are indicated by different colors in the pictures. The exception statuses include:

- Alarms
- Interlockings
- Masked events
- Faults.

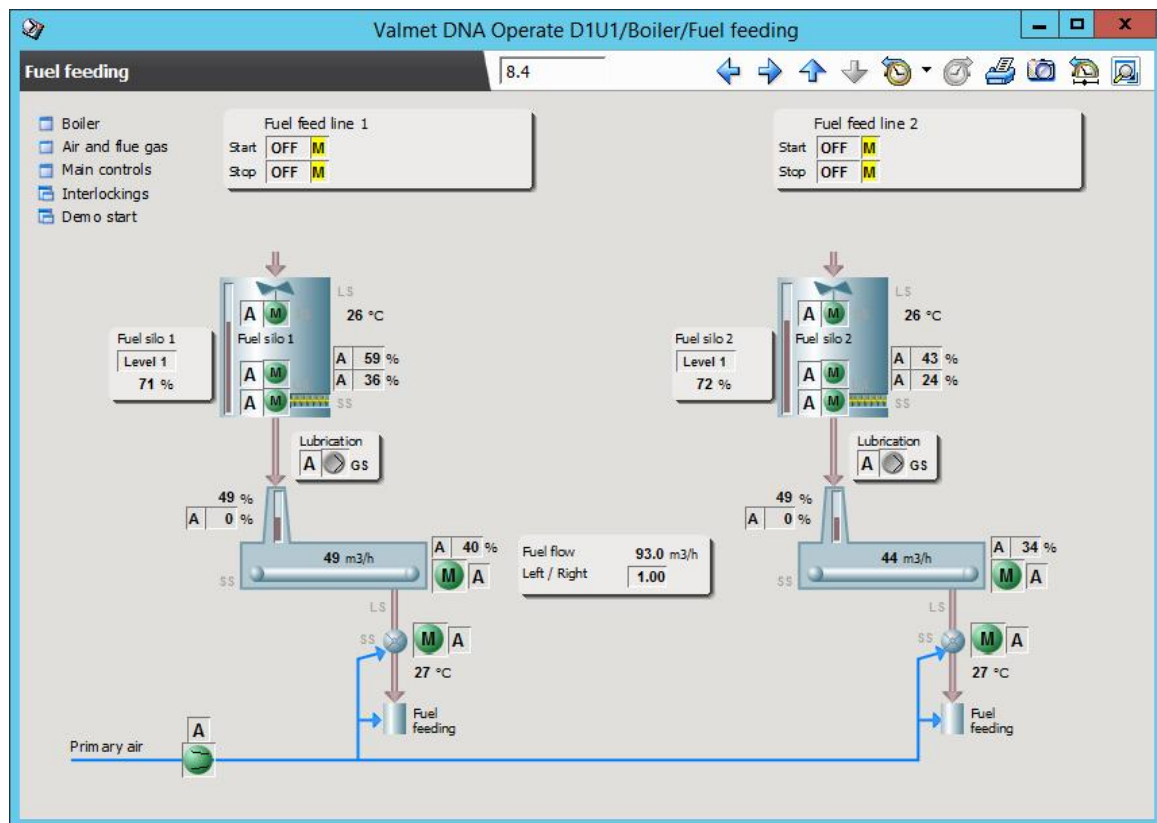
2.4 Status Colors and Font Sizes

In the normal status, the colors are clear but smooth. For normal statuses, a further distinction can be made between active and passive statuses. The active status means that a process or equipment is running. The passive status, on the other hand, means that it has been stopped. Active statuses may have been highlighted by **green** background color. Exceptional statuses which require operator's immediate attention are highlighted in bright background colors, such as **red** and **violet**. **Yellow** background color is used in loop windows as a general attention color to highlight states which differ from the normal status.

NOTE! Green and gray can have other meanings as well.

Flashing equipment or process calls operator's attention to target in alarm or fault situations. Flashing may also indicate a change in object's status.

Since in control rooms people often view the pictures together and while standing, the font sizes for buttons and loops, for example, have been defined so that the text is legible even when read from a longer distance, for example, when standing behind another person.



In the normal status, the colors for symbols and bars are clear and smooth.

2.5 Presenting Normal Statuses

2.5.1 Symbols

The symbols are used to present the different parts of a process and their statuses. An example of a motor is presented below. When the motor's status changes, the change is reflected in the symbol. The color and shape of the symbol will change. The motor has four normal run statuses, which are indicated by changes in the motor symbol as follows:



The motor is stopped.



The motor is starting. The symbol is flashing.



The motor is running.



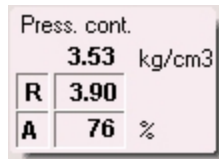
The motor is stopping. The symbol is flashing.

2.5.2 Numerical Values

In DNA Operate, numerical values are used to present measurements and setpoint values, for example. The unit used in the measurement and a text indicating the type of measurement are often displayed in connection with the numerical value.



Measurement value and unit of fuel heat power

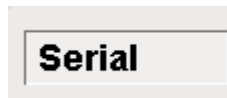


Controller is on automatic control (A = Automatic) and its setpoint value is read from another controller (R = Remote).

2.5.3 Texts

Various texts are used to provide information about the statuses of equipment and processes, and the options available for each equipment or process. Different colors are often used to indicate whether a particular status is active or passive. The texts are always application-specific.

Button texts are status indicators the user can operate on. The operations are described in more detail in the section [Command Buttons](#).



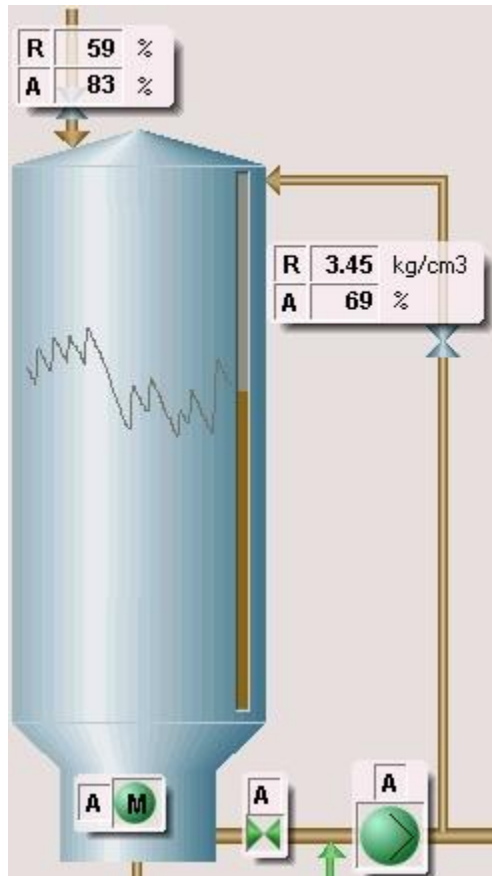
Button text indicating the refiners are connected in series



Button text indicating the refiners are connected in parallel

2.5.4 Bars

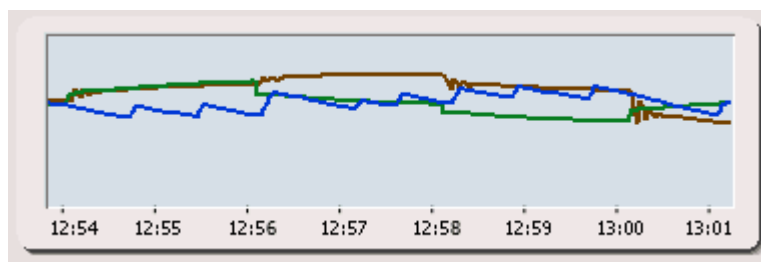
Bar height or length is used to indicate the measurement magnitude. Bars can be either vertical or horizontal. In connection with the bar, there can be a scale from which the exact value can be read.



Bar indicating the level in a chest

2.5.5 Diagrams

Diagrams can be used to present, for example, history trends for measurements and the measured profiles. More complex diagrams can also be edited if necessary. You can, for example, zoom and scale them, and select particular measurements for a closer inspection.



Refiner trend diagram

2.6 Presenting Exceptional Statuses

2.6.1 Alarms

Alarm status means that the process is in an undesired state, and thus requires actions from the operator. The alarm is indicated with **red** color.

For each alarm, the following information is displayed:

- Red color indicates that the alarm is on.
- The hue of the red color indicates the priority of the alarm: high, medium, or low.
- Flashing of the alarm color indicates whether the alarm is acknowledged.

Priority indicates the urgency or severity of the alarm. High priority means that the alarm requires the operator's immediate attention. Correspondingly, low priority means that the alarm is not that urgent even though some measures are needed.

The acknowledgement status of the alarm indicates whether the operator has marked the alarm acknowledged or not. If the alarm color flashes, the operator has not acknowledged the alarm.

The pictures display all the current alarm statuses for the loop. A motor, for example, might have several alarms active at the same time. More detailed information about the alarm can be gained by opening the loop window related to the target.

NOTE! If the loop has several alarms active at the same time, the alarm with the highest priority will be displayed.

Examples of Alarms

In symbols which display the status of the process or equipment, the alarm is indicated with red background color:



In numerical values, the alarm is indicated with red background color. Note the text color which has been changed for better legibility:



In bars, the alarm is indicated with red bar color:



2.6.2 Interlockings

Interlocking status means that a piece of equipment cannot be temporarily operated due to the present state of the process.

The interlocking status is indicated with **blue background**.



2.6.3 Masked Events

When an event has been masked, the status differs from the normal but is not alarming as such. The masked event status is indicated with **turquoise background**.

Examples of Masked Events

Numerical value:

R	2.64	m3/min
A	54	%

Symbol:



Bar:



2.6.4 Faults

Fault status means that the information displayed on the screen is unreliable. A fault can be, for example, a malfunction in the field equipment or a system error in data transmission. Fault statuses usually occur during implementation and service.

The fault status is indicated with **red background**.

Examples of Faults

Symbol:



Numerical value:



Text:

Parallel

Bar:



Button:



2.6.5 Shelved Alarms

Device status is normal during the shelved alarm. Shelved alarm status is indicated with a functional background color, which is defined palette-specifically. The default color is orange.

Examples of Shelved Alarms

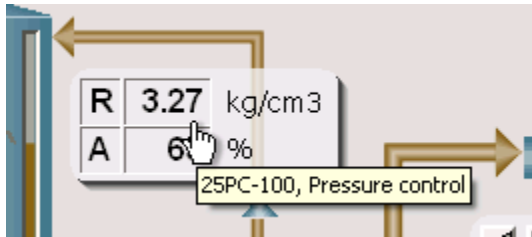
Numerical value:

L	0	%
M	0	%

2.7 How to Perform Operations

2.7.1 Basic Principles

Processes are controlled by operating the objects (controls, motors, etc.) in DNA Operate pictures. The operations are mainly performed by pointing to the target and then pressing either mouse button. Operable objects appear embedded. When pointing to the target in pictures, a tooltip is displayed on the screen. It shows the tag identifier and also the name of the target, if defined.



An example of a tooltip when pointing a target.

Clicking the left mouse button on the target executes its primary operation. Depending on the target, this either promptly performs the operation or opens the dialog connected with it.

Clicking the right mouse button opens the action menu which contains commands for viewing more information on the target and executing secondary operations.

2.7.2 Operations Performed Directly in Pictures

Commands displayed in the picture are executed immediately when clicked. These kinds of commands include:

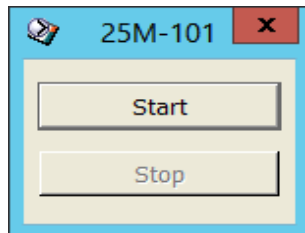
- Button commands
- Options in drop-down lists
- Data entered in text fields



Refiners connected in parallel can be changed to serial connection with a single click.

2.7.3 Operations Opening a Dialog

Clicking the target may open a dialog for performing the operation. Dialogs close automatically after executing the command.



The dialog of a motor closes after the command button has been clicked.

2.7.4 Performing Operations in Pictures and Dialogs

Command Buttons

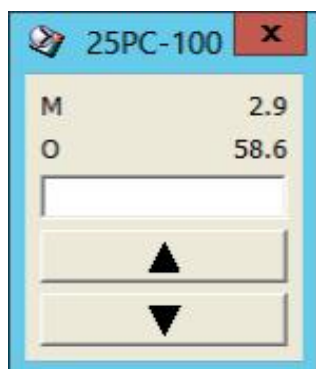
Once you click a button, the related command is executed immediately. For example, the dialog pictured below controls a valve. When the Open button is clicked, the valve which had been closed will open immediately. The Close button is not available and it appears dimmed. When the valve is open, the Close button will be enabled and it will close the valve when clicked.



Open/Close dialog for a valve

Spin Buttons

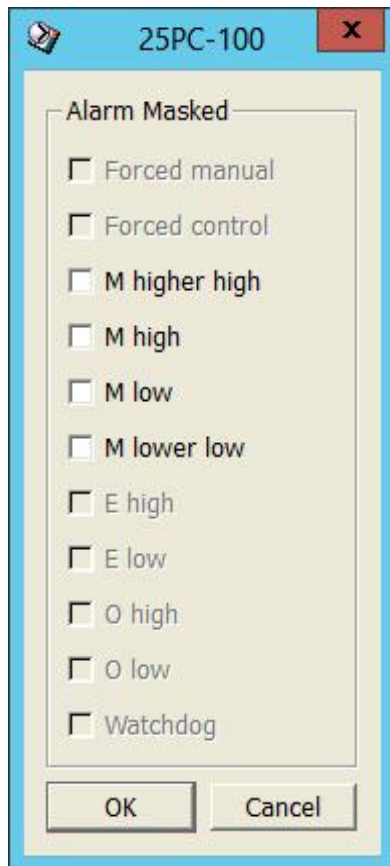
Click the spin button arrows to increase or decrease the values. Each click sends the updated information directly into the system.



Value changes are updated in the system in real time.

Check Boxes

Dialogs may contain check boxes. An option in a dialog is selected if its check box is checked. Check the option by clicking the check box next to it and uncheck the option by clicking the check box again. Several check boxes can be selected at a time.



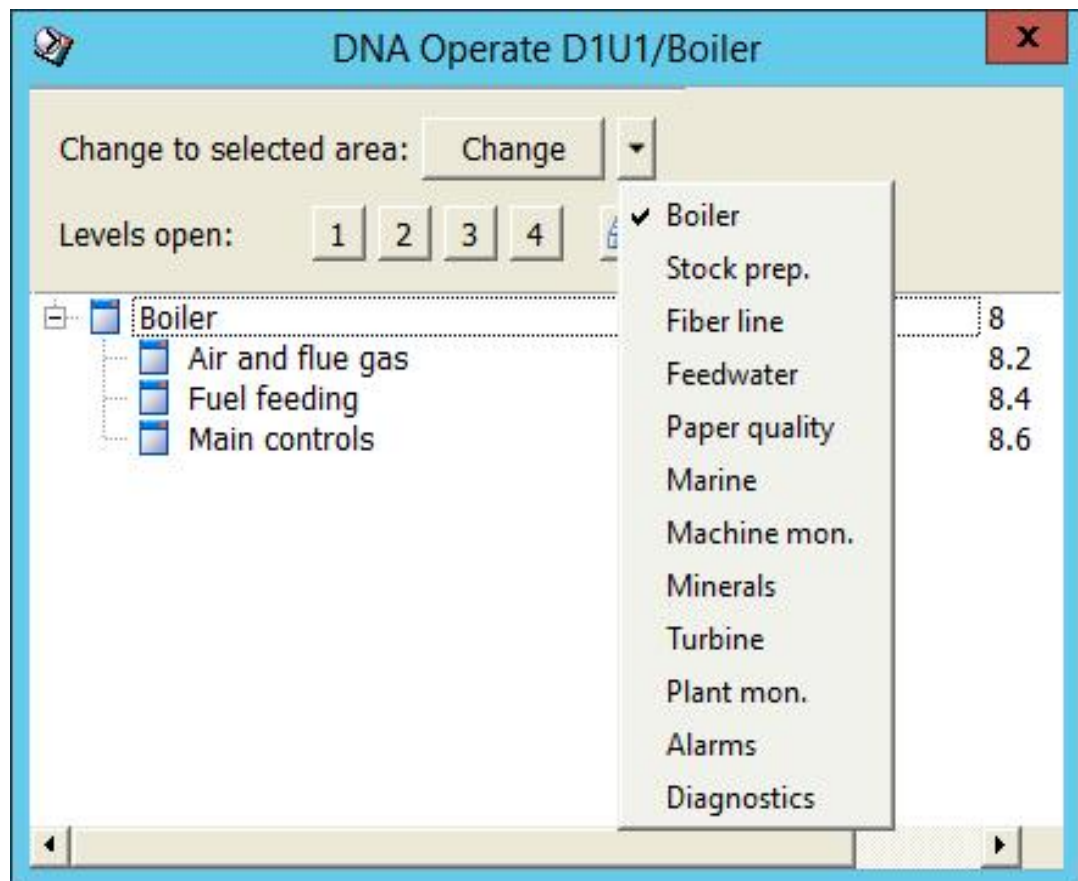
In the event mask dialog, certain events can be prevented from triggering events.

Drop-Down Lists

Once opened, a drop-down list displays a list of options one of which can be selected at a time. It is used, for example, in the picture browser for area selection.

When you want to make a selection from a drop-down menu, proceed as follows:

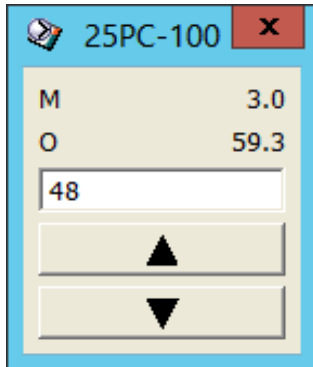
1. Click the arrow ▼ to open the drop-down list.
2. Select the desired option on the list. The list will close and the selected option will be activated.



Click the arrow in the picture browser to view the list of selectable areas.

Input Fields

You can enter the desired value as numbers or text directly in an input field. Input fields have normally white background. If the background is dimmed, the input field is disabled.



Input field of a dialog.

When you want to enter a new value in an input field, proceed as follows:

1. Click the old value to activate the input field.
2. Select either of the following:
 - Type the new value to replace the activated (highlighted) old value.

OR

- Click the input field again to edit the contents that are currently displayed in the field. You can then, for example, press the Backspace or Delete key to delete characters.
4. Press ENTER to accept your changes.

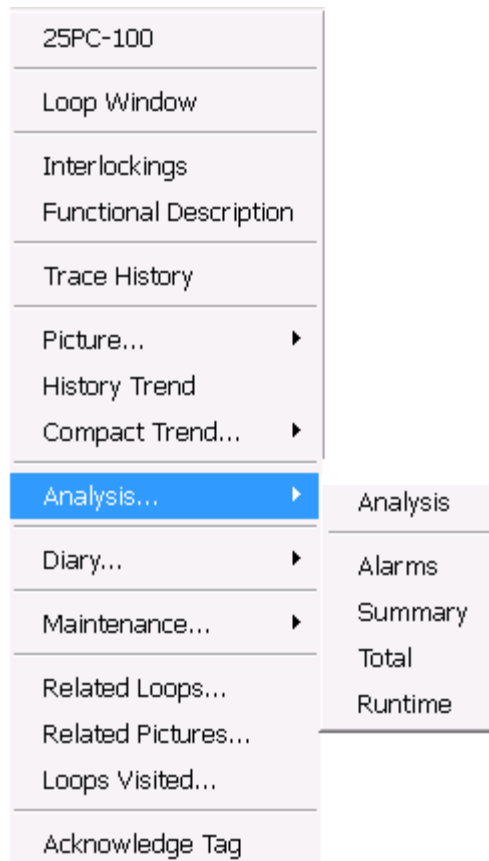
Action Menu

To open the action menu, right-click an item in a picture window. From the action menu, you can view details of the target and perform operations that are not available in the dialogs.

The action menu contains general operations that are the same throughout the desktop, and operations that pertain to the target only. The contents of the action menu might vary in different applications. The action menu may also consist of only general operations. The operations that are not available for the target or which are currently disabled, appear dimmed.

To operate the target through the action menu, proceed as follows:

1. Right-click the target (for example, motor symbol). The action menu will be displayed next to the target.
2. Point to the desired operation or submenu.
3. Click the desired operation.



Action menu may contain several submenus.

3. Controller User Manual

3.1 Introduction

Controller Component

The controller component displays all the basic information of a single controller. With the controller component, you can operate the controller and, when necessary, open the loop window, which displays more detailed information on the controller.

In the controller component, the following information is displayed:

- [Measurement value and unit](#)
- [Setpoint value](#)
- [Output value and unit](#)
- [Control mode \(M/A/F\)](#)
- [Selected setpoint \(L/R/C\)](#)
- [Alarm status](#)
- [System fault status](#)
- Other information, such as tag identifier and/or name in plain text

In the controller component, you can perform the following operations:

- [Select control mode](#)
- [Select setpoint](#)
- [Change the value of the selected setpoint](#)
- [Change output value](#)

Loop Window

You can open the loop window through the action menu of the controller component. With the loop window, you can perform the same operations as with the controller component and, additionally, you can also view more detailed information on the states and parameters of the controller.

In addition to the operations common with the controller component, you can perform the following operations in the loop window:

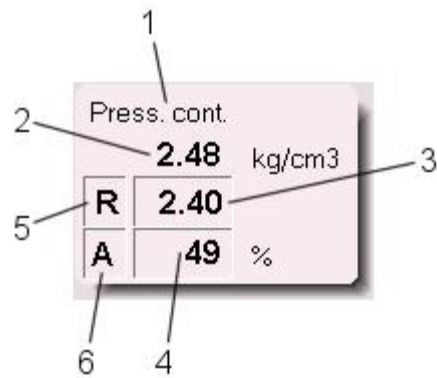
- [Change the view of the loop window](#)
- [Change all setpoint values](#)
- [Change controller parameters](#)
- [Change event limits](#)
- [Change event masks](#)

3.2 Contents of the Controller

3.2.1 Controller Component

Elements

The controller component consists of the following elements:



1. Tag identifier and/or name
2. [Measurement value and unit](#)
3. [Setpoint value](#)
4. [Output value and unit](#)
5. [Selected setpoint](#)
6. [Control mode](#)

Displayed Values

The following numerical values are displayed on the controller component:

- Measurement value
- Setpoint value
- Output value

The measurement value displays the value measured from the process. The controller attempts to keep this value the same as the setpoint value.

The setpoint value displays the current [selected setpoint](#) value. The controller uses this value for adjusting the process.

The output value is used by the controller to control the actuator. This is typically a valve position, for example.

The number of the displayed numerical values may vary depending on the controller component configuration.

Control Mode

The control mode indicates the mode of the controller. Modes are displayed as follows:

A	Automatic control (the output value is determined by the controller)
M	Manual control (the output value can be set by the user)
F	Forced control (the controller is regulated to predetermined value)
f	Forced manual control (the controller is forced to manual control mode)

Selected Setpoint

The currently selected setpoint indicates which setpoint value is used for adjustment. The three alternative setpoints are:

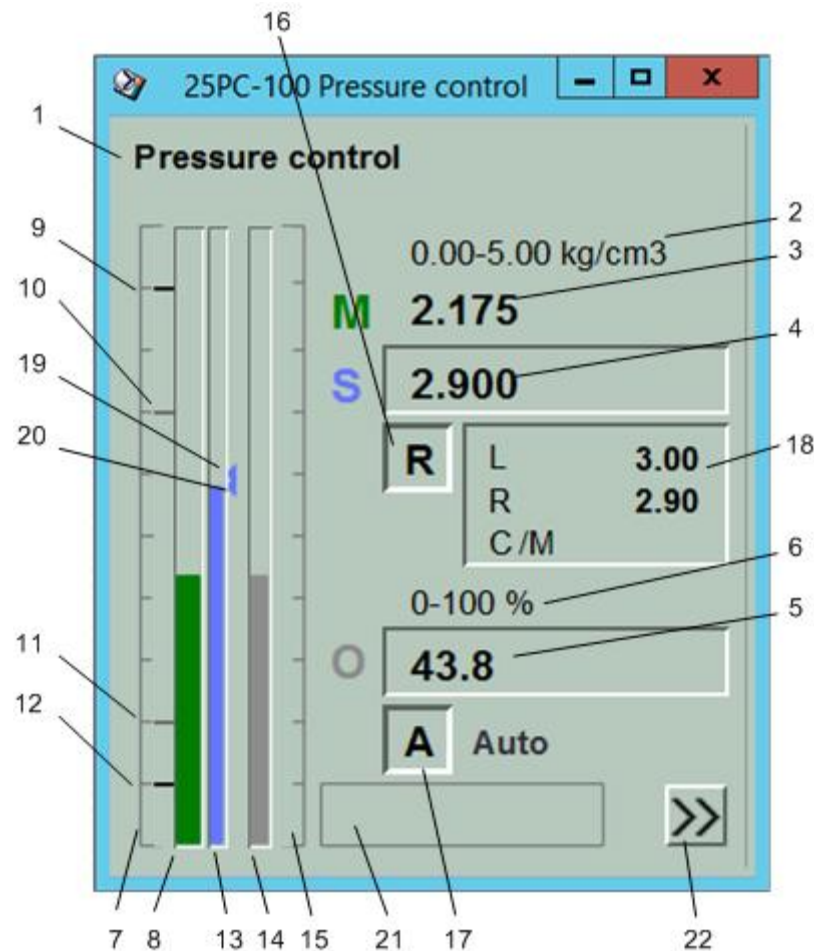
L	Local user-defined setpoint
R	Remote setpoint received from, for example, another controller
C	Computer setpoint

The value of the local setpoint L can be modified while the values of R and C are usually fixed.

3.2.2 Loop Window

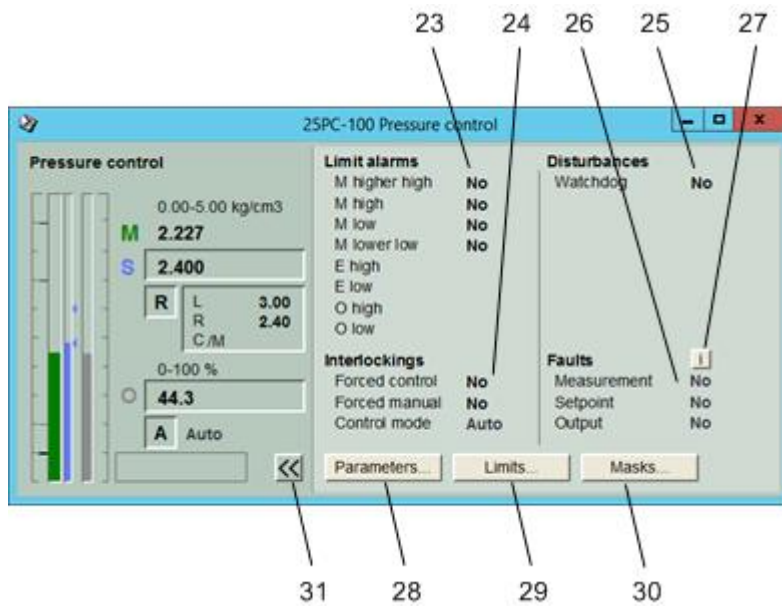
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for parameter operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. Measuring range and unit
3. **Measurement value** in numerical format
4. Value of the **selected setpoint** in numerical format
5. **Output value** in numerical format
6. Unit of output value
7. Scale for measurement and setpoint value bars
8. Measurement value in graphical format
9. Higher high limit of measurement
10. High limit of measurement
11. Low limit of measurement
12. Lower low limit of measurement
13. Value of the selected setpoint in graphical format
14. Output value in graphical format
15. Scale for output value bar
16. **Selected setpoint**
17. **Control mode**
18. Setpoint values
19. Local setpoint marker
20. Remote setpoint marker
21. Disturbance field
22. **Unfold button**

In addition, the expanded view displays the following:



- 23. Limit alarms
- 24. Interlockings
- 25. Disturbances and their specifiers
- 26. Faults
- 27. Fault window button
- 28. Parameter dialog button
- 29. Limit dialog button
- 30. Event mask dialog button
- 31. Fold button

Displayed Values

Measurement, setpoint and output values are displayed in the loop window in both numerical and graphical format (bars). The higher high and the lower low limits of the measurement value are marked with black lines on the scale next to the measurement bar. The high and low limits have been correspondingly marked with gray lines.

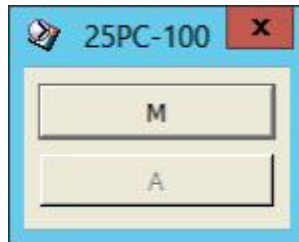
All defined setpoint values are displayed in numerical format in a separate field and in graphical format (blue triangles) next to the setpoint value bar.

3.3 Operations

3.3.1 How to Change the Control Mode

When you want to change the control mode, proceed as follows:

1. Click the control mode symbol in the controller component or in the loop window. The following dialog will appear on the desktop:

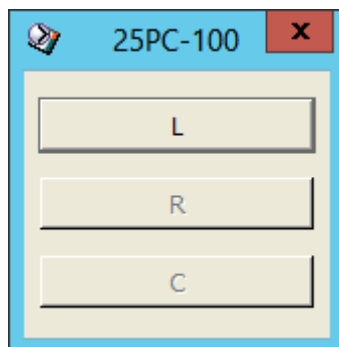


2. Click the desired control mode button. The button of the currently active control mode appears dimmed. The dialog will close and the controller will switch to the selected control mode.

3.3.2 How to Change the Setpoint Selection

When you want to change the setpoint selection, proceed as follows:

1. Click the setpoint symbol in the controller component or in the loop window. The following dialog will appear on the desktop:



2. Click the desired setpoint button. The button of the currently selected setpoint appears dimmed. The dialog will close and the controller will switch to use the selected setpoint.

The C setpoint cannot be selected unless the controller is driven by a process computer.

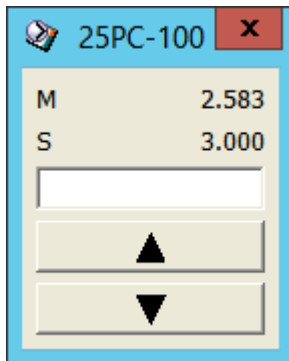
3.3.3 How to Change the Setpoint Value

By entering a new setpoint value, you instruct the controller to adjust the measurement to the given value. You can change the setpoint value of a controller manually if it has been allowed for the tag in question.

How to Change the Value of the Selected Setpoint

When you want to change the value of the selected setpoint, proceed as follows:

1. Click the setpoint value of the controller in the controller component or in the loop window. The following dialog will appear on the desktop:



2. Type the new setpoint value in the input field, and then press Enter to update the value in the system. Alternatively, you can click the arrow buttons to increase or decrease the percentage value. Each change in value is directly updated in the system. The value of a step is implementation-specific.

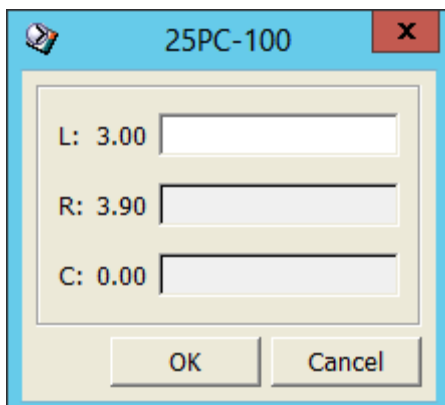
When necessary, click the  button to close the dialog.

How to Change Setpoint Values

In the loop window, you can change not only the selected but all setpoint values.

When you want to change setpoint values, proceed as follows:

1. Click the setpoint value field in the loop window. The setpoint value dialog will open on the desktop:



2. Click the input field of the setpoint value you want to change.
3. Enter the desired value.
4. Click the OK button. The dialog will close and the new value will be updated to the system.

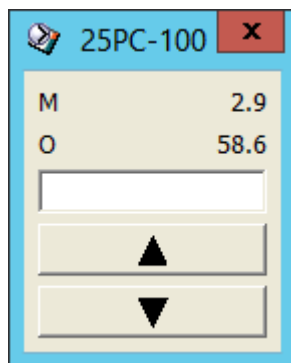
NOTE! Setpoint value modification may have been prevented in controller component configuration. If the input field is dimmed, the setpoint value is not operable.

3.3.4 How to Change the Output Value

The output value of the controller can be changed manually if it has been permitted in controller component configuration. In practice this may mean, for example, adjusting the valve position from the control room. The controller must be in manual control mode and it cannot be in forced or local control. There is, however, one exception: A controller can be loop-specifically configured to permit manual changes in output value even when in automatic control.

When you want to change the output value, proceed as follows:

1. Click the output value of the controller in the controller component or in the loop window. The following dialog will appear on the desktop:



2. Type the new output value in the input field, and then press Enter to update the value in the system. Alternatively, you can click the arrow buttons to increase or decrease the percentage value. Each change in value is directly updated in the system. The value of a step is implementation-specific.

When necessary, click the  button to close the dialog.

3.3.5 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the controller component.
2. On the appearing action menu, select Loop Window. The loop window will open on the desktop.

3.3.6 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

When you want to fold the loop window, proceed as follows:

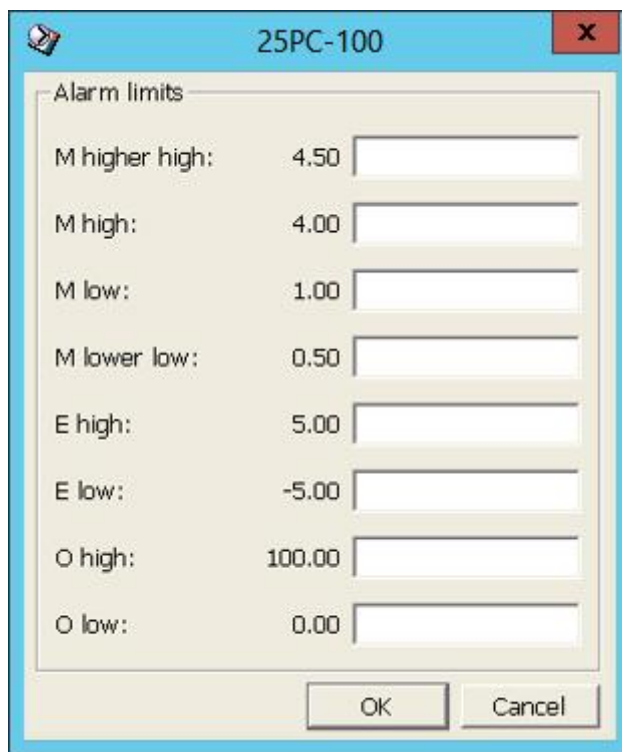
- Click the << button in the loop window.

3.3.7 How to Change the Event Limits

In the Alarm limits dialog, you can change the limit values of the events. When any of these values is exceeded, an alarm is triggered unless it has been disabled in the [Alarm masks dialog](#).

When you want to change the event limit values, proceed as follows:

1. Click the Limits... button in the loop window. The Alarm limits dialog will open on the desktop:



M higher high	Measurement higher high limit
M high	Measurement high limit
M low	Measurement low limit
M lower low	Measurement lower low limit
E high	Error high limit
E low	Error low limit
O high	Output value high limit
O low	Output value low limit

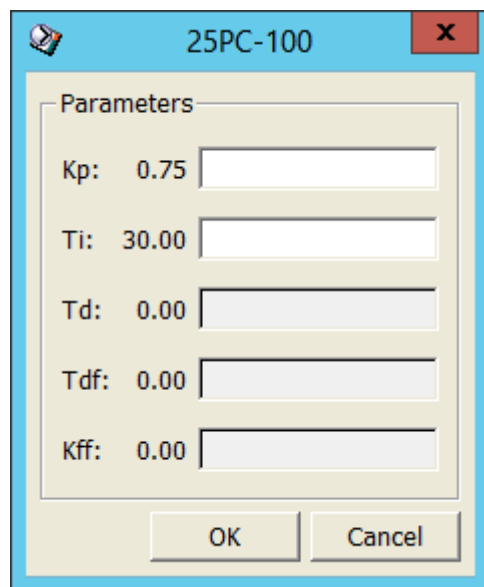
If the input field of an event limit is dimmed, the limit value is not operable.

2. Click the input field of the event limit you want to change.
3. Enter the desired value.
4. Click the OK button. The dialog will close and the new value will be updated to the system.

3.3.8 How to Change the Parameters

When you want to change the parameter values of the controller, proceed as follows:

1. Click the Parameters... button in the loop window. The Parameters dialog will open on the desktop.



Kp	Proportional gain
Ti	Integral action time
Td	Derivative action time
Tdf	Derivative action time filter constant
Kff	Feedforward channel gain

If the input field of a parameter is dimmed, the parameter value is not operable.

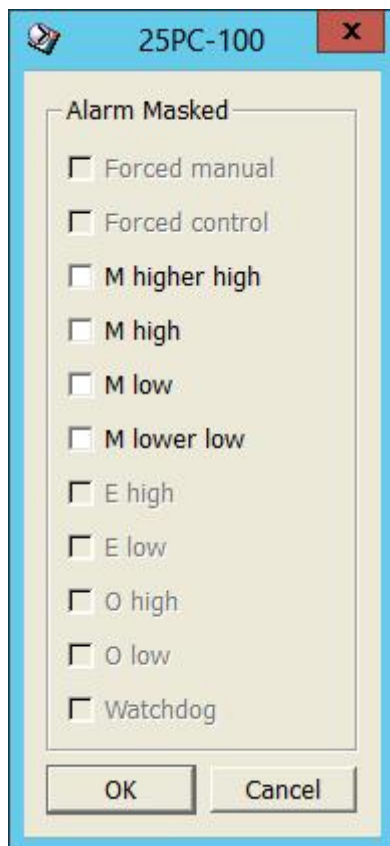
2. Click the input field of the parameter you want to change.
3. Enter the desired value.
4. Click the OK button. The dialog will close and the new value will be updated to the system.

3.3.9 How to Change the Event Masks

In the Alarm masks dialog, you can toggle event masks and thus temporarily prevent events from triggering alarms. Some events are status-related (for example, disturbances) while others are value-related and their limits can be changed in the [Alarm limits dialog](#).

When you want to toggle the event masks, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop.



Forced manual	Forced manual control
Forced control	Forced control
M higher high	Measurement higher high limit
M high	Measurement high limit
M low	Measurement low limit
M lower low	Measurement lower low limit
E high	Error high limit
E low	Error low limit
O high	Output value high limit
O low	Output value low limit
Watchdog	Watchdog disturbance

Dimmed event masks are not operable.

2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.

3. Click the OK button. The dialog will close and the changes will be updated to the system.

3.3.10 Error Messages

The following error messages may occur in controller operations:

- *The loop is not operable.*
- *The entered value is above the maximum limit.*
- *The entered value is below the minimum limit.*
- *The operation is disabled because the loop is in automatic mode.*
- *The operation is disabled because the loop is in local mode.*
- *The operation is disabled because forced control is active.*
- *The operation is disabled because forced manual is active.*
- *Switching from automatic mode to manual mode is disabled.*
- *Switching from manual mode to automatic mode is disabled.*
- *The operation is disabled, because the setpoint is following the measurement.*
- *You are not authorized to perform the action.*

3.4 Status Information

3.4.1 Alarms

Controller alarm indicates that the controller is in an undesired state. The alarm is indicated by the red background color of the measurement value and the measurement bar. In the expanded loop window, the alarm is also indicated by the changed status (*Active*) of the event which has caused the alarm. The Active status is highlighted in red. The background color blinks if the alarm has not yet been acknowledged.

The controller may have any of the following alarms:

Limit alarms	· Measurement value alarm · Error signal alarm · Output signal alarm
Interlockings	· Forced control alarm · Forced manual control alarm
Disturbances	· Watchdog alarm

3.4.2 Limit Alarms

Limit alarms indicate that the controller's measurement, error, or output value has exceeded the allowed higher or lower alarm limit. There are eight alarm limits:

M higher high	Measurement higher high limit
M high	Measurement high limit
M low	Measurement low limit
M lower low	Measurement lower low limit
E high	Error high limit
E low	Error low limit
O high	Output value high limit
O low	Output value low limit

3.4.3 Interlockings

The purpose of an interlocking is to prevent incorrect operations and to force the controller into a required state when necessary. In case of disturbances, for example, the purpose of the interlockings is to put the process into a secured state. Interlockings are caused by other control loops or sequence controls.

There are two interlockings:

- Forced control, which forces the controller to a specific output value
- Forced manual control, which forces the controller into manual control mode

Both interlockings may trigger an alarm.

3.4.4 Disturbances

A disturbance indicates a fault in the control loop. It triggers an alarm and, when necessary, a shift to forced control. When a disturbance occurs, a message of it will be displayed in the [dedicated field](#) in the loop window. The disturbance specifier is displayed below the status text.

A disturbance can have any of the following specifiers:

- Local control
- Actuator fault + forced man.
- Setpoint fault
- Measurement fault
- Meas. fault + forced contrl.
- Meas. fault + forced man.

3.4.5 Faults

A controller fault indicates that part of the presented information is not reliable. In the controller component, a fault is indicated with the violet background color of the measurement value. In the loop window, a fault is indicated with the violet background color of the object connected with the faulty signal.

The controller may have any of the following system faults:

- Measurement value fault
- Setpoint fault
- Output signal fault

When a fault occurs, a series of fault numbers (1 – 7) highlighted in yellow is displayed in the loop window in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.

Faults	
Measurement	Fault 45
Setpoint	Fault 45
Output	Fault 45

The Fault window can be opened by clicking the  button.

25PC-100			
Fault	Measurement	Setpoint	Output
1 External fault:	No	No	No
2 Inaccurate signal:	No	No	No
3 Not under system control:	No	No	No
4 Invalid data:	Active	Active	Active
5 Old data:	Active	Active	Active
6 Derived fault:	No	No	No
7 Simulation:	No	No	No

4. Counter User Manual

4.1 Introduction

Counter Component

Counter component presents the numerical value of the counter. With the component, you can [set new values](#) for the counter if it has been enabled for the position in question, and also open the loop window which displays more detailed information on the counter.

In the counter component, the following information is displayed:

- Counter value
- Unit of counter
- [Alarm status](#)
- [System fault status](#)
- Other information, such as tag identifier and/or name in plain text

Loop Window

You can open the loop window through the action menu of the counter component. With the loop window, you can set new values similarly to the counter component and, additionally, view and modify other information of the counter.

In addition to setting a new value, you can perform the following operations in the loop window:

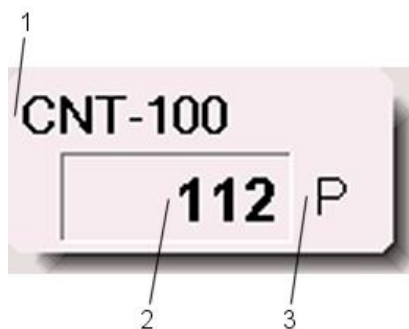
- [Change event masks](#)
- [View signal faults](#)

4.2 Contents of the Counter

4.2.1 Counter Component

Elements

The counter component consists of the following elements:

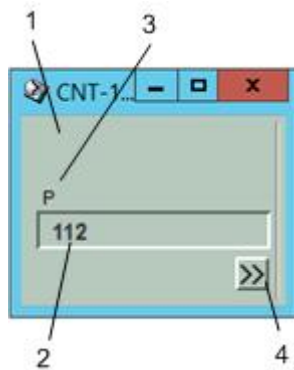


1. Name and/or tag identifier
2. Counter value
3. Unit of measure

4.2.2 Loop Window

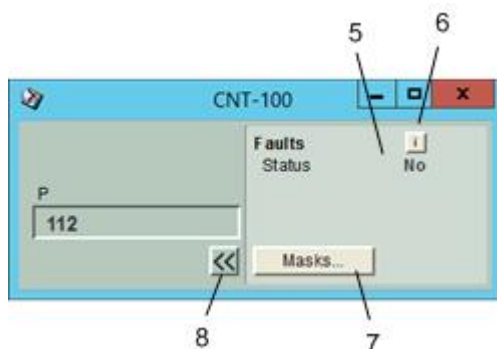
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for parameter operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. Counter value
3. Unit of measure
4. [Unfold button](#)

In addition, the expanded view displays the following:



5. [Faults](#)
6. Fault window button
7. [Event mask](#) window button
8. [Fold button](#)

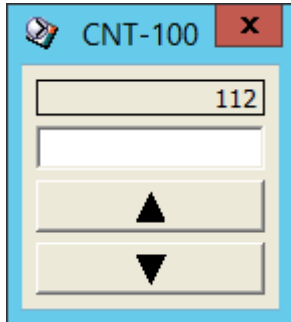
4.3 Operations

4.3.1 How to Change the Counter Value

You can change the counter value if this has been enabled for the tag in question.

When you want to enter a new value for the counter, proceed as follows:

1. Click the counter value in the counter component or loop window. The operation dialog will open on the desktop:



2. Add a new value by clicking the arrow buttons. The new value is updated into the system with each click.

OR

Click the value entry field and type a new value. Update the new value into the system by pressing Enter.

If necessary, click the  button to close the dialog.

4.3.2 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the counter component.
2. On the appearing action menu, select Loop Window. The loop window will open on the desktop.

4.3.3 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

When you want to fold the loop window, proceed as follows:

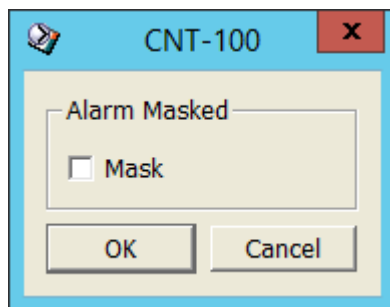
- Click the << button in the loop window.

4.3.4 How to Change the Event Masks

In the Alarm masks dialog, you can toggle event masks and thus temporarily prevent events from triggering alarms.

When you want to toggle the event masks, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop.



2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the changes will be updated into the system.

4.3.5 Error Messages

The following error messages may occur in counter operations:

- *The loop is not operable.*
- *The entered value is outside the limits or otherwise invalid.*
- *You are not authorized to perform the action.*

4.4 Status Information

4.4.1 Alarms

Counter alarm is indicated with the red background color of the numerical value. The background color will blink unless the alarm has been acknowledged.

4.4.2 Faults

A fault indicates that part of the presented information is not reliable. The fault is indicated with the violet background color of the numerical value.

When a fault occurs, a series of fault numbers (1 – 7) highlighted in yellow is displayed in the loop window in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.



The Fault window can be opened by clicking the  button.

Fault	Status
1 External fault	No
2 Inaccurate signal	No
3 Not under system control	No
4 Invalid data	Active
5 Old data	Active
6 Derived fault	No
7 Simulation	No

5. Magnetic Valve User Manual

5.1 Introduction

Magnetic Valve Component

The magnetic valve component displays all the basic information of a single valve. With the valve component, you can operate the valve and, when necessary, open the loop window, which displays more detailed information on the valve.

In the magnetic valve component, the following information is displayed:

- [Valve position](#)
- [Control mode](#)
- [Operation prohibition](#)
- [Alarm status](#)
- [System fault status](#)
- Other information, such as tag identifier and/or name in plain text

In the magnetic valve component, you can perform the following operations:

- [Open/close the valve](#)
- [Change control mode](#)

Loop Window

You can open the loop window through the action menu of the valve component. With the loop window, you can perform the same operations as with the valve component and, additionally, you can also view more detailed information on the states and parameters of the valve.

In addition to the operations common with the valve component, you can perform the following operations in the loop window:

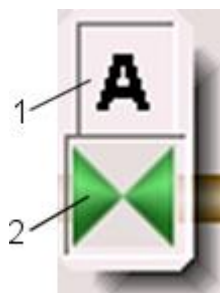
- [Disable/enable operations](#)
- [Change limit switch mask](#)
- [Change event masks](#)
- [View faults](#)

5.2 Contents of the Magnetic Valve

5.2.1 Magnetic Valve Component

Elements

The magnetic valve component consists of the following elements:



1. Control mode
2. Magnetic valve symbol

In addition, the valve component may have title texts, such as name and/or tag identifier.

Position of the Magnetic Valve

The magnetic valve has four normal positions which manifest themselves in the valve symbol as follows:



Valve is closed.



Valve is opening. The symbol blinks.



Valve is open.



Valve is closing. The symbol blinks.

Control Mode

The control mode indicates the mode of the magnetic valve. Modes are displayed in uppercase letters as follows:

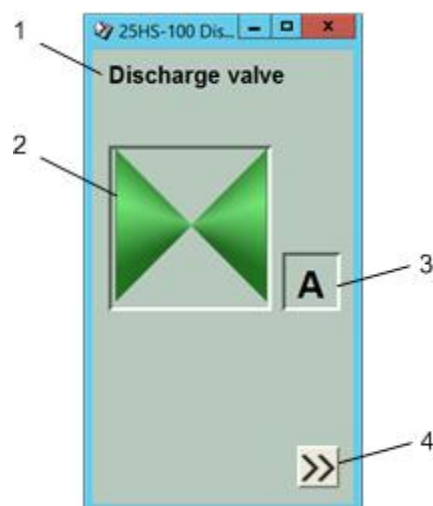
A	Automatic control (the valve is opened and closed by the system)
M	Manual control (the valve is opened and closed by the user)
L	Local control (the valve is controlled locally in the field)

NOTE! The change of the control mode may have been disabled or there may be only two modes available.

5.2.2 Loop Window

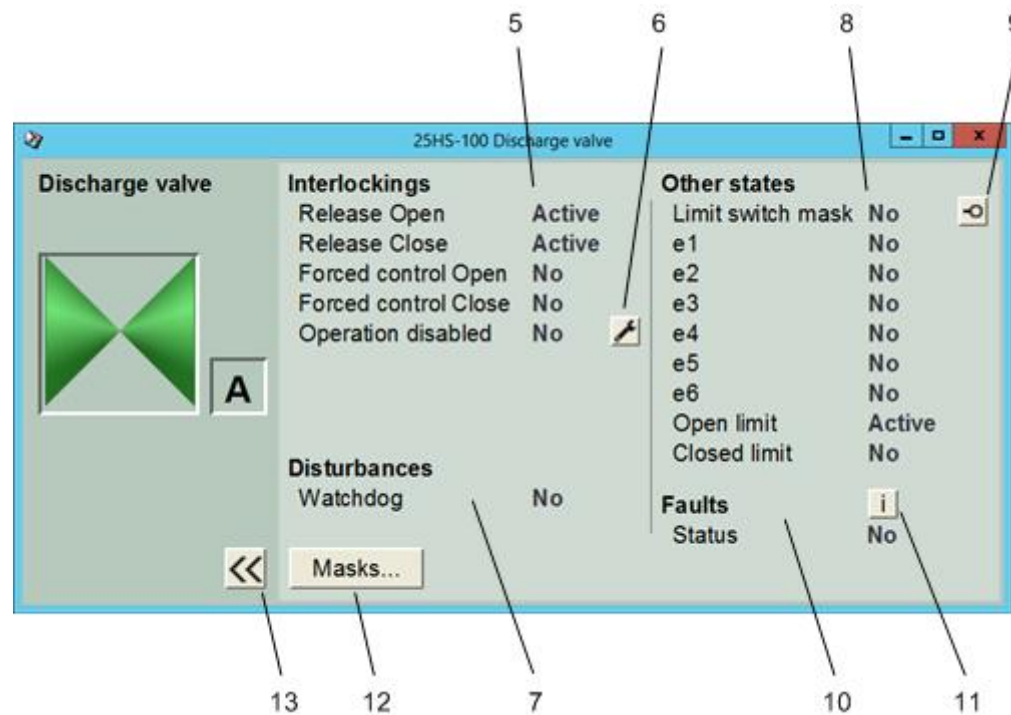
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. Magnetic valve symbol
3. Control mode
4. Unfold button

In addition, the expanded view displays the following:



- 5. [Interlockings](#)
- 6. [Operability](#) dialog button
- 7. [Disturbances](#)
- 8. [Other states](#)
- 9. [Limit switch mask](#) dialog button
- 10. [Faults](#)
- 11. Fault window button
- 12. [Event mask](#) dialog button
- 13. [Fold](#) button

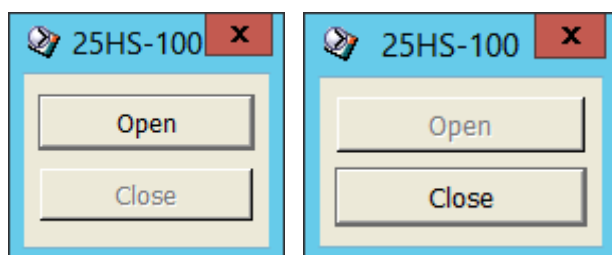
5.3 Operations

5.3.1 How to Open or Close the Magnetic Valve

The magnetic valve can be opened or closed from the valve component or loop window if the valve is in manual control and its operations have not been disabled.

When you want to open or close the valve, proceed as follows:

1. Click the valve symbol in the valve component or loop window. The valve operation dialog will appear on the desktop:



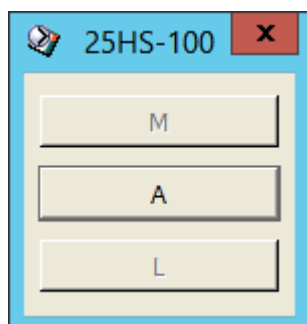
The command button corresponding to the current valve position appears dimmed.

2. Open the valve by clicking the Open button or close the valve by clicking the Close button. The dialog will close and the position of the valve will change accordingly.

5.3.2 How to Change the Control Mode

When you want to change the control mode for the magnetic valve, proceed as follows:

1. Click the control mode symbol in the magnetic valve component or in the loop window. The control mode dialog will appear on the desktop:



The command button corresponding to the currently active control mode appears dimmed. Local control (L) may have been prevented for the tag in question in which case its button is also dimmed.

2. Click the desired control mode button. The dialog will close and the valve will switch to the selected control mode.

5.3.3 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the magnetic valve component.
2. On the appearing action menu, select Loop Window. The loop window will open on the desktop.

5.3.4 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

When you want to fold the loop window, proceed as follows:

- Click the << button in the loop window.

5.3.5 How to Disable or Enable Operations

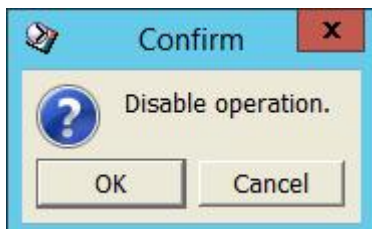
Magnetic valve operations may be temporarily disabled, for example, during service. Operation prohibition is set and removed in the loop window of the valve. As long as the operations are disabled, a tool symbol will be displayed on top of the valve symbol.



NOTE! Temporary disabling only prevents the operations executed in the magnetic valve component or loop window. It does not prevent the magnetic valve to be controlled by other means, e.g. sequence, group control, or other automatic feature.

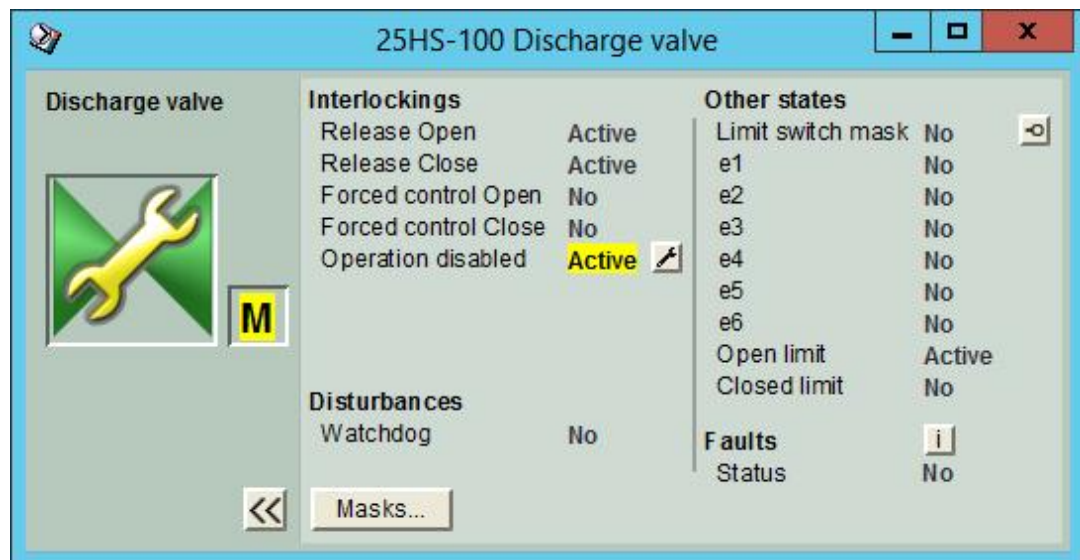
When you want to temporarily disable the operations, proceed as follows:

1. Click the  button in the loop window. The confirmation dialog will appear on the desktop:



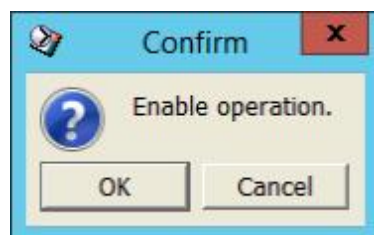
2. Click the OK button.

The dialog will close and the tool symbol will appear on top of the valve symbol to indicate that operations have been disabled. The state of the Operation disabled interlocking will also change to *Active*.



Operations are enabled in a similar method:

1. Click the  button in the loop window. The confirmation dialog will appear on the desktop:



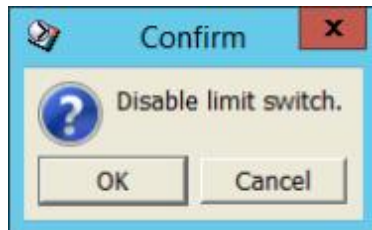
2. Click the OK button. The dialog will close, the tool symbol will disappear from the valve symbol, and the state of the Operation disabled interlocking will return to *No*.

5.3.6 How to Change the Limit Switch Mask

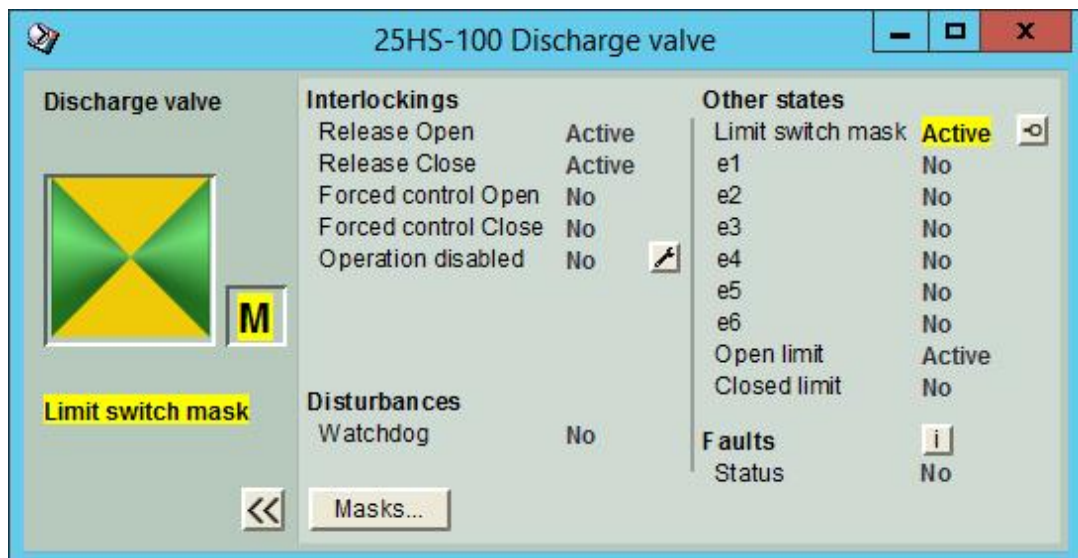
A faulty limit switch can be temporarily passivated in order to prevent it from hindering normal valve operation. The limit switch mask is an optional feature which can be set or removed in the loop window of the magnetic valve.

When you want to set the limit switch mask, proceed as follows:

1. Click the  button in the loop window. The limit switch mask dialog opens on the desktop:

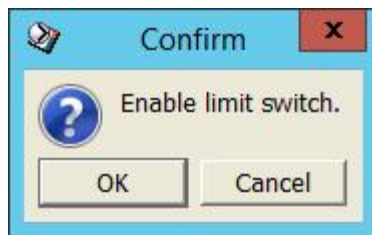


2. Click the OK button. The dialog will close, the background color of the valve symbol will change to yellow, and the Limit switch mask state will change to *Active*. In addition, a separate state message will appear below the valve symbol in the loop window.



The limit switch mask is removed in a similar method:

1. Click the  button in the loop window. The limit switch mask dialog opens on the desktop:



2. Click the OK button. The dialog will close, the yellow background color will disappear, and the state information will restore to normal.

5.3.7 How to Change the Event Masks

In the Alarm masks dialog, you can toggle event masks and thus temporarily prevent events from triggering alarms.

When you want to toggle the event masks, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop:



Listed events refer to the status information displayed in the loop window. Statuses that have not been configured in the application appear dimmed.

2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the changes will be updated to the system.

5.3.8 Error Messages

The following error messages may occur in magnetic valve operations:

- *The loop is not operable.*
- *The operation is disabled because the loop is in automatic mode.*
- *The operation is disabled because the loop is in local mode.*
- *The operation is disabled because a release is missing.*
- *The operation is disabled because forced control is active.*
- *Switching to automatic or local mode is disabled.*
- *Switching to manual or local mode is disabled.*
- *Switching from local mode to system's control is disabled.*
- *The operation is disabled because forced manual is active.*
- *The operation is disabled because an external disturbance is active.*
- *You are not authorized to perform the action.*

5.4 Status Information

5.4.1 Alarms

Magnetic valve alarm indicates that the valve is in an undesired state. The alarm is indicated with the red background color on the valve symbol. In the expanded loop window, the alarm is also indicated by the changed status (*Active*) of the event which has caused the alarm. The Active status is highlighted in red. The background color blinks if the alarm has not yet been acknowledged.

The valve may have the following alarms:

Interlockings	· Forced control open · Forced control close
Disturbances	· Watchdog
Other states	· External input

5.4.2 Interlockings

The purpose of an interlocking is to prevent incorrect operations and to force the magnetic valve into a required state when necessary. In case of disturbances, for example, the purpose of the interlockings is to put the process into a secured state. Interlockings are caused by other control loops or sequence controls.

There are five interlockings:

Release Open	Open release is missing.
Release Close	Close release is missing.
Forced control Open	Valve is forced to open.
Forced control Close	Valve is forced to close.
Operation disabled	Valve operation has been disabled.

Forced control Open and Forced control Close may also cause an alarm.

5.4.3 Disturbances

A disturbance indicates a fault in the magnetic valve or in its controls. It triggers an alarm and, when necessary, a shift to forced control. The disturbance specifier is displayed below the status text.

A disturbance can have any of the following specifiers:

- Open limit disappeared
- Close limit disappeared
- Open control time exceeded
- Close control time exceeded
- Jammed when opened
- Jammed when closed
- Open limit fault
- Close limit fault
- Open control fault
- Close control fault
- Open status faulty
- Close status faulty

5.4.4 Other States

With the exception of Limit switch mask, other states consist of the plant-specific external inputs. All inputs may not be in use. An external input may cause a disturbance. The name of the input refers to the true origin of the disturbance.

5.4.5 Faults

A fault indicates that part of the presented information is not reliable. The fault is indicated with the violet background color of the magnetic valve symbol.

A magnetic valve can only have a status-related system fault.

When a fault occurs, a series of fault numbers (1 - 7) highlighted in yellow is displayed in the loop window in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.



The Fault window can be opened by clicking the  button.

Fault	Status
1 External fault	No
2 Inaccurate signal	No
3 Not under system control	No
4 Invalid data	Active
5 Old data	Active
6 Derived fault	No
7 Simulation	No

6. Measurement User Manual

6.1 Introduction

Measurement Component

Measurement component presents measured and set values in numerical format. With the component, you can [set new values](#) for the measurement if it has been enabled for the position in question, and also open the loop window which displays more detailed information on the measurement.

In the measurement component, the following information is displayed:

- Measured value
- Unit of measure
- [Alarm status](#)
- [Fault status](#)
- Other information, such as tag identifier and/or name in plain text

Loop Window

You can open the loop window through the action menu of the measurement component. With the loop window, you can set new values similarly to the measurement component and, additionally, view and modify other information of the measurement.

In addition to setting a new value, you can perform the following operations in the loop window:

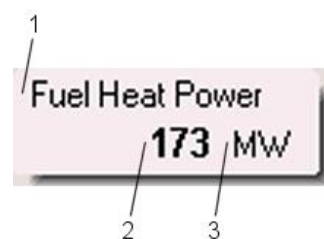
- [Change event delays](#)
- [Change event limits](#)
- [Change event masks](#)
- [View faults](#)

6.2 Contents of the Measurement

6.2.1 Measurement Component

Elements

The measurement component consists of the following elements:

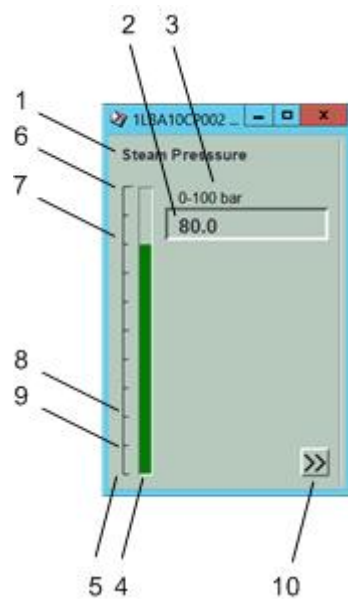


1. Name and/or tag identifier
2. Measured value
3. Unit of measure

6.2.2 Loop Window

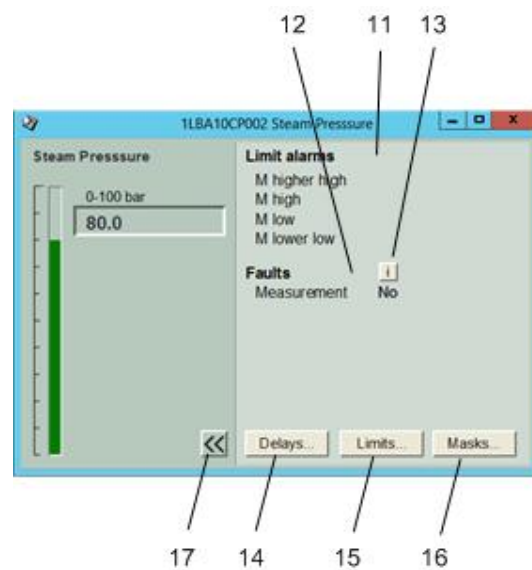
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for parameter operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. Measured/set value in numerical format
3. Measurement/setting range and unit of measure
4. Measured/set value in graphical format
5. Measurement bar scale
6. Higher high limit of measurement
7. High limit of measurement
8. Low limit of measurement
9. Lower low limit of measurement
10. [Unfold button](#)

In addition, the expanded view displays the following:



- 11. Limit alarms
- 12. Faults
- 13. Fault window button
- 14. Alarm delay window button
- 15. Event limit window button
- 16. Event mask window button
- 17. Fold button

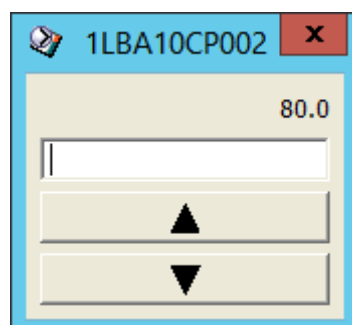
6.3 Operations

6.3.1 How to Change the Measurement Value

You can change the measurement value if this has been enabled for the tag in question.

When you want to enter a new value for the measurement, proceed as follows:

1. Click the measured numerical value in the measurement component or loop window. The operation dialog will open on the desktop:



2. Add a new value by clicking the spin buttons. The new value is updated into the system with each click.

OR

Click the value entry field and enter a new value with the keyboard. Update the new value into the system by pressing the line feed key.

If necessary, click the  button to close the dialog.

6.3.2 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the measurement component.
2. On the appearing action menu, select Loop Window . The loop window will open on the desktop.

6.3.3 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

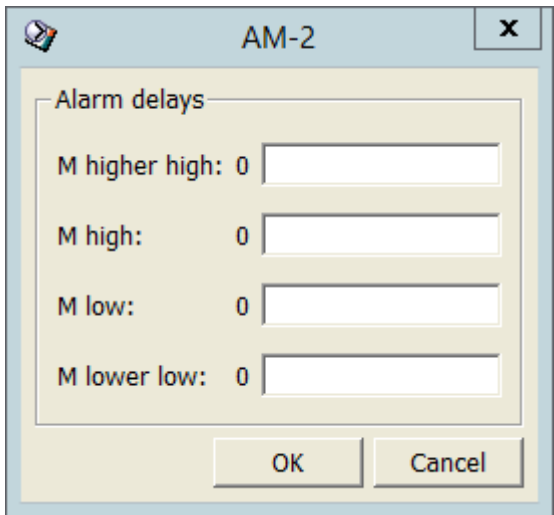
When you want to fold the loop window, proceed as follows:

- Click the << button in the loop window.

6.3.4 How to Change the Event Delays

When you want to change the event delays, proceed as follows:

1. Click the Delays... button in the loop window. The Alarm delays dialog will open on the desktop.



M higher high	Delay of the higher high limit
M high	Delay of the high limit
M low	Delay of the low limit
M lower low	Delay of the lower low limit

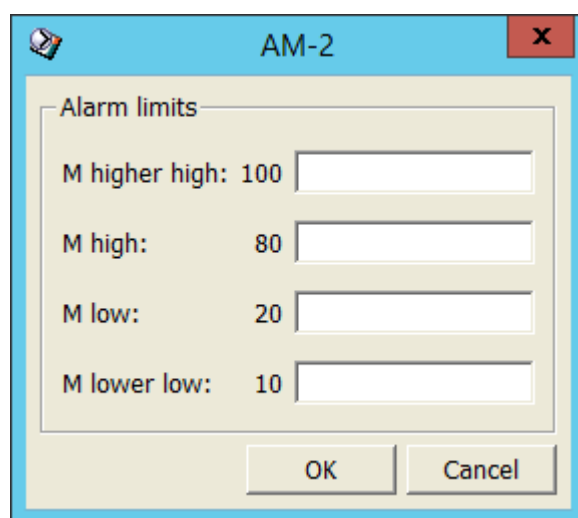
2. Click the input field of the desired delay and enter a new value with the keyboard. The value of the delay is given in seconds.
3. Click the OK button. The dialog is closed and the changes are updated into the system

NOTE! Event delay modification may be prevented in the configuration of the application.

6.3.5 How to Change the Event Limits

When you want to change the event limits, proceed as follows:

1. Click the Limits... button in the loop window. The Alarm limits dialog will open on the desktop.



M higher high	Higher high limit of the measured value/setpoint
M high	High limit of the measured value/setpoint
M low	Low limit of the measured value/setpoint
M lower low	Lower low limit of the measured value/setpoint

2. Click the input field of the desired limit and enter a new value with the keyboard.
3. Click the OK button. The dialog will close and the changes are updated into the system.

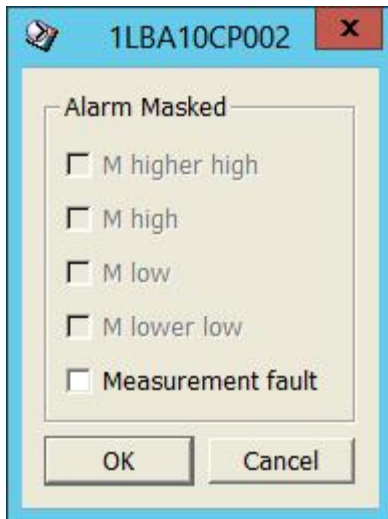
NOTE! Event limit modification may be prevented in the configuration of the application.

6.3.6 How to Change the Event Masks

In the Alarm masks dialog, you can toggle event masks and thus temporarily prevent events from triggering alarms.

When you want to toggle the event masks, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop.



M higher high	Higher high limit of the measurement value
M high	High limit of the measurement value
M low	Low limit of the measurement value
M lower low	Lower low limit of the measurement value
Measurement fault	Measurement (input) fault

2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the changes will be updated into the system.

6.3.7 Error Messages

The following error messages may occur in measurement operations:

- *The loop is not operable.*
- *The entered value is outside the limits or otherwise invalid.*
- *You are not authorized to perform the action.*

6.4 Status Information

6.4.1 Alarms

Measurement alarm indicates that the measurement is in a undesired state. The measurement alarm is indicated by the red background color of the measurement value and the measurement bar. In the expanded loop window, the alarm is also indicated by the changed status (Active) of the event which has caused the alarm. The Active status is highlighted in red. The background color blinks if the alarm has not yet been acknowledged.

The measurement can have only [limit alarms](#).

6.4.2 Limit Alarms

Limit alarms indicate that the measured value has exceeded the set alarm limit. There are four alarm limits:

M higher high	Higher high limit of the measurement value
M high	High limit of the measurement value
M low	Low limit of the measurement value
M lower low	Lower low limit of the measurement value

6.4.3 Faults

Measurement fault indicates that part of the presented information is not reliable. In the measurement component, a fault is indicated with the violet background color of the measurement value. In the loop window, a fault is indicated with the violet background color of the object connected with the faulty signal.

Measurement can have only measurement signal related faults.

In the loop window, a series of fault numbers (1 – 7) highlighted in yellow is displayed in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.



The Fault window can be opened by clicking the button.

Fault	Measurement
1 External fault	No
2 Inaccurate signal	No
3 Not under system control	No
4 Invalid data	Active
5 Old data	Active
6 Derived fault	No
7 Simulation	No

7. Motor User Manual

7.1 Introduction

This guide can be applied to pumps, fans, and switches as well.

Motor Component

The motor component displays all the basic information of a single motor. With the motor component, you can operate the motor and, when necessary, open the loop window, which displays more detailed information on the motor.

In the motor component, the following information is displayed:

- [Running status](#)
- [Control mode](#)
- [Operation prohibition](#)
- [Alarm status](#)
- [System fault status](#)
- Other information, such as tag identifier and/or name in plain text

In the motor component, you can perform the following operations:

- [Start/stop the motor](#)
- [Select control mode](#)

Loop Window

You can open the loop window through the action menu of the motor component. With the loop window, you can perform the same operations as with the motor component and, additionally, you can also view more detailed information on the states and parameters of the motor.

In addition to the operations common with the motor component, you can perform the following operations in the loop window:

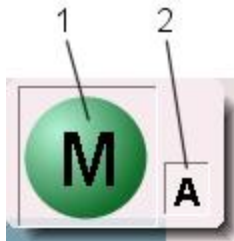
- [Disable/enable operations](#)
- [Change event masks](#)
- [View faults](#)

7.2 Contents of the Motor

7.2.1 Motor Component

Elements

The motor component consists of the following elements:



1. [Running status symbol](#)
2. [Control mode](#)

In addition, the motor component may have title texts, such as tag identifier and/or name.

Running Status

The motor has four normal running statuses which manifest themselves in the motor symbol as follows:



Motor is stopped.



Motor is starting. The symbol is flashing.



Motor is running.



Motor is stopping. The symbol is flashing.

NOTE! Pump, fan, and switch symbols are used when applicable. Other colors and symbols may also be used in plant-specific solutions to present different statuses.

Control Mode

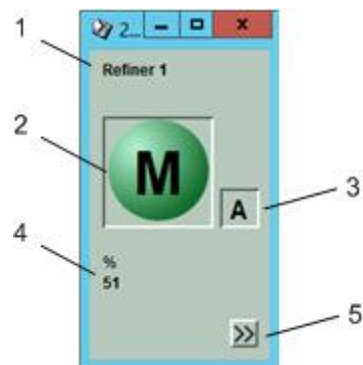
The control mode indicates the mode of the motor. Modes are displayed in uppercase letters as follows:

A	Automatic control (the motor is started and stopped by the system)
M	Manual control (the motor is started and stopped by the user)
L	Local control (the motor is controlled locally in the field)

7.2.2 Loop Window

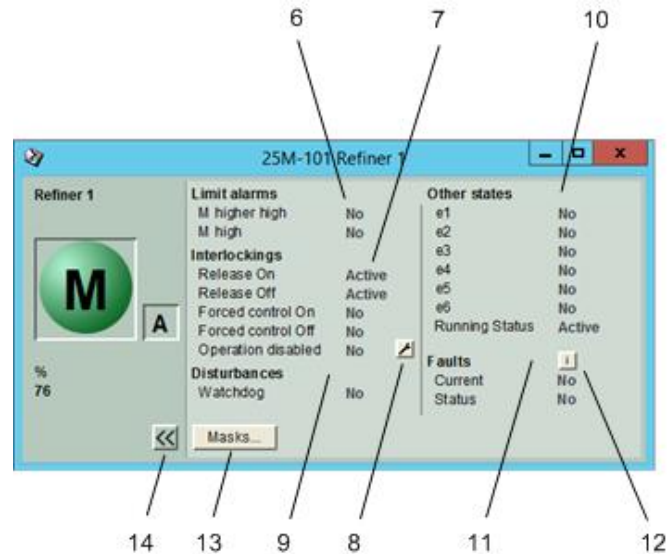
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. [Running status symbol](#)
3. [Control mode](#)
4. Measured current value
5. [Unfold button](#)

In addition, the expanded view displays the following:



6. Limit alarms
7. Interlockings
8. Operability dialog button
9. Disturbances
10. Other states
11. Faults
12. Fault window button
13. Event mask dialog button
14. Fold button

7.3 Operations

7.3.1 How to Start or Stop the Motor

The motor can be started or stopped from the motor component or loop window if it is in manual control and its operations have not been disabled.

When you want to start or stop the motor, proceed as follows:

1. Click the running status symbol in the motor component or loop window. The motor operation dialog will appear on the desktop:



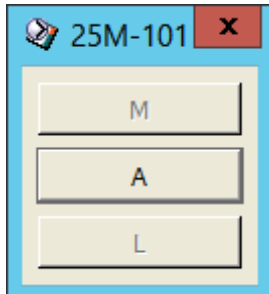
The command button corresponding to the current running status appears dimmed.

2. Start the motor by clicking the Start button or stop the motor by clicking the Stop button. The dialog will close and the status of the motor will change accordingly.

7.3.2 How to Change the Control Mode

When you want to change the control mode for the motor, proceed as follows:

1. Click the control mode symbol in the motor component or in the loop window.
The following dialog will appear on the desktop:



The command button corresponding to the currently active control mode appears dimmed. Local control (L) may have been prevented for the tag in question in which case its button is also dimmed.

2. Click the desired control mode button. The dialog will close and the motor will switch to the selected control mode.

7.3.3 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the motor component.
2. On the appearing action menu, select Loop Window . The loop window will open on the desktop.

7.3.4 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

When you want to fold the loop window, proceed as follows:

- Click the << button in the loop window.

7.3.5 How to Disable and Enable Operations

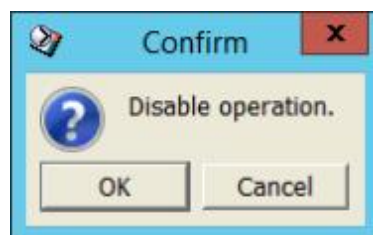
Motor operations may be temporarily disabled, for example, during service. Operation prohibition is set and removed in the loop window of the motor. As long as the operations are disabled, a tool symbol will be displayed on top of the running status symbol.



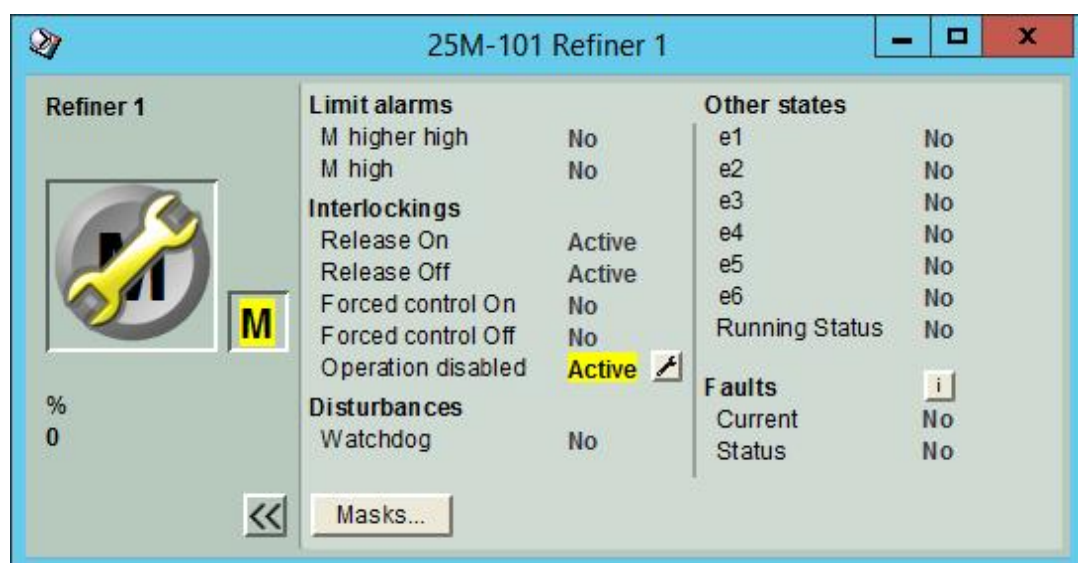
NOTE! Temporary disabling only prevents the operations executed in the motor component or loop window. It does not prevent the motor to be controlled by other means, e.g. sequence, group control, or other automatic feature.

When you want to temporarily disable the operations, proceed as follows:

1. Click the  button in the loop window. The confirmation dialog will appear on the desktop:

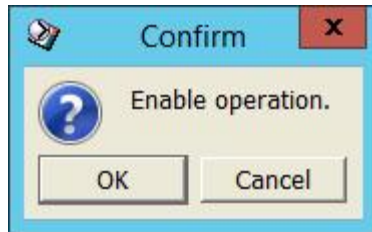


2. Click the OK button. The dialog will close and the tool symbol will appear on top of the running status symbol to indicate that operations have been disabled. The state of the Operation disabled interlocking will also change to *Active*.



Operations are enabled in a similar method:

1. Click the  button in the loop window. The confirmation dialog will appear on the desktop:



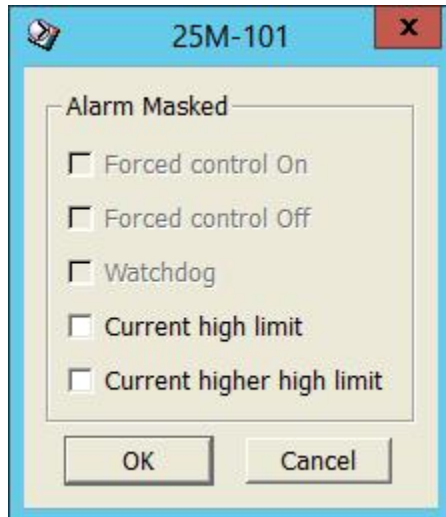
2. Click the OK button. The dialog will close, the tool symbol will disappear from the running status symbol, and the state of the Operation disabled interlocking will return to No.

7.3.6 How to Change the Event Masks

In the Alarm masks dialog, you can toggle event masks and thus temporarily prevent events from triggering alarms.

When you want to toggle the event masks, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop.



Abbreviations refer to the status information displayed in the loop window. Plant-specific statuses that have not been configured appear dimmed and unmarked.

2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the changes will be updated to the system.

7.3.7 Error Messages

The following error messages may occur in motor operations:

- *The loop is not operable.*
- *The operation is disabled because the loop is in automatic mode.*
- *The operation is disabled because the loop is in local mode.*
- *The operation is disabled because a release is missing.*
- *The operation is disabled because forced control is active.*
- *Switching to automatic or local mode is disabled.*
- *Switching to manual or local mode is disabled.*
- *Switching from local mode to system's control is disabled.*
- *The operation is disabled because forced manual is active.*
- *The operation is disabled because an external disturbance is active.*
- *You are not authorized to perform the action.*

7.4 Status Information

7.4.1 Alarms

Motor alarm indicates that the motor is in an undesired state. The alarm is indicated with the red background color on the running status symbol. In the expanded loop window, the alarm is also indicated by the changed status (*Active*) of the event which has caused the alarm. The Active status is highlighted in red. The background color blinks if the alarm has not yet been acknowledged.

The motor may have the following alarms:

Limit alarms	· Current exceeds the higher high limit · Current exceeds the high limit
Interlockings	· Forced control On · Forced control Off
Disturbances	· Watchdog disturbance
Other states	· External input

7.4.2 Limit Alarms

Limit alarms indicate that the motor current has exceeded the allowed alarm limit. There are two alarm limits:

M higher high	Measured current exceeds the higher high limit.
M high	Measured current exceeds the high limit.

7.4.3 Interlockings

The purpose of an interlocking is to prevent incorrect operations and to force the motor into a required state when necessary. In case of disturbances, for example, the purpose of the interlockings is to put the process into a secured state. Interlockings are caused by other control loops or sequence controls.

There are five interlockings:

Release On	On release is missing.
Release Off	Off release is missing.
Forced control On	Motor is forced to run.
Forced control Off	Motor is forced to stop.
Operation disabled	Motor operation has been disabled.

7.4.4 Disturbances

A disturbance indicates a fault in the motor or in its controls. It triggers an alarm and, when necessary, a shift to forced control. The disturbance specifier is displayed below the status text.

A disturbance can have any of the following specifiers:

- Disturbance start
- Disturbance stop
- Failed start
- Failed stop
- Faulty status data
- On control fault
- Off control fault

7.4.5 Other States

Other states consist of the plant-specific external inputs. All inputs may not be in use. An external input may cause a disturbance. The name of the input refers to the true origin of the disturbance.

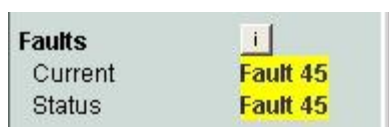
7.4.6 Faults

A fault indicates that part of the presented information is not reliable. It is indicated with the violet background color of the running status symbol.

A motor can have the following system faults:

- Current
- Status

When a fault occurs, a series of fault numbers (1 – 7) highlighted in yellow is displayed in the loop window in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.



The Fault window can be opened by clicking the button.

Fault	Current Status	
1 External fault	No	No
2 Inaccurate signal	No	No
3 Not under system control	No	No
4 Invalid data	Active	Active
5 Old data	Active	Active
6 Derived fault	No	No
7 Simulation	No	No

8. Motor Actuator User Manual

8.1 Introduction

Motor Actuator Component

The motor actuator component displays all the basic information of a single motor actuator. With the motor actuator component, you can operate the actuator and, when necessary, open the loop window, which displays more detailed information on the actuator.

In the motor actuator component, the following information is displayed:

- [Actuator position](#)
- [Control mode](#)
- Position measurement value
- [Operation prohibition](#)
- [Alarm status](#)
- [System fault status](#)
- Other information, such as tag identifier and/or name in plain text

In the motor actuator component, you can perform the following operations:

- [Control the actuator](#)
- [Select control mode](#)
- [Control position measurement](#)

Loop Window

You can open the loop window through the action menu of the motor actuator component. With the loop window, you can perform the same operations as with the actuator component and, additionally, you can also view more detailed information on the states and parameters of the motor actuator.

In addition to the operations common with the motor actuator component, you can perform the following operations in the loop window:

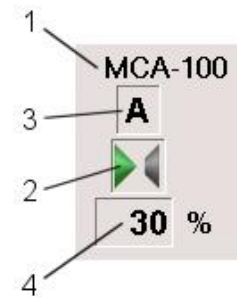
- [Disable/enable operations](#)
- [Acknowledge watch disturbances](#)
- [Acknowledge torque disturbances](#)
- [Change event masks](#)
- [View faults](#)

8.2 Contents of the Motor Actuator

8.2.1 Motor Actuator Component

Elements

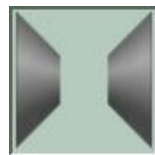
The motor actuator component consists of the following elements:



1. Name and/or tag identifier
2. [Motor actuator symbol](#)
3. [Control mode](#)
4. Motor actuator position in numerical format

Motor Actuator Position

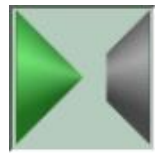
The motor actuator has five normal running statuses which manifest themselves in the actuator symbol as follows:



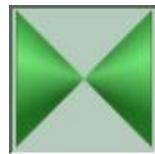
Actuator is closed.



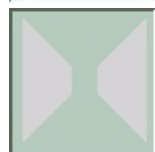
Actuator is opening. The symbol is flashing.



Actuator is in intermediate position.



Actuator is open.



Actuator is closing. The symbol is flashing.

The position of the actuator is also presented in numerical format on the scale of 0 (fully closed) - 100 (fully open) below the actuator symbol.

Control Mode

The control mode indicates the mode of the motor actuator. Modes are displayed in uppercase letters as follows:

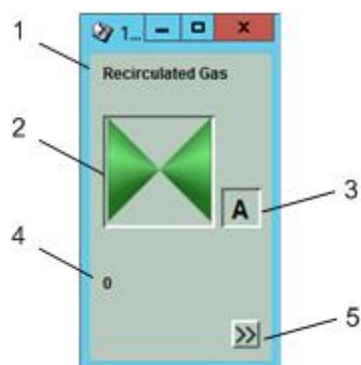
A	Automatic control (the motor actuator is opened and closed by the system)
M	Manual control (the motor actuator is opened and closed by the user)
L	Local control (the motor actuator is controlled locally in the field)

NOTE! The change of the control mode may have been disabled or there may be only two modes available.

8.2.2 Loop Window

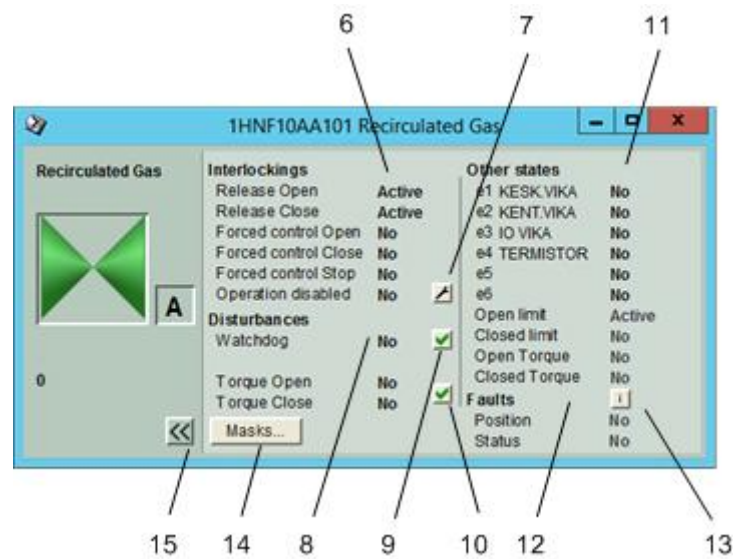
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. [Motor actuator symbol](#)
3. [Control mode](#)
4. Actuator position in numerical format
5. [Unfold button](#)

In addition, the expanded view displays the following:



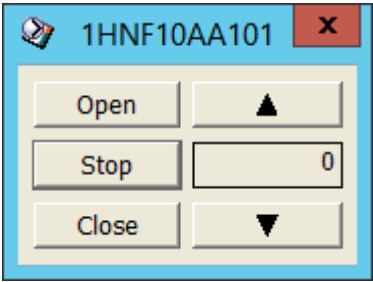
- 6. [Interlockings](#)
- 7. [Operability](#) dialog button
- 8. [Disturbances](#)
- 9. [Watchdog disturbance](#) dialog button
- 10. [Torque disturbance](#) dialog button
- 11. [Other states](#)
- 12. [Faults](#)
- 13. [Fault window](#) button
- 14. [Event mask](#) dialog button
- 15. [Fold](#) button

8.3 Operations

8.3.1 How to Control the Motor Actuator

When you want to control the motor actuator, proceed as follows:

1. Click the motor actuator symbol in the actuator component or in the loop window. The operation dialog will open on the desktop:



Open	Runs the actuator to the open position predefined in the application
Close	Runs the actuator to the close position predefined in the application
Stop	Intercepts the actuator after the Open or Close button has been clicked
▲ ▼	Runs the actuator in the corresponding direction (see picture)
0	Motor actuator position in numerical format

With the spin buttons, you can run the actuator in the desired direction one click at a time which causes the actuator to change its position by a step defined in the application. Alternatively, you can click and hold the spin button which causes the actuator to change its position as long as you hold the button.

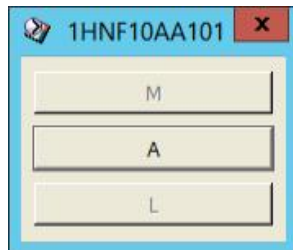
2. Perform the desired operation.
3. Finally, click the  button to close the operation dialog.

NOTE! It may not be possible to run the actuator to the intermediate position. In this case the operation dialog only contains the Open and Close buttons.

8.3.2 How to Change the Control Mode

When you want to change the control mode for the motor actuator, proceed as follows:

1. Click the control mode or the setpoint symbol in the actuator component or in the loop window. The control mode dialog will open on the desktop:



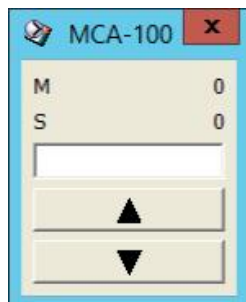
The command button corresponding to the currently active control mode appears dimmed. Local control (L) may have been prevented for the tag in question in which case its button is also dimmed.

2. Click the desired control mode button. The dialog will close and the motor actuator will switch to the selected control mode.

8.3.3 How to Change the Position Setpoint Value

When you want to change the position setpoint value, proceed as follows:

1. Click the position measurement value in the actuator component or in the loop window. The setpoint value dialog will open on the desktop:



M	Measured value
S	Setpoint value

2. Enter a new setpoint value by clicking the spin buttons. The new value is updated into the system with each click.

OR

Click the value entry field and enter a new value with the keyboard. Update the new value into the system by pressing the line feed key.

If necessary, click the  button to close the dialog.

8.3.4 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the motor actuator component.
2. On the appearing action menu, select Loop Window . The loop window will open on the desktop.

8.3.5 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

When you want to fold the loop window, proceed as follows:

- Click the << button in the loop window.

8.3.6 How to Disable and Enable Operations

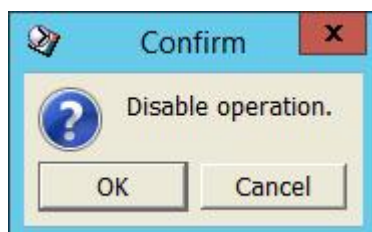
Motor actuator operations may be temporarily disabled, for example, during service. Operation prohibition is set and removed in the loop window of the actuator. As long as the operations are disabled, a tool symbol will be displayed on top of the motor actuator symbol.



NOTE! Temporary disabling only prevents the operations executed in the motor actuator component or loop window. It does not prevent the motor actuator to be controlled by other means, e.g. sequence, group control, or other automatic feature.

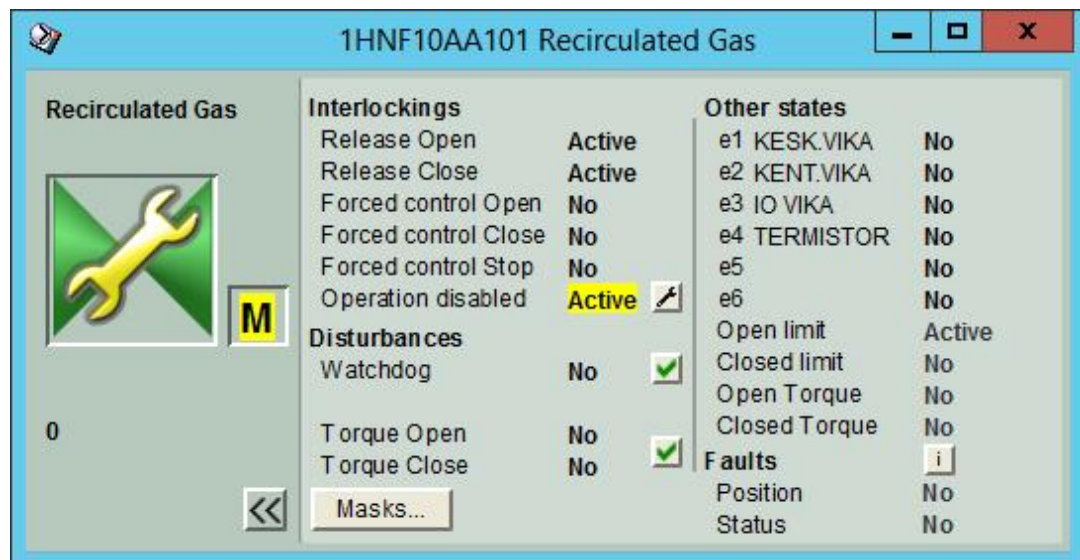
When you want to temporarily disable the operations, proceed as follows:

1. Click the  button in the loop window. The confirmation dialog will appear on the desktop:



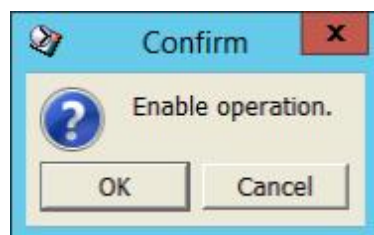
2. Click the OK button.

The dialog will close and the tool symbol will appear on top of the motor actuator symbol to indicate that operations have been disabled. The state of the Operation disabled interlocking will also change to *Active*.



Operations are enabled in a similar method:

1. Click the  button in the loop window. The confirmation dialog will appear on the desktop:



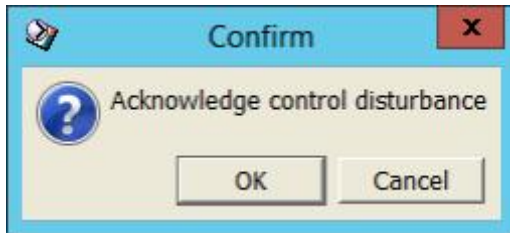
2. Click the OK button. The dialog will close, the tool symbol will disappear from the motor actuator symbol, and the state of the Operation disabled interlocking will return to *No*.

8.3.7 How to Acknowledge Disturbances

When you want to acknowledge watchdog or torque disturbances, proceed as follows:

Watchdog Disturbance

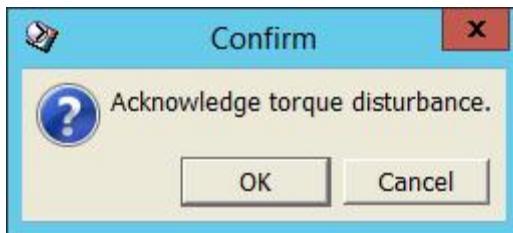
1. Click the watchdog disturbance button  in the loop window. The watchdog disturbance acknowledgement dialog will open on the display:



2. Click the OK button. The dialog will close and the disturbance is acknowledged.

Torque Disturbance

1. Click the torque disturbance button  in the loop window. The torque disturbance acknowledgement dialog will open on the display:



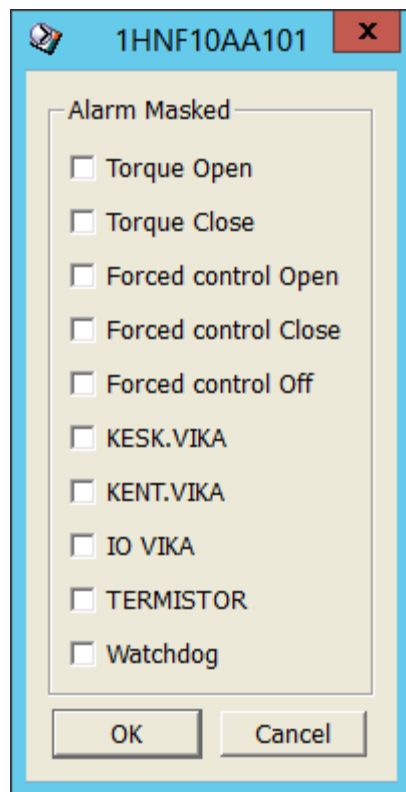
2. Click the OK button. The dialog will close and the disturbance is acknowledged.

8.3.8 How to Change the Event Masks

In the Alarm masks dialog, you can toggle event masks and thus temporarily prevent events from triggering alarms.

When you want to toggle the event masks, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop.



Statuses that have not been configured in the application appear dimmed and unmarked.

2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the changes will be updated to the system.

8.3.9 Error Messages

The following error messages may occur in motor actuator operations:

- *The loop is not operable.*
- *The operation is disabled because the loop is in automatic mode.*
- *The operation is disabled because the loop is in local mode.*
- *The operation is disabled because forced control is active.*
- *Switching to automatic or local mode is disabled.*
- *Switching to manual or local mode is disabled.*
- *Switching from local mode to system's control is disabled.*
- *The operation is disabled because an external disturbance is active.*
- *The operation is disabled because the torque disturbance is active.*
- *You are not authorized to perform the action.*

8.4 Status Information

8.4.1 Alarms

Motor actuator alarm indicates that the actuator is in an undesired state. The alarm is indicated with the red background color on the actuator symbol. In the expanded loop window, the alarm is also indicated by the changed status (*Active*) of the event which has caused the alarm. The Active status is highlighted in red. The background color blinks if the alarm has not yet been acknowledged.

The motor actuator may have any of the following alarms:

Interlockings	· Forced control open · Forced control close · Forced control stop
Disturbances	· Watchdog · Torque open · Torque close
Other States	· External input

8.4.2 Interlockings

The purpose of an interlocking is to prevent incorrect operations and to force the motor actuator into a required state when necessary. In case of disturbances, for example, the purpose of the interlockings is to put the process into a secured state. Interlockings are caused by other control loops or sequence controls.

There are six interlockings:

Release On	On release is missing.
Release Off	Off release is missing.
Forced control Open	Motor actuator is forced to open.
Forced control Close	Motor actuator is forced to close.
Forced control Stop	Motor actuator is forced to stop.
Operation disabled	Motor actuator operation has been disabled.

Forced control Open, Forced control Close, and Forced control Stop may also cause an alarm.

8.4.3 Disturbances

A disturbance indicates a fault in the motor actuator or in its controls. It triggers an alarm and, when necessary, a shift to forced control. A motor actuator can have either a watchdog or torque disturbance.

Watchdog Disturbance

The disturbance specifier is displayed below the watchdog status text.



A watchdog specifier can have any of the following specifiers:

- *Open status faulty*
- *Close status faulty*
- *Motor to open faulty*
- *Motor to close faulty*
- *Open limit fault*
- *Close limit fault*
- *Open torque faulty*
- *Close torque faulty*
- *Open torque memory faulty*
- *Close torque memory faulty*
- *Position faulty*
- *Control command faulty*
- *Release open faulty*
- *Release close faulty*
- *Force open faulty*
- *Force close faulty*
- *Force stop faulty*
- *External control e1 faulty*
- *External control e2 faulty*
- *External control e3 faulty*
- *External control e4 faulty*
- *External control e5 faulty*
- *External control e6 faulty*
- *Open control fault*
- *Close control fault*
- *Motor to open doesn't start*

- *Motor to close doesn't start*
- *Motor to open doesn't stop*
- *Motor to close doesn't stop*
- *Motor to open unexpected start*
- *Motor to close unexpected start*
- *Motor to open unexpected stop*
- *Motor to close unexpected stop*
- *Open limit doesn't get off*
- *Close limit doesn't get off*
- *Open limit unexpectedly off*
- *Close limit unexpectedly off*
- *Open limit unexpectedly on*
- *Close limit unexpectedly on*
- *Open torque doesn't get off*
- *Close torque doesn't get off*
- *Open torque unexpectedly off*
- *Close torque unexpectedly off*
- *Open torque unexpectedly on*
- *Close torque unexpectedly on*
- *Position doesn't change*
- *Position changes unexpectedly*
- *Position changes in wrong direction*
- *Time to open exceeded*
- *Time to close exceeded*
- *Open control unexpectedly off*
- *Close control unexpectedly off*
- *Open control unexpectedly on*
- *Close control unexpectedly on*
- *Force control collision*
- *External control collision*
- *Incorrect parameter value*
- *Open and close collision*

Torque Disturbance

There are two torque disturbances:

- Torque Open
- Torque Close

Torque disturbances do not have specifiers.

8.4.4 Other States

Other states consists of the plant-specific external inputs. All inputs may not be in use. An external input may cause a disturbance. The name of the input refers to the true origin of the disturbance.

8.4.5 Faults

A fault indicates that part of the presented information is not reliable. In the motor actuator component, a fault is indicated with the violet background color of the position numerical value and motor actuator symbol. In the loop window, a fault is indicated with the violet background color of the object connected with the faulty signal.

A motor actuator can have the following system faults:

- Position
- Status

When a fault occurs, a series of fault numbers (1 – 7) highlighted in yellow is displayed in the loop window in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.



The Faults window can be opened by clicking the  button.

Fault	Status	Measurement
1 External fault	No	No
2 Inaccurate signal	No	No
3 Not under system control	No	No
4 Invalid data	Active	Active
5 Old data	Active	Active
6 Derived fault	No	No
7 Simulation	No	No

9. Sequence User Manual

9.1 Introduction

Sequence can be used to start or stop a process in multiple steps. The steps of the sequence can contain transition conditions which must be met before moving the next step, or pauses which the operator must acknowledge before the sequence can proceed.

Sequence Component

With the sequence component, you can execute all the basic operations of the sequence and, when necessary, open the loop window for more advanced operations.

In the sequence component, the following information is displayed:

- [Running status](#)
- Active step
- Starting and pause step
- [Control mode](#)
- Other information, such as name in plain text

In the sequence component, you can perform the following operations:

- [Start/stop the sequence](#)
- [Select control mode](#)
- [Acknowledge pause step](#)

Loop Window

You can open the loop window through the action menu of the sequence component. With the loop window, you can perform all sequence operations and also view more detailed information on the states and parameters of the sequence.

In addition to the operations common with the sequence component, you can perform the following operations in the loop window:

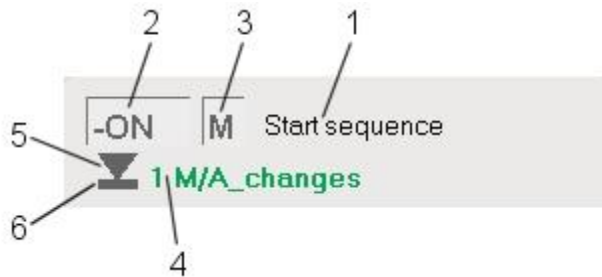
- [Shift to the next step](#)
- [Shift to another step](#)
- [Restart the sequence](#)
- [Set and clear starting step](#)
- [Set and clear pause step](#)
- [Change event masks](#)
- [Modify execution parameters](#)
- [View watchdog failures](#)
- [View signal faults](#)

9.2 Contents of the Sequence

9.2.1 Sequence Component

Elements

The sequence component consists of the following elements:



1. Name text
2. Sequence running status
3. Control mode
4. Step text
5. Symbol of starting step
6. Symbol of pause step

Running Status

The sequence has four running statuses:

ON	Sequence is running
OFF	Sequence is stopped
-ON	Sequence is in pause step
-OFF	Sequence is stopped and in pause step

Control Mode

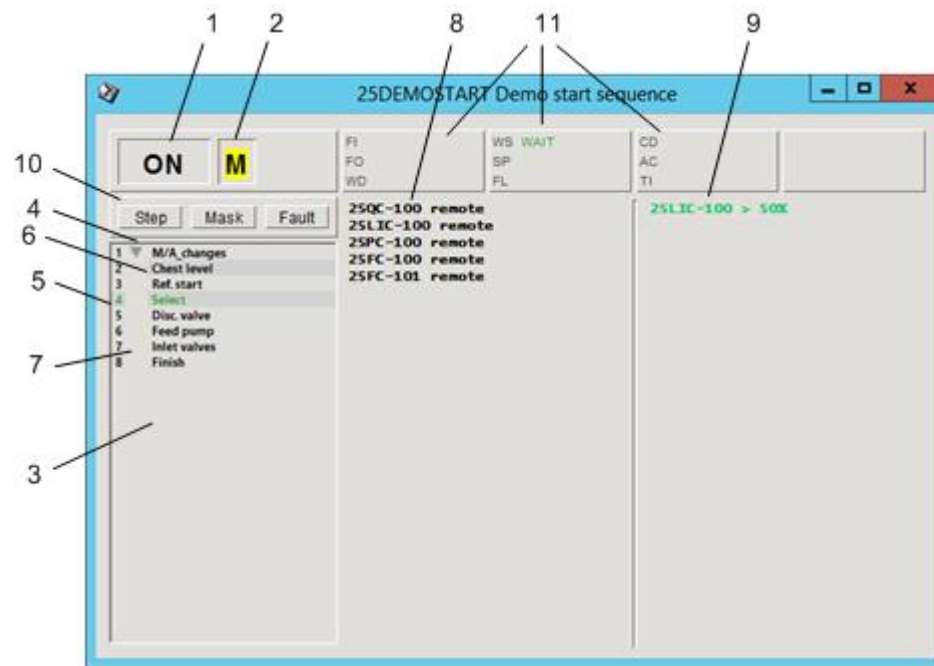
The control mode indicates the mode of the sequence. Modes are displayed in uppercase letters as follows:

A	Automatic control (the sequence is controlled by another loop)
M	Manual control (the sequence is controlled by the operator)

9.2.2 Loop Window

Elements

The loop window of a sequence consists of the following elements:



1. [Sequence running status](#)
2. [Control mode](#)
3. List of steps
4. Symbol of starting step
5. Active step
6. Selected step
7. Symbol of pause step
8. Actions executed in the selected step
9. Transition conditions set for the selected step
10. [Command buttons](#)
11. [Status information area](#)

Command Buttons

The loop window contains the following command buttons:

Step	Opens the step operation dialog
Mask	Opens the event mask and execution parameter dialog
Fault	Opens the signal fault and watchdog failure dialog

In addition, the control mode and running status symbols also function as command buttons.

Status Information Area

The sequence has the following fixed status information:

FI	Forced control ON
FO	Forced control OFF
WD	Watchdog failure
WS	Wait state
SP	Sequence speed-up
FL	Fault
CD	Condition check
AC	Action execution
TI	Time-out interruption

Fixed statuses have the following corresponding descriptions:

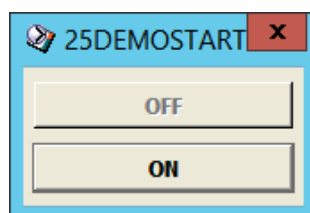
FORC ON	Forced control started
FORC OFF	Forced control stopped
DISTURB	Watchdog disturbance
WAIT	Step in wait state
SPEED UP	Sequence speed-up set
<fault number>	Number refers to the list in Fault field
NO COND	Step conditions are not checked
NO ACTION	Step action is not executed
TIME INT	Time-out interrupted

9.3 Operations

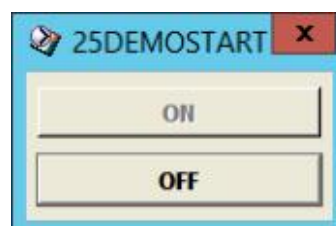
9.3.1 How to Start or Stop the Sequence

When you want to start or stop the sequence, proceed as follows:

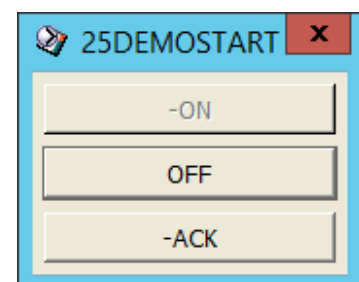
1. Click the running status symbol of the sequence. The operation dialog, the contents of which will depend on the running status, will appear on the desktop:



Sequence is stopped



Sequence is running



Sequence is in pause step

Dimmed buttons are not operable.

2. Click the desired button. The dialog will close and the running status of the sequence will change according to your selection.

NOTE! The running status can only be changed if the sequence is in manual control.

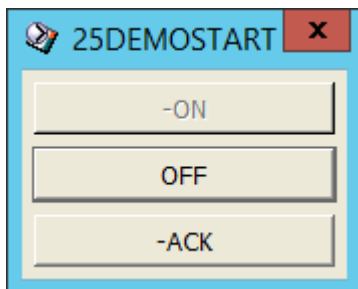
9.3.2 How to Acknowledge a Pause Step

If the sequence is stopped in a pause step, it will not proceed before the pause step has been acknowledged. The pause step is indicated by the corresponding running status (-ON) and the symbol of the pause step (-) in front of the step text.

The pause step cannot be acknowledged if the sequence is not running.

When you want to acknowledge a pause step, proceed as follows:

1. Click the running status symbol in the sequence component or in the loop window. The following dialog will open on the desktop:



2. Click the -ACK button to acknowledge the pause step. The dialog will close and the sequence will proceed to the next step.

9.3.3 How to Change the Control Mode

When you want to change the control mode for the sequence, proceed as follows:

1. Click the control mode symbol in the sequence component or in the loop window. The following dialog will appear on the desktop:



2. Click the desired control mode button. The command button corresponding to the currently active control mode appears dimmed. The dialog will close and the sequence will switch to the selected control mode.

9.3.4 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the sequence component.
2. On the appearing action menu, select Loop Window. The loop window will open on the desktop.

9.3.5 How to Shift to the Next Step

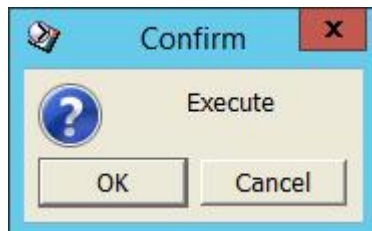
When the sequence is running, you can move forward in it one step at a time.

Proceed as follows:

1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



When the Confirm execution radio button is selected, a separate dialog is opened for confirming the command execution if any of the command buttons is clicked.



2. Click the Next button and confirm the execution (if set). The sequence will proceed to the next step.

9.3.6 How to Shift to Another Step

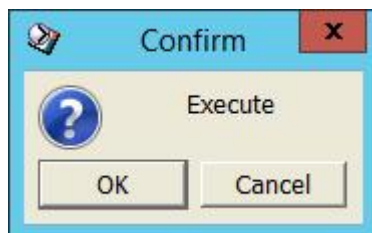
When the sequence is running, you can shift to another step in the sequence.

Proceed as follows:

1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



When the Confirm execution radio button is selected, a separate dialog is opened for confirming the command execution if any of the command buttons is clicked.



2. Click the target step number in the loop window. The selected step number will be highlighted in darker shade.

2	Chest level
3	Ref. start
4	Select

3. Click the Selected button and confirm the execution (if set). The sequence will shift to the selected step.

2	Chest level
3	Ref. start
4	Select

9.3.7 How to Restart the Sequence

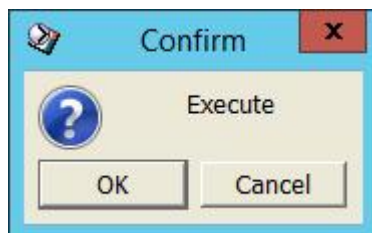
When the sequence is running, you can force it to restart from the starting step.

Proceed as follows:

1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



When the Confirm execution radio button is selected, a separate dialog is opened for confirming the command execution if any of the command buttons is clicked.

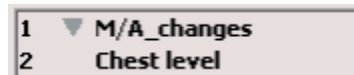


2. Click the Reset button and confirm the execution (if set). The sequence will be restarted from the starting step.

9.3.8 How to Set a Starting Step

You can freely set a starting step for the sequence. When the sequence is run, it will start from the set step. A starting step can be set even if the sequence is not running.

The symbol of the starting step is a triangle standing on its apex.

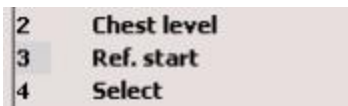


When you want to set a starting step, proceed as follows:

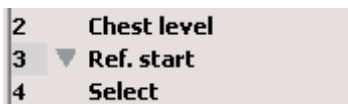
1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



2. Click the desired step number in the loop window. The selected step number will be highlighted in darker shade.



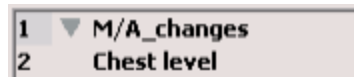
3. Click the Start button. The selected step will be marked as the new starting step.



9.3.9 How to Clear a Starting Step

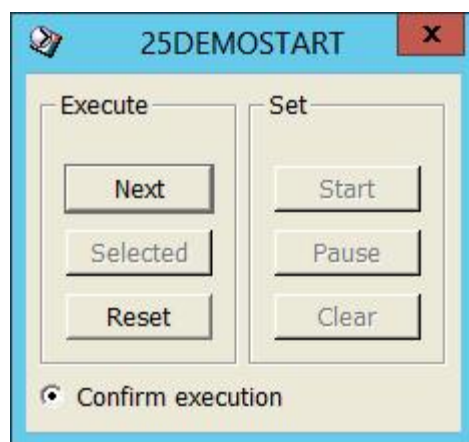
The starting step of the sequence can be cleared. The sequence needs not to be running during the operation.

The symbol of the starting step is a triangle standing on its apex.

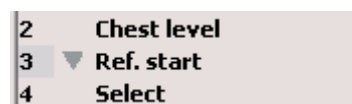


When you want to clear a starting step, proceed as follows:

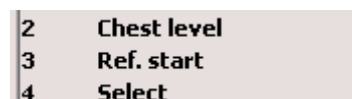
1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



2. Click the number of the starting step in the loop window. The selected step number will be highlighted in darker shade.



3. Click the Clear button. The starting step symbol will be cleared on the selected step.



NOTE! Clicking the Clear button will also remove a pause step from the selected step, if applicable.

9.3.10 How to Set a Pause Step

You can set one or more pause steps for the sequence. When the sequence enters a pause step, it will stop until the pause step has been acknowledged. A pause step can be set even if the sequence is not running.

The symbol of a pause step is an underline character.

2	<u>Chest level</u>
3	Ref. start
4	Select

When you want to set a pause step, proceed as follows:

1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



2. Click the desired step number in the loop window. The selected step number will be highlighted in darker shade.

2	Chest level
3	<u>Ref. start</u>
4	Select

3. Click the Pause button. The selected step will be marked as a pause step.

2	Chest level
3	<u>Ref. start</u>
4	Select

9.3.11 How to Clear a Pause Step

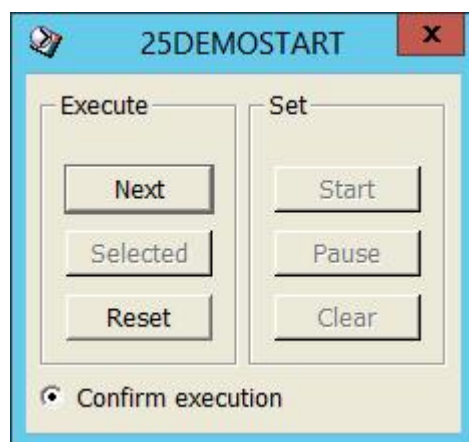
The pause step of the sequence can be cleared. The sequence needs not to be running during the operation.

The symbol of a pause step is an underline character.

2	<u>Chest level</u>
3	Ref. start
4	Select

When you want to clear a pause step, proceed as follows:

1. Click the Step button in the sequence loop window. The following dialog will open on the desktop:



2. Click the number of the pause step in the loop window. The selected step number will be highlighted in darker shade.

2	Chest level
3	<u>Ref. start</u>
4	Select

3. Click the Clear button. The pause step symbol will be cleared on the selected step.

2	Chest level
3	Ref. start
4	Select

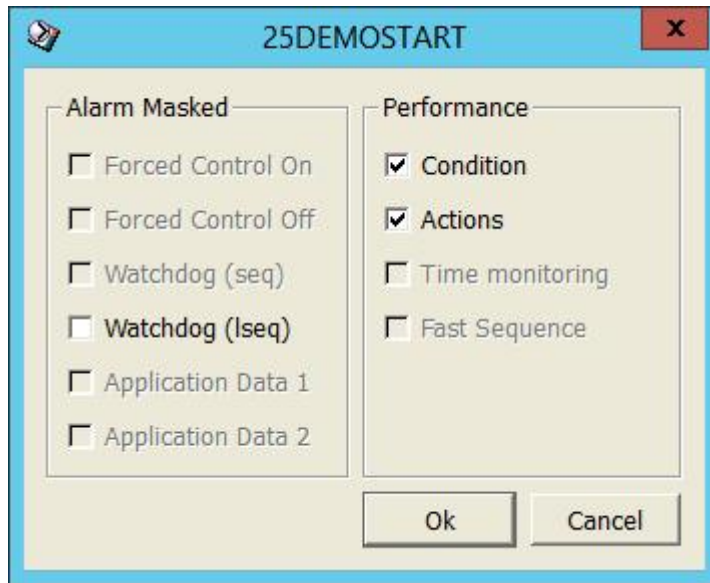
NOTE! Clicking the Clear button will also remove a starting step from the selected step, if applicable.

9.3.12 How to Change the Event Masks

In the Alarm mask field, you can toggle event masks and thus temporarily prevent events from triggering alarms.

When you want to toggle the event masks, proceed as follows:

1. Click the Mask button in the loop window. The following dialog will open on the desktop:



The Alarm mask field contains check boxes for the following event masks:

FI	Forced control ON
FO	Forced control OFF
WD / s	Watchdog failure
WD / I	Watchdog failure
AP1	Application-specific data
AP2	Application-specific data

Dimmed event masks cannot be changed.

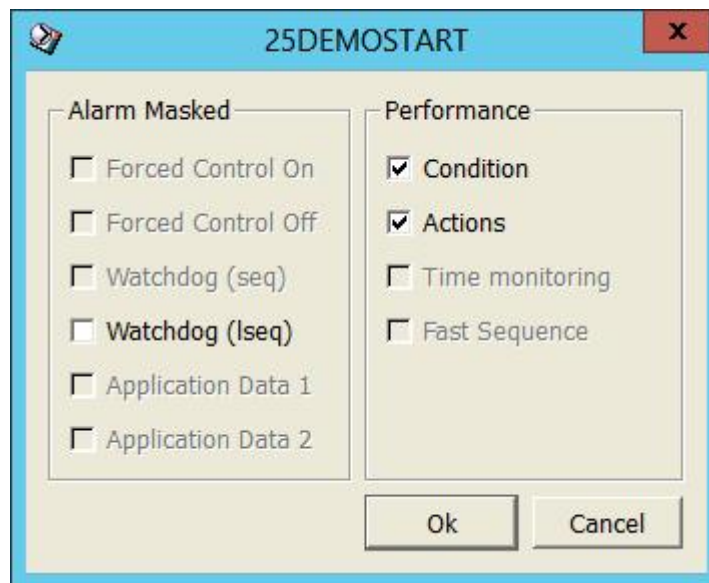
2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the changes will be updated to the system.

9.3.13 How to Modify the Execution Parameters

In the Performance field, you can modify the parameters that control the sequence execution.

Proceed as follows:

1. Click the Mask button in the loop window. The following dialog will open on the desktop:



The Performance field contains option buttons for the following parameters:

Condition	Determines whether transition conditions are checked after step execution. If the conditions are not checked, the sequence will not automatically proceed to the next step.
Actions	Determines the execution of the step-specific actions
Time	Determines the interruption or continuity of the time-out. The parameter can be used to prevent a known fault from causing an unnecessary alarm without interrupting the timing.
Sequence	Determines the sequence speed-up (min -> s or s -> min)

NOTE! Parameter operations may have been prevented.

2. Modify the target parameter by clicking the corresponding option button.
3. Click the OK button. The dialog will close and the changes will be updated to the system.

9.3.14 Error Messages

The following error messages may occur in sequence operations:

- *Error: no oper*

Operations are disabled.

- *Error: autom.*

The sequence is in an automatic control mode.

- *Error: forced c*

The sequence is in a forced-control mode.

- *Error: M–A forb*

The sequence is in a manual control mode, and switching to an automatic mode is disabled.

- *Error: A–M forb*

The sequence is in an automatic control mode, and switching to a manual mode is disabled.

- *Error: no authorization*

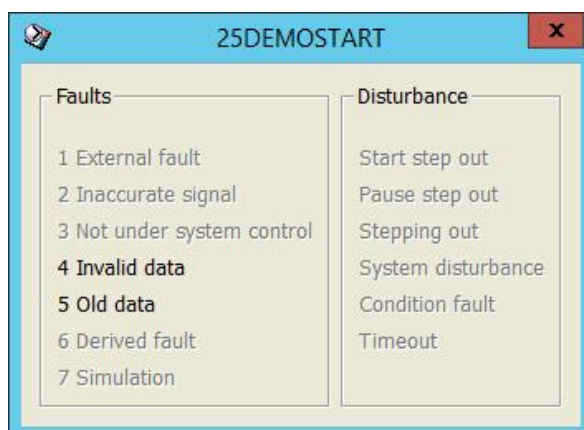
The operator has no rights for the operation.

9.4 Disturbances and Faults

9.4.1 Disturbances

The DISTURB text shown after the WD status on the status information area indicates that a watchdog failure has occurred.

More detailed information on the cause of the watchdog failure can be viewed by clicking the Fault button in the loop window.

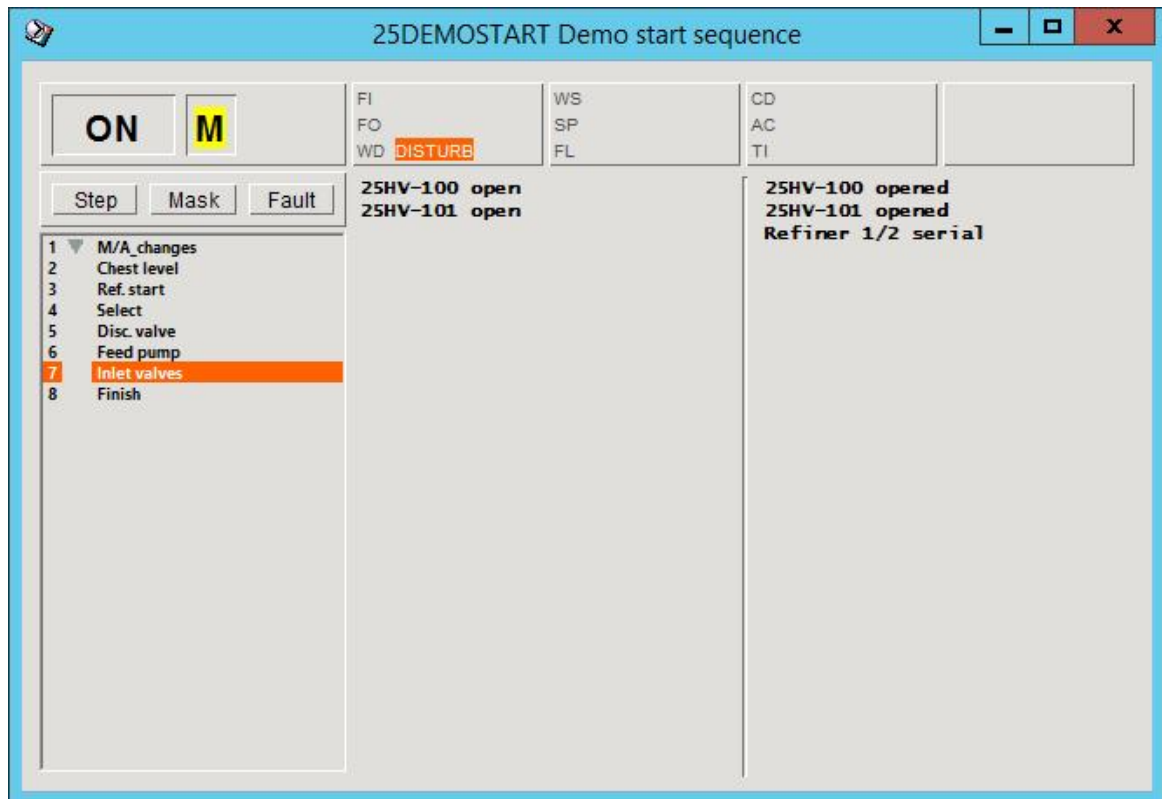


9.4.2 Alarms

In the sequence component, the alarm is indicated by the red background color of the step text.



In the loop window, the alarm is indicated by the red background color of the step text, related status description, and related action.



The sequence may have the following alarming states:

- Forced control ON
- Forced control OFF
- Watchdog failure
- Application-specific alarms

9.4.3 Signal Faults

A signal fault indicates that part of the presented information is not reliable. The signal fault is indicated with the violet background color of the running status symbol.

During a signal fault, a series of fault numbers (1 – 7) is displayed after the FL status on the status information area. The numbers refer to the faults listed in the Fault field. For example, series 45 indicates that faults 4 and 5 are active.

Fault specifiers can be viewed by clicking the Fault button in the loop window.

Faults	Disturbance
1 External fault	Start step out
2 Inaccurate signal	Pause step out
3 Not under system control	Stepping out
4 Invalid data	System disturbance
5 Old data	Condition fault
6 Derived fault	Timeout
7 Simulation	

10. State Information User Manual

10.1 Introduction

State Information Component

The state information component displays the state of a device or function in text format. There can be two (for example, on / off) or more (storage / circulation / cooking / out) states. The [state of a device or function can be changed](#) by operating the state information component.

Loop Window

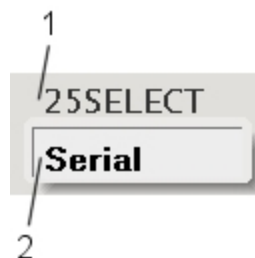
You can open the loop window through the action menu of the state information component. With the loop window, you can [change the state](#) of a device or function similarly to the state information component and, additionally, [change the event mask](#) of the state information and [view the faults](#) related to state information.

10.2 Contents of the State Information

10.2.1 State Information Component

Elements

The state information component consists of the following elements:

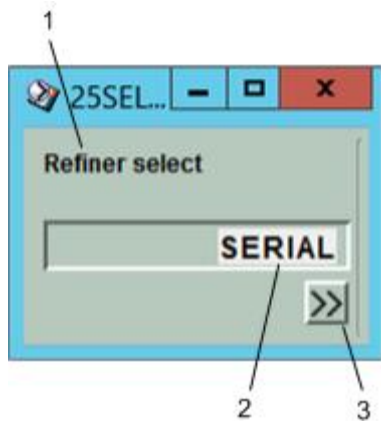


1. Name and/or tag identifier
2. State information

10.2.2 Loop Window

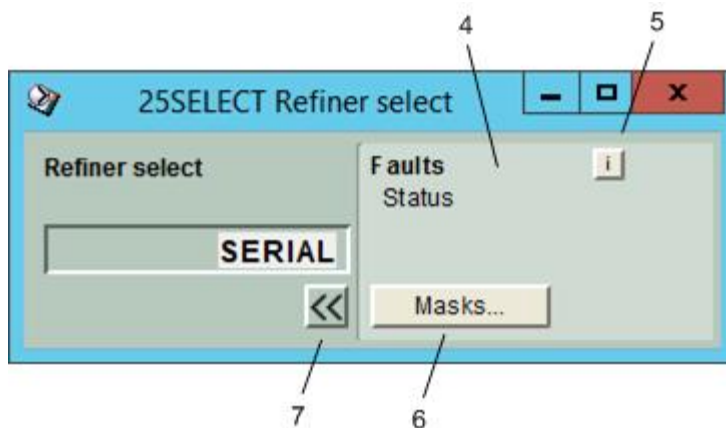
Elements

The loop window has two possible views: basic view and expanded view. The basic view displays all the information needed in normal everyday operation. The expanded view displays additional information on the state of the loop and also contains the command buttons for parameter operations. The loop window is opened on the desktop in basic view.



1. Name and/or tag identifier
2. State information
3. [Unfold button](#)

In addition, the expanded view displays the following:



4. [Faults](#)
5. Fault window button
6. [Event mask](#) dialog button
7. [Fold button](#)

10.3 Operations

10.3.1 How to Change the State

When you want to change the state of a device or function, proceed as follows:

1. Click the text field of the state information. The operation dialog will open on the desktop:



The button presenting the current state is dimmed.

2. Select the desired state by clicking the corresponding button. The dialog will close and the target will change to the selected state.

NOTE! Changing the state may have been prohibited in the application.

10.3.2 How to Open the Loop Window

When you want to open the loop window, proceed as follows:

1. Right-click the state information component.
2. On the appearing action menu, select Loop Window. The loop window will open on the desktop.

10.3.3 How to Toggle the View of the Loop Window

When you want to unfold the loop window, proceed as follows:

- Click the >> button in the loop window.

When you want to fold the loop window, proceed as follows:

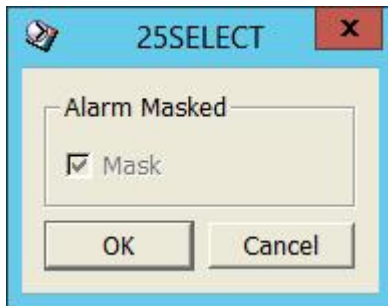
- Click the << button in the loop window.

10.3.4 How to Change the Event Mask

In the Alarm masks dialog, you can toggle the event mask and thus temporarily prevent an event from triggering an alarm.

When you want to toggle the event mask, proceed as follows:

1. Click the Masks... button in the loop window. The Alarm masks dialog will open on the desktop.



2. To enable or disable an event mask, check or uncheck the corresponding check box by clicking it.
3. Click the OK button. The dialog will close and the change will be updated to the system

10.3.5 Error Messages

The following error messages may occur in state information operations:

- *The loop is not operable.*
- *You are not authorized to perform the action.*

10.4 Status Information

10.4.1 Alarms

State information alarm is indicated in both the status information component and the loop window with the red background color on the text field. The background color blinks if the alarm has not yet been acknowledged.

10.4.2 Faults

A fault indicates that part of the presented information is not reliable. The fault is indicated with the violet background color on the state information text field.

When a fault occurs, a series of fault numbers (1 – 7) highlighted in yellow is displayed in the loop window in the Faults field next to the faulty signal. The numbers refer to the faults listed in the Fault window. For example, series 45 indicates that faults 4 and 5 are active.



The Fault window can be opened by clicking the  button.

Fault	Status
1 External fault	No
2 Inaccurate signal	No
3 Not under system control	No
4 Invalid data	Active
5 Old data	Active
6 Derived fault	No
7 Simulation	No

11. Trend Picture User Manual

11.1 Introduction

Part of the information in DNA Operate is presented as trend graphs with which, for example, chest levels, temperatures, pressures, or running statuses of motors can be represented. By examining the changes in trend graphs, you can predict the state of the process in the near future. You can also follow how a modification or disturbance affects the process and furthermore analyze the relationships between different measurements.

Trends located in pictures are used in normal process monitoring. They can be found in process diagrams or separate trend pictures.

Trends are mainly operated through the action menu. Trend-related action menu is opened by right-clicking the graph area.

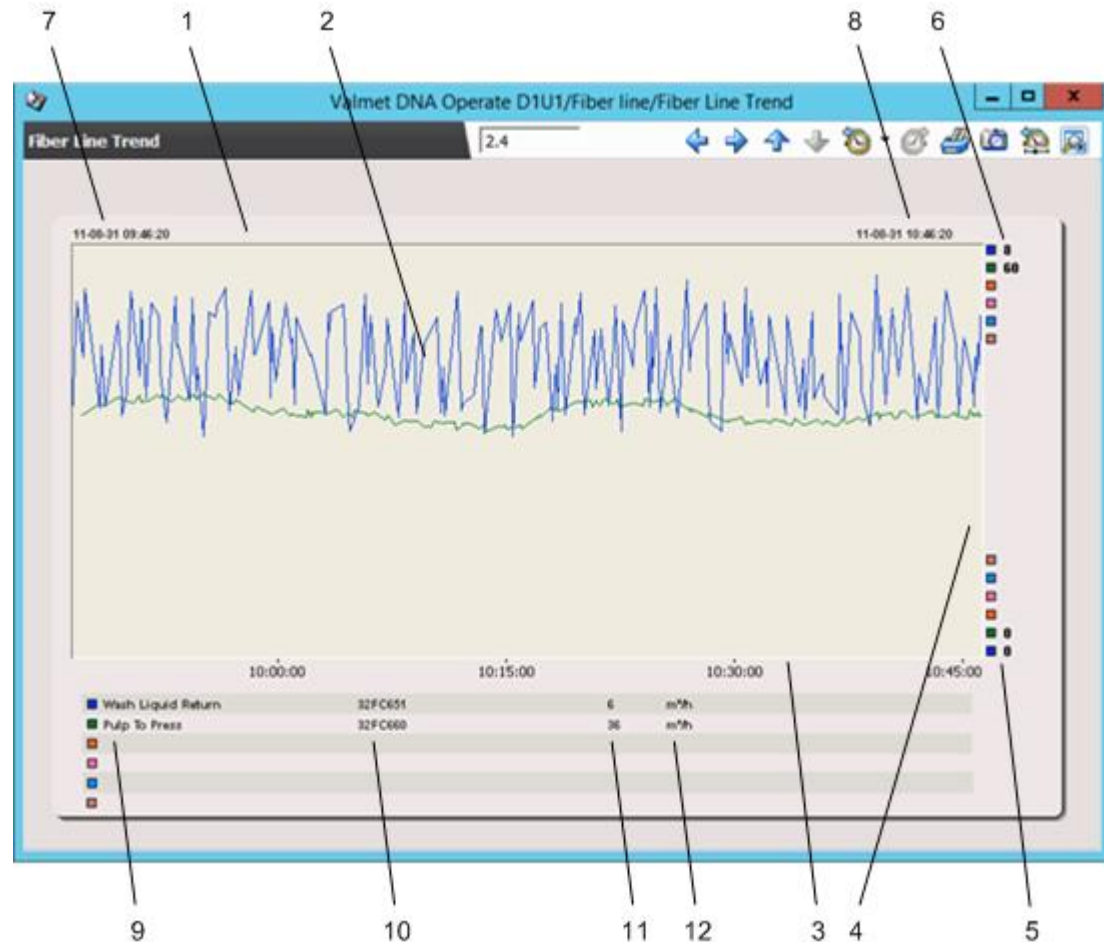
In trends, you can perform the following operations:

- [Open the loop window of the tag presented in the trend.](#)
- [Use a hairline to display an instantaneous value.](#)
- [Copy trend data](#) to the clipboard.
- [Scale the trend.](#)
- [Zoom the trend.](#)
- [View the statistics.](#)
- [Modify the trend properties.](#)
- [Add a tag to the trend.](#)
- [Save a trend analysis tool to the hard disk.](#)
- [Open the saved trend analysis tool from the hard disk.](#)

11.2 Trend Picture

11.2.1 Elements of a Trend Picture

A trend picture consists of the following elements:



1. [Coordinates](#)
2. [Graphs](#)
3. [X-axis](#) (span)
4. [Y-axis](#) (scales)
5. Minimum values of graph scales
6. Maximum values of graph scales
7. Starting point of time span
8. End point of time span
9. Tag name or description
10. Tag identifier
11. Instantaneous value
12. Signal unit

There are several variations of trends. For example, the background color may have been modified, the scales may be missing, and the number of the presented graphs may vary.

11.2.2 Contents

Coordinates

A maximum of ten graphs can be presented on the coordinates of a trend picture. The graphs are displayed on the graph area of the coordinates. [Zooming](#) and hairline operations can be executed directly on the graph area.

Graphs

A trend graph has four colors:

- Graph color
- Fault color
- Alarm color for low and high limits
- Alarm color for lower low and higher high limits

Graph color is used when the displayed values are within normal limits. Fault color is reserved for unreliable and faulty data. Alarm colors are used when graph values exceed the given limits.

Available graph types are:

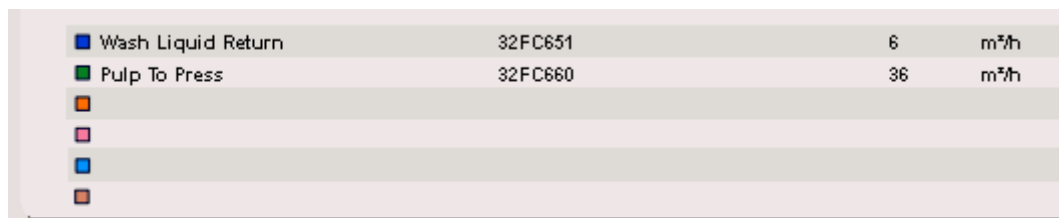
- Breakline
- Column

Available line types are:

- Continuous line
- Broken line
- Dash-and-dot line
- Dotted line
- Filled area

Texts

Graph-related information is displayed in graph color.



Wash Liquid Return	32FC651	6	m³/h
Pulp To Press	32FC660	36	m³/h

The graph-related information consists of:

- Tag name or description
- Tag identifier
- Instantaneous value
- Signal unit

Coordinates-related information consists of the minimum and maximum values of the graph scales, which are displayed in graph colors, and the starting and end point (usually presented with date and time stamp) of the coordinates span.

X-axis

One of the following properties may be displayed on the X-axis:

- Absolute time stamp presented with one of the following units:
 - Year
 - Month
 - Day
 - Hour
 - Minute
 - Second
 - Millisecond
- Relative time stamp presented with one of the following units:
 - Day
 - Hour
 - Minute
 - Second
 - Millisecond
- Common texts of events

X-axis may also be left empty.

Y-axis

Graph scales are displayed on the Y-axis. By default, the scale lines of the axis are shown with a spacing of 20%.

11.3 Operations

11.3.1 How to View Statistics

The statistics of the trend graphs can be viewed in the Statistics window. The statistical data is calculated from the graphs presented in the trend.

When you want to open the Statistics window, proceed as follows:

- Right-click the trend and select Statistics... on the action menu. The Statistics window will open on the desktop.

		Specifier	Unit	Min	Max	Ave	Std	Points
Fw-Tank Pressure Control	■	me	kPa	250	329	282	22	225
Fw-Tank Pressure Control	■	spa	kPa	250	350	284	28	225

1. Tag name or description
2. Graph color
3. Signal specifier
4. Signal unit
5. Minimum value of trend graph
6. Maximum value of trend graph
7. Average value of trend graph
8. Standard deviation of trend graph
9. Number of points used in statistics calculations

No operations can be executed in the Statistics window.

11.3.2 How to Open the Loop Window

You can open the loop window of a tag presented in a trend directly from the trend picture.

Proceed as follows:

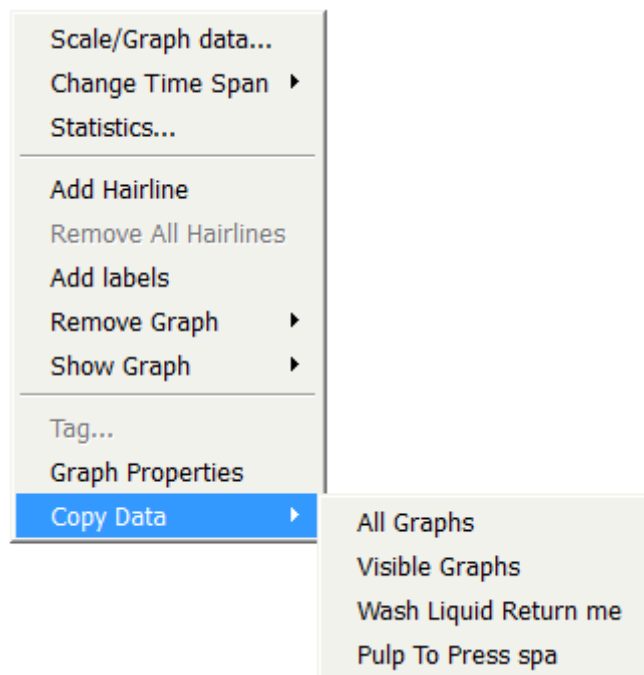
- Right-click the tag identifier of the target graph and select Loop Window on the action menu. The loop window will open on the desktop.

11.3.3 How to Copy Trend Data

Trend data can be copied to the clipboard and pasted to other applications in table format.

Proceed as follows:

- Right-click the trend and select the desired data on the submenu of the Copy Data command.



You can choose to copy the data of all graphs, visible graphs, or a single graph.

11.3.4 Zooming

How to Zoom In

By zooming you can crop an interesting area of the trend picture to be viewed, and the changes in graph values can then be examined more closely.

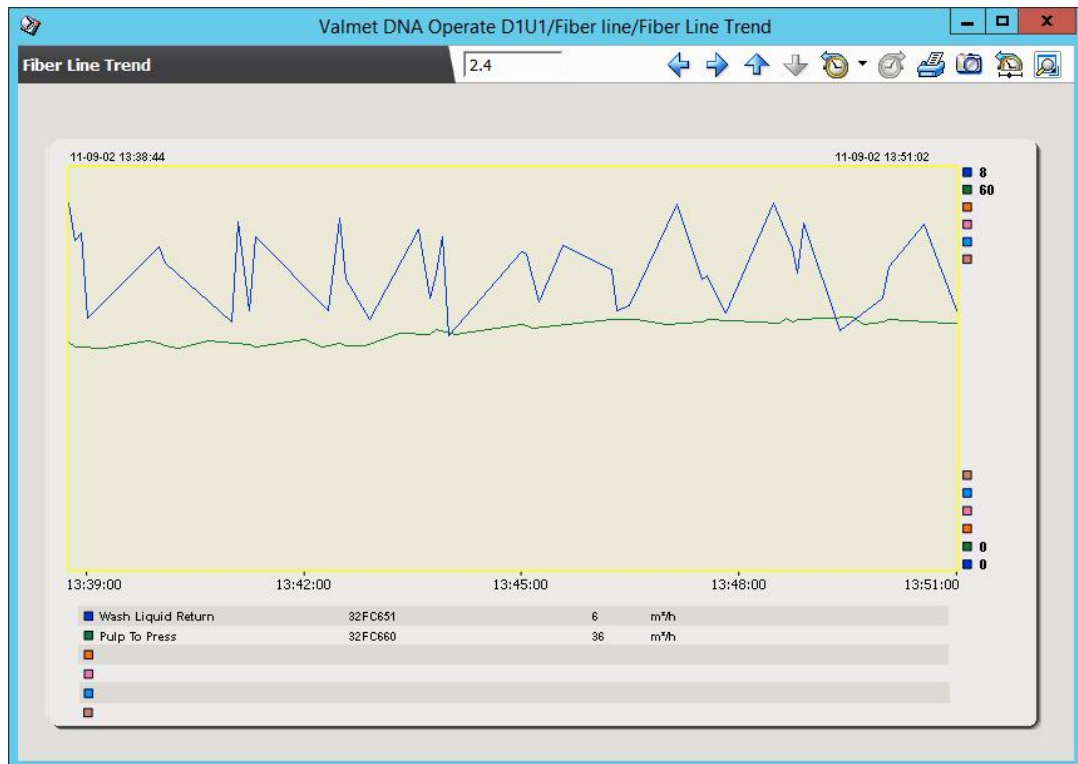
When you want to zoom in a trend picture, proceed as follows:

1. In the coordinates of a trend picture, move the pointer to the corner of the area you want to zoom.
2. Click and hold the left mouse button while moving the pointer to outline an area. The values of the trend picture X-axis (time) and Y-axis scales are determined by the outlined area.

If you want to keep the values of the Y-axis scale unchanged while zooming, hold the Shift key down while moving the pointer to outline an area.

During outlining the value of X-axis at cursor point is displayed in the picture as a tooltip (eg. 09:45:36).

3. Release the mouse button. The outlined area will be opened in the trend picture and the borders of the coordinates will be highlighted in yellow attention color to indicate that the trend picture is zoomed in.



The zoomed trend picture can be scrolled with the keyboard shortcuts. The values of the X-axis can be scrolled with the Ctrl + arrow left/right keys and the values of the Y-axis with the Ctrl + arrow up/down keys.

It is possible to zoom in a trend picture several times in a row.

Note that you cannot save or take a snapshot of a zoomed trend picture because when saving, the zoomed trend picture always returns to the situation before zooming.

How to Zoom Out

When you want to zoom out a trend picture, proceed as follows:

- Right-click the zoomed trend picture and select Reset zoom on the action menu.

OR

- Press the ESC key.

Zooming is cancelled automatically after a while since the last zooming.

11.3.5 Scaling

How to Open the Scale Window

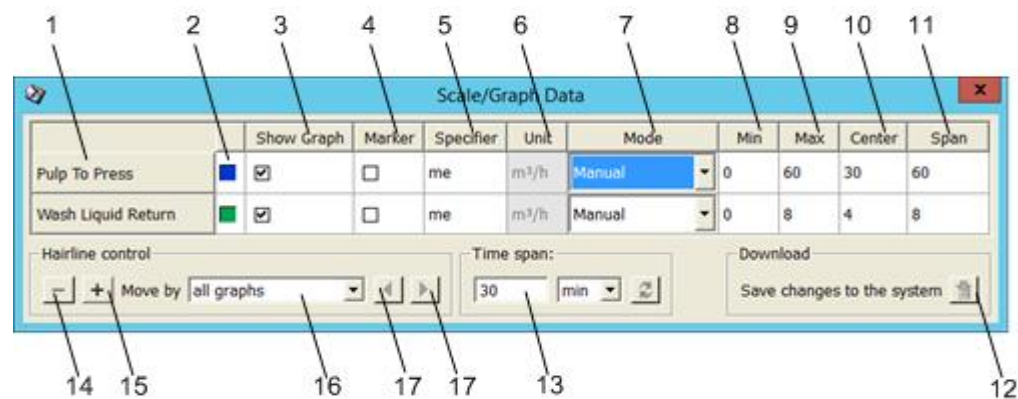
You can change the scaling mode and the values of the trend graph scales in the Scale /Graph Data window.

When you want to open the Scale/Graph Data window, proceed as follows:

- Right-click the trend and select Scale /Graph Data... on the action menu.
The Scale/Graph Data window will open on the desktop.

Elements of the Scale Window

The Scale /Graph Data window consists of the following elements:



- Tags presented in the trend
- Graph color
- Check box for displaying/hiding the graph
- Check box for displaying point markers
- Signal specifier
- Signal unit
- Scaling mode
- Scale minimum value
- Scale maximum value
- Scale center value
- Scale span
- Button for saving changes to the system
- Time span length

Buttons related to hairline operations:

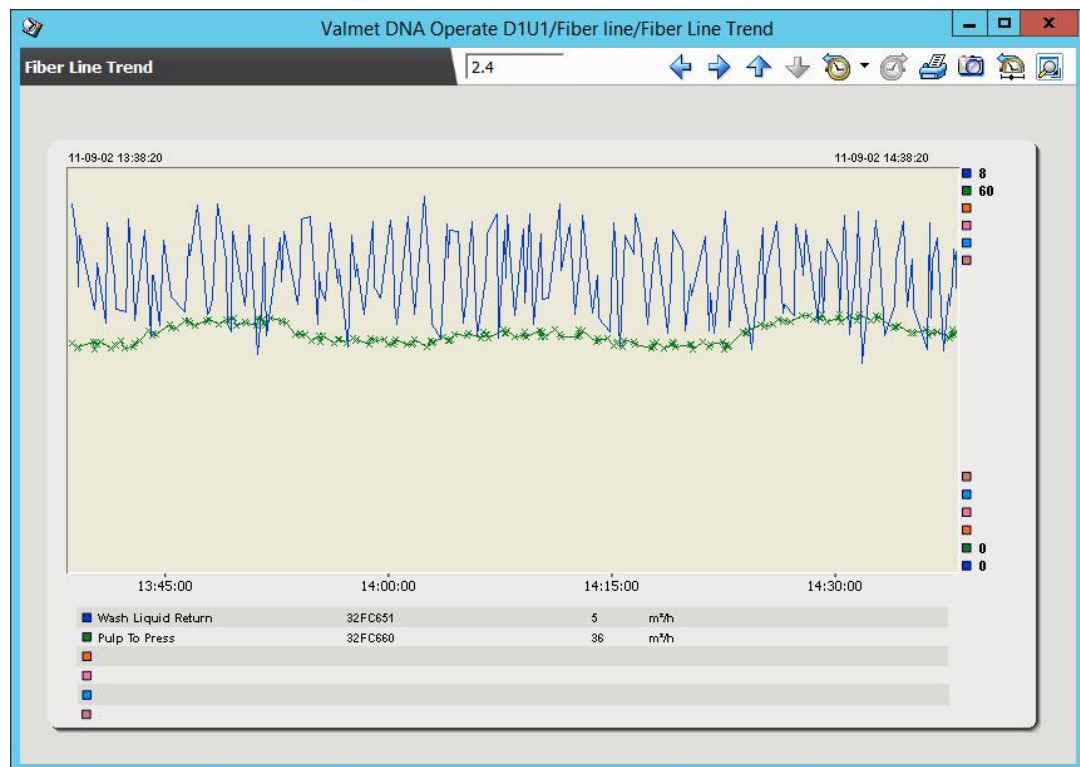
- Remove button
- Add button
- Selecting the graph/graphs
- Arrow buttons for moving the selected hairline

How to Display Value Points

The value points of a graph can be showed with markers. The option is set separately for each graph.

Proceed as follows:

- Click to select the Marker check box of the target graph in the Scale/Graph Data window. A check mark will appear in the box and the value points of the selected graph will be displayed in the trend.



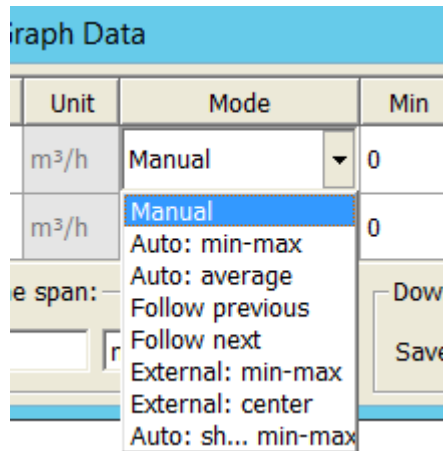
How to Change the Scaling Mode

The scaling mode of a trend graph can be changed in the Scale /Graph Data window. The following modes are available:

Manual	The minimum and maximum value or the center and span of the scale can be set manually.
Auto: min-max	Scale is automatically determined by the extreme values of the graph.
Auto: average	Scale is determined by an average value calculated from the graph data.
Follow previous	Scale is copied from the previous signal.
Follow next	Scale is copied from the next signal.
External: min-max	Scale is determined by the extreme values gained from external connection data.
External: center	Scale is determined by the center point gained from external connection data.
Auto: shared min-max	Scale is calculated by the extreme values of multiple curves and scaled according to the largest value.

When you want to change the scaling mode of a trend graph, proceed as follows:

1. In the Scale/Graph Data window, click the Mode field of the target graph. The list of the scaling modes will open below the field.



2. Select the desired scaling mode. The menu will close and the scaling mode of the target graph will change accordingly.

How to Change Scale Values

The scale of a trend graph can be operated in connection with the following scaling modes:

Manual

In the manual scaling mode, you can operate the Min, Max, Center, and Span values and thus affect the graph format in the trend.

Proceed as follows:

1. In the Scale/Graph data window, click the value you want to operate.
2. Delete the old value with the Backspace key. You can also move the cursor in the text box by clicking the desired location inside the box.
3. Enter a new value and press the Enter key. The graph scale will be updated accordingly in the trend.

NOTE! Changing one value may affect the other scale values of the graph as well. For example, if you modify the Span value, the Min and Max values will also change.

Auto: average

In average scaling, an average value is calculated from the graph data visible in the trend and set as the Center value. Min, Max and Span values can be operated. Changing any of them will affect the other two correspondingly. Values are operated as described above in the Manual chapter.

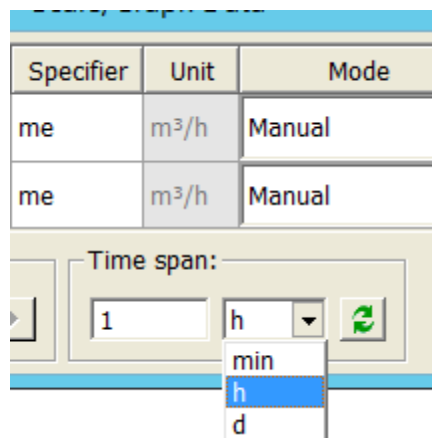
External: center

In external center scaling, the Center value of the scale is gained from external connection data. Min, Max and Span values can be operated. Changing any of them will affect the other two correspondingly. Values are operated as described above in the Manual chapter.

How to Change the Length of the Time Span

When you want to change the length of the time span, proceed as follows:

1. In the Scale/Graph Data window, in the Time span section, write the value in the field and select a unit from the list.



OR

Right-click the trend coordinates area and select Change Time Span on the action menu. The list of the available time spans will open.

2. Select the desired time span. The list will close and the time span of the trend picture will change accordingly.

The Time span box appears dimmed if the trend picture data is not retrieved from the history database.

11.3.6 Hairline Operations

How to Add a Hairline

The values of individual points on a trend graph can be examined by adding hairlines in desired locations. The values of the marked points are displayed in the Scale /Graph Data window.

When you want to add a hairline to a trend, proceed as follows:

- Click the desired spot in the trend. A hairline will appear in that location.

OR

Right-click the desired spot in the trend and select Add Hairline on the action menu. A hairline will appear in that location.

OR

Click the  button in the Scale /Graph Data window. A hairline will appear in the middle of trend in the center of the displayed span.

You can add any number of hairlines to a trend.

How to Move a Hairline

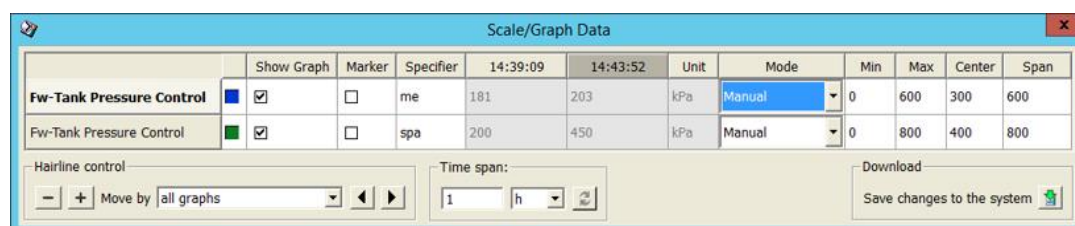
You can move a hairline in a trend by dragging it or by operating it through the Scale /Graph Data window.

When you want to move a hairline by dragging it in a trend, proceed as follows:

1. Move the pointer on top of the target hairline until its appearance changes to a hand.
2. Press and hold the left mouse button to grab the hairline.
3. Drag the hairline to the desired location in the trend.
4. Release the mouse button.

When you want to move the hairline through the Scale/Graph Data window to achieve better precision, proceed as follows:

1. Unless the Scale/Graph Data window is already open, right-click the trend and select Scale /Graph Data... on the action menu.



2. In the Scale/Graph Data window, click the time stamp of the target hairline.

OR

Click the target hairline in the trend.

The selected hairline appears black in the trend.

3. Select the graph by the points of which you want to move the hairline in the Move by field.

By default, the hairline moves by all graphs which means that when either arrow button is clicked, the selected hairline will move to the nearest point on any graph in the arrow button direction.

If after selecting a graph you want to reselect all graphs, just select all graphs in the field.

4. Move the selected hairline to the desired location in the trend by clicking the arrow buttons.

How to Lock the Hairline Position

You can lock the hairline position in the trend. The values measured at the hairline position will be constantly updated in the Scale / Graph Data window.

When you want to lock the hairline position, proceed as follows:

1. Move the pointer on top of the target hairline until its appearance changes to a hand.
2. Right-click the hairline and select Position locked on the action menu. The hairline will lock at its current position.

If you open the action menu on the locked hairline, the Position locked command will be checked to indicate the locking.

How to Release a Locked Hairline

When you want to release a locked hairline, proceed as follows:

1. Move the pointer on top of the target hairline until its appearance changes to a hand.
2. Right-click the hairline and select Position locked on the action menu. The hairline will be released and the check mark will disappear from the action menu.

How to Remove a Hairline

You can remove hairlines from a trend one at a time either directly in the trend picture or in the Scale / Graph Data window.

When you want to remove hairlines one at a time, proceed as follows:

1. Move the pointer on top of the target hairline until its appearance changes to a hand.
2. Right-click the hairline and select Remove hairline on the action menu. The hairline will be removed from the trend and its data will disappear in the Scale/Graph Data window.

OR

1. Unless the Scale/Graph Data window is already open, right-click the trend and select Scale / Graph Data... on the action menu.
2. Select the target hairline by clicking its time stamp field in the Scale/Graph Data window.

OR

Click the target hairline in the trend.

The selected hairline will appear black.

3. Click the  button in the Scale/Graph Data window. The selected hairline will be removed from the trend and its data will disappear from the Scale/Graph Data window.

Alternatively, you can remove all hairlines from a trend at one go.

Proceed as follows:

- Right-click the trend and select Remove All Hairlines on the action menu. All hairlines displayed in the trend will be removed and their data will disappear from the Scale/Graph Data window.

How to Add Labels

Locked hairlines can be added to the beginning and the end of the visible trend.

Proceed as follows:

- Right-click the trend and select Add labels on the action menu. Locked hairlines will appear to the beginning and the end of the visible trend.

11.3.7 Trend Property Modifications

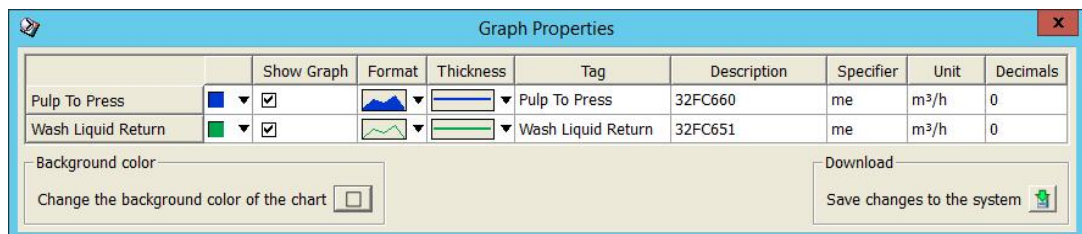
How to Open the Properties Window

You can execute the following operations in the Graph Properties window:

- Select graph color
- Hide and restore a graph
- Select graph format
- Select graph thickness
- Edit the text properties of a graph
- Select trend background color
- Select the time span of the trend.

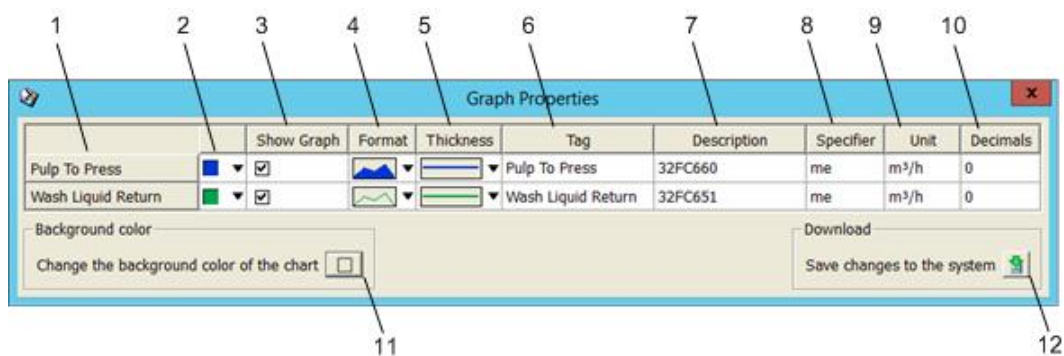
When you want to open the Graph Properties window, proceed as follows:

- Right-click the trend and select Graph Properties... on the action menu. The Graph Properties window will open on the desktop.



Elements of the Properties Window

The Graph Properties window consists of the following elements:

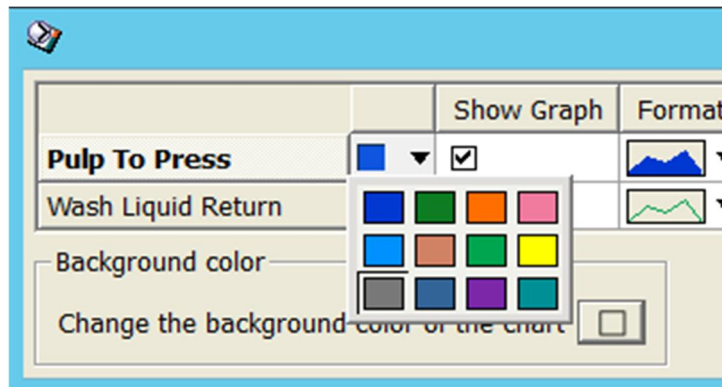


1. Tags presented in the trend
2. Graph color
3. Check box for displaying/hiding the graph
4. Graph format
5. Graph thickness
6. Tag identifier
7. Tag name or description
8. Signal specifier
9. Signal unit
10. Number of decimals used in instantaneous and graph scale values
11. Trend background color
12. Button for saving changes to the system

How to Change Graph Color

When you want to change graph color, proceed as follows:

1. In the Graph Properties window, click the arrow button next to the color box of the target graph. The graph color menu will open below the arrow button.



2. Click the desired color box on the menu. The menu will close and the color of the target graph will change to match your selection.

How to Hide and Restore a Graph

You can hide a graph displayed in a trend and later restore it back.

When you want to hide a graph, proceed as follows:

- Click to clear the Show Graph check box of the target graph in the Scale /Graph Data or Graph Properties window. The check mark will disappear and the graph will be removed from the trend.

The graph is restored correspondingly by selecting the Show Graph check box.

OR

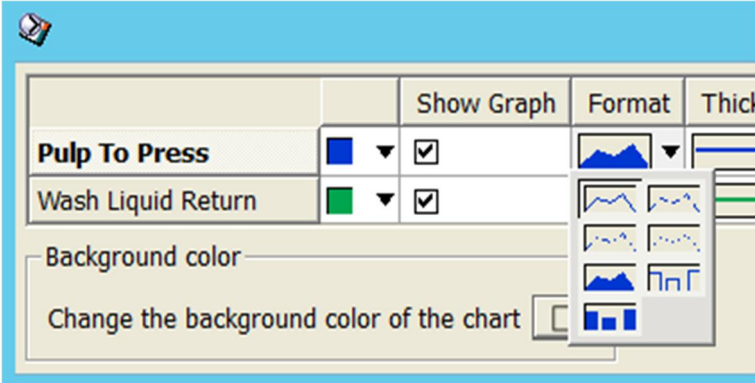
- Right-click the trend and select the target graph on the submenu of the Show Curve > command. The graph will be removed from the trend and the corresponding check mark will disappear from the submenu.

Repeat the operation to restore the graph.

How to Change Graph Format

When you want to change the graph format, proceed as follows:

1. In the Graph Properties window, click the arrow button next to the Format column of the target graph. The graph format menu will open below the arrow button.

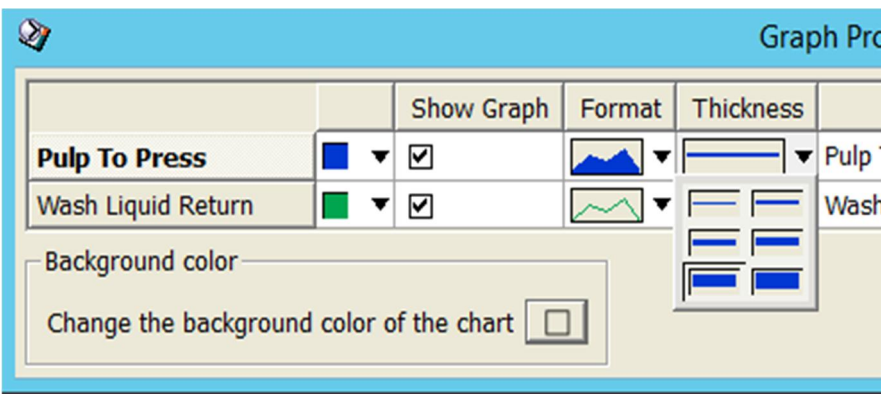


2. Click the desired format box on the menu. The menu will close and the format of the target graph will change to match your selection.

How to Change Graph Thickness

When you want to change the graph thickness, proceed as follows:

1. In the Graph Properties window, click the arrow button next to the Thickness column of the target graph. The graph format menu will open below the arrow button.



2. Click the desired format box on the menu. The menu will close and the thickness of the target graph will change to match your selection.

How to Edit Text Properties

You can modify the text properties of the trend graphs in the Graph Properties window. The modifiable properties include name text (Tag), tag description (Description), signal type (Specifier), signal unit (Unit), and the number of decimals used in instantaneous and graph scale values (Decimals). Any modifications will be updated in the data presented in the trend.

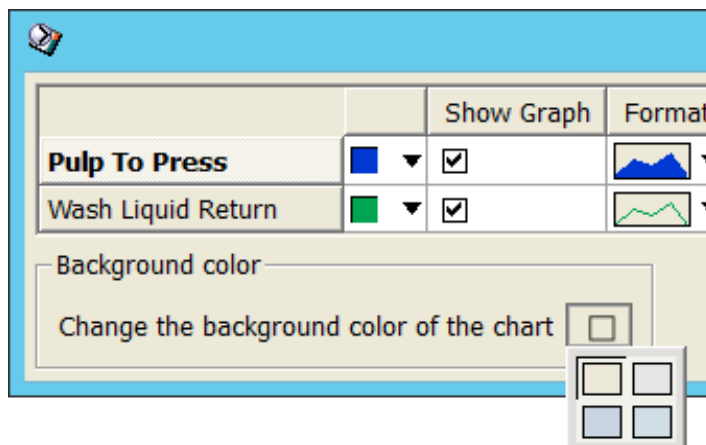
When you want to edit the text properties of a graph, proceed as follows:

1. In the Graph Properties window, click the text field whose contents you want to edit.
2. Edit the text.
3. Confirm the changes by pressing the Enter key or by clicking another text field.

How to Change Background Color

When you want to change the background color of a trend, proceed as follows:

1. In the Graph Properties window, click the Background button. The background color menu will open below the button.



2. Click the desired background color box on the menu. The menu will close and the background color of the trend will change to match your selection.

11.3.8 Tag Selection

How to Open the Tag Window

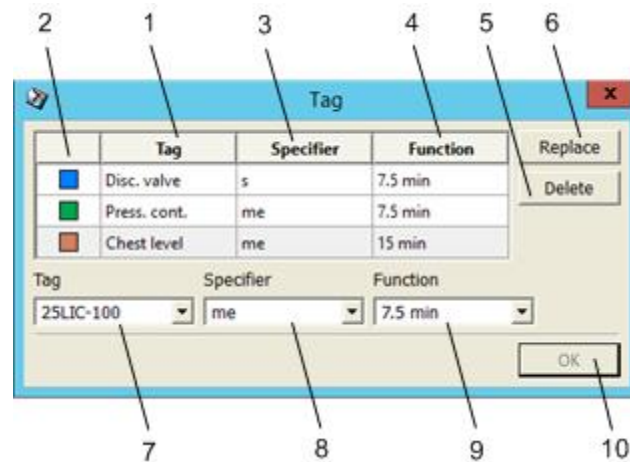
In some trend pictures you can freely select the tags presented in the trend as graphs. Tag selection and signal attribute modifications take place in the Tag window.

When you want to open the Tag window, proceed as follows:

- Right-click the trend and select Tag... on the action menu. The Tag window will open on the desktop.

Elements of the Tag Window

The Tag window consists of the following elements:



1. Tag name or description
2. Graph color
3. Signal specifier
4. Additional attribute
5. Delete button
6. Replace button
7. Tag drop-down list box
8. Specifier drop-down list box
9. Function drop-down list box
10. OK button

How to Select a Tag

When you want to select a tag to add a graph to a trend, proceed as follows:

1. In the Tag window, click the row to which you want to add or replace a tag.
2. Click the arrow button in the Tag box and select the desired tag on the appearing list.

OR

Click the Tag list box, type the tag identifier and press Enter.

3. Click the arrow button in the Specifier list box and select the desired specifier on the appearing list.
4. Click the arrow button in the Function list box and select the additional attribute on the appearing list.
5. Click the Replace button to save the row.
6. Click the OK button. The Tag window will close and the changes will be updated in the trend.

The changes you have made in the Tag window will not be updated in the trend until you have clicked the OK button. You can thus make several changes to the trend in one go.

Clicking the  button closes the window while canceling all the changes you have made.

How to Edit Signal Attributes

When you want to edit signal attributes, proceed as follows:

1. In the Tag window, click the row the attributes of which you want to edit.
2. Click the arrow button in the Tag list box and, on the appearing list, select the tag displayed on the target row.

OR

Click the Tag box, type the tag identifier displayed on the target row and press Enter.

3. Make the desired changes by selecting new values in the Specifier and Function list boxes.
4. Click the Replace button to save your changes.
5. Click the OK button. The Tag window will close and the changes will be updated in the trend.

The changes you have made in the Tag window will not be updated in the trend until you have clicked the OK button. You can thus make several changes to the trend in one go.

Clicking the  button closes the window while canceling all the changes you have made.

How to Delete a Tag

When you want to delete a tag from the trend, proceed as follows:

1. In the Tag window, click the row whose tag you want to delete.
2. Click the Delete button. The selected row will be cleared of data.
3. Click the OK button. The Tag window will close and the changes will be updated in the trend.

The changes you have made in the Tag window will not be updated in the trend until you have clicked the OK button. You can thus make several changes to the trend in one go.

Clicking the  button closes the window while canceling all the changes you have made.

11.3.9 How to Save a Trend Picture to the Hard Disk

When you have added the desired tags to a trend picture, it is possible to save the trend picture to the hard disk for later use.

When you want to save the trend picture to the hard disk, proceed as follows:

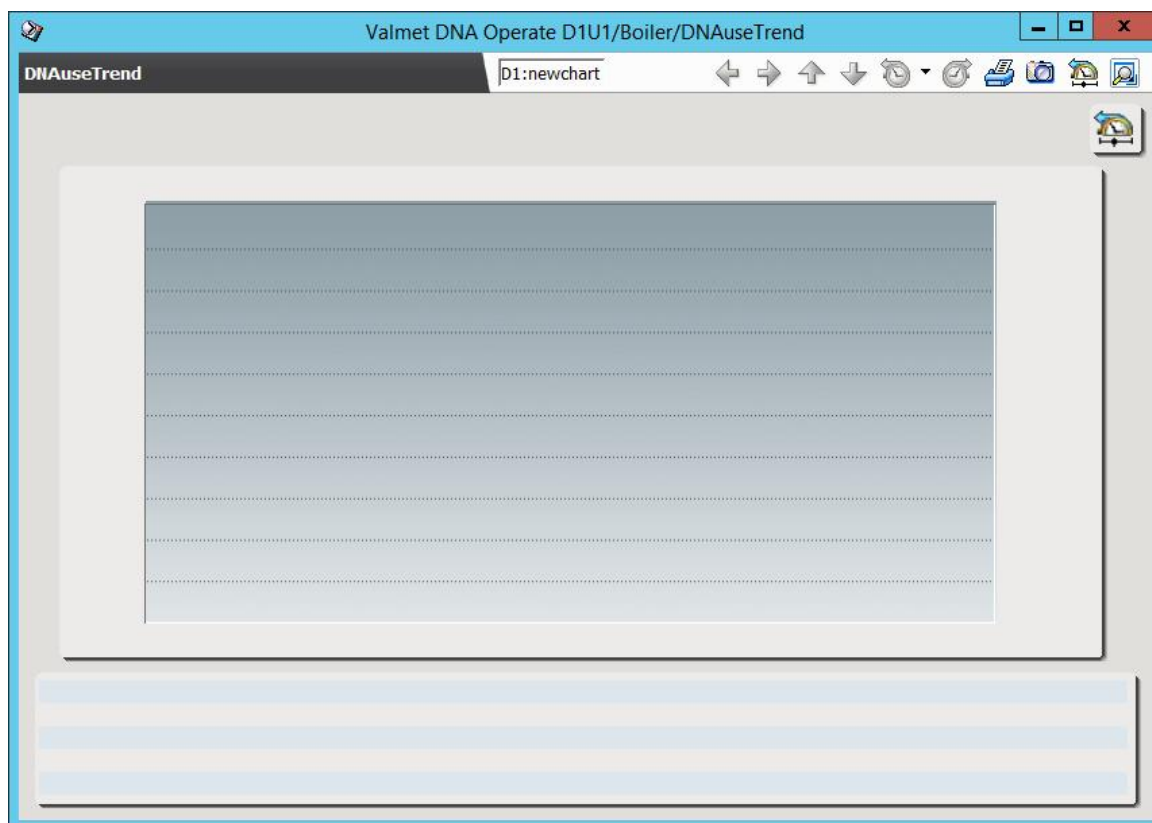
- Right-click the trend picture to be saved (however, not on the coordinates area) and select **Save As...** on the submenu of the **Picture...** command.

For more information on this function, see chapter [Save As...](#)

11.4 Trend Analysis Tool

11.4.1 Properties

Trend analysis tool is a particular case of the basic trend picture. All the normal operations of the trend picture with the exception of tag selection through the Tag window can be applied to it as well. The difference between the trend picture and the trend analysis tool is that the latter does not contain any pre-configured tags. Instead, the trend analysis tool opens on the desktop as an empty template to which graphs can be added by dragging and dropping tags from picture windows, tag manager or the history tag manager.



11.4.2 How to Open the Trend Analysis Tool

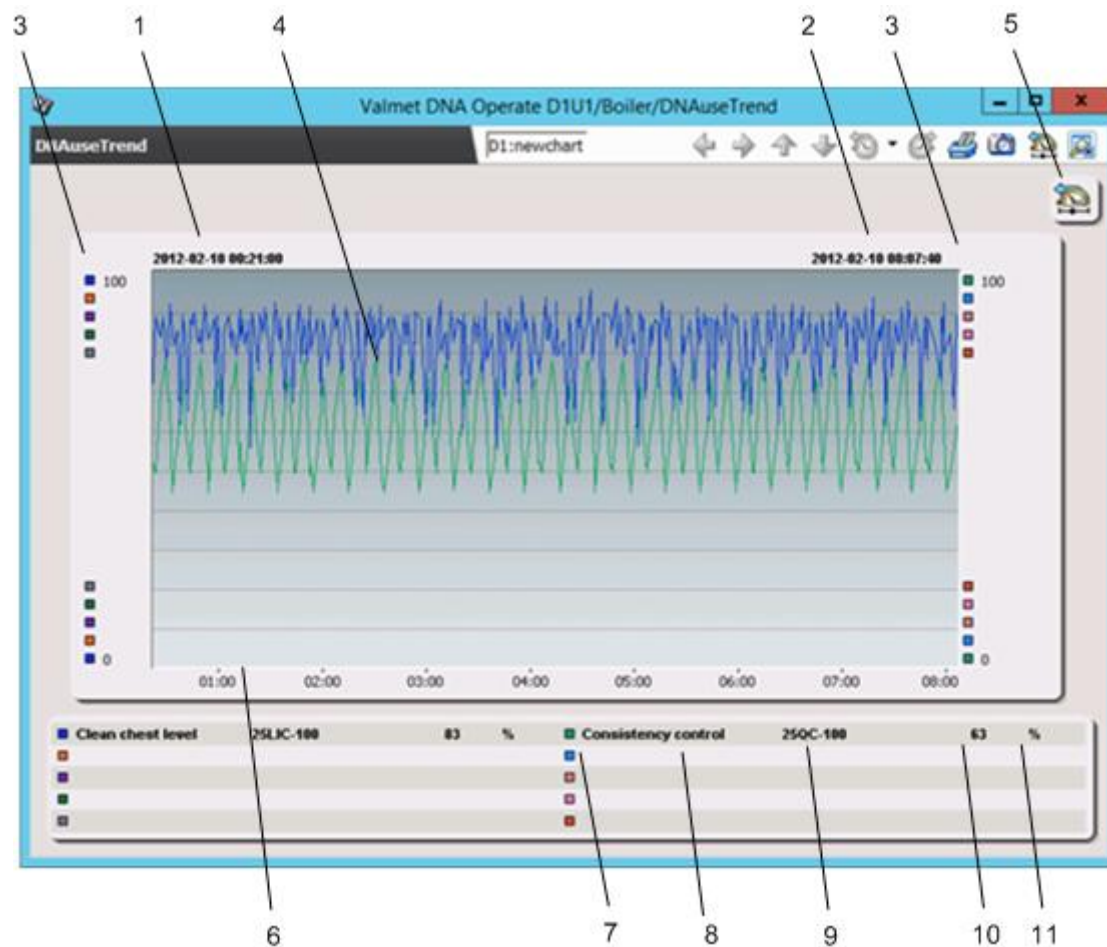
The trend analysis tool is opened from the toolbar of the control panel.

Proceed as follows:

- Click the trend analysis button  on the control panel toolbar. The trend analysis tool will open on the desktop in a separate window.

11.4.3 Elements of the Trend Analysis Tool

The trend analysis tool consists of the following elements:



- Starting point of time span
- End point of time span
- Graph scales
- Coordinates
- History mode button
- Time span
- Graph colors
- Tag name or description
- Tag identifier
- Instantaneous value
- Signal unit

11.4.4 Tag Addition

How to Add a Tag from a Picture Window

A tag presented in a picture window can be easily added to the trend analysis tool.

Proceed as follows:

1. Grab the target tag specifier in a picture window.
2. Drag the tag to the trend analysis tool.
3. Drop the tag on the coordinates area. The graph and data of the tag will be added to the trend analysis tool.

How to Add a Tag from the History Tag Manager

[History tag manager](#) is an application executed from a picture window with which you can search for tags by tag identifier, description or type. You can then drag tags from the search results to the trend analysis tool for examination.

How to Copy a Tag to the Trend Analysis Tool

Proceed as follows:

1. Grab the target tag on the search result list.
2. Drag the tag to the trend analysis tool.
3. Drop the tag on the coordinates area. The graph and data of the tag will be added to the trend analysis tool.

11.4.5 How to Save a Trend Analysis Tool to the Hard Disk

When you have added the desired tags to the trend analysis tool, it is possible to save the trend analysis tool to the hard disk for later use.

When you want to save the trend analysis tool to the hard disk, proceed as follows:

- Right-click the trend analysis tool to be saved (however, not on the coordinates area) and select Save As... on the submenu of the Picture... command.

For more information on this function, see chapter [Save As...](#)

11.4.6 How to Open a Trend Analysis Tool from the Hard Disk

When you want to open trend analysis tools saved on the hard disk, proceed as follows:

- Right-click a picture window and select **Open Picture...** on the submenu of the **Picture...** command. The saved trend analysis tool will open in the topmost picture window on the desktop.

For more information on this function, see chapter [How to use the Action Menu -> Open picture...](#) .

11.4.7 How to Enter History Mode

When you want to analyze trend history, proceed as follows:

- Click the history mode button  in the trend analysis tool. The timeline dialog will open.

For more information on this function, see chapter [History Mode -> How to Enter History Mode](#).

11.5 History Trend

11.5.1 Properties

History trend is another particular case of the basic trend picture. All the normal operations of the trend picture with the exception of tag selection through the Tag window can be applied to it as well. The difference between the trend picture and the history trend is that the latter always contains certain pre-configured graphs depending on the selected tag type. History trend window also contains the Save

button  for saving the history trend to the hard disk for later use.

11.5.2 How to Open the History Trend

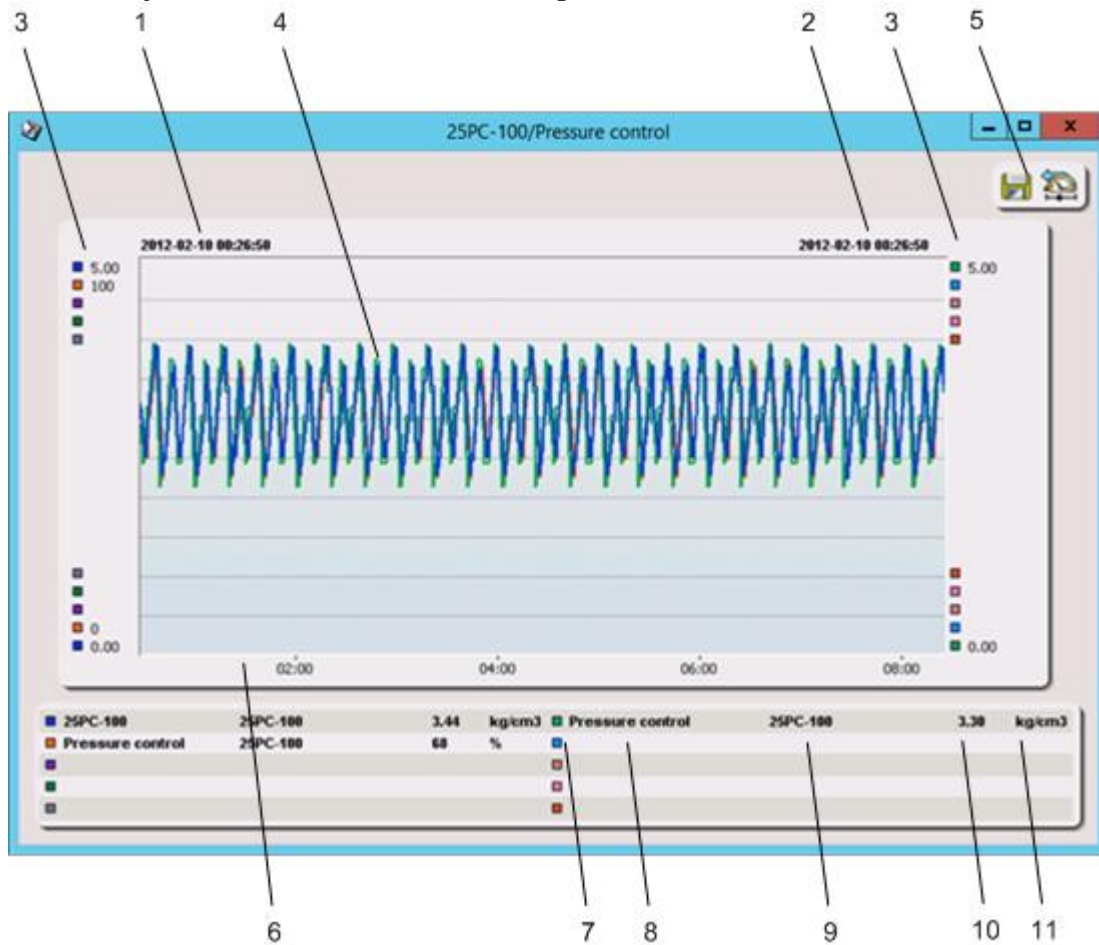
The history trend is opened from the action menu.

Proceed as follows:

- Right-click the desired tag in the DNA Operate picture and select **History Trend** on the action menu. The history trend will open on the desktop in a separate window.

11.5.3 Elements of the History Trend

The history trend consists of the following elements:



1. Starting point of time span
2. End point of time span
3. Graph scales
4. Coordinates
5. Save and History mode buttons
6. Time span
7. Graph colors
8. Tag name or description
9. Tag identifier
10. Instantaneous value
11. Signal unit

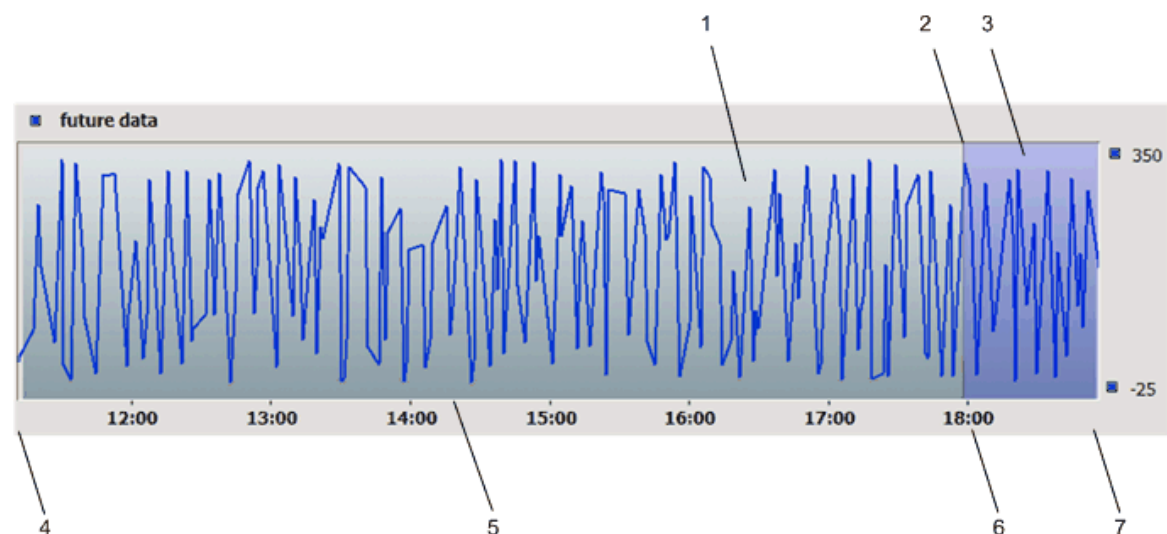
11.6 Time Shifting in a Trend Picture

In a trend picture, it is possible to present trend data which ends to the current time (default situation), time in the past or time in the future. If all these alternatives are in use at the same time, then the end point of the X-axis (time span) is chosen according to the latest time (future). Note that if the time span includes time from the future, then it shows less time from the past. Future data is calculated based on the past data (history data) and produced by a dedicated forecasting application.

The future data area in the trend picture is shown with a different background color than the rest of the trend area, and the current time is marked with a separator line. The length of the future data area has been defined when trends and their graphs were configured. Note that if you remove all the graphs which contain future data, the future data area disappears from the trend picture.

Future data can be presented in pictures also in numerical values. You can drag and drop these values to a trend picture. If you drag and drop a numerical value presenting future data to a trend picture, the future data area appears in the trend picture if not shown earlier.

When a trend picture presents trend data which ends to a time in the past, then you cannot see current time values on the X-axis but the end point of the time span represents past time.



The trend which contains future data consists of the following elements:

1. History data area
2. Separator line for current time
3. Future data area
4. Starting point of the time span
5. X-axis (time span)
6. Current time
7. End point of the time span

12. Profile User Manual

12.1 Introduction

This guide describes the profile component, which is typically used for displaying transverse paper web measurement values, such as paper web thickness, paper moisture, or paper basis weight. You can also use the profile component for controlling transverse process variables and actuators, entering setpoint values and controlling actuator statuses in a paper machine.

Profiles can be found in process diagrams or in separate profile pictures.

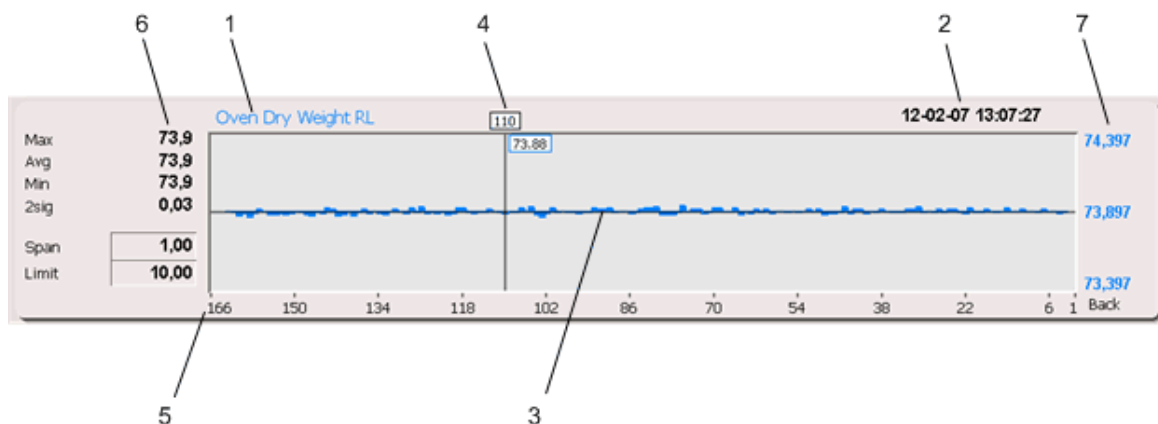
Profiles are mainly operated through the action menu. The profile-related action menu is opened by right-clicking the graph area.

In profiles, you can perform the following operations:

- Use a hairline to display an instantaneous value
- Scale the profile
- Zoom the profile
- View the statistics
- Change profile graph values

12.2 Elements of a Profile

A profile consists of the following elements:



1. Name text
2. Date and time
3. Profile graphs
4. Hairline
5. Points on the profile
6. Profile values
7. Maximum, average and minimum graph scale values

There are several variations of profiles. For example, the scale values may be missing and the number of the presented graphs may vary.

12.3 Operations

12.3.1 How to View Statistics

The statistics of the profile graphs can be viewed in the Statistics window. The statistical data is calculated from the graphs presented in the profile.

When you want to open the Statistics window, proceed as follows:

- Right-click the profile and select Statistics... on the action menu. The Statistics window will open on the desktop.

		Min	Max	Ave	Std	Points
Current		73.86	73.94	73.90	0.014	161
Oven Dry Weight RL		73.85	73.93	73.89	0.015	161

1. Graphs presented in the profile
2. Graph color
3. Minimum value of profile graph
4. Maximum value of profile graph
5. Average value of profile graph
6. Standard deviation of profile graph
7. Number of points used in statistics calculations

No operations can be executed in the Statistics window.

12.3.2 Zooming

How to Zoom In

By zooming you can crop an interesting area of the profile to be viewed, and the changes in graph values can be examined more closely.

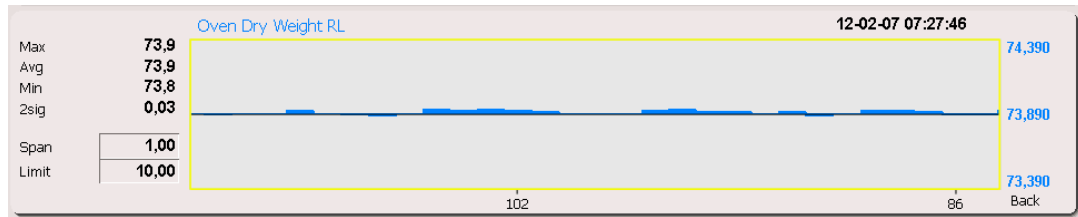
When you want to zoom in a profile, proceed as follows:

1. In a profile, move the pointer to the corner of the area you want to zoom.
2. Click and hold the left mouse button while moving the pointer to outline an area. The values of the profile X-axis (time) and Y-axis scales are determined by the outlined area.

If you want to keep the values of the Y-axis scale unchanged while zooming, hold the Shift key down while moving the pointer to outline an area.

During outlining the value of X-axis at cursor point is displayed in the picture as a tooltip (eg.).

3. Release the mouse button. The outlined area will be opened in the profile and the borders of the coordinates will be highlighted in yellow attention color to indicate that the profile is zoomed in.



The zoomed profile can be scrolled with the keyboard shortcuts. The values of the X-axis can be scrolled with the Ctrl + arrow left/right keys and the values of the Y-axis with the Ctrl + arrow up/down keys.

It is possible to zoom in a profile several times in a row.

Note that you cannot save or take a snapshot of a zoomed profile because when saving, the zoomed profile always returns to the situation before zooming.

How to Zoom Out

When you want to zoom out a profile, proceed as follows:

- Right-click the zoomed profile and select Reset zoom on the action menu.

OR

- Press the ESC key.

Zooming is cancelled automatically after a while since the last zooming.

12.3.3 Scaling

How to Open the Scaling Window

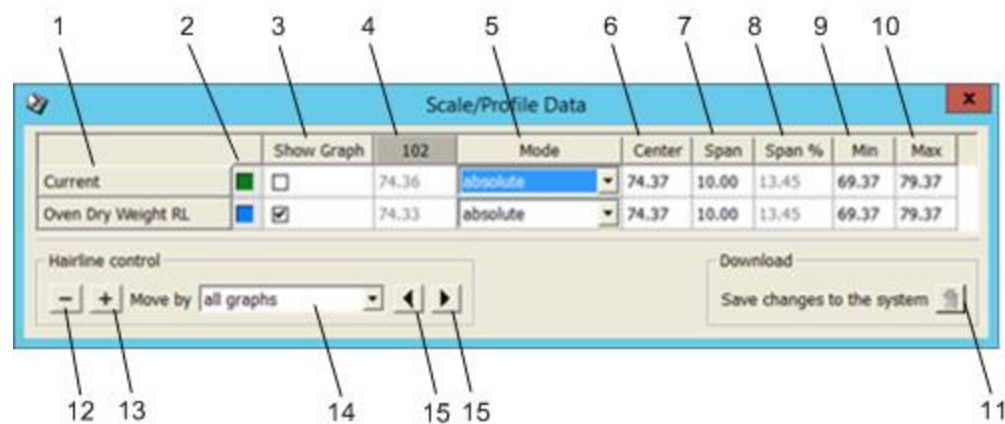
In the scaling window, you can change the scaling modes and the scale values of the profile graphs.

When you want to open the Scale/Profile Data window, proceed as follows:

- Right-click the profile and select Scale/Profile Data... on the action menu. The Scale/Profile Data window will open on the desktop.

Elements of the Scaling Window

The Scale /Profile Data window consists of the following elements:



1. Graphs presented in the profile
2. Graph color
3. Check box for displaying/hiding the graph
4. Hairline location and the corresponding graph value
5. Scaling mode
6. Scale center value
7. Scale span
8. Scale span as a percentage value of the center
9. Scale minimum value
10. Scale maximum value
11. Button for saving changes to the system

Buttons related to hairline operations:

12. Remove button
13. Add button
14. Selecting the graph/graphs
15. Arrow buttons for moving the selected hairline

How to Change the Scaling Mode

The scaling mode of a profile graph can be changed in the Scale /Profile Data or Change Profile window. The following modes are available:

Absolute	Center and span are absolute values. The center is located on the median.
Relative	Center is absolute value and located on the median. Span is given as a percentage of the center value.

When you want to change the scaling mode of a profile graph, proceed as follows:

1. In the Scale/Profile Data or Change Profile window, click the mode field of the target graph. The list of the scaling modes will open below the field.
2. Select the desired scaling mode. The menu will close and the scaling mode of the target graph will change accordingly.

How to Change Scale Values

The Center, Min and Max values of a profile graph can be changed regardless of the selected scaling mode. In addition, the Span value can be changed when the absolute scaling mode has been selected. Similarly, the Span % value can be changed when the relative scaling mode has been selected. Value modifications affect the graph format in the profile.

Proceed as follows:

1. In the Scale / Profile Data or Change Profile window, click the value you want to operate.
2. Delete the old value with the Backspace key. You can also move the cursor in the text box by clicking the desired location inside the box.
3. Enter a new value and press the Enter key. The graph scale will be updated accordingly in the profile.

NOTE! Changing one value may affect the other scale values of the graph as well. For example, if you modify the Span value, the Min and Max values will change accordingly.

NOTE! If both the center and the span value of a profile graph is received from the process control server, they cannot be operated.

12.3.4 How to Hide and Restore a Graph

You can hide a graph displayed in a profile and later restore it back.

When you want to hide a graph, proceed as follows:

- Right-click the profile and select the target graph on the submenu of the Show Graph > command. The graph will be removed from the profile and the corresponding check mark will disappear from the submenu.

Repeat the operation to restore the graph.

OR

- Click to clear the Show Graph check box of the target graph in the Scale / Profile Data or Change Profile window. The check mark will disappear and the graph will be removed from the profile.

The graph is restored correspondingly by selecting the Show Graph check box.

12.3.5 Hairline Operations

How to Add a Hairline

The values of individual points on a profile graph can be examined by adding hairlines in desired locations. The values of the marked points are displayed in the Scale / Profile Data and Change Profile window.

The screenshot shows the 'Scale/Graph Data' window with a table of data and hairline controls.

	Show Graph	Marker	Specifier	Unit	Mode	Min	Max	Center	Span
Basis Wgt RL	☒	☐	me	g/m2	Manual	0.0	250.0	125.0	250.0
Moisture RL	☒	☐	me	%	Manual	0.0	20.0	10.0	20.0
Total Ash RL	☒	☐	me	%	Manual	0.00	30.00	15.00	30.00

Below the table, there is a 'Hairline control' section with a '-' button, a '+' button, a 'Move by' dropdown menu set to 'all graphs', and left/right arrow buttons. To the right, there is a 'Download' section with a 'Save changes to the system' button and a save icon.

When you want to add a hairline to a profile, proceed as follows:

- Click the desired spot in the profile. A hairline will appear in that location and the Change Profile window may open on the desktop.

OR

Right-click the desired spot in the profile and select Add Hairline on the action menu. A hairline will appear in that location and the Change Profile window may open on the desktop.

OR

Click the  button in the Scale/Profile Data or Change Profile window. A hairline will appear in the location which was clicked when the window was opened.

How to Move a Hairline

You can move a hairline in a profile by dragging it or by operating it in the Scale / Profile Data or Change Profile window.

When you want to move a hairline by dragging it in a profile, proceed as follows:

1. Move the pointer on top of the target hairline until its appearance changes to a hand.
2. Press and hold the left mouse button to grab the hairline.
3. Drag the hairline to the desired location in the profile.
4. Release the mouse button.

When you want to move the hairline by operating it in the Scale/Profile Data or Change Profile window to achieve better precision, proceed as follows:

1. Open either window by right-clicking the profile and selecting Scale / Profile Data... or Change Profile... on the action menu.
2. In the appearing window, click the column title of the target hairline.

OR

Click the target hairline in the profile.

The selected hairline will appear black.

3. Click Move by field to select the graph by the points of which you want to move the hairline.

By default, the hairline moves by all graphs which means that when either arrow button is clicked, the selected hairline will move to the nearest point on any graph in the arrow button direction.

If after selecting a graph you want to reselect all graphs, just select all graphs in the field.

4. Move the selected hairline to the desired location in the profile by clicking the arrow buttons.

How to Remove a Hairline

You can remove hairlines from a profile one at a time either directly in the profile itself or in the Scale / Profile Data or Change Profile window.

When you want to remove hairlines one at a time, proceed as follows:

1. Move the pointer on top of the target hairline until its appearance changes to a hand.
2. Right-click the hairline and select Remove hairline on the action menu. The hairline will be removed from the profile and its data will disappear in the window.

OR

1. Right-click the profile and select Scale / Profile Data... or Change Profile... on the action menu.
2. In the appearing window, click the column title of the target hairline.

OR

Click the target hairline in the profile.

The selected hairline will appear black.

3. Click the  button. The selected hairline will be removed from the profile and its data will disappear from the window.

Alternatively, you can remove all hairlines from a profile at one go.

Proceed as follows:

- Right-click the profile and select Remove All Hairlines on the action menu. All hairlines displayed in the profile will be removed and their data will disappear from the Scale / Profile Data or Change Profile window.

12.3.6 Changing Profile Values

How to Open the Change Profile Window

The values of the data points on a profile graph can be changed in the Change Profile window.

When you want to open the Change Profile window, proceed as follows:

- Click the profile.

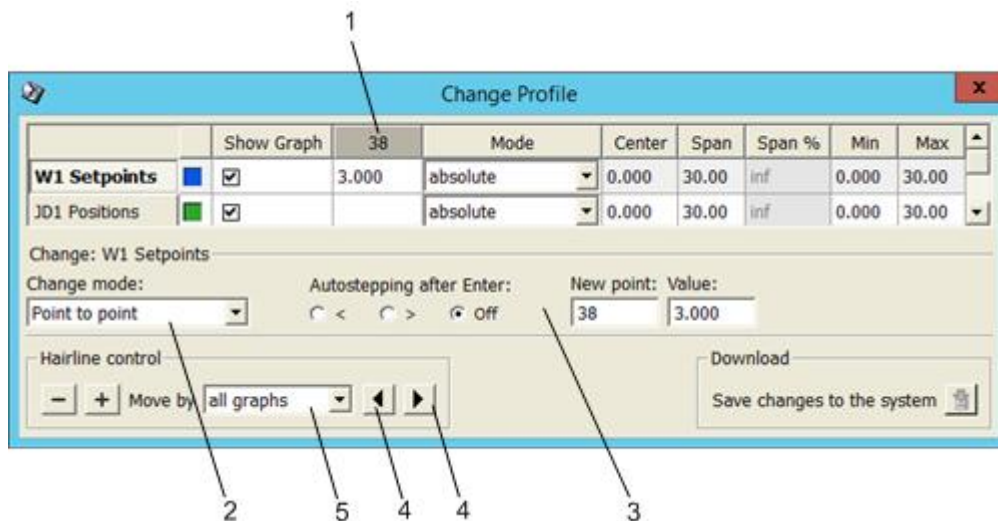
OR

Right-click the target point on the profile graph and select Change Profile... on the action menu.

A hairline will appear in that location and the Change Profile window will open on the desktop.

Elements of the Change Profile Window

The Change Profile window contains the following elements related to profile modification:



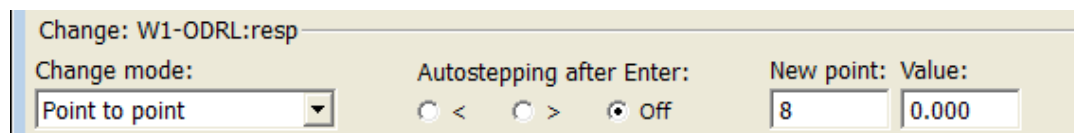
1. Hairline location and the corresponding graph value
2. Drop-down list for selecting the change mode
 - Point by point
 - Same value to points
 - Incr/Decr points
 - Linearize
 - Stretch
 - Draw
3. The rest of the contents in the Change field changes according to the selected change mode.
4. Arrow buttons for moving the selected hairline
5. Selecting the graph/graphs

How to Change Data Point Values Point by Point

The values of the individual points on a profile graph can be changed one by one.

Proceed as follows:

1. In the Change Profile window, click the Change mode drop-down list and select Point by point.



2. Click the column title of the hairline you wish to operate.

OR

Click the target hairline in the profile.

The selected hairline is displayed in black.

3. Click the New point box, type a new value, and press the Enter key.

OR

Move the hairline onto the target point as described in the chapter [How to Move a Hairline](#).

4. Click the value to be changed in the hairline column, or click the Value box.
5. Enter a new value and press the Enter key. Negative values can be used. The graph will be updated accordingly in the profile.
6. If you want to modify more data point values, move the hairline to the next target point as described in step 3.

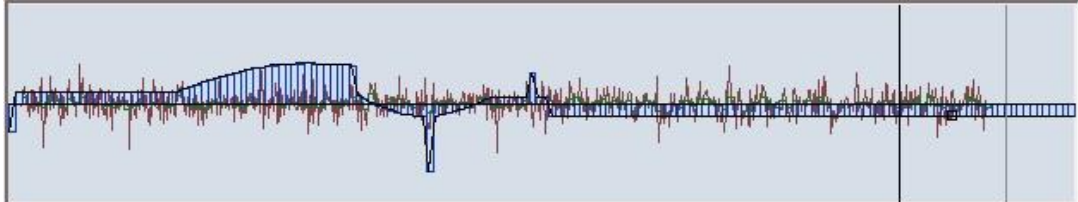
Alternatively, you can select the stepping direction by clicking the desired Autostepping option button. In this case, the hairline is moved in the selected direction to the next data point on the [selected graph](#) when the Enter key is pressed to update the new value.

<	Next data point left
>	Next data point right
Off	No autostepping

How to Set the Same Value to Consecutive Data Points

When you want to set the same value to several consecutive data points, proceed as follows:

1. In the Change Profile window, click the Change mode drop-down list and select Same value to points. Two hairlines and a black shadow profile will appear in the profile.

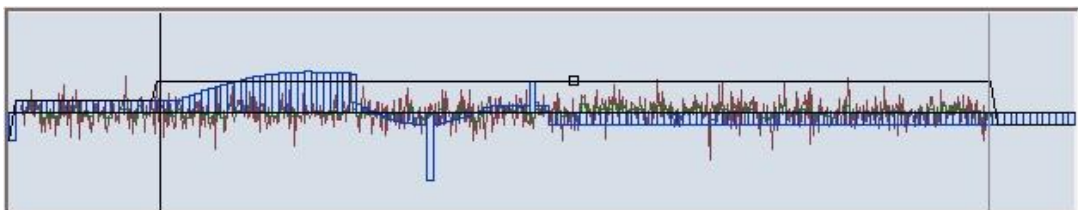


The value of the data point marked with the right-hand hairline is determined by the value of the data point marked with the left-hand hairline. Outside the defined area, the shadow profile follows the graph [specified in the Move by field](#).

2. To define the start point, select either of the following:
 - Move the hairline onto the target point as described in the chapter [How to Move a Hairline](#).
 - Click the Start point box, type a new value, and press the Enter key. The location of the hairline in the profile will change accordingly.

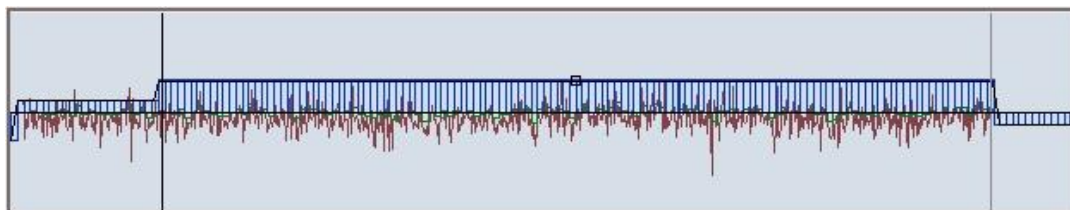
Change: W1-ODRL:nrs8			
Change mode:	Start point:	End point:	Value:
Same value to points ▼	14	8	0.000
			Accept

3. Define the End point similarly.
4. Click the Value box, type a new value, and press the Enter key. The shadow profile displayed in the defined area will be updated to match the given value. Negative values can be used.

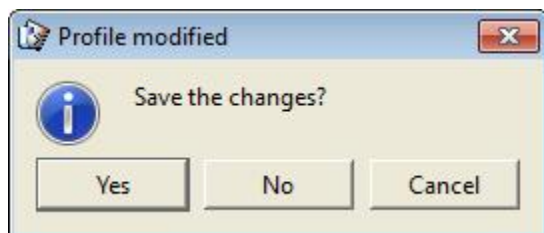


Alternatively, you can grab the box displayed in the shadow profile between the hairlines and drag the area to the desired location and value in the profile. The width of the area remains as it is.

- Click the Accept button. The profile will be updated to match the changes.



NOTE! If the Change profile window is closed before the changes have been accepted, the dialog displayed below will be opened on the desktop.

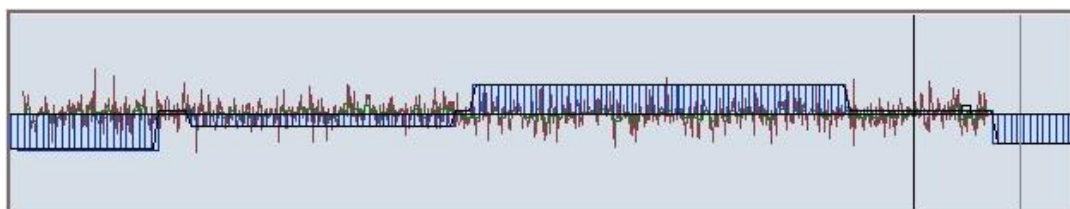


Click Yes to save your changes, No to ignore the changes, or Cancel to leave the Change profile window open on the desktop.

How to Increase or Decrease Data Point Values

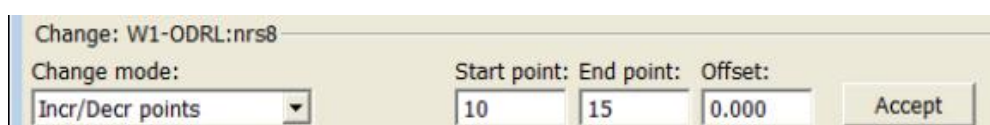
When you want to increase or decrease data points values in the selected area of the profile, proceed as follows:

- In the Change Profile window, click the Change mode drop-down list and select Incr/Decr points. Two hairlines and a black shadow profile will appear in the profile.



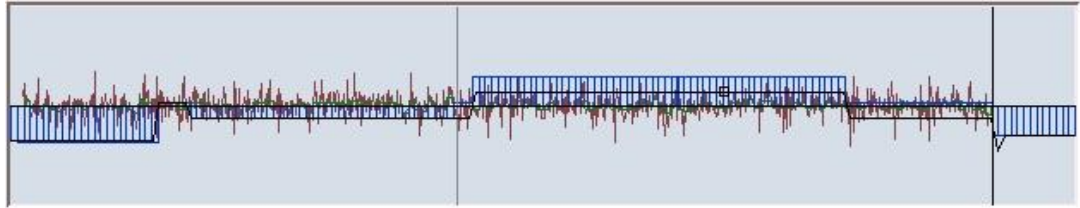
The shadow profile follows the graph [specified in the Move by field](#).

- To define the start point, select either of the following:
 - Move the hairline onto the target point as described in the chapter [How to Move a Hairline](#).
 - Click the Start point box, type a new value, and press the Enter key. The location of the hairline in the profile will change accordingly.



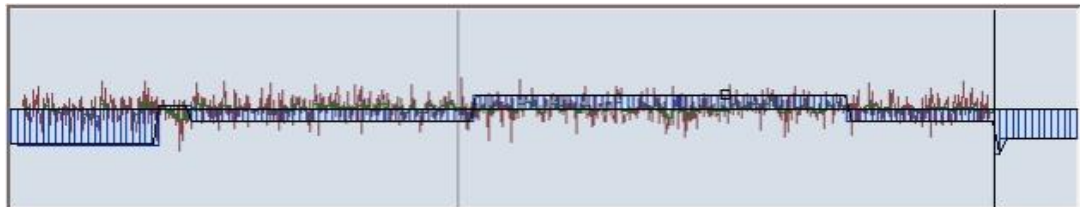
- Define the End point similarly.

4. Click the Offset box, type the value of change, and press the Enter key. Positive value will increase and negative value decrease the data point values of the shadow profile in the selected area.

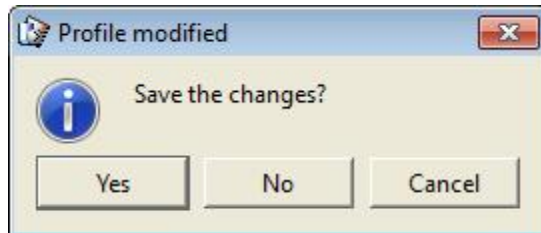


Alternatively, you can grab the box displayed in the shadow profile between the hairlines and drag the area to the desired location and value in the profile. The width of the area remains as it is.

5. Click the Accept button. The profile will be updated to match the changes.



NOTE! If the Change profile window is closed before the changes have been accepted, the dialog displayed below will be opened on the desktop.



Click Yes to save your changes, No to ignore the changes, or Cancel to leave the Change profile window open on the desktop.

How to Linearize a Profile Graph

When you want to linearize a profile graph in the selected area, proceed as follows:

1. In the Change Profile window, click the Change mode drop-down list and select Linearize. Two hairlines and a black shadow profile will appear in the profile.



In the defined area, the shadow profile is displayed as a linear line drawn between the points marked with the hairlines. Outside the defined area, the shadow profile follows the graph [specified in the Move by field](#).

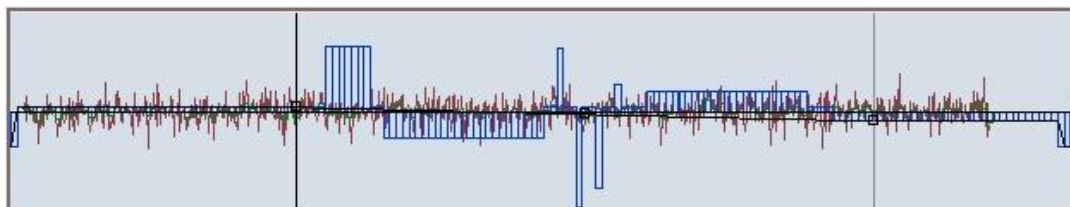
2. To define the start point, select either of the following:
 - Move the hairline onto the target point as described in the chapter [How to Move a Hairline](#).
 - Click the Start point box, type a new value, and press the Enter key. The location of the hairline in the profile will change accordingly.

Change: W1-ODRL:nrs8					
Change mode:	Start point:	End point:	Start value:	End value:	
Linearize	6	11	0.315	0.000	Accept

3. Define the End point similarly.
4. Click the Start value box, type a new value, and press the Enter key. The shadow profile displayed in the defined area will be updated to match the given value. Negative values can be used.

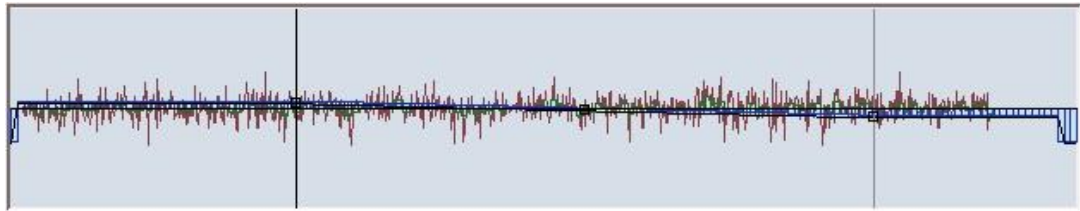
Alternatively, you can grab the box displayed on the hairline and drag it to the desired location in the profile.

5. Define the End value similarly. Negative values can be used.

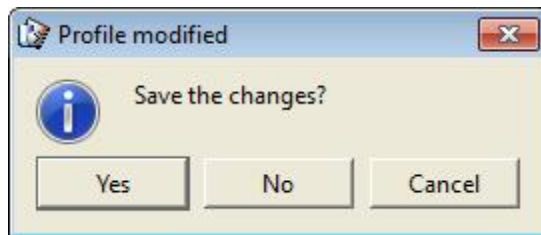


You can also grab the box displayed in the shadow profile between the hairlines and drag the area to the desired location in the profile. The width of the area remains as it is.

- Click the Accept button. The profile will be updated to match the changes.



NOTE! If the Change profile window is closed before the changes have been accepted, the dialog displayed below will be opened on the desktop.

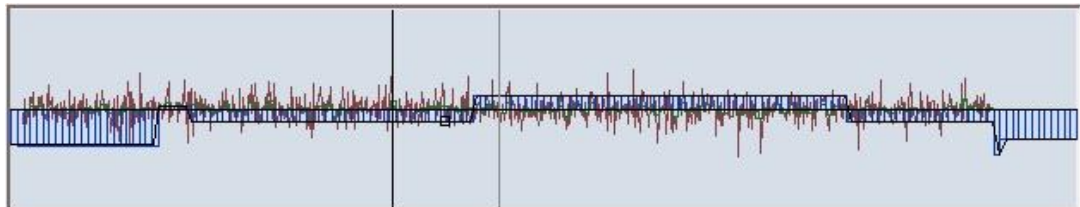


Click Yes to save your changes, No to ignore the changes, or Cancel to leave the Change profile window open on the desktop.

How to Stretch a Profile Graph

When you want to stretch a profile graph in the selected area, proceed as follows:

- In the Change Profile window, click the Change mode drop-down list and select Stretch. Two hairlines and a shadow profile will appear in the profile.



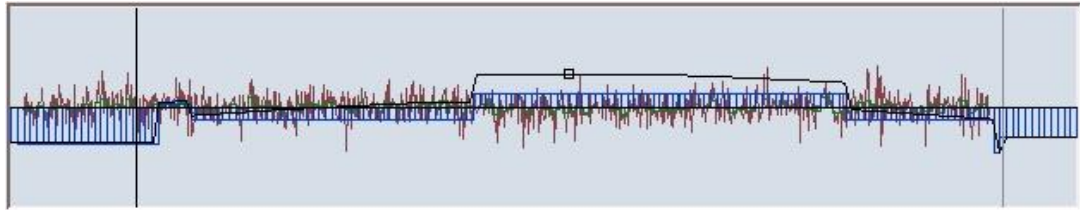
The shadow profile follows the graph [specified in the Move by field](#).

- To define the start point, select either of the following:
 - Move the hairline onto the target point as described in the chapter [How to Move a Hairline](#).
 - Click the Start point box, type a new value, and press the Enter key. The location of the hairline in the profile will change accordingly.

Change: W1-ODRL:nrs8			
Change mode:	Start point:	End point:	Offset:
Stretch	7	12	0.000
			Accept

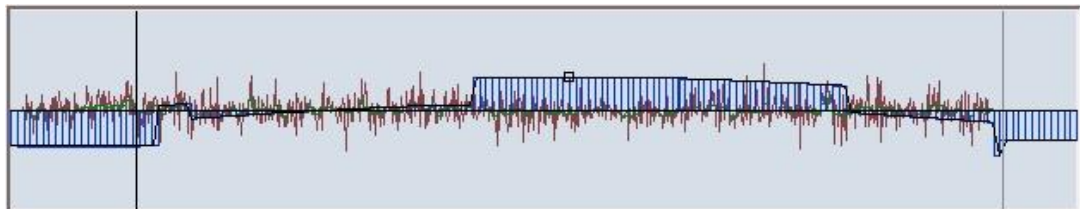
- Define the End point similarly.
- Click the Offset box, type the value of change, and press the Enter key.

Positive value will increase and negative value decrease the data point values of the shadow profile in the selected area. The values of the start and end point will not change, but in the defined area the magnitude of the change grows linearly so that its vertex (i.e. the value given in the Offset box) is reached at the midpoint.

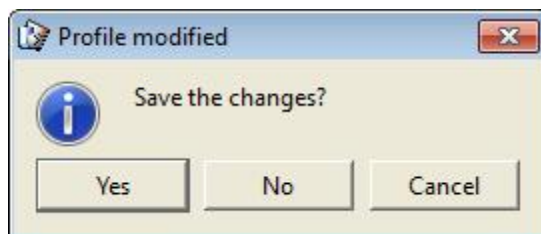


Alternatively, you can grab the box displayed in the shadow profile between the hairlines and drag the area to the desired location and value in the profile. The width of the area remains as it is.

5. Click the Accept button. The profile will be updated to match the changes.



NOTE! If the Change profile window is closed before the changes have been accepted, the dialog displayed below will be opened on the desktop.

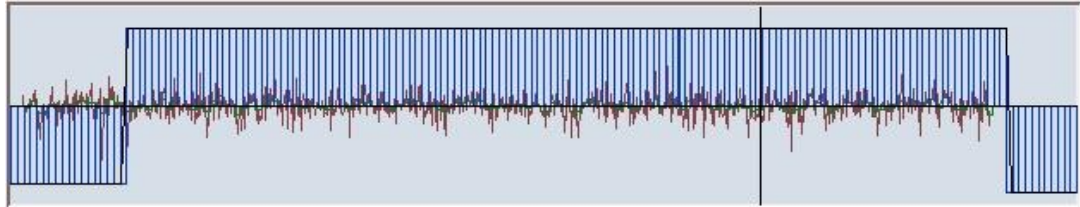


Click Yes to save your changes, No to ignore the changes, or Cancel to leave the Change profile window open on the desktop.

How to Draw a Profile Graph

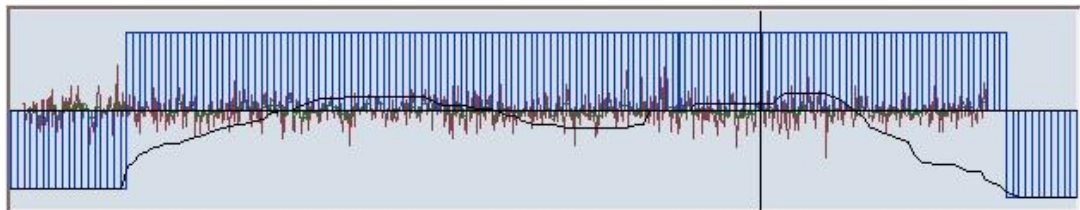
When you want to draw a freeform profile graph, proceed as follows:

1. In the Change Profile window, click the Change mode drop-down list and select Draw. A black shadow profile will appear in the profile. The hairlines displayed in the profile remain as they were.

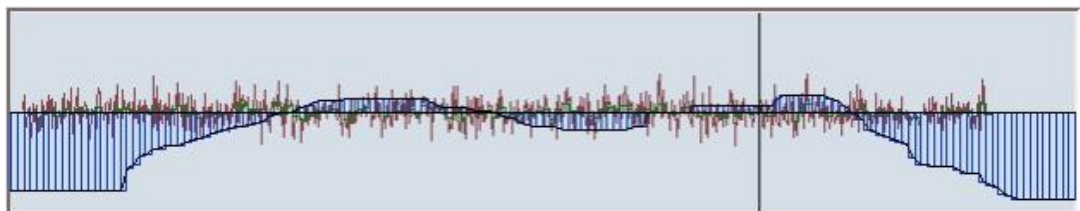


The shadow profile follows the graph [specified in the Move by field](#).

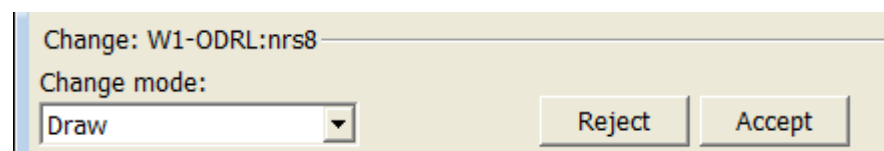
2. Move the pointer to the profile. Its shape changes to a cross.
3. Point to the location you want to start drawing from and press down the left mouse button.
4. Draw the desired profile slowly while holding the mouse button.
5. Release the button when you wish to stop drawing.



6. Click the Accept button. The profile will be updated to match the changes.



If you are not satisfied with the shadow profile you drew, click Reject to restore the original profile.



12.3.7 How to Save a Profile to the Hard Disk

When you want to save a profile to the hard disk, proceed as follows:

- Right-click the profile to be saved (however, not on the coordinates area) and select Save As... on the submenu of the Picture... command.

For more information on this function, see chapter [Save As...](#)

13. Functional Descriptions and Interlockings

13.1 Introduction

Functional descriptions and interlockings are opened with the corresponding action menu commands. They contain useful information on the loop operations as a part of the process. To be more specific, functional descriptions display general information on the loop behavior, whereas interlockings are updated dynamically according to the state of the process which makes them an important asset in troubleshooting. Both functional descriptions and interlockings open in a new browser window when first opened from the action menu.

Depending on the implementation, interlockings may feature the commenting option where users can enter their own comments on the related topic. These comments are available for all users and can be later edited.

13.2 Functional Descriptions

13.2.1 How to Open a Functional Description

When you want to open a functional description, proceed as follows:

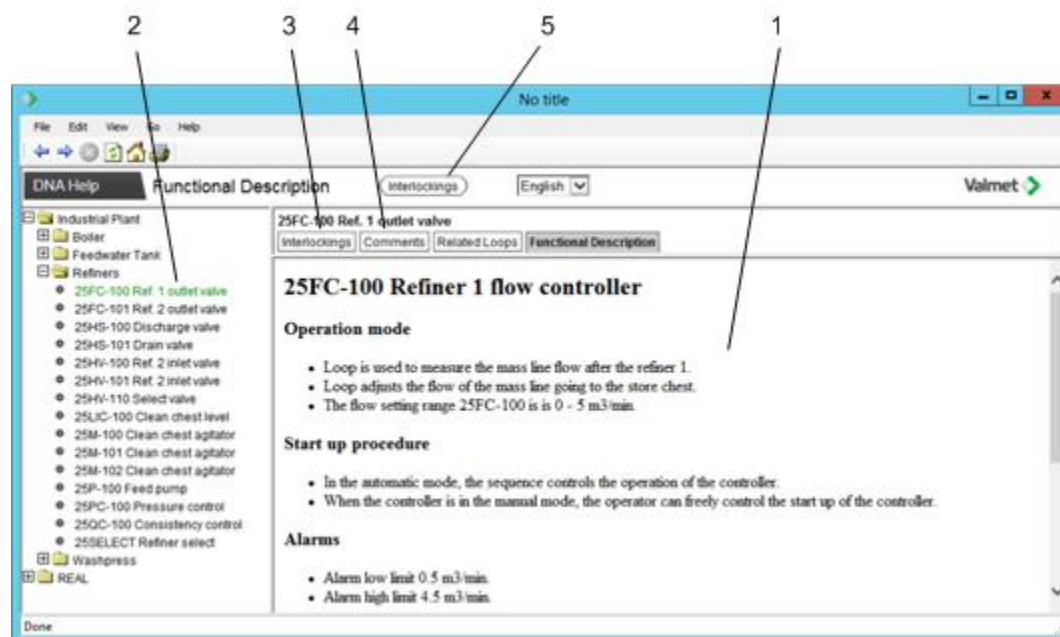
1. Right-click the target in a picture window.
2. On the appearing action menu, select Functional Description. The functional description of the target loop will open on the display in a dedicated browser window.

OR

- On the Interlocks or Comments tab, click the Functional Description tab. The browser window will update to display the loop-related functional description.

13.2.2 Function of the Functional Descriptions

A functional description usually displays a compact description of the loop as a part of the process. In some implementations, all loops related to a process may share one common functional description. Functional descriptions usually contain information on the function of the loop, connections to other loops, starting procedure, alarms, and implementation in Valmet DNA. From the functional descriptions it is easy to switch to examine the interlockings or comments of the loop. Functional descriptions open in a dedicated browser window or in the same browser window with the interlockings, depending on the implementation.



1. Functional description
2. Loops of the process area
3. Interlockings of the target loop (Interlocks tab)
4. Comments of the target loop (Comments tab)
5. Button for opening the Interlocks window

13.3 Interlockings

13.3.1 How to Open Interlockings

When you want to examine interlockings, proceed as follows:

1. Right-click the target in a picture window.
2. On the appearing action menu, select Interlockings. The interlockings of the target loop will open on the display in a dedicated browser window.

OR

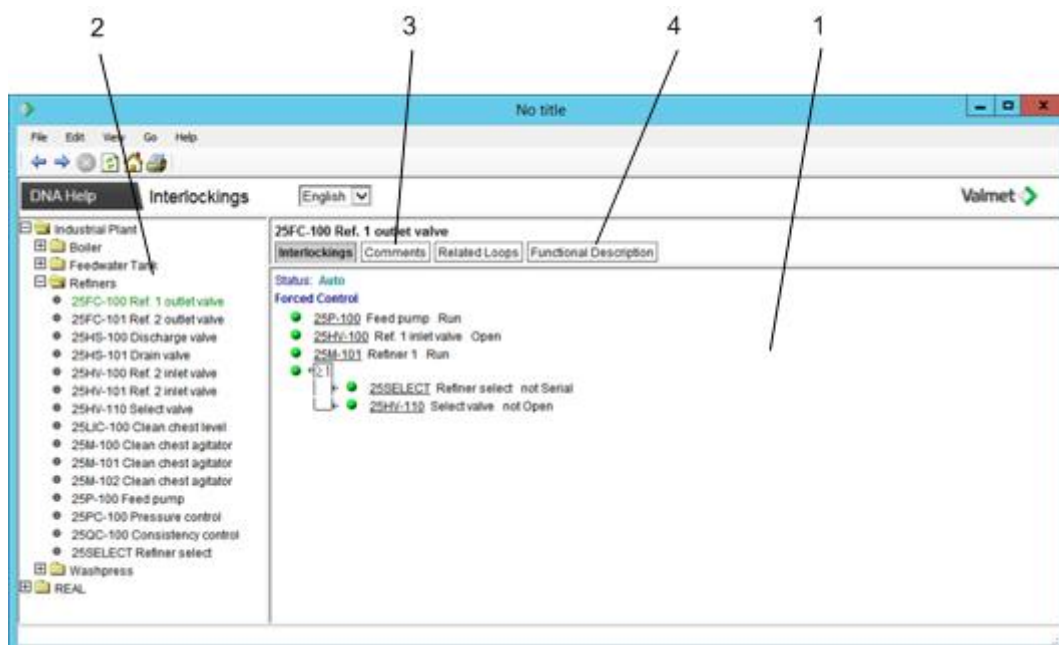
- If you want to open the Interlocks window from the Functional Description tab, click the Interlocks button. The browser window will display the Interlocks window of the target loop.

OR

- On the Functional Description or Comments tab, click the Interlocks tab. The browser window will update to display the loop-related interlockings.

13.3.2 Function of the Interlockings

The interlockings of the loop are dynamically presented in a dedicated browser window. The Interlocks window displays the state of the loop and any interlocking connections to other loops. The Interlocks window also includes tabs to functional descriptions and comments.



1. State of the loop and interlockings
2. Loops of the process area
3. Comments of the target loop (Comments tab)
4. Functional descriptions of the target loop (Functional Description tab)

13.3.3 Interlocking Symbols

Depending on the implementation, the following symbols may be used on the interlocking pages:

	Normal state, no interlockings
	Loop that triggered the first interlocking
	Cause of interlocking
	Information button (normal state)
	Information button (interlocking)
	Fault instructions

13.4 Operations

13.4.1 How to Change the Loop

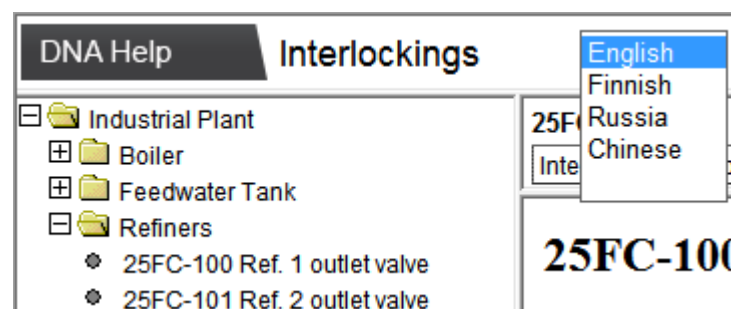
You can change the loop displayed in the functional description or interlockings window directly in the browser window in question.

Proceed as follows:

- Click the desired loop on the process area loop list. The loop data will be updated in the browser window.

13.4.2 How to Change the Language

Depending on the implementation, the functional descriptions or interlockings may be available in two languages. In this case it is possible to change the presentation language with the language buttons available on the left edge of the browser window:

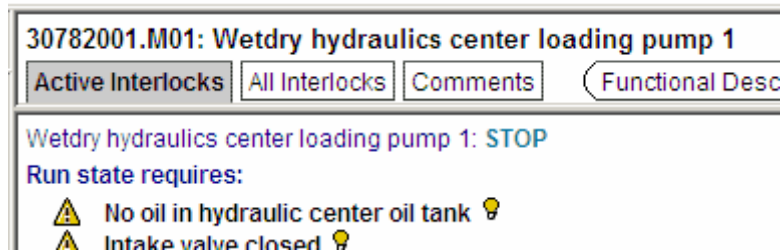


When you want to change the presentation language, proceed as follows:

- Click the desired language button. The contents of the browser window will be updated to use the selected language.

13.4.3 How to Change the View in Machine Control Interlockings

In some implementations it is possible to change the interlocking view. Views are presented as tabs in the browser window:



1. Active Interlocks view (only active interlocking rows displayed)
2. All Interlocks view (all interlocking rows displayed)

When you want to change the interlocking view, proceed as follows:

- Click the desired view button. The contents of the browser window will be updated to display the selected view.

13.4.4 Comments

How to Open a Comment Page

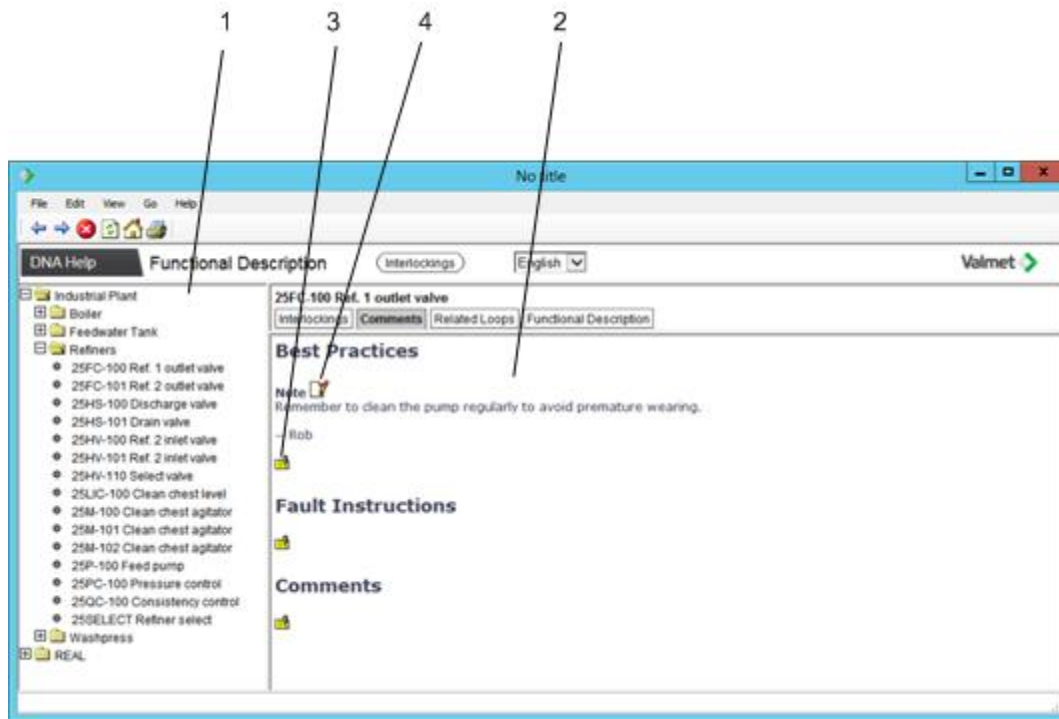
Comments are displayed on a dedicated tab. Optionally, comments may have been left out completely.

When you want to examine the comments, proceed as follows:

- Click the Comments tab on the Functional Description or Interlocks tab of a loop. The contents of the browser window will be updated to display the loop-related comments.

Function of the Comments

Comment pages provide the operating and maintenance personnel with a forum in which they can read, add, or edit loop-related practices, fault instructions, or comments. Valuable empirical knowledge can thus be shared by all users alike.

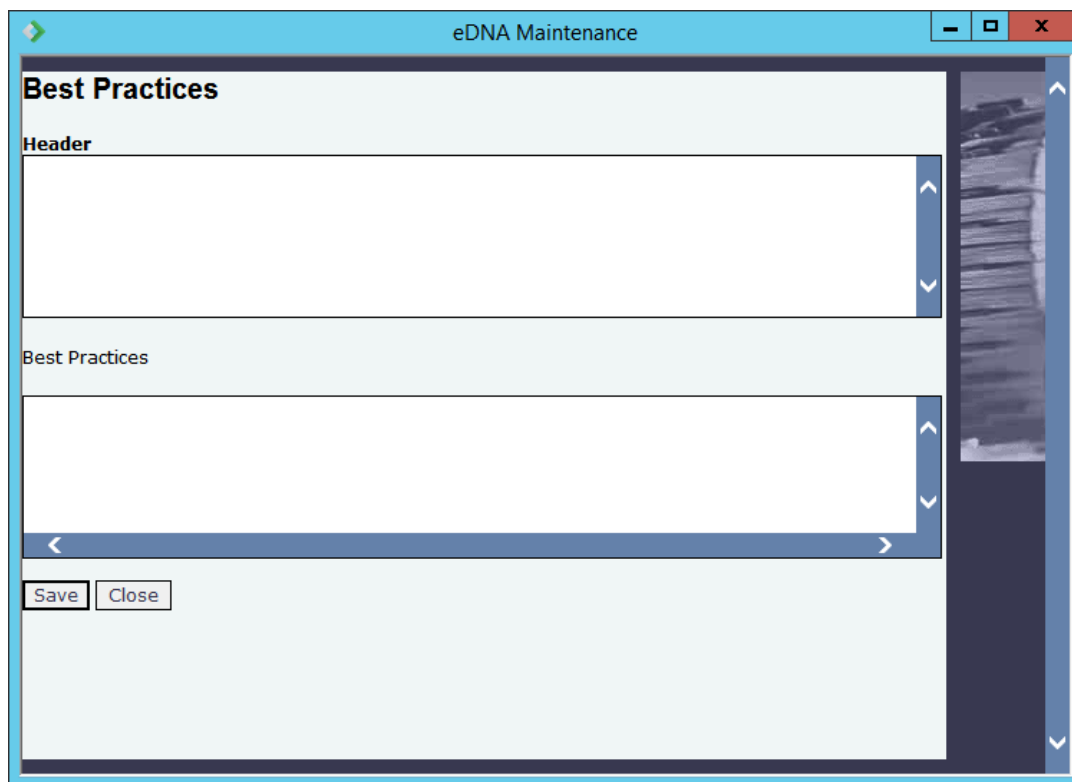


1. Loops of the process area
2. Comment and its given header
3. [Add Comment button](#)
4. [Edit Comment button](#)

How to Add a Comment

When you want to add a comment, proceed as follows:

1. On a comment page, click the Add Comment button  under the desired category. The Add Comment window will open on the display:



2. Click the Header field and enter the header for your comment.
3. Click the content field of the comment (in this case, Best Practices) and enter your comment.
4. Click the Save button to save your comment. The window will close and your message will appear on the Comments tab.

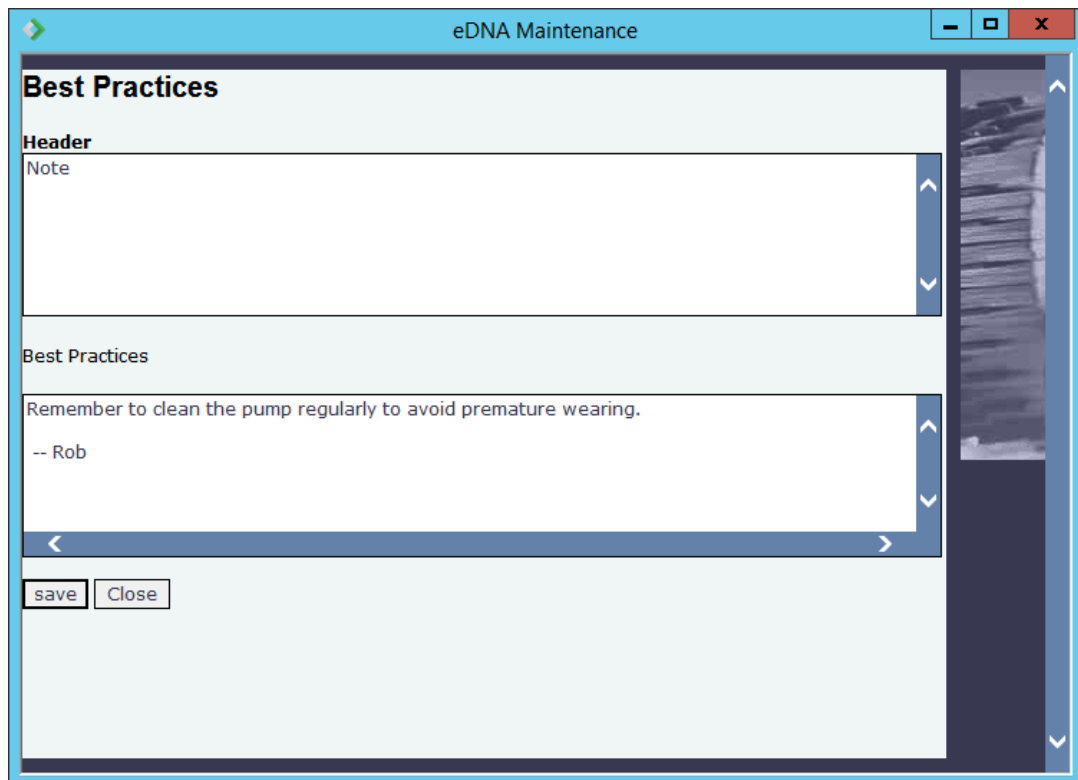
If required, you can abort the operation by clicking the Close button.

How to Edit a Comment

Any comment entered on a comment page can be edited later.

When you want to edit a comment, proceed as follows:

1. On a comment page, click the Edit Comment button  next to the header of the comment you want to edit. The pre-filled Add Comment window will open on the display:



The screenshot shows a software window titled "eDNA Maintenance". Inside, there's a section titled "Best Practices". Under this section, there's a "Header" area with a "Note" field. Below the "Note" field, there's another "Best Practices" section containing a comment: "Remember to clean the pump regularly to avoid premature wearing." followed by "-- Rob". At the bottom of the window, there are two buttons: "save" and "Close". The window has a standard Windows-style title bar with minimize, maximize, and close buttons.

2. Click the target field and edit its content to your liking.
3. Click the Save button to save your changes. The window will close and the changes you made will be updated on the Comments tab.

If required, you can abort the operation by clicking the Close button.